eProjects WO#: 1783178

FY26 MAINTENANCE DREDGING AT:
RESERVE BASIN
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA

DESIGNED BY:

NAVFAC MID-ATLANTIC - PDC CORE
Hydrographic Team (PDC45C)
9080 VIRGINIA AVE
NAVAL STATION NORFOLK, BLDG CEP-188
NORFOLK, VIRGINIA 23511

Date: November 5, 2025

SPECIFICATION PREPARED BY: James S. Georgo, P.E.

Project Manager:	Steven A. McKechnie,P.E.
Chief Engineer:	Jonathan W. Jarrett, P.E.

SPECIFICATIONS APPROVED BY:

Lead Discipline Team Manager:

James S. Georgo, P.E.

Digitally signed by GEORGO.JAMES.S.1392763089
Date: 2025.11.05 11:19:18 -05'00'

James S. Georgo, P.E. Date

For Commander, NAVFAC MID-ATLANTIC:
Design Production Director
Capital Improvements Core

DONAHUE.JAMES.E.1290785860

Digitally signed by DONAHUE.JAMES.E.1290785860
Date: 2025.11.05 14:58:39 -05'00'

James E. Donahue, R.A. Date

PROJECT TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DIVISION 01 - GENERAL REQUIREMENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

-- End of Project Table of Contents --

DOCUMENT 00 01 15

LIST OF DRAWINGS
02/24

PART 1 GENERAL

1.1 SUMMARY

This section lists the drawings for the project pursuant to contract clause "DFARS 252.236-7001, Contract Drawings and Specifications."

1.2 CONTRACT DRAWINGS

Contract drawings are as follows:

SHEET NO.	NAVFAC#	SHEET TITLE
V-001	12902788	TITLE SHEET
V-002	12902789	DETAIL SHEET
V-101	12902790	RESERVE BASIN - SOUTH

-- End of Document --

SECTION 01 11 00

SUMMARY OF WORK
02/24, CHG 1: 05/25

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Dredging Plan; G

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

The work includes mechanical dredging of mud and silt material, using a Sealed Environmental Clamshell Bucket, along with incidental related work in accordance with the drawings, specifications and all permits and other regulatory agency terms and conditions at Pier 4 West, Philadelphia Navy Yard, Philadelphia, PA. IF HARD MATERIAL IS ENCOUNTERED OR DEBRIS REMOVAL IS REQUIRED THEN AND STANDARD CLAMSHELL BUCKET MAY BE UTILIZED. ONCE HARD MATERIAL AND/OR DEBRIS HAS BEEN REMOVED THE CONTRACTOR SHALL SWITCH BACK TO USING THE SEALED ENVIRONMENTAL CLAMSHELL BUCKET.

The following dredge material disposal sites shall be utilized as described below:

A) Dredge Areas RB-A, RB-C AND RB-D: Dredged materials shall be transported by bottom dumping barge or scow and disposed within the Weeks Marine Whites Rehandling Basin, located in Logan Township, Gloucester County, New Jersey OR transported by barge and hydraulically unloaded at the Waste Management Corporation Biles Island disposal facility located in Morrisville, Bucks County, Pennsylvania.

B) Dredge Area RB-A 7/8: Dredged materials shall be transported by barge and hydraulically unloaded at the Waste Management Corporation Biles Island disposal facility located in Morrisville, Bucks County, Pennsylvania.

26th Street Bridge Lifts:
Contractor shall coordinate all 26th street bridge lifts in advance, with the owner (NAVY YARD PIDC), as necessary for dredging in Area RB-D and to facilitate contractor vessel and barge traffic. See Appendix "C" for PIDC Bridge Lift Policy. Contact JoeGoldschmidt, Joe.Goldschmidt@cbre.com

1.2.2 Location

The work is located within the Philadelphia Navy Yard Reserve Basin, generally shown on the plans. Order of work shall be closely coordinated with the government as required to schedule ship movements necessary for contractor to gain access to dredging areas.

FINAL DREDGING LIMITS SHALL BE DETERMINED BY THE GOVERNMENT USING THE PRE-DREDGING SURVEY -30' CONTOUR.

1.3 LOCATION OF UNDERGROUND UTILITIES

Contact local utility companies prior to commencing with dredging work to mark and/or verify ALL existing submarine utilities including, but not limited to electric, gas, telephone, data, etc...,within sufficient time required. Any existing utilities location and depth shown on the contract drawings are approximate and shall be verified prior to dredging. Contractor shall hire a qualified third party surveying firm to conduct, at a minimum, sub-bottom profiling and magnetometer survey of the dredging areas in order to verify all existing utilities location and depth. Contractor shall coordinate with the local utility companies as necessary for utility location verification. All coordination with utility companies and location work shall be borne by the contractor.
Contact: Jeremy Rigau

Existing 15 KV Power Line (shown on plans), originally installed laying on the seafloor, shall be field located using contractor's divers in order to verify current location and depth below current seafloor surface. Power line horizontal location shall be marked by attaching buoys @ max. 50' spacing. Survey buoy locations using RTK GPS and furnish the government X and Y Coordinates in PA State Plane Coordinate system. Existing cable depth at each buoy, (below existing seafloor to top of cable) shall be measured by contractor's divers. Depth recordings shall be to measured to at least the nearest 0.50'. CONTRACTOR SHALL SECURE AND PROTECT POWER LINE DURING DREDGING. MAX. DREDGE DEPTH @ ELEC. CABLE SHALL BE FIELD DETERMINED BASED ON AVAILABLE SLACK IN CABLE. CONTRACTOR SHALL COORDINATE WITH GOVERNMENT AND DOMINION ENERGY PRIOR TO COMMENCING WITH DREDGING

Utility Company Contacts:

Dominion Energy:
Manager Base Privatization
Philadelphia Navy Yard
Dominion Energy
4747 South Broad Street
Building B-101, Suite LL40
Philadelphia, PA. 19112

Cell: 757-408-5948
Office:804-775-5007
email: JEREMY.D.RIGAU@dominionenergy.com

Philadelphia Gas Works
Jubayer Ahmed
Engineer II & III Distribution
Philadelphia Gas Works | 5000 Summerdale Ave.

Philadelphia, PA 19124

Phone: (267) 835-1132 | Fax: (215) 684-6984
email: Jubayer.Ahmed@pgworks.com

1.4 SALVAGE MATERIAL AND EQUIPMENT

Items designated by the Contracting Officer to be salvaged remain the property of the Government. Segregate, itemize, deliver and off-load the salvaged property at the [Government designated] storage area.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 14 00

WORK RESTRICTIONS
11/22, CHG 5: 05/25

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel,Dredge Plans; G

Dredge plans shall show progressive positioning of barges, scows, tender boats, etc

SD-07 Certificates

1.2 SPECIAL SCHEDULING REQUIREMENTS

a. The entire reserve basin shall remain in operation during the entire dredgingproject including maintaining navigation in the access channel. The Contractor shall conduct his operations so as to cause the least possible interference with normal operations of the activity.

b. Contractor shall coordinate with the Contracting Officer prior to start of dredging to determine order of work. The Government reserves the right to change the order of work at any time. Order of work shall be closely coordinated with the government as required to schedule ship movements necessary for contractor to gain access to dredging areas.

1.3 CONTRACTOR ACCESS AND USE OF PREMISES

1.3.1 Activity Regulations

Ensure that Contractor personnel employed on the Activity become familiar with and obey Activity regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE) in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Ensure all Contractor equipment, including delivery vehicles, are clearly identified with their company name.

1.3.1.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.3.1.2 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer will identify designated smoking areas.

1.3.2 Working Hours

Normal duty hours for work shall be 24 hours a day Sunday through Saturday.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES
11/20, CHG 5: 02/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 1110-1-8	(2021) Engineering and Design -- Construction Equipment Ownership and Operating Expense Schedule

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

[Schedule of Prices]; G

1.3 [SCHEDULE OF PRICES]

1.3.1 Data Required

Within 15 calendar days of notice of award, prepare and deliver to the Contracting Officer a Schedule of Prices (construction contract) as directed by the Contracting Officer. Provide a detailed breakdown of the contract price, giving quantities for each of the various kinds of work, unit prices, and extended prices therefore in accordance with the bid schedule. Costs shall be summarized and totals provided for each construction category.

1.3.2 Payment Schedule Instructions

Payments will not be made until the [Schedule of Prices] has been submitted to and accepted by the Contracting Officer.

1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause DFARS 252.236-7000 Modification Proposals-Price Breakdown, and where actual ownership and operating costs of construction equipment cannot be determined from Contractor accounting records, base equipment use rates upon the applicable provisions of the

EP 1110-1-8.

1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

1.5.1 Content of Invoice

Requests for payment will be processed in accordance with the Contract Clause FAR 52.232-27 Prompt Payment for Construction Contracts and FAR 52.232-5 Payments Under Fixed-Price Construction Contracts. Invoices not completed in accordance with contract requirements will be returned to the Contractor for correction of the deficiencies. Include the documents listed below in the requests for payment.

a. The Contractor's invoice, on NAVFAC Form 7300/30 furnished by the Government, showing in summary form, the basis for arriving at the amount of the invoice. Form 7300/30 must include certification by Quality Control (QC) Manager as required by the Contract.

b. The [Estimate for Voucher/ Contract Performance Statement on NAVFAC Form 4330/54 furnished by the Government. Use NAVFAC Form 4330, unless otherwise directed by the Contracting Officer, on NAVFAC Contracts when a Monthly Estimate for Voucher is required.][Earned Value Report from the cost-loaded NAS.]

c. Contractor's Monthly Estimate for Voucher and Contractors Certification (NAVFAC Form 4330) with Subcontractor and supplier payment certification. Other documents, including but not limited to, that need to be received prior to processing payment include the following submittals as required. These items are still required monthly even when a pay voucher is not submitted.

d. Monthly Work-hour report.

e. Updated Construction Progress Schedule and tabular reports required by the contract.

f. Contractor Safety Self Evaluation Checklist.

g. Updated submittal register.

h. Solid Waste Disposal Report.

i. Certified payrolls.

j. Updated testing logs.

k. Other supporting documents as requested.

1.5.2 Submission of Invoices

Monthly invoices and supporting forms for work performed through the anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

1.6 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests by the Contractor which comply with the requirements of this section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

1.6.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this Contract will, at the discretion of the Contracting Officer, be subject to reductions and suspensions permitted under the FAR and agency regulations including the following in accordance with FAR 32.103 Progress Payments Under Construction Contracts:

a. Reasonable deductions due to defects in material or workmanship;

b. Claims which the Government may have against the Contractor under or in connection with this Contract;

c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and

d. Failure to maintain accurate "as-built" or record drawings in accordance with FAR 52.236.21 and the contract.

e. Failure to submit accurate monthly schedule updates in accordance with the contract.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS
11/20, CHG 5: 02/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability, during the entire period of performance under this contract. Provide other insurance coverage as required by [State]law.

1.3 SUPERVISION

1.3.1 Superintendent Qualifications

Provide project superintendent with a minimum of [10][_____] years experience in construction with at least [5][_____] of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

For projects where the superintendent is permitted to also serve as the Quality Control (QC) Manager as established in Section01 45 00 QUALITY CONTROL, the superintendent must have qualifications in accordance with that section.

1.3.2 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the Contract, then that individual must also have fluent English communication skills.

1.3.3 Duties

The project superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Red

Zone meetings, partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.3.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time or excess costs or damages by the Contractor.

1.4 PRECONSTRUCTION MEETING

prior to commencing any work at the site, coordinate with the Contracting Officer a time and place to meet for the Preconstruction Meeting. The purpose of this meeting is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, base-access, outage requests, hot work permits, schedule requirements, quality control, schedule of prices or earned value report, shop drawings, submittals, cybersecurity, prosecution of the work, government acceptance, final inspections and contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing.

1.4.1 Attendees

Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and major subcontractors.

1.5 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start planning the turnover process at the Pre-Construction Conference meeting with a discussion of the Red Zone process and convene at regularly scheduled Red Zone Meetings beginning at approximately 75 percent of project completion. Include the following in the facility Turnover effort:

1.5.1 Red Zone Checklist

a. Contracting Officer's Technical Representative (COTR) will provide the Contractor a copy of the Red Zone Checklist template.

b. Prior to 75 percent completion, modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. Submit the modified Red Zone Checklist to the Contracting Officer. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary.

1.5.2 Meetings

a. Conduct regular Red Zone Meetings beginning at approximately 75 percent project completion, or three to six months prior to Beneficial Occupancy Date (BOD), whichever comes first.

b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.

c. Using the Red Zone Checklist as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.

d. During the Red Zone Meetings discuss with the COTR any upcoming activities that require Government involvement.

e. Maintain the Red Zone Checklist by documenting the actual completion dates as work is completed and update the Red Zone Checklist with revised planned completion dates as necessary to match progress. Distribute copies of the current Red Zone Checklist to attendees at each Red Zone Meeting.

1.6 MOBILIZATION

Mobilize to the jobsite within [30 calendar days of GOVERNMENT ISSUANCE OF NOTICE TO PROCEED. Mobilize is defined as having equipment AND having a physical presence of at least one person from the contractor's team on the jobsite.

]PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 31 23.13 20

ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM (eCMS)
08/23, CHG 1: 02/24

PART 1 GENERAL

1.1 CONTRACT ADMINISTRATION

Utilize the Naval Facilities Engineering Systems Command's (NAVFAC's) Electronic Construction and Facility Support Contract Management System (eCMS) for the transfer, sharing, and management of electronic technical submittals and documents. The web-based eCMS is the designated means of transferring technical documents between the Contractor and the Government. Paper media or email submission, including originals or copies, of the documents are not permitted unless identified within the contract.

All government contracting specialist/officer, legal, and command communications will remain the same.

1.2 USER PRIVILEGES

The Contractor's key staff may be provided access to eCMS. Contact the COR for eCMS account access. Project roles and system roles will be established to control each user's menu, application, and software privileges, including the ability to create, edit, or delete objects. Additional project roles may be assigned for workflow. The COR makes the final decision on roles for the project. User's ability to view, edit documents may be lowered at the discretion of the COR.

Only one eCMS user account is required regardless of the number of user's projects. Notify the COR within seven calendar days if a contractor user is no longer associated with company or project so they can remove them from any open record and inactivate them from the project.

1.2.1 eCMS Subcontractor Users

If the contractor's user is a subcontractor, the subcontractor must be registered under the name of their company and email. For example it is common for contractors to contract Quality Control Managers. The QC Manager's account should be under their company's name and email reducing the number of eCMS accounts required.

1.2.2 Users with Multiple Roles

Users may have multiple roles associated with their account within eCMS. Roles are used in workflow. When a user is added to the project, they will be assigned the default role when the user was created. Contact the COR to change or add roles to the user for the project.

1.2.3 Loss of Privilege

Users may lose privilege to access eCMS at the discretion of the Contracting Officer. The eCMS is a collaborative system that allows flexibility of use and does not restrict all inappropriate user actions. User activities are logged into eCMS in visible and background data collection. Users found to use eCMS in an inappropriate action may have their eCMS access revoked. Examples include, but not limited to, fraudulent representations, sharing user accounts with others, and changing approved records without the consent of the COR. Depending on the severity of the infraction, the users can lose eCMS access for a period of time, permanently for the project, or lose eCMS access for any project. The contractor may appeal the suspension in writing to the contracting officer within 14 calendar days of notice. The appeal must identify the infraction, supporting information, and steps to ensure the infraction will not happen in the future.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contractor's Personnel; G

For Division 1 government approved Pre-Construction submittals, combine into a single Pre-Construction Submittal Package. Annotated with SD Type of SD-01. Pre-Construction submittal package approval date will be used as a KPI.

1.4 SYSTEM REQUIREMENTS AND CONNECTIVITY

1.4.1 General

NAVFAC eCMS requires a web-browser (platform-neutral) and Internet connection. For best results, recommend using browser in InPrivate/Incognito mode; Internet speeds greater than 40mbps when uploading files, computers with high RAM and Solid State Drives, "White List" eCMS website, Zip or Split files for better uploading. Non-NAVFAC Users are not to use VPN when using eCMS per NAVFAC IT.

The use of eCMS is required by the Contractor and all associated costs and time necessary to utilize eCMS will be borne by the Contractor with no allowance for time extensions and at no additional cost to the government.

1.4.2 Contractor Personnel List

Within 20 calendar days of contract award, provide to the Contracting Officer a list of Contractor's personnel who will have the responsibility for the transfer, sharing and management of electronic submittals, RFIs, daily reports, and other files and will require access to the eCMS. Project personnel roles which must be filled as applicable in the eCMS, include, at a minimum, the Contractor's Project Manager (KTR-PM), Superintendent (KTSUPT), Quality Control (QC) Manager (KTR-QC), Principal (KTR-PRIN), and Site Safety and Health Officer (KTR-SSHO). Notify the COR immediately of any personnel changes to the project. The Contracting Officer reserves the right to perform a security check on all potential users.

Internal design deliberations under a D-B contract are managed outside of eCMS. Besides the D-B DOR Representative, D-B design team members are neither required nor encouraged to create user accounts in eCMS. Users are highly recommended to take eCMS live or online training prior to use. See https://www.navfac.navy.mil/Business-Lines/Design-and-Construction/About-Us/Design-and-Construction-Documents/Electronic-Construction-Management-System-eCMS/ for eCMS useful information

Provide the following information:

Company Name

Name (First, Last)

Email Address

Project Role (CQM, SSHO, Superintendent, CM, PM, Principal)

Existing or New eCMS User

1.5 SECURITY CLASSIFICATION

In accordance with Department of Navy guidance, all military construction contract data are unclassified, unless specified otherwise by a properly designated Original Classification Authority (OCA) and in accordance with an established Security Classification Guide (SCG). Refer to the project's OCA when questions arise about the proper classification of information.

In conformance with the Freedom of Information Act (FOIA), DoD INSTRUCTION 5200.48 CONTROLLED UNCLASSIFIED INFORMATION (CUI), and DoD requirements, any unclassified project documentation uploaded into the eCMS must be designated either "U - UNCLASSIFIED" (U) or "CUI - CONTROLLED UNCLASSIFIED INFORMATION" (CUI).NAVFAC eCMS must only be used for the transaction of unclassified information associated with construction projects. Controlled Unclassified Identification (CUI) documents may be loaded into eCMS with the appropriate markings.

1.5.1 Markings on CUI Documents

Contractor's proprietary information, or documents determined by the originator in accordance with CUI guidance, should be marked CUI. Proprietary information not marked CUI can be released under the Freedom of Information Act (FOIA). Apply the appropriate markings before any document is uploaded into eCMS. Markings are not required on Unclassified U documents.

1.6 eCMS UTILIZATION

Establish, maintain, and update data and documentation in the eCMS throughout the duration of the contract. Utilize eCMS to transfer all submittals, RFIs, daily reports, and other files required by contract to be forwarded to the government.

Full eCMS use is required. All Submittals/Information to use eCMS Modules including, but not limited to, RFIs, Daily Reports, Meeting Minutes, Communications, Issues, Punch Lists, Checklists, and Flysheets, unless otherwise directed by the COR or Contracting Officer.

The eCMS as an Electronic File Repository is authorized for projects under $1M. Upload all files using the appropriate eCMS module or as directed by the COR. Bulk uploads of RFIs, Submittals, Daily Reports, Issues, Meeting Minutes or Communications records into eCMS modules are allowed. For RFIs, group files by specification number, RFP Part number, or sheet number. For

Submittals, group files by specification number. For Daily Reports, group by date range(s), Issues by date range(s), Meeting Minutes by date range(s), and Communications by Type.

1.6.1 Restricted Information

Personally Identifiable Information (PII) transmittal such as credit card, driver's license, passport, social security, and payroll number are not permitted in eCMS. Name, address, and email are permitted. Pre-negotiation information such as cost estimates that require formal negotiations are not allowed. For example, proposed changes over the SAP level of $250k require formal negotiations. Cost estimates for LEAN, ULTRA LEAN, and Design Changes under the SAP level are at the discretion of the COR's or Contract Specialist/Officer's direction. The eCMS must only be used for the transaction of unclassified information associated with construction projects. Controlled Unclassified Identification (CUI) documents may be loaded into eCMS with the appropriate markings. Uploading of files directly into the Documents folder is not allowed. All documents must be uploaded using an eCMS module.

1.6.2 Naming Convention for Files

Titles of files uploaded are to be descriptive of the purpose and content of the file. For example RFI_ROOF_Leak.doc or for submittals, SUB_LIGHT_FIXTURE.pdf. Titles of file to be uploaded must only contain uppercase letters, lowercase letters, numbers, hyphens (-), underscores (_) and periods (.). Use of any other characters is not allowed and may create an error. When practicable, adding the record number to the title is desired. For example RFI_XYZ12345_ROOF_Leak.doc. Uploading files with the same title will create a new revision in eCMS. Original revision is Rev 0, the first revision is Rev 1. Uploaded files are to use the default file location regardless of the module used unless directed by the COR.

Table 1 also identifies which eCMS application is to be used in the transmittal of data (these are subject to change based on the latest software configuration).

Table 1 - Project Documentation Types

SUBJECT/NAME	**REMARKS**	**eCMS APPLICATION**
As-Built Drawings	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Building Information Modeling (BIM)	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Construction Permits	Refer to rules of the issuing activity, state or jurisdiction	Submittals

SUBJECT/NAME	REMARKS	eCMS APPLICATION
Construction Schedules (Activities and Milestones)		Submittals
Construction Schedules		Submittals
Construction Schedules (3-Week Look ahead)	Import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline	Meeting Minutes
DD 1354 Transfer of Real Property	When applicable, required for final billing.	Submittals
Daily Production Reports	Provide weather conditions, crew size, man-hours, equipment, and materials information	Daily Report
Daily Quality Control (QC) Reports	Provide QC Phase, Definable Features of Work Identify visitors	Daily Report
Designs and Specifications	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Environmental Notice of Violation (NOV), Corrective Action Plan	Refer to rules of the issuing activity, state or jurisdiction	Submittals
Environmental Protection Plan (EPP)		Submittals
Invoice (Supporting Documentation)	Applies to supporting documentation only. Invoices are submitted in Wide-Area Workflow (WAWF)	Submittals
Jobsite Documentation, Bulletin Board, Labor Laws, SDS	Redact any PII information when loaded into eCMS	Submittals
Meeting Minutes		Meeting Minutes

SUBJECT/NAME	REMARKS	eCMS APPLICATION
Modification Documents	Provide final modification documents for the project. Upload into Modifications RFPs folder	Communications
Operations & Maintenance Support Information (OMSI/eOMSI), Facility Data Worksheet	1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager 2. Design reviews will be performed in existing "Dr Checks"	Submittals
Photographs	Subject to base/installation restrictions	Submittals
QCM Initial Phase Checklists		Meeting Minutes or Checklists
QCM Preparatory Phase Checklists		Meeting Minutes or Checklists
Quality Control Plans		Submittals
QC Certifications		Submittals
QC Punch List		Punch Lists
Red-Zone Checklist		Punch List or Checklists
Rework Items List		Punch Lists
Request for Information (RFI) Post-Award		RFIs
Safety Plan		Submittals
Safety - Activity Hazard Analyses (AHA)		Submittals
Safety - Mishap Reports		Daily Report

SUBJECT/NAME	**REMARKS**	**eCMS APPLICATION**
Shop Drawings	Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager	Submittals
Storm Water Pollution Prevention (Notice of Intent - Notice of Termination)	Refer to rules of the issuing activity, state or jurisdiction	Submittals
Submittals and Submittal Register		Submittals
Testing Plans, Logs, and Reports		Submittals
Training/Reference Materials		Submittals
Training Records (Personnel)	Redact any PII information if storing in eCMS	Submittals
Utility Outage/Tie-In Request/Approval		Submittals
Warranties/BOD Letter		Submittals
Quality Assurance Reports		Checklists (Government initiated)
Non-Compliance Notices		Non-Compliance Notices (Government initiated)
Other Government-prepared documents		GOV ONLY
Letters to government contracting, claims, REAs, and other contracting officer communications	eCMS is not the primary tool to use in contracting officer communications. eCMS can only store documents or letters after the submission to the contracting officer is made.	Communications

SUBJECT/NAME	REMARKS	eCMS APPLICATION
All Othere Documents	Refer to FOIA guidelines and contact the FOIA official to determine whether exemptions exist	As applicable

1.6.3 RFIs Module

Create contractor RFIs using eCMS RFIs module. The contractor must confirm the numbering convention with the COR if different than eCMS default.

If the government (GOV) Response has "No" Cost or Schedule Impact, this reply is given with the expressed understanding that it does not constitute a basis for any change in the amount or time of subject contract. Information provided in this response does not authorize work not currently included in the contract. If GOV Response is "Yes" or "Potentially" then this response may require a change to the contract. If the contractor disagrees with either the government's No Cost determination or No Schedule impact determination, or both, the contractor has 14 calendar days to notify the COR and Contracting Officer in writing.

1.6.4 Submittals Module

Create contractor submittals using eCMS Submittals module. The contractor must confirm the numbering convention with the COR if different than eCMS default.

1.6.5 Submittal Packages Module

Create submittal packages using the eCMS Submittal Packages module in lieu of or in addition to Related Objects. Submittal Packages track completion of the packaged submittals and is used in NAVFAC HQ's KPIs.

1.6.6 Communications Module

Create communications using the eCMS Communications module. The Communications module is used to create or document communications that are not a part of other eCMS modules. Use of Communications module will memorialize information into an eCMS record file. The following are Types of Communications:

Email

Memo to File

Face to Face

Telephone

Web Collaboration

Photos

Other Documents

Other

Unless directed by the COR, upload documents or files that do not have a corresponding eCMS module. Choose "Photos" Type for Photos and "Other Documents" for all other documents.

1.6.7 Issues Module

Create or respond to issues using the eCMS Issues module. Respond to CPARS issues using the Issues module.

1.6.8 Meeting Minutes Module

Create or respond to Meeting Minutes using the eCMS Meetings module.

Document required contractual meetings. Dates of meetings are used in NAVFAC KPIs. Minimum meetings in eCMS include the following:

Post Award Kickoff (PAK)

Pre-construction (Pre Con)

Initial and Preparatory Three Phases of Control

Quality Control (QC)

1.6.9 Potential Change Items Module

Not used.

1.6.10 Daily Report Module

Create Daily Reports using the eCMS Daily Report Module. The contractor must confirm the numbering convention with the COR if different than eCMS default.

1.6.11 Punchlists Testing Logs (Legacy)

Punchlist Testing Logs is a legacy program that is being replaced by the Punch Lists Module. This module is to be used for reference of past projects. Use the Punch Lists Module for all future work.

1.6.12 Punch Lists Module

The eCMS Punch Lists module is useful more than just for Punchlists. The module includes the capability of batch editing, create items from Optical Character Recognition (OCR) plans, assign tasks and track completion of individual items.

Create the following using the Punch Lists module:

Rework Items List

DFOW List

Punch-Out Inspection

Pre-Final Punchlist Inspection

Final Punchlist Inspection

Testing Logs

1.6.13 FWD UltraLean COAR RFP Module

Not used.

1.6.14 Non-Compliance Notices (NCN) Module

Respond to Non-Compliance Notices listed in the Non-Compliance Notices module.

1.6.15 Checklists

Use Checklist listed in the contractor's eCMS menu and as directed by the COR. Checklists capture data and is used in dashboards and KPIs.

1.6.15.1 Partnering Team Health Survey Checklist

Contractor must use the eCMS checklist to document the partnering team health survey. Partnering Team Health Survey is in accordance with the Partnering Specification of this contract.

1.6.16 Flysheets

Use Flysheets listed in the contractor's eCMS menu, if available, and as directed by the COR. Flysheets allow the contractor to print out information from other systems and upload into eCMS. The eCMS will use OCR to capture the information as data. Flysheets capture data and used in dashboards and KPIs.

1.6.17 eCMS Outage

In the case where eCMS is unavailable for 8 hours or more, paper or email may be used in the interim to maintain project schedule.

Once the system is operational, all final records are required to be recreated using the appropriate module. Subject/title of the record to include the type of record i.e., RFI/Submittal/Daily Report/Communication/Other, the identification number(s), and the statement "Processed Outside of eCMS". Example, "RFI 001 Processed Outside of eCMS".

1.6.18 User Account Activity

NAVFAC eCMS captures user data and activities that are directly related to the user's account. The user agrees through the use of eCMS, their account activities will be captured and can be displayed on eCMS printed reports.

1.7 QUALITY ASSURANCE

Requested Government response dates on Submittals must be in accordance with the terms and conditions of the Contract unless previously agreed by the COR. Requesting response dates earlier than the required review and response time, without concurrence by the Government COR, may be cause for rejection.

Incomplete submittals will be rejected without further review and must be resubmitted. Required Government response dates for resubmittals must reflect the date of resubmittal, not the original submittal date.

All submittals and associated attachments must be transmitted to the Government via the COR. Transmittals are no longer required when using eCMS since approval status is tracked on the submittal. Transmittal forms can be attached to submittals if approved by the COR. Submittals requiring government approval are "Transmitted For" "Approval". Submittals requiring contractor approved submittals, including those designated for Contractor Quality Control approval or Information Only, are "*Transmitted For" "Information Only" in the Submittal Module. Provide and sign the QC certification or approving statement on the attachment per submittal specification section. When Submittal Packages are required, use eCMS Submittal Packages after creating individual submittals. Importing Submittals from the Submittal Register is optional. Contact the COR for the data conversion requirements.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 32 16.00 20

SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES
08/18, CHG 1: 08/20

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Baseline Construction Schedule; G

SD-07 Certificates

Monthly Updates

1.2 PRE-CONSTRUCTION SCHEDULE REQUIREMENT

Prior to the start of work, prepare and submit to the Contracting Officer a Baseline Construction Schedule in the form of a Bar Chart Schedule in accordance with the terms in Contract Clause FAR 52.236-15 Schedules for Construction Contracts, except as modified in this contract. The approval of a Baseline Construction Schedule is a condition precedent to:

a. The Contractor starting demolition work or construction stage(s) of the contract.

b. Processing Contractor's invoice(s) for construction activities/items of work.

c. Review of any schedule updates.

Submittal of the Baseline Construction Schedule, and subsequent schedule updates, is understood to be the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents, represents the Contractor's plan on how the work will be accomplished, and accurately reflects the work that has been accomplished and how it was sequenced (as-built logic).

1.3 SCHEDULE FORMAT

1.3.1 Bar Chart Schedule

The Bar Chart must, as a minimum, show work activities, submittals, Government review periods, material/equipment delivery, utility outages, on-site construction, inspection, testing, and closeout activities. The Bar Chart must be time scaled and generated using an electronic spreadsheet program.

1.3.2 Schedule Submittals and Procedures

Submit Schedules and updates in hard copy and on electronic media that is acceptable to the Contracting Officer. Submit an electronic back-up of the project schedule in an import format compatible with the Government's scheduling program.

1.4 SCHEDULE MONTHLY UPDATES

Update the Construction Schedule at monthly intervals or when the schedule has been revised. Keep the updated schedule current, reflecting actual activity progress and plan for completing the remaining work. Submit copies of purchase orders and confirmation of delivery dates as directed by the Contracting Officer.

a. Narrative Report: Identify and justify the following:

(1) Progress made in each area of the project;

(2) Longest Path: Include printed copy on 11 by 17 inch paper, landscape setting;

(3) Date/time constraint(s), other than those required by the contract;

(4) Listing of changes made between the previous schedule and current updated schedule including: added or removed activities, original and remaining durations for activities that have not started, logic (sequence, constraint, lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading.

(5) Any decrease in previously reported activity Earned Amount;

(6) Pending items and status thereof, including permits, changes orders, and time extensions;

(7) Status of Contract Completion Date and interim milestones;

(8) Current and anticipated delays (describe cause of delay and corrective actions(s) and mitigation measures to minimize);

(9) Description of current and future schedule problem areas.

For each entry in the narrative report, cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

1.5 3-WEEK LOOK AHEAD SCHEDULE

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Construction Schedule. Key the work plans to activity numbers when a NAS is required and update each week to show the planned work for the current and following two-week period. Additionally, include upcoming outages, closures, preparatory meetings, and initial meetings. Identify critical path activities on the Three-Week Look Ahead Schedule. The detail work plans are to be bar chart type schedules, maintained separately from the Construction Schedule on an electronic spreadsheet program and printed on 8-1/2 by 11 inch sheets as directed by the Contracting Officer. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work. Deliver three hard copies and one electronic file of the 3-Week Look Ahead Schedule to the Contracting

Officer no later than 8 a.m. each Monday, and review during the weekly CQC Coordination or Production Meeting.

1.6 CORRESPONDENCE AND TEST REPORTS:

Correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) must reference Schedule Activities that are being addressed. Test reports (e.g., concrete, soil compaction, weld, pressure) must reference Schedule Activities that are being addressed.

1.7 ADDITIONAL SCHEDULING REQUIREMENTS

Any references to additional scheduling requirements, including systems to be inspected, tested and commissioned, that are located throughout the remainder of the Contract Documents, are subject to all requirements of this section.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site.

Government approved Division 01 preconstruction submittals that are required prior to or commencing with the start of work must be submitted within 30 calendar days of contract award unless specified elsewhere in the specifications.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Construction Schedule; G

)

Submittal Register

Schedule Of Prices

Accident Prevention Plan

Dredging Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-11 Closeout Submittals

Final Post-Dredge Survey as specified in Spec Section 35 20 23.

1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.1.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.3 SUBMITTAL CLASSIFICATION

1.3.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.3.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.4 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL

As soon as practicable after award of contract, and before procurement or fabrication, forward to the NAVFAC PME Construction Office, 4921 South Broad Street, Philadelphia, PA 19112

[1.5 PREPARATION

1.5.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels to the office of the approving authority using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

1.5.2 Identifying Submittals

The Contractor's [Quality Control Manager] must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

a. Project title and location

b. Construction contract number

c. Dates of the drawings and revisions

d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.

e. Section number of the specification by which submittal is required

f. Submittal description (SD) number of each component of submittal

g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission

h. Product identification and location in project.

1.5.3 Submittal Format

1.5.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3.2 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.3.2.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.3.2.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.2.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will [not] be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.3.3 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.4 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit [two][three] sets of administrative submittals.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Attachment A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Column (f): Lists the approving authority for each submittal. Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

1.8.5.1 Government Review Action Codes

"A" - "Approved as submitted"

"AN" - "Approved as noted"

"RR" - "Disapproved as submitted"; "Completed"

"NR" - "Not Reviewed"

"RA" - "Receipt Acknowledged"

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission [of a variation submittal] will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a [transmittal letter][variation submittal]. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

[When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.]

[The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk.]"

Specifically point out variations from contract requirements in a [transmittal letter][variation submittal]. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of [14] [_____] working days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. [Allow an additional [_____] working days for review and approval of submittals for [food service equipment] [and] [refrigeration and HVAC control systems]].

a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within [_____] working days after the date of submission.

d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least [15] [_____] working days for submittals for QC manager approval and [20] [_____] working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.

e. For submittals requiring review by a Government fire protection engineer, allow a review period, beginning when the Government receives the submittal from the QC organization, of [30][_____] working days for return of the submittal to the Contractor.

1.10.1 Reviewing, Certifying, and Approving Authority

The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification sections, a notation "G" following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government Approving authority

1.10.2 Constraints

Conform to provisions of this section, unless explicitly stated otherwise for submittals listed or specified in this contract.

Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.

When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.

Approval of a separate material, product, or component does not imply approval of the assembly in which the item functions.

1.10.3 QC Organization Responsibilities

a. Review submittals for conformance with project design concepts and compliance with contract documents.

b. Process submittals based on the approving authority indicated in the submittal register.

(1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.

(2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.

c. Ensure that material is clearly legible.

d. Stamp each sheet of each submittal with a QC certifying statement or an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.

(1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to the Contracting Officer with the following certifying statement:

"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number [_____] is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.

Certified by Submittal Reviewer _____________________, Date _______
(Signature when applicable)

Certified by QC Manager _____________________________, Date ______"
(Signature)

(2) When approving authority is the QC manager, the QC manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."

"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number [_____] is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is approved for use.

Certified by Submittal Reviewer ______________________, Date ______
(Signature when applicable)

Approved by QC Manager _______________________________, Date _____"
(Signature)

e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.

f. Update the submittal register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.

g. Retain a copy of approved submittals and approved samples at the project site.

h. For "S" submittals, provide a copy of the approved submittal to the Government Approving authority.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

a. Note the date on which the submittal was received from the QC manager.

b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.

c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.

b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.

c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.

d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained with in each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
05/24, CHG 1: 05/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2020) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1	(2014; R 2019) American National Standard for Industrial Head Protection

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards

29 CFR 1919	Gear Certification
29 CFR 1926	Safety and Health Regulations for Construction

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Dive Operations Plan; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Contractor Safety Self-Evaluation Checklist

License Certificates

Portable Gauge Operations Planning Worksheet; G

Third Party Certification of Floating Cranes and Barge-Mounted Mobile Cranes

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth day of each month for the previous month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment.

1.6 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the preconstruction conference. Complete the checklist monthly and submit with each request for payment voucher. This submission is required monthly even when a payment voucher is not requested. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Contractor Safety Self-Evaluation checklist can be found on the Whole Building Design Guide website at www.wbdg.org/dod/ufgs/ufgs-01-35-26

1.7 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and the following[federal, state, and local][host nation] laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7.1 Subcontractor Safety Requirements

For this Contract, neither Contractor nor any subcontractor may enter into Contract with any subcontractor that fails to meet the following requirements. The term subcontractor in this and the following paragraphs means any entity holding a Contract with the Contractor or with a subcontractor at any tier.

1.7.1.1 Experience Modification Rate (EMR)

Subcontractors on this Contract must have an effective EMR less than or equal to 1.10, as computed by the National Council on Compensation Insurance (NCCI) or if not available, as computed by the state agency's rating bureau in the state where the subcontractor is registered, when entering into a subcontract agreement with the Prime Contractor or a subcontractor at any tier. The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable EMR range cannot be achieved. Relaxation of the EMR range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain the certified EMR ratings for all subcontractors on the project and make them available to the Government at the Government's request.

1.7.1.2 OSHA Days Away from Work, Restricted Duty, or Job Transfer (DART) Rate

Subcontractors on this Contract must have a DART rate, calculated from the most recent, complete calendar year, less than or equal to 3.4 when entering into a subcontract agreement with the Prime Contractor or a subcontractor at any tier. The OSHA Dart Rate is calculated using the following formula:

(N/EH) x 200,000

Where:

N = number of injuries and illnesses with days away, restricted work, or job transfer

EH = total hours worked by all employees during most recent, complete calendar year

200,000 = base for 100 full-time equivalent workers (working 40-hours per week, 50 weeks per year)

The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable OSHA Dart rate range cannot be achieved for a particular subcontractor. Relaxation of the OSHA DART rate range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain self-certified OSHA DART rates for all subcontractors on the project and make them available to the Government at the Government's request.

1.8 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.8.1 Site Safety and Health Officer (SSHO)

1.8.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract.

1.8.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.8.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.8.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

(1) Are designated on an ENG Form 6282.

(2) Meet minimum training and experience requirements identified in EM 385-1-1.

(3) Execute roles and responsibilities identified in EM 385-1-1.

1.8.3 Additional Requirements

The Level Two orThre SSHOmay also serve as the Quality Control Manager. The SSHO must not serve as the Superintendent.

1.8.4 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1 for this project.

1.8.5 Dredging Contract Site Safety and Health Officer(s)(SSHOs) Requirements

1.8.5.1 Dredging SSHO Personnel Requirements

a. Provide a minimum of one primary Level One SSHO assigned for the primary shift.

Note: Hopper Dredges with U.S. Coast Guard, credentialed crews the prime contractor may designate a Level Two SSHO in lieu of having a Level one SSHO onboard.

b. For a project involving multiple work shifts, provide a minimum of a Level One SSHO for each additional shift.

c. For individual dredging projects the prime contractor will designate additional Level Three SSHOs at locations where the primary Level One SSHO is not located (example: on dredge, tug, material placement site).

Examples of one dredging project site is reflected in each of the following:

(1) a mechanical dredge, tug(s) and scow(s), scow route, and material placement site; or

(2) a hydraulic pipeline dredge, attendant plant, and material placement site; or,

(3) a hopper dredge (include land-based material placement site - if applicable.)

d. Designated SSHOs must be present at the project site, located so that they have full mobility and reasonable access to all major work operations and must be available during their shift for immediate verbal consultation and notification.

e. Designated Level One and Two SSHOs must have direct report authority to a senior project (or corporate) management official.

f. Designated Level Three SSHOs must report potential safety and occupational health hazards, incidents, and concerns to the Level One or Two SSHO on shift.

g. Level One and Level Two SSHOs for dredging must have a minimum of 3 years experience in one of the following areas:

(1) Supervising/managing dredging activities.

(2) Supervising/managing marine construction activities.

(3) Supervising/managing land-based construction activities.

(4) Work managing safety programs or processes.

(5) Conducting hazard analyses and developing controls in activities or environments with similar hazards.

1.8.6 Qualified Trainer Requirements

Individuals qualified to instruct the 40-hour contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer as defined in the EM 385-1-1, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction; Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.

Instructors are required to:

a. Prepare class presentations that cover construction-related safety requirements.

b. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least 5 years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.

c. Update training course materials whenever an update of the EM 385-1-1 becomes available.

d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.

e. Request, review and incorporate student feedback into a continuous course improvement program.

1.8.7 Preconstruction Conference

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.

c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.9 ACCIDENT PREVENTION PLAN (APP)

1.9.1 Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the

Prime Contractor must provide an interpreter.

[Submit the APP to the Contracting Officer 15 calendar days prior to the date of the preconstruction conference for acceptance.]Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

1.9.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.9.3 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including the following:

1.9.3.1 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

]1.9.3.2 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

]1.9.3.3 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

1.9.3.4 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

]1.9.3.5 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 [DEMOLITION] [AND] [DECONSTRUCTION] and referenced sources. Include engineering survey as applicable.

1.10 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review at least [15][_____] working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.10.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.10.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.13 NOTIFICATIONS AND REPORTS

1.13.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table
Accident Reporting Required Timeline

Accident Type	**Notify KO or COR**	**Complete Final Accident Report on ENG 3394 and provide to KO or COR**
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over $600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.13.2 Accident Reports

a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Complete and submit an accident investigation report in ESAMS within 7 days for accidents as defined by EM 385-1-1. Complete the investigation report within 30 days. Accidents must include a written report submitted as an attachment in ESAMS using the following outline:

(1) Summary description to include:

(a) process

(b) findings

(c) outcomes

(2) Root Cause

(3) Direct Factors

(4) Indirect and Contributing Factors

(5) Corrective Actions

(6) Recommendations

All accidents are reportable, regardless of whether or not it is recordable.

a. Near Misses: For Navy Projects, complete the applicable documentation in NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Near miss reports are considered positive and proactive Contractor safety management actions.

b. Conduct an accident investigation for any load handling equipment accident (including rigging accidents) to establish the root cause(s) of the accident. Complete the Load Handling Equipment (LHE) Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

1.13.4.1 Certificate of Compliance

Provide a Certificate of Compliance for LHE utilized to perform work under this Contract in accordance with Section 16-7 of the EM 385-1-1. Post completed certificate on the crane.

[1.13.5 Third Party Certification of Floating Cranes and Barge-Mounted Mobile Cranes

Floating cranes and barge-mounted mobile cranes used to perform work under the terms of this Contract must be certified in accordance with 29 CFR 1919 by an OSHA accredited person prior to submitting the required Lift Plan. Include proof of certification with the initial Lift Plan submission.

]1.14 DIVE SAFETY REQUIREMENTS

Develop a Dive Operations Plan, AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation. Submit these documents to the District Dive Coordinator (DDC) via the Contracting Officer, for review and approval at least [15][_____] working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

1.15 [SEVERE STORM PLAN][SEVERE WEATHER PLAN FOR MARINE ACTIVITIES (SWPMA)]

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

a. Secure outside equipment and materials and place materials that could be damaged in protected areas.

b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.

c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

PART 3 EXECUTION

2.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

a. Head Protection that meets ANSI/ISEA Z89.1

b. Long Pants

c. Appropriate Safety Footwear

d. Appropriate Class Reflective Vests

2.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

2.1.2 Hazardous Material Use

Each hazardous material must receive approval from the Contracting Office or their designated representative prior to being brought onto the job site or prior to any other use in connection with this Contract. Allow a minimum of 10 working days for processing of the request for use of a hazardous material.

2.1.3 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

2.1.4 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR Part 1910.1000). If material(s) that can be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations, stop portions of work that would disturb the material and notify the Contracting Officer immediately. Within [14][_____] calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

2.2 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the[Installation representative][Public Utilities representative]. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

2.3 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1.

2.3.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

2.3.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

2.3.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

[]2.4 USE OF EXPLOSIVES

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

ELECTRICAL

Perform electrical work in accordance with EM 385-1-1.

2.5 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the[Contracting Officer][Commanding Officer]. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.

2.6 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local[and Host Nation] requirements applicable to where work is being performed.

2.7 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

2.8 Grounding

Ground electrical circuits, equipment and enclosures in accordance with [NFPA 70][and][IEEE C2] to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

2.9 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
05/24

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: https://www.assp.org/

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: https://www.astm.org/

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 and 501 Hoes Lane
Piscataway, NJ 08854-4141
Ph: 732-981-0060 or 800-701-4333
Fax: 732-981-9667
E-mail: onlinesupport@ieee.org
Internet: https://www.ieee.org/

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)
1101 Wilson Blvd, Suite 1425
Arlington, VA 22209-1762
Ph: 703-525-1695
Fax: 703-528-2148
Internet: https://safetyequipment.org/

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park

Quincy, MA 02169-7471
Ph: 800-344-3555
Fax: 800-593-6372
Internet: https://www.nfpa.org

U.S. ARMY CORPS OF ENGINEERS (USACE)
CRD-C DOCUMENTS available on Internet:
http://www.wbdg.org/ffc/army-coe/standards
Order Other Documents from:
Official Publications of the Headquarters, USACE
E-mail: hqpublications@usace.army.mil
Internet: http://www.publications.usace.army.mil/
or
https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272
Internet: https://www.archives.gov/
Order documents from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

WASHINGTON STATE ADMINISTRATIVE CODE (WAC)
Legislative Information Center
Cheri Randich, Manager
110 Legislative Building
Olympia, WA 98504-0600
Ph: 360-786-7573
E-mail: support@leg.wa.gov
Internet: https://app.leg.wa.gov/wac/

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION 01 45 00

QUALITY CONTROL
08/23, CHG 2: 05/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1077	(2024) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E329	(2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
ASTM E543	(2021) Standard Specification for Agencies Performing Non-Destructive Testing

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Bid Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the Preconstruction Conference, before start of construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor and the Government. Provide a copy of the signed minutes to all attendees[and incorporate into the relevant QC Plan]. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

a. Waste Management Plan.

b. Procedures for noise and acoustics management.

c. Environmental Protection Plan.

d. Environmental regulatory requirements.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager,[Assistant QC Manager,][QC Specialists,][Environmental Manager,] and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent,[the QC Specialists,] and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

a. Review the minutes of the previous meeting.

b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.

c. Review the status of submittals and Request For Information (RFIs).

d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY LOOK AHEAD in Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS) and all documentation required for that work.

e. Review Testing Plan and Log including status of tests performed since last QC Meeting.

f. Resolve QC and production problems. Discuss status of pending change orders.

g. Address items that may require revising the QC Plan.

h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.

i. Review environmental requirements and procedures.

j. Review Environmental Management Plan.

k. Review Waste Management Plan.

l. Review the status of training completion.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than [30][_____] days after Contract Award, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.

b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.

c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.

d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.

e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.

f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to[the Assistant QC Manager and] all other QC Specialists or quality control representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.

g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.

h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.

i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.

j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.

k. Reporting procedures, including proposed reporting formats.

l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.

m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination and Mutual Understanding Meeting.

n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.

o. [o][p]. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.

p. [p][q]. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.

q. [q][r]. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

1.5.3 Acceptance of the Quality Control (QC) Plan

The Contracting Officer's acceptance of the Contractor QC Planis required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Preliminary Construction Work Authorized Prior to Acceptance

The only construction work that is authorized to proceed prior to the acceptance of the QC Plan is mobilization of storage and office trailers, temporary utilities, and surveying with specific prior approval of the Contracting Officer.

1.5.5 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Quality Control (QC) Manager

1.6.1.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program

[, and to serve as the [Level two][Level three] Site Safety and Health Officer (SSHO) as detailed in Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS].[In addition to implementing and managing the QC program, the QC Manager may perform the duties of Project Superintendent.][The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract.] The QC Manager must attend the partnering meetings, QC Plan Meetings, Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of control[except for those phases of control designated to be performed by QC Specialists], perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by[the QC Specialists,] testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities.

1.6.1.2 Qualifications

The QC Manager must be an individual with a minimum of [5][10][_____] years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least [2][4][_____] years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

[The QC Manager must possess at least a Bachelor's degree in engineering, architecture, or construction management, with[a current professional engineer registration and] a minimum of [10][_____] years construction experience as a Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least [2][4][_____] years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.1.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager Alternate QC Manager, Superintendent, QC Specialist, and DQCM must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

[1.6.1.4 Quality Control Requirements for Multiple Work Sites

Provide separate full-time QC Managers at the work sites listed below. Provide a separate full-time QC Manager at the following work sites:

a. [INDICATE WORK SITE LOCATION]

b. [INDICATE WORK SITE LOCATION]

c. [INDICATE WORK SITE LOCATION]

The QC Managers for the work sites listed above are each responsible for implementing and managing the QC Program at the designated work site. They must perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the QC Managers at each site are listed in the paragraph QUALIFICATIONS.

]1.6.2 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The period of absence may not exceed 2 weeks at one time, and not more than 30 workdays during a calendar year. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

[1.6.4 Assistant Quality Control (QC) Manager Duties and Qualifications

Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The qualification requirements for the Assistant QC Manager must be [_____]. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

[Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The Assistant QC Manager must be on the work site during supplemental work shifts[beyond the regular shift] and perform the duties of the QC Manager during such supplemental shift work. The qualification requirements for the Assistant QC Manager must be [FILL IN BASED ON NATURE AND COMPLEXITY OF JOB]. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.

The Assistant QC Manager must be capable of reading, writing, and conversing fluently in the English language.

1.6.5 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who[may perform production related duties but must be allowed sufficient time to perform][must have no duties other than] their assigned quality control duties. These individuals or specialized technical companies[are directly employed by the Prime Contractor and cannot be employed by a supplier or subcontractor on this project][are employees of the Prime or subcontractor]. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix		
1. Area	**2-1. Qualification** **2-2. Experience**	**3-1. Area of Responsibility**
[Civil]	[2-1. Graduate Civil Engineer or Construction Manager with [a professional engineer registration [in the state of [_____]], and] 2-2. 2 years experience in the type of work being performed on this project or technician with 5 years related experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final inspection.]
[Mechanical]	[2-1. Graduate Mechanical Engineer with [a professional engineer registration [in the state of [_____]], and] 2-2. 2 years experience or person with 5 years of experience supervising mechanical features of work in the field with a construction company]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final inspection.]

[Electrical]	[2-1. Graduate Electrical Engineer with [a professional engineer registration [in the state of [_____]], and] 2-2. 2 years related experience or person 5 years of experience supervising electrical features of work in the field with a construction company]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final inspection.]
[Structural]	[2-1. Graduate Civil Engineer (with Structural Track or Focus) or Construction Manager with [a professional engineer registration [in the state of [_____]], and] 2-2. 2 years experience or person 5 years of experience supervising structural features of work in the field with a construction company]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final inspection.]
[Architectural]	[2-1. Graduate Architect with [a registered architect [in the state of [_____]], and] 2-2. 2 years experience or person with 5 years related experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final
[Environmental]	[2-1. Graduate Environmental Engineer with [a professional engineer registration [in the state of [_____]], and] 2-2. 3 years experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final
[Submittals]	[2-1. Submittal Clerk 2-2. 1 year experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final

[Occupied Family Housing]	[2-1. Person, customer relations type, 2-2. Coordinator experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final
[Concrete, Pavements and Soils]	[2-1. Materials Technician 2-2. 2 years experience for the appropriate area]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final
[Testing, Adjusting and Balancing (TAB) Personnel]	[2.1. TAB Team Field Leader must be a member of AABC or an experienced technician of the firm certified by the NEBB 2-2. 3 years experience immediately preceding this Contract]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final inspection.]
[Roofing Manufacturer's Technical Representative]	[2-1. Installation and testing of roofing systems, Section 07 53 23 ETHYLENE-PROPYLENE-DIENE-MONOMER ROOFING 2-2. 5 years experience minimum]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [Full-time] during installation and including final
[Mechanical Inspector, International Code Council (ICC) Certified]	[2-1. Installation and testing of boilers, Section 23 52 49.00 20 STEAM BOILERS AND EQUIPMENT (500,000 - 18,000,000 BTU/HR) 2-2. 5 years experience minimum]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [Minimum three times a week during installation and full-time during testing] during installation and
[Geotechnical]	[2-1. [_____] 2-2. [_____] years of experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final

[Telecommunication	[2-1. [_____] 2-2. [_____] years of experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final
[Low-Voltage Specialties/ESS]	[2-1. [_____] 2-2. [_____] years of experience]	[3-1. Area of Responsibility: [_____] 3-2. Frequency of QC surveillance and inspection will be [_____] during installation and including final
[Underwater QC Team]	[Note: See paragraph UNDERWATER QC TEAM]	[Note: See paragraph UNDERWATER QC TEAM]
[Fire Protection QC Specialist (FPQC)]	[Note: See paragraph FIRE PROTECTION QC SPECIALIST (FPQC)]	[Note: See paragraph FIRE PROTECTION QC SPECIALIST (FPQC)]
[Building Envelope QC Specialist]	[2-1. Roofing Manufacturer's Technical Representative 2-2. 5 years minimum with roofing system used]	[3-1. Area of Responsibility: Installation and testing of roofing. 3-2. Frequency of QC surveillance and inspection will be once a week during installation, once a week during flashing installation and full-time during roof testing.]
[_____]	[_____]	[_____]

1.6.5.1 Underwater QC Team

Provide Underwater QC (UWQC) Team at the work site to perform underwater surveillance and inspection for the Contractor. The UWQC Team divers must have current commercial diver's license, with a minimum of 5 years experience with underwater inspection. The personnel make-up of the UWQC team must comply with EM 385-1-1, OSHA, and local requirements for Contract diving operations. Comply with all the applicable safety requirements of EM 385-1-1, OSHA, and local requirements for Contract diving operations. The UWQC lead diver must be thoroughly familiar with the design plans and specifications to sufficiently understand the engineering aspects of the underwater construction and to be able to recognize and document potential problem areas, such as improperly constructed or defective areas. Provide all necessary equipment to conduct surveillance and inspection services, including diver's equipment, dive boat, communication equipment, and photographic or video equipment or both. Diver(s) must be equipped to maintain two-way communication with QC personnel during diving operations. Prepare and submit a report including photographs or videos or both with the QC report after each dive. Frequency of underwater surveillance and inspection will be [_____] during installation and including final inspection. The UWQC Team must be an independent third party hired directly by the Prime Contractor, and must not be involved with the design, preparation of Contract, or installation of work.

1.6.5.2 Fire Protection QC Specialist (FPQC)

Provide a Fire Protection Quality Control Specialist (FPQC) within the QC organization to perform quality control related activities as specified herein on fire protection and life safety systems installed under this Contract.

1.6.5.2.1 Qualifications

The FPQC must have the following qualifications:

a. Be a registered Professional Engineer (P.E.) licensed by a Licensing Board in the United States, the District of Columbia, Guam or Puerto Rico, having passed the National Council of Examiners for Engineering and Surveying (NCEES) examination specifically in the discipline of Fire Protection Engineering.

b. Have a minimum of 5 years of Fire Protection Engineering experience on projects of similar relevance and complexity to the fire protection work specified under this Contract.

c. Other than the contractual obligations with the Prime Contractor, the FPQC must have no other business relationship (i.e., employee, owner, partner, operating officer, distributor, salesman, technical representative, family relationship, or financial investment) with the Prime Contractor or subcontractors.

d. Be employed by an independent engineering firm or company. The firm may identify multiple, to a maximum of five, licensed Fire Protection Engineers for the performance of the duties under this Contract but must submit the names and qualifications for Government approval for all individuals identified prior to them performing any work under this Contract. These individuals may not be substituted without prior approval from the Contracting Officer.

1.6.5.2.2 Responsibilities

FPQC duties and responsibilities:

a. Assist in the development of the QC Plan including the Testing Plan and Log and executing the three phases of control for work involving the installation and testing of fire protection and life safety systems as an extension of the QC Manager.

b. Participate in project QC Meetings. Participate in Preparatory and Initial Phase meetings and perform and Follow-up inspections for work involving the installation and testing of fire protection and life safety systems.

c. Review and certify that all submittals pertaining to fire protection and life safety systems are complete and accurate prior to submission to the Government for approval. The FPQC Specialist is responsible for ensuring submittals are complete and accurate and all corrections have been made prior to submission to the Government. The Government reserves the right to reject any submittal that has not first been reviewed and certified by the FPQC and so marked, in writing, attesting to such review and completeness of the submittal.

d. The Government reserves the right to reject any submittal or construction that is not in compliance to Contract. Government reviews do not relieve the Contractor responsibility for providing adequate quality control measures and do not constitute or imply acceptance of Contract variation.

e. Perform construction surveillance in accordance with the Schedule of Fire Protection System Inspections. Construction surveillance includes but is not limited to performing periodic on-site inspections during construction at specified milestones, performing a pre-final inspection of installed systems and witnessing functional testing; and participating and documenting in an on-site final acceptance inspection of fire protection and life safety systems with the Government FPE.

f. Document inspection results on a FPQC report prepared each day inspections are performed. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the FPQC Specialist.[[In person inspection][and][Remote inspections] must be documented via [video (.mp4)][or][photo (.jpeg)][or both]. Video/photographic documentation must include before and after conditions and physical measurements.] Forward the FPQC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the FPQC must be documented on a FPQC daily report.

1.6.5.2.3 Schedule of Fire Protection System Inspections

A schedule, prepared by the Fire Protection DOR, which lists each of the required visual inspections and observations required by the FPQC. The schedule is included at the end of this UFGS section.

1.6.6 Submittal Reviewer[s] Duties and Qualifications

Provide[a] Submittal Reviewer[s], other than the QC Manager, qualified in the discipline[s] being reviewed, to review and certify that the submittals meet the requirements of this Contract prior to certification or approval by the QC Manager.

Each submittal must be reviewed by an individual with 10 years of construction experience.

Each submittal must be reviewed by a registered architect or professional engineer.

Each of the following submittals must be reviewed by [an] individual[s] meeting the qualifications/experience specified below:

Qualification / Experience in Submittal Discipline	Submittals to be reviewed:	
	Section No	Submittal
[_____]	[_____]	[_____]

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least 2 business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by[the QC Specialists,] the Project Superintendent, and the foreman responsible for the DFOW. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFOW:

a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the prepatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

b. Review the Contract drawings.

c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.

d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.

e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.

f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.

g. Review coordination of product/material delivery to designated prepared areas to execute the work.

h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.

i. Check to assure that all materials and equipment have been tested, submitted, and approved.

j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOW. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.

k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with[the QC Specialists,] the Project Superintendent, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the[daily CQC Report and in the] Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFOW:

a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.

b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.

c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

d. Resolve any workmanship issues.

e. Ensure that testing is performed by the approved laboratory.

f. Check work procedures for compliance with the APP and the appropriate

AHA to ensure that applicable safety requirements are met.

g. Review project specific work plans (i.e., HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.

b. Maintain the quality of workmanship required.

c. Ensure that testing is performed by the approved laboratory.

d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.

e. Do not build upon nor conceal non-conforming work.

f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least 2 weeks prior to the start of the preparatory and initial phases.

1.8.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.

b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.

c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.

d. The Contractor is responsible for including those items identified by the Contracting Officer.

e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.

f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.

g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site or within [_____] miles. Perform the following activities and record and provide the following data:

a. Verify that testing procedures comply with Contract requirements.

b. Verify that facilities and testing equipment are available and comply with testing standards.

c. Check test instrument calibration data against certified standards.

d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.

e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this

Contract.

1.9.1 Accreditation Requirements

Construction materials testing laboratories must be accredited by a laboratory accreditation authority and must submit a copy of the Certificate of Accreditation and Scope of Accreditation. The laboratory's scope of accreditation must include the appropriate ASTM standards (ASTM E329, ASTM C1077, ASTM D3666, ASTM D3740, ASTM E543) listed in the technical sections of the specifications. Laboratories engaged in Hazardous Materials Testing must meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the Corporate Office.

1.9.2 Laboratory Accreditation Authorities

Laboratory Accreditation Authorities include the National Voluntary Laboratory Accreditation Program (NVLAP) administered by the National Institute of Standards and Technology at https://www.nist.gov/nvlap, the American Association of State Highway and Transportation Officials (AASHTO) Accreditation Program at http://www.aashtoresource.org/aap/overview, International Accreditation Services, Inc. (IAS) at https://www.iasonline.org/, U.S. Army Corps of Engineers Materials Testing Center (MTC) at https://www.erdc.usace.army.mil/Media/Fact-Sheets/Fact-Sheet-Article-View/Article/476661/materials-testing-center/, the American Association for Laboratory Accreditation (A2LA) program at https://a2la.org/, the Washington Association of Building Officials (WABO) at https://www.wabo.org/ (Approval authority for WABO is limited to projects within Washington State), and the Washington Area Council of Engineering Laboratories (WACEL) at https://www.wacel.org/lab-accreditation-and-inspection-agency-audit-programs/laboratory-accreditation-program/(Approval authority by WACEL is limited to projects within Facilities Engineering Command (FEC) Washington geographical area).

1.9.3 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

1.9.4 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager. Furnish a summary report of field tests at the end of each month, in accordance with paragraph DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER.

1.9.5 Test Reports and Monthly Summary Report of Tests

Furnish the signed reports, certifications, and a summary report of field tests at the end of each month to the Contracting Officer. Attach a copy of the summary report to the last daily CQC Report of each month.[Provide a copy of the signed test reports and certifications to the Operation and Maintenance Support Information (OMSI) preparer for inclusion into the OMSI documentation, in accordance with Sections 01 78 23 OPERATION AND MAINTENANCE DATA and 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW).]

1.10 COMPLETION INSPECTIONS

1.10.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.10.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred by the Contracting Officer in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.10.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.11.3 Invoice Certification

Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that as-built drawings are current, coordinated and attesting that the work for which payment is requested, including stored material, complies with Contract requirements.

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

1.12.1 Construction Documentation

Prepare a Construction Production Report and a Contractor Quality Control Report for each day that work is performed. Attach the Construction Production Report to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities.
Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

a. The name and area of responsibility of the Contractors and any subcontractors.

b. Operating plant/equipment with hours worked, idle, or down for repair.

c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.

d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.

e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.

f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.

g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.

h. Offsite surveillance activities, including actions taken.

i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.

j. Instructions given/received and conflicts in plans and specifications.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

1.12.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.

b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.

c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.

d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES. Monthly Summary Report of Tests:[Submit the report as an electronic attachment to the CQC Report at the end of each month.][Mail or hand-carry the original attached to the last QC Report of the month.]

e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.

f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.

g. [g][h][i]. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.6 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.12.7 As-Built Drawings

The QC Manager must ensure the as-built drawings, required by Section 01 78 00 CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document commences with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the Contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager[or QC Specialist assigned to an area of responsibility] must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
11/20, CHG 2: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

1.3 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Section 01 Emergency Planning and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.

b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.

c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.

d. Condition ONE. (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.4 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.4.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.4.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.4.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.4.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.4.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.4.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.4.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.4.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. Or if there is a Government agency that's responsible, identify that agency.

1.4.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.4.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.4.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

PART 2 PRODUCTS [NOT USED]

PART 3 EXECUTION [NOT USED]

-- End of Section --

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1200	Hazard Communication
40 CFR 64	Compliance Assurance Monitoring
40 CFR 112	Oil Pollution Prevention
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 273	Standards for Universal Waste Management
40 CFR 279	Standards for the Management of Used Oil

40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 355	Emergency Planning and Notification
40 CFR 403	General Pretreatment Regulations for Existing and New Sources of Pollution
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 172.101	Hazardous Material Regulation-Purpose and Use of Hazardous Material Table
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

WASHINGTON STATE ADMINISTRATIVE CODE (WAC)

WAC-173-303-573	Standards for Universal Waste Management
WAC-173-303-573(2)	Standards for Universal Waste Management - Batteries
WAC-173-303-573(3)	Standards for Universal Waste Management - Mercury-containing Equipment
WAC-173-303-573(5)	Standards for Universal Waste Management - Lamps

1.2 DEFINITIONS

1.2.1 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.2 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.3 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.4 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with 40 CFR 261.

1.2.5 Hazardous Materials

Hazardous material is any material that: Is defined in 49 CFR 171, listed in 49 CFR 172, regulated as a hazardous material in accordance with 49 CFR 173; or requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.1200; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D. Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this section for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.6 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibits a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D, or meets a state, local, or host nation definition of a hazardous waste.

1.2.7 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with other state and local requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in other state regulations or local ordinances; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.8 Regulated Waste

Regulated waste are solid wastes that have specific additional federal, state, or local controls for handling, storage, or disposal.

1.2.9 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.10 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.10.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 2.5-inch particle size that is: a manufactured object; broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.10.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.10.3 Material Not Regulated As Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.10.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 261.

1.2.10.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires.

1.2.10.6 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.10.7 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.11 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks meeting the definition of "waters of the United States". Surface discharges from construction sites are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, state, and local laws and regulations.

1.2.12 Wastewater

Wastewater is the used water and solids that flow through a sanitary sewer to a treatment plant.

1.2.12.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.13 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

1.2.14 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.15 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.2.16 Location Specific Universal Waste

Any of the following dangerous waste that are subject to the universal waste requirements of WAC-173-303-573: Batteries as described in WAC-173-303-573(2)); Lamps as described in WAC-173-303-573(5); Mercury-containing equipment as described in WAC-173-303-573(3).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Regulatory Notifications; G

Environmental Manager Qualifications; G

Environmental Protection Plan; G

Solid Waste Management Permit; G

Spill Prevention Control And Countermeasure (SPCC) Plan; G

SD-06 Test Reports

Monthly Solid Waste Disposal Report; G

SD-07 Certificates

ECATTS Certificate Of Completion; G

Employee Training Records; G

SD-11 Closeout Submittals

Regulatory Notifications; G

Assembled Employee Training Records; G

Disposal Documentation for Hazardous and Regulated Waste; G

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)

1.4.1.1 Personnel Requirements

The Environmental Manager is responsible for environmental compliance on projects. The Environmental Manager, must complete applicable ECATTS training modules (installation specific or general) prior to starting respective portions of on-site work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete applicable ECATTS training within 14 days of assignment to the project.

1.4.1.2 Certification

Submit an ECATTS certificate of completion for personnel who have completed the required ECATTS training. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Assessment, Training, and Tracking System, by logging on to https://environmentaltraining.ecatts.com/. Obtain the password for registration from the Contracting Officer.

1.4.1.3 Refresher Training

This training has been structured to allow contractor personnel to receive credit under this contract and to carry forward credit to future contracts. Ensure the Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work. Some training modules are tailored for specific state regulatory requirements; therefore, Contractors working in multiple states will be required to retake modules tailored to the state where the contract work is being performed.

1.4.2 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS).

Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the special environmental requirements listed here and attached at the end of this section.

1.6 QUALITY ASSURANCE

1.6.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.6.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with federal, state and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of regulatory notifications to the Contracting Officer prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.6.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP) or equipment local requirement. Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

1.6.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating contractor compliance with federal, state, local, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer.

1.6.5 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. FAR 52.242-14 Suspension of Work provides that a suspension, delay, or interruption of work due to the fault or negligence of the Contractor allows for no adjustments to the contract for time extensions or equitable adjustments. In addition to a suspension of work, the Contracting Officer may use additional authorities under the contract or law.

1.7 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 15 days after Contract award and not less than 10 days before the preconstruction meeting. Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of any applicable federal, state, and local environmental protection laws and regulations. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version onsite.

The EPP includes, but is not limited to, the following elements:

1.7.1 General Overview and Purpose

1.7.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as spill control plan, solid waste management plan,contaminant prevention plan, Hazardous, Toxic and Radioactive Waste (HTRW) Plan Non-Hazardous Solid Waste Disposal Plan .

1.7.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

1.7.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

1.7.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.7.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.7.2 General Site Information

1.7.2.1 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site.

Show where any fuels, hazardous substances, solvents, or lubricants will be stored. Provide a spill plan to address any releases of those materials.

1.7.2.2 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.7.3 Management of Natural Resources

a. Land resources
b. Replacement of damaged landscape features
c. Temporary construction
D. Fish and wildlife resources

1.7.4 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste.

Control and disposal of hazardous waste.

This item consist of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan when within an installation. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan as applicable.

As a minimum, include the following:

a. List of the types of hazardous wastes expected to be generated

b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated

c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications

d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)

e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)

f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions (40 CFR 268)

g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar

h. Used oil management procedures in accordance with 40 CFR 279; Hazardous waste minimization procedures

i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.

1.7.5 Prevention of Releases to the Environment

Procedures to prevent releases to the environment

Notifications in the event of a release to the environment

1.7.6 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.8 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7 Permits and Responsibilities. Notify the Government of all equipment that may require permits or special approvals that the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7 Permits and Responsibilities.

See Appendix A of this specification for copies of Permits that have been secured by the Government to support this project.

For permits obtained by the Government, whether or not required by the permit, the Contractor is responsible for conforming to all permit requirements and performing all quality control inspections of the work in progress, and submit notifications and certifications to the applicable regulatory agency via the Contracting Officer.

1.9 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.10 SOLID WASTE MANAGEMENT PERMIT

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.10.1 Monthly Solid Waste Disposal Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office as applicable, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.2 Waters of the United States

Do not enter, disturb, destroy, or allow discharge of contaminants into waters of the United States. The protection of waters of the United States shown on the drawings in accordance with paragraph LICENSES AND PERMITS is the Contractor's responsibility. Authorization to enter specific waters of the United States identified does not relieve the Contractor from any obligation to protect other waters of the United States within, adjacent to, or in the vicinity of the construction site and associated boundaries.

3.3 PROTECTION OF CULTURAL RESOURCES

3.3.1 Archaeological Resources

If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

3.4 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with 40 CFR 64 and state air emission and performance laws and standards.

3.4.1 Burning

Burning is prohibited on the Government premises.

3.4.2 Odors

Control odors from construction activities. The odors must be in compliance with state regulations and local ordinances and may not constitute a health hazard.

3.5 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.5.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

3.5.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	cubic yards, tons
C&D Debris Recycled	cubic yards, tons
C&D Debris Composted	cubic yards, tons
Total C&D Debris Generated	cubic yards, tons
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	cubic yards, tons

3.6 WASTE MANAGEMENT AND DISPOSAL

3.6.1 Solid Waste Management

3.6.1.1 Project Solid Waste Disposal Documentation Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation or Contractor certification must include the receiver's tax identification number and business, EPA or state registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.6.1.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with 40 CFR 260, state, and local requirements for solid waste disposal. A Subtitle D RCRA permitted landfill is the minimum acceptable offsite solid waste disposal option. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Solid waste disposal offsite must comply with most stringent local, state, and federal requirements, including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with 49 CFR 173.

3.6.2 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer and Installation Hazardous Waste Manager.

3.6.2.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with federal, state, and local regulations, including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, and 40 CFR 268.

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with 49 CFR 173 and 49 CFR 178. Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government. Prior to removal of any hazardous waste from Government property, hazardous waste manifests must be signed by personnel from the Installation Environmental Office. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in 40 CFR 372-SUBPART D.

3.6.2.2 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with 40 CFR 262 and applicable state or local regulations. Individual waste streams will be limited to 55 gallons of accumulation (or one quart for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90-day or 180-day, as appropriate, accumulation point be established. Submit a request in writing to the Contracting Officer.
Attach a Waste Determination form for the expected waste streams. Allow 10 working days for processing this request. Additional compliance requirements (e.g., training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.6.2.3 Hazardous Waste Disposal

3.6.2.3.1 Responsibilities for Contractor's Disposal

Provide hazardous waste manifest to the Installations Environmental Office for review, approval, and signature prior to shipping waste off Government property.

3.6.2.3.1.1 Services

Provide service necessary for the final treatment or disposal of the hazardous material or waste in accordance with 40 CFR 260 - 40 CFR 279, local, and state, laws and regulations, and the terms and conditions of the Contract within 60 days after the materials have been generated. These services include necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal or transportation, include manifesting or complete waste profile sheets, equipment, and compile documentation).

3.6.2.3.1.2 Samples

Obtain a representative sample of the material generated for each job done to provide waste stream determination.

3.6.2.3.1.3 Analysis

Analyze each sample taken and provide analytical results to the Contracting Officer. See paragraph WASTE DETERMINATION DOCUMENTATION.

3.6.2.3.1.4 Labeling

During waste accumulation label all containers in accordance with 40 CFR 262. Prior to offering a waste for off-site transport, determine the Department of Transportation's (DOT's) proper shipping names for waste in accordance with 49 CFR 172 (each container requiring disposal) and demonstrate to the Contracting Officer how this determination is developed and supported by the sampling and analysis requirements contained herein. Label all containers of hazardous waste with the words "Hazardous Waste" or other words to describe the contents of the container in accordance with 40 CFR 262 and applicable state or local regulations.

3.6.2.3.2 Contractor Disposal Turn-In Requirements

Hazardous waste generated must be disposed of in accordance with the following conditions to meet installation requirements:

a. Drums must be compatible with waste contents and drums must meet DOT requirements for 49 CFR 173 for transportation of materials.

b. Band drums to wooden pallets.

c. No more than three 55 gallon drums or two 85 gallon over packs are to be banded to a pallet.

d. Band using 1-1/4 inch minimum band on upper third of drum.

e. Provide label in accordance with 49 CFR 172.101.

f. Leave 3 to 5 inches of empty space above volume of material.

3.6.2.4 Disposal Documentation for Hazardous and Regulated Waste

Contact the Contracting Officer for the facility RCRA identification number that is to be used on each manifest.

Submit a copy of the applicable EPA and or state permit(s), manifest(s), or license(s) for transportation, treatment, storage, and disposal of hazardous and regulated waste by permitted facilities. Hazardous or toxic waste manifests must be reviewed, signed, and approved by the Contracting Officer before the Contractor may ship waste. To obtain specific disposal instructions, coordinate with the Installation Environmental Office. Refer to location special requirements for the Installation Point of Contact information.

3.6.3 Releases/Spills of Oil and Hazardous Substances

3.6.3.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Installation Fire Department, the Installation Command Duty Officer, the Installation Environmental Office, the Contracting Officer.

Submit verbal and written notifications as required by the federal (40 CFR 300.125 and 40 CFR 355), state, local regulations and instructions. Provide copies of the written notification and documentation that a verbal notification was made within 20 days. Spill response must be in accordance with 40 CFR 300 and applicable state and local regulations. Contain and clean up these spills without cost to the Government.

3.6.3.2 Clean Up

Clean up hazardous and non-hazardous waste spills. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor- responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.6.4 Wastewater

3.6.4.1 Disposal of Wastewater

Disposal of wastewater must be as specified below.

3.6.4.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related waste water off-Government property in accordance with 40 CFR 403, state, regional, and local laws and regulations.

3.7 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this contract. Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with 40 CFR 261 and state requirements.

3.7.1 Contractor Hazardous Material Inventory Log

Submit the "Contractor Hazardous Material Inventory Log"(found at: https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables), which provides information required by (EPCRA Sections 312 and 313) along with corresponding SDS, to the Contracting Officer at the start and at the end of construction (30 days from final acceptance), and update no later than January 31 of each calendar year during the life of the contract. Keep copies of the SDSs for hazardous materials onsite. At the end of the project, provide the Contracting Officer with copies of the SDSs, and the maximum quantity of each material that was present at the site at any one time, the dates the material was present, the amount of each material that was used during the project, and how the material was used.

The Contracting Officer may request documentation for any spills or releases, environmental reports, or off-site transfers.

3.8 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

3.9 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with EPA 40 CFR 112, and other federal, state, regional, and local laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of the United States. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided.

3.9.1 Used Oil Management

Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE

DISPOSAL.

3.9.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 5 inches freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 1320 gallons will be used onsite (only containers with a capacity of 55 gallons or greater are counted), provide and implement a Spill Prevention Control and Countermeasure (SPCC) plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.10 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.11 CHLORDANE

Evaluate excess soils and concrete foundation debris generated during the demolition of housing units or other wooden structures for the presence of chlordane or other pesticides prior to reuse or final disposal.

3.12 SOUND INTRUSION

Make the maximum use of low-noise emission products, as certified by the EPA. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and then only during the designated times. .

Keep construction activities under surveillance and control to minimize environment damage by noise. Comply with the provisions of the State of Virginia rules.

3.13 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

-- End of Section --

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.1.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.1.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.1.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.1.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.1.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.1.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.1.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.1.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.1.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.2 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.3 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.3.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.3.2 Oversight

The Environmental Manager, as specified in Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS, is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project.

1.3.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.3.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.3.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste steams not listed:

a. Land Clearing Debris

b. Asphalt

c. Masonry and CMU

d. Concrete

e. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)

f. Wood (nails and staples allowed)

g. Glass

h. Paper

i. Plastics (PET, HDPE,PVC,LDPE,PP,PS, Other)

j. Gypsum

k. Non-hazardous paint and paint cans

l. Carpet

m. Ceiling Tiles

n. Insulation

o. Beverage Containers

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; S

1.5 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00 QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

a. Preconstruction meeting.

b. Regular Quality Control meetings.

c. Work safety meeting (if applicable).

1.6 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 15 days after notice to proceed. Revise and resubmit Construction Waste Management Plan as necessary, in order for construction to begin. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.

b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.

c. Description of the regular meetings to be held to address waste management.

d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.

e. Name of landfill and incinerator to be used.

f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.

g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.

h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.

i. Description of the means by which materials identified in item (g) above will be protected from contamination.

j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).

k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.

Distribute copies of the waste management plan to each subcontractor, Environmental Manager, and the Contracting Officer.

1.7 RECORDS (DOCUMENTATION)

1.7.1 General

Maintain records to document the types and quantities of waste generated and diverted though re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.7.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.8 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide Final Construction Waste Diversion Report 60days prior to the Beneficial Occupancy Date (BOD).

1.9 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS. Separate materials by one of the following methods described herein:

1.9.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.9.2 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.10 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.10.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is allowed on the Installation. Consider the use of surplus industrial supply broker services, who match entities with reusable or repurpose industrial materials with entities with need of such materials.

1.10.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.10.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used. -- End of Section --

SECTION 35 20 23

DREDGING
04/06

PART 1 GENERAL

1.1 DEFINITION

Hard material is defined as material requiring blasting or the use of special equipment for economical removal, and includes boulders or fragments too large to be removed in one piece by the dredge.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

None Required

1.3 MATERIAL TO BE REMOVED

The material to be removed is silt and mud.

1.3.1 Hard Material

The removal of hard material is not anticipated. Should the Government direct in writing that hard material be removed, the work shall be performed and an adjustment in the contract price or time for completion, or both, will be made in accordance with "FAR 52.236-2, Differing Site Conditions." If hard material is to be removed, blasting will not be permitted.

1.4 ARTIFICIAL OBSTRUCTIONS AND DEBRIS

While the Government has no knowledge of miscelaneous debris such as, but not limited to, concrete debris, metal bands, pallets, pieces of broken cable, rope, fire hose, and broken piles any miscelaneous debris encountered within the dredging area shall be removed and legally disposed as specified below. The Government has no knowledge of existing wrecks, wreckage, or other material of such size or character as to require the use of explosives or special or additional plant for its economical removal. Prior to dredging, the Contractor shall rake the dredge areas and shall remove debris encountered. All Debris encountered within the dredged area, both before and during dredging operation, shall be removed from the water. Disposal shall be the responsibility of the Contractor and disposal shall be outside the limits of government property. The contractor shall be compensated for debris removal and upland disposal pursuant to unit price quoted in bid schedule.

1.5 QUANTITY OF MATERIAL-BASE BID

The estimated quantity for bidding purposes and for application of the "FAR

52.212-11, Variation in Estimated Quantity" shall be as shown on Sheet V-002 of the project drawings. The quantities listed are estimates only. Within the limits of available funds, complete the work specified whether the quantities involved are greater or less than those estimated.

1.6 OVERDEPTH DREDGING

To cover unavoidable inaccuracies of dredging processes, material actually removed to a depth of 2 feet below the depth specified and within the dredging limits will be measured and paid for at full contract price.

1.7 SIDE SLOPES

Dredging on side slopes shall follow, as closely as practicable, the lines indicated or specified. A 1 foot allowance will be made for dredging beyond the indicated or specified side slopes, except as provided herein. The amount of material excavated from side slopes will be determined by either cross-sections or computer, or both.

1.8 PERMIT

The Contractor shall comply with conditions and requirements of the Corps of Engineers Permit and other State or Federal permits. (See Appendix "B")

1.9 CHARGES

The Contractor shall pay all charges imposed by the dredged material disposal facilities.

1.10 [Enter Appropriate Subpart Title Here]

1.11 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain during the life of the contract, environmental protective measures. Also, provide environmental protective measures required to correct conditions, such as oil spills or debris, that occur during the dredging operations. Comply with Federal, State, and local regulations pertaining to water, air, and noise pollution.

1.12 BASIS FOR BIDS

Payment will be at the contract unit price per cubic yard, multiplied by total cubic yards of acceptable dredging. Base bids on total cubic yards of dredging indicated. Include a bid unit price per cubic yard of dredging based on the quantity as specified or indicated. If the Contracting Officer requires an increase or a decrease in total volume of dredging, the contract price will be adjusted in accordance with the "FAR 52.211-18, Variation in Estimated Quantity." Dredging conditions specified and indicated describe conditions which are known. However, the Contractor is responsible for other conditions encountered which are not unusual when compared to conditions recognized in the dredging business as usual in dredging activities such as those required under this contract.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 INSPECTION

Inspect the work, keep records of work performed, and ensure that gages, targets, ranges, and other markers are in place and usable for the intended purpose. Furnish, at the request of the Contracting Officer, boats, boatmen, laborers, and materials necessary for inspecting, supervising, and surveying the work. When required, provide transportation for the Contracting Officer and inspectors to and from the disposal area and between the dredging plant and adjacent points on shore.

3.2 CONDUCT OF DREDGING WORK

3.2.1 Order of Work

The Contractor shall coordinate with the Contracting Officer prior to start of dredging to determine order of work. The Government reserves the right to change the order of work at any time.

3.2.2 Interference with Navigation

Minimize interference with the use of channels and passages. The Contractor is responsible for shifting or moving of dredges or the interruption of dredging operations to accommodate the movement of vessels and floating equipment, if necessary. Adhere to Coast Guard Regulations for passing vessels

3.2.2.1 Compensation for Interruption of Operations

If dredging operations are interrupted due to the government's movement of vessels or floating equipment, an adjustment in the contract price or time for completion, or both, will be made as provided by the contract. The Contracting Officer will notify the Contractor 72 hours prior to ship movements that will affect dredging operations.

3.2.3 Lights

Each night, between sunset and sunrise and during periods of restricted visibility, provide lights for floating plants, pipelines, ranges, and markers. Also, provide lights for buoys that could endanger or obstruct navigation. When night work is in progress, maintain lights from sunset to sunrise for the observation of dredging operations. Lighting shall conform to United States Coast Guard requirements for visibility and color.

3.2.4 Ranges, Gages, and Lines

Furnish, set, and maintain ranges, buoys, and markers needed to define the work and to facilitate inspection. Establish and maintain gages in locations observable from each part of the work so that the depth may be determined. Suspend dredging when the gages or ranges cannot be seen or followed. The Contracting Officer will furnish, upon request by the Contractor, survey lines, points, and elevations necessary for the setting of ranges, gages, and buoys.

3.2.5 Plant

Maintain the plant, scows, coamings, barges, pipelines, and associated equipment to meet the requirements of the work. Remove dredged material placed due to leaks and breaks.

3.2.6 Disposal of Excavated Material

Provide for safe transportation and disposal of dredged materials. Transport and dispose of dredged material The following dredge material disposal sites shall be utilized as described below:

A) Dredge Areas RB-A, RB-C AND RB-D: Dredged materials shall be transported by bottom dumping barge or scow and disposed within the Weeks Marine Whites Rehandling Basin, located in Logan Township, Gloucester County, New Jersey OR transported by barge and hydraulically unloaded at the Waste Management Corporation Biles Island disposal facility located in Morrisville, Bucks County, Pennsylvania.

B) Dredge Area RB-A 7/8: Dredged materials shall be transported by barge and hydraulically unloaded at the Waste Management Corporation Biles Island disposal facility located in Morrisville, Bucks County, Pennsylvania.

All dredged material disposal shall be done in strict accordance with all permits and other pertinant regulatory approvals and in strict accordance with the disposal facility's requirements. The deposit of dredged materials in unauthorized places is forbidden. Comply with rules and regulations of local port and harbor governing authorities.

3.2.6.1 [Enter Appropriate Subpart Title Here]
3.2.6.2 [Enter Appropriate Subpart Title Here]

3.2.7 Method of Communication

Provide a system of communication between the dredge crew and the crew at the disposal area. A portable two-way radio is acceptable.

3.2.8 Salvaged Material

Anchors, chains, firearms, and other articles of value, which are brought to the surface during dredging operations, shall remain or become the property of the Government and shall be deposited on shore at a convenient location near the site of the work, as directed.

3.2.9 Safety of Structures

The prosecution of work shall ensure the stability of piers, bulkheads, and other structures lying on or adjacent to the site of the work, insofar as structures may be jeopardized by dredging operations. Repair damage resulting from dredging operations, insofar as such damage may be caused by variation in locations or depth of dredging, or both, from that indicated or permitted under the contract.

3.2.10 Plant Removal

Upon completion of the work, promptly remove plant, including ranges, buoys, piles, turbidity curtain and other markers or obstructions.

3.3 MEASUREMENT

The Contractor shall employ a government approved independent licensed

Hydrographic Surveying Company with at least 10 years experience specializing in Bathymetric Surveys and underwater data and object identification. All survey work shall be performed in accordance with the current version the US Army Corps Hydrographic Surveying Manual EM 1110-2-1003 and shall be performed under the direction of a State Licensed Professional Surveyor or Civil Engineer. Additionally, all surveys shall be performed in accordance with requirement specified in Para 3.3.1 below.

3.3.1 Method of Measurement

The material removed will be measured by cubic yard in place, by means of soundings taken before and after dredging. The drawings represent existing conditions based on current available information, but will be verified and corrected, if necessary, by soundings taken before dredging in each project area. Soundings shall be recorded using a 200 - 220 kHz Multi-Beam Sonar System using an inertial navigation system, as approved by the Government.

The surveying company shall utilize the same multi-beam sonar system and surveying vessel throughout the entire project.

The government reserves the right to be present when such soundings are made.

All raw field surveying data files necessary for post-field processing shall be furnished to the government in HYPACK/HYSWEEP HSX file format. Additionally, all associated correction files shall be furnished to the government including, but not limited to, sound velocity profiles, POS-PAC SBET data files, tide correction files, MB Sonar system offset data, etc... The government shall process all raw field survey files using HYPACK/HYSWEEP Editor, generating TIN model surfaces as follows:

Cell Size: 5' x 5'
Sounding Position: Center of Cell
Cell Statistics: Average

All volume calculations shall be performed by the government and will be determined via TIN to Advanced Channel Plan methodology using HYPACK MAX. Government generated survey XYZ data and volume calculations shall be the official survey data for execution of this contract.

3.3.2 Surveys During Progress of Work

Contract depth will be determined by soundings or sweepings taken behind the dredge as work progresses. The Contractor shall take progress soundings or sweepings; contractor shall be able to produce drawings with soundings at any time during dredging operations if requested by the contracting officer.

3.3.3 Monthly Estimates

Monthly estimates of work completed will be based on the result of soundings taken during the progress of the work. Deductions will be made for dredging and disposal not in accordance with the specifications.

3.4 FINAL EXAMINATION AND ACCEPTANCE

As soon as practicable after the completion of areas, which in the opinion of the Contracting Officer, will not be affected by further dredging operations, each area will be examined by the Government by utilizing the post-dredge survey(s). Remove shoals and lumps by dragging the bottom or by dredging. However, if the bottom is soft and the shoal areas form no material obstruction to navigation, removal may be waived at the discretion of the Contracting Officer. When areas are found to be in a satisfactory condition, the work therein will be accepted as complete. Final estimates will be subject to deductions or correction of deductions previously made because of excessive overdepth, dredging outside or authorized areas, or disposal of material in an unauthorized manner.

-- End of Section --

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION: PNY Reserve Basin FY26 Maintenence Dredging

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION / GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION: ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY: DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 14 00	SD-01 Preconstruction Submittals														
			List of Contact Personnel,Dredge Plans; G Dredge plans shall show progressive positioning of barges, scows, tender boats, etc														
		01 20 00	SD-01 Preconstruction Submittals														
			[Schedule of Prices]	1.3	G												
		01 31 23.13 20	SD-01 Preconstruction Submittals														
			List of Contractor's Personnel	1.4.2	G												
		01 32 16.00 20	SD-01 Preconstruction Submittals														
			Baseline Construction Schedule	1.2	G												
			SD-07 Certificates														
			Monthly Updates	1.4													
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.9.1	G												
			Dive Operations Plan	1.14	G												
			SD-06 Test Reports														
			Accident Reports	1.13.2	G												
			LHE Inspection Reports	1.13.3													
			Monthly Exposure Reports	1.5	G												
			SD-07 Certificates														
			Activity Hazard Analysis (AHA)	1.10	G												
			Certificate of Compliance	1.13.4													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION: PNY Reserve Basin FY26 Maintenence Dredging

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION: ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY: DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Contractor Safety Self-Evaluation	1.6														
			Checklist															
			License Certificates															
			Portable Gauge Operations		G													
			Planning Worksheet															
			Third Party Certification of	1.13.5														
			Floating Cranes and															
			Barge-Mounted Mobile Cranes															
		01 45 00	SD-01 Preconstruction Submittals															
			Contractor Quality Control (CQC)	1.5.2	G													
			Plan															
		01 50 00	SD-01 Preconstruction Submittals															
			Contractor Computer	1.4.1.4	G													
			Cybersecurity Compliance															
			Statements															
			Contractor Temporary Network	1.4.6	G													
			Cybersecurity Compliance															
			Statements															
		01 57 19	SD-01 Preconstruction Submittals															
			Regulatory Notifications	1.6.2	G													
			Environmental Manager	1.6.4	G													
			Qualifications															
			Environmental Protection Plan	1.7	G													
			Solid Waste Management Permit	1.10	G													
			Spill Prevention Control And	3.9.2	G													
			Countermeasure (SPCC) Plan															

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
PNY Reserve Basin FY26 Maintenence Dredging

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION: ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY: DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19	SD-06 Test Reports														
			Monthly Solid Waste Disposal	1.10.1	G												
			Report														
			SD-07 Certificates														
			ECATTS Certificate Of	1.4.1.2	G												
			Completion														
			Employee Training Records		G												
			SD-11 Closeout Submittals														
			Regulatory Notifications	1.6.2	G												
			Assembled Employee Training		G												
			Records														
			Disposal Documentation for	3.6.2.4	G												
			Hazardous and Regulated Waste														
		01 74 19	SD-01 Preconstruction Submittals														
			Construction Waste Management	1.6	G												
			Plan														
			SD-11 Closeout Submittals														
			Final Construction Waste	1.8	S												
			Diversion Report														
		35 20 23	SD-02 Shop Drawings														
			None Required														

eProjects WO#: 1783178

Appendix "A"

Sediment Data

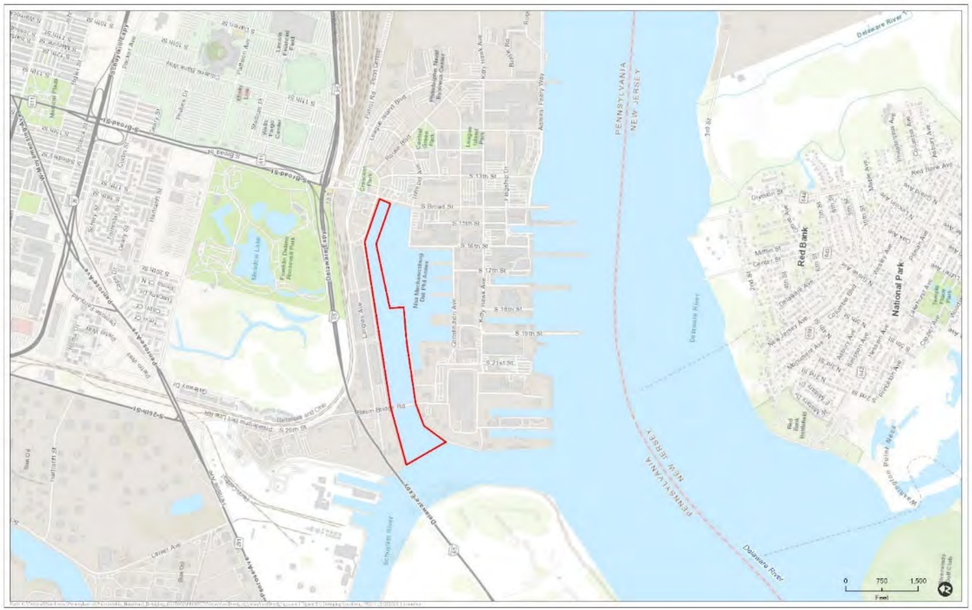
FY26 MAINTENANCE DREDGING AT:
RESERVE BASIN
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA

EXCERPT FROM

EVALUATION OF DREDGED MATERIAL

PHILADELPHIA NAVY YARD
RESERVE BASIN MAINTENANCE DREDGING
PHILADELPHIA, PENNSYLVANIA

FINAL



Prepared for:

Department of the Navy
Naval Facilities Engineering Command (NAVFAC)
Mid-Atlantic Division (MIDLANT)
Norfolk, Virginia 23511

Prepared by:

EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, Maryland 21031
(410) 584-7000

Under Contract to:
HDR, Inc.
4144 Hermitage Point
Virginia Beach, Virginia 23455

August 2025

EVALUATION OF DREDGED MATERIAL

PHILADELPHIA NAVY YARD
RESERVE BASIN MAINTENANCE DREDGING
PHILADELPHIA, PENNSYLVANIA

FINAL REPORT

AUGUST 2025

Prepared for:
Department of the Navy
Naval Facilities Engineering Command (NAVFAC)
Mid-Atlantic Division (MIDLANT)
Norfolk, Virginia 23511

Prepared by:
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle
Suite 400
Hunt Valley, Maryland 21031
(410) 584-7000

Under Contract to:
HDR, Inc.
4144 Hermitage Point
Virginia Beach, Virginia 23455

EXECUTIVE SUMMARY

Naval Support Activity Philadelphia Navy Yard (“Philadelphia Navy Yard”) in Philadelphia, Pennsylvania, provides operationally ready, secure shore infrastructure and installation management for military and civil staff and provides a home base for multiple commands, services, and agencies serving warfighter operations.

The U.S. Navy proposes to perform maintenance dredging of the entrance to, and portions of, the Reserve Basin at the Philadelphia Navy Yard, located on the Schuylkill River at the confluence with the Delaware River (Figure ES-1). Current usage of the Reserve Basin is for mooring of decommissioned vessels. There is no active dismantling of ships in the area.

The proposed maintenance dredging area is approximately 20 acres and is divided into two subareas: Area A (Figure ES-2) and Area B (Figure ES-3). Both areas require maintenance dredging to a depth of -30 feet (ft) mean lower low water (MLLW) plus 2 ft of overdepth allowance, for a total depth characterization and permitting depth of -32 ft MLLW. Approximately 100,000 cubic yards (cy) of material (95,800 cy from Area A and 4,200 cy from Area B) will require dredging and disposal. The project will be contracted by the U.S. Navy and the dredging is expected to occur in mid to late 2025.

The goal of this project was to collect the data necessary to characterize the existing physical and chemical attributes of the material proposed for maintenance dredging and to determine the suitability of the material for available placement options. Available placement options include Whites Rehandling Basin and Confined Disposal Facility (CDF) in Logan Township, New Jersey or Biles Island Upland Disposal Facility in Fall’s Township, Pennsylvania (Figure ES-4).

The project consisted of collecting sediment cores from 15 locations to a depth of -32 ft MLLW using vibracoring technology; collecting site water/elutriate preparation water; conducting analytical testing of bulk sediment composites, site water, and modified elutriates; and evaluating test results with respect to criteria for identified placement options.

ES.1 AVAILABLE PLACEMENT OPTIONS

The placement/disposal options are detailed as follows:

Whites Rehandling Basin and CDF

Whites Rehandling Basin and CDF is located in Logan Township, New Jersey. The facility is owned and operated by Weeks Marine and consists of a partially enclosed open water basin and an adjacent 2,400-acre upland area that includes two upland placement cells (Cell 1 and Cell 2). Dredged material is mechanically dredged, transported, and placed in the rehandling basin using bottom dump scows. The rehandling basin has a capacity of approximately 1.8 million cubic yards. Every 2 to 3 years, material is hydraulically dredged from the rehandling basin and pumped to the upland containment cells. Cells 1 and 2 are approximately 150 and 325 acres in size. As the material dewaters, settles, and consolidates, the effluent is discharged through a permitted outfall from Cell 2 into the rehandling basin.

The New Jersey Department of Environmental Protection (NJDEP) requires testing of bulk sediment to evaluate placement at Whites Rehandling Basin. Bulk sediment data were compared to New Jersey non-residential soil remediation standards (New Jersey Administrative Code 7:26D).

Biles Island CDF

The Biles Island CDF is a 500-acre mining site along the Delaware River just south of Trenton, New Jersey, and Morrisville, Pennsylvania. The facility is owned and operated by Waste Management. It is divided into five cells and is located approximately 37 miles upstream of the Reserve Basin maintenance dredging area. Cell 3 is mainly used for dredging projects in close proximity and for material from downstream that does not pass criteria for Whites Rehandling Basin. Cell 3 is adjacent to the Delaware River and designated for materials from private dredging projects. Material designated for Cell 3 is hydraulically pumped from a scow over the berm into the cell. There is no discharge from this facility.

Analytical testing data were generated in accordance with the screening and evaluation protocols in the Upland Testing Manual (U.S. Army Corps of Engineers 2003). Bulk sediment data were compared to Commonwealth of Pennsylvania non-residential soil standards. Modified elutriate data were compared to Commonwealth of Pennsylvania Water Quality Standards. Additional testing required for this location included Toxicity Characteristic Leaching Procedure (TCLP) and comparison of leachate concentrations to criteria as per 40 Code of Federal Regulations (CFR) Part 261.24. Radioactivity data in aqueous media (modified elutriates) were compared to the Delaware River Basin Commission (DRBC) stream quality objectives as specified in DRBC Water Quality Regulations, 18 CFR Part 410, 4 December 2013 and Federal Drinking Water Standards Radionuclides Rule, 66 Federal Register 76708, 7 December 2000: Volume 65, No. 236. Per- and polyfluoroalkyl substances (PFAS) in aqueous media (site water and modified elutriates) were compared to the Commonwealth of Pennsylvania Maximum Contaminant Levels (MCLs) per 25 PA. CODE CH. 109. Safe Drinking Water PFAS MCL Rule 14 January 2023, as well as to the US Environmental Protection Agency (USEPA) final recommended acute aquatic life criteria and benchmarks for select PFAS (USEPA 2024a).

ES.2 SEDIMENT TESTING

A total of 15 sampling locations were targeted for the sampling and testing program as specified by NJDEP and the Pennsylvania Department of Environmental Protection (Figures ES-2 and ES-3). Individual locations were sub-sampled for volatile organic compounds (VOCs) and total petroleum hydrocarbons (TPH) before being composited. Cores collected from adjacent locations in Area A were used to create a total of six sample composites for the specified analytical testing. The three cores collected from Area B were analyzed as individual whole core composites.

The testing requirements for each placement option are provided in Table ES-1. Concentrations of detected analytes in bulk sediment samples were compared to State of New Jersey non-residential soil standards, Commonwealth of Pennsylvania non-residential soil standards, and TCLP regulatory criteria (40 CFR 261.24).

Results of the bulk sediment evaluation determined the following:

Physical Characteristics and Total Organic Carbon (TOC). Grain size analyses indicated that the sediments were predominantly fine grained throughout dredging Area A (ranging from 75.4 to 96.4 percent silt+clay) and were a mix and fines (ranging from 73.9 to 81.7 silt+clay) in dredging Area B along the edge of Wharf N. Percent solids in the sediments ranged from 32 to 44.4 percent. TOC ranged from 3.8 to 4.8 percent in Area A and from 6.2 to 9.5 percent in the Area B samples.

Metals. Arsenic concentrations exceeded the State of New Jersey non-residential soil standard (19 milligrams per kilogram [mg/kg]) in one composite for dredging Area A (A-07/08; 28 mg/kg) and in each of the three samples in dredging Area B (ranging from 22 to 49 mg/kg). Arsenic concentrations did not exceed Commonwealth of Pennsylvania non-residential soil standards. The lead concentration for one location in Area B (B-4) exceeded both Commonwealth of Pennsylvania and State of New Jersey non-residential soil standards (1,000 and 800 mg/kg, respectively), with a concentration of 5,900 mg/kg.

Polycyclic Aromatic Hydrocarbons (PAHs). Each of the 18 tested PAHs were detected in the Area A and Area B sediment samples. None of the individual PAH concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards. Total PAH concentrations (non-detect [ND] = reporting limit [RL]) ranged from 2.14 to 8.65 mg/kg in dredging Area A from 13.3 to 26.9 mg/kg in dredging Area B.

Polychlorinated Biphenyl (PCB) Congeners and Homologs. Total PCB congeners (ND=0) and total PCB homologs ranged from 0.0097 to 0.19 mg/kg in Area A and ranged from 0.092 to 0.3 mg/kg in Area B. None of the total PCB congener or total PCB homolog concentrations exceeded the State of New Jersey non-residential soil standard of 1.1 mg/kg.

PCB Aroclors. Two PCB Aroclors (PCB-1248 and PCB-1260) were detected in the Reserve Basin sediment samples. None of the detected Aroclor concentrations exceeded the Commonwealth of Pennsylvania non-residential soil standards (46 mg/kg for both PCB-1248 and PCB-1260). Total PCB Aroclor concentrations (ND=RL) ranged from 0.024 to 0.365 mg/kg in Area A and from 0.24 to 0.79 mg/kg in Area B. None of the total PCB Aroclor concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards (1.1 and 46 mg/kg, respectively).

Dioxins/Furan Congeners: 2,3,7,8-TCDD was detected in four composite samples from Area A and in one duplicate composite sample; each concentration was estimated below the laboratory RL. Estimated 2,3,7,8-TCDD concentrations did not exceed Commonwealth of Pennsylvania non-residential soil standards. The dioxin toxicity equivalency quotient (ND=RL) ranged from 0.000021 to 0.000242 mg/kg.

Semivolatile Organic Compounds (SVOCs). Five of the 46 tested SVOCs (benzoic acid, bis(2-ethylhexyl)phthalate, dibenzofuran, methylphenol 3&4, and phenol) were detected in the sediment composites, with the majority of concentrations estimated below the laboratory RL. None of the detected SVOCs exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

VOCs. Four VOCs (2-butanone, chlorodibromomethane, chloroform, and dichlorobromomethane) were detected in the sediment samples. Many of the detected concentrations were estimated below the laboratory RL. None of the detected VOCs exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

Chlorinated Pesticides. Eleven of the 22 tested chlorinated pesticides were detected in the Reserve Basin sediment samples. None of the detected concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

TPH. TPH – gasoline range organics (C6-C10) was detected in samples from four of the 12 locations in Area A (ranging from 1.4 to 76 mg/kg) and in samples from each of the three locations in Area B (ranging from 12 to 15 mg/kg). TPH – diesel range organics (C10-C20) was detected in samples from four of the 12 locations in Area A (ranging from 14 to 900 mg/kg) and in samples from each of the three locations in Area B (ranging from 210 to 1,000 mg/kg). TPH – oil range organics (C20-C34) was detected in samples from nine of the 12 locations in Area A (ranging from 15 to 1,200 mg/kg) and in samples from each of the three locations in Area B (ranging from 370 to 2,400 mg/kg). Overall, the highest concentrations of TPH were detected in samples from A-07, B-2, and B-4.

Radioactivity. Gross alpha was detected in five of the six composite samples from Area A (ranging from 7.2 to 25.6 picocuries per gram [pCi/g]) and in one sample in Area B (B-4) at a concentration of 8.9 pCi/g. Gross beta was detected in in five of the six composite samples from Area A [ranging from 7.2 to 25.6 pCi/g] and in one sample in Area B (B-4) at a concentration of 8.9 pCi/g. each of sediment samples in Area A(ranging from 8.05 to 22.4 pCi/g) and in two of the samples in Area B (ranging from 5.37 to 5.57 pCi/g).

PFAS. Four of the six tested PFAS were detected in the Reserve Basin sediments. None of the detected concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

TCLP. Eight constituents (arsenic, barium, cadmium, chromium, lead, selenium, silver, and chloroform) were detected in at least one of the leachates for Reserve Basin sediment samples. None of the detected concentrations exceeded TCLP criteria.

ES.3 MODIFIED ELUTRIATE TESTING

Modified elutriates were prepared using sediment composites and site water from the Reserve Basin maintenance dredging area. Concentrations of metals and organic constituents in the full strength modified elutriates were compared to the Commonwealth of Pennsylvania maximum/continuous surface water quality standards. Results of the modified elutriate analyses include analyte concentrations in both the total and dissolved fractions. Total concentrations of organics were compared to applicable water quality standards. Either dissolved or total concentrations of metals were compared to applicable water quality standards based on the fraction specified for each metal in the Commonwealth of Pennsylvania standards.

Results of the modified elutriate testing indicated:

- Aluminum exceeded the Commonwealth of Pennsylvania maximum standard (750 micrograms per liter [µg/L]) in each of the total fraction elutriates. The Commonwealth of Pennsylvania maximum standard was exceeded by factors ranging from 3.3 to 62.7.

- Cobalt exceeded the Commonwealth of Pennsylvania continuous criterion (19 µg/L) in six of the nine total fraction elutriates but did not exceed the maximum criterion of 95 µg/L. Cobalt in the total fraction exceeded the continuous criterion by factors ranging from 1.2 to 2.9.

- Three individual PAHs [acenaphthene, phenanthrene, and benzo(a)anthracene] were detected at concentrations that exceeded the Commonwealth of Pennsylvania continuous standard in the total fractions of the modified elutriate. Acenaphthene concentrations in the total fraction of the modified elutriate at sample location B-5 (28 µg/L) exceeded the Commonwealth of Pennsylvania continuous standard (17 µg/L) but did not exceed the Pennsylvania maximum standard (83 µg/L). The phenanthrene concentration (10 µg/L) in the same sample (B-5) exceeded both Commonwealth of Pennsylvania continuous (1 µg/L) and maximum (5 µg/L) standards. Benzo(a)anthracene exceeded the Commonwealth of Pennsylvania continuous standard (0.1 µg/L) in six samples of the total fraction of the modified elutriate by factors ranging from 1.1 to 3.5.

- Total PCB concentrations (ND=0, sum of congeners) in the total fractions of the modified elutriates ranged from 199 to 1,378 nanograms per liter (ng/L) and exceeded the Commonwealth of Pennsylvania continuous standard (14 ng/L) by factors ranging from 14 to 98. Total PCBs calculated based on the sum of homologs had comparable results, ranging from 200 to 1,400 ng/L.

- Total PCB Aroclor concentrations (ND=0) ranged from 0 to 0.29 µg/L and exceeded the Commonwealth of Pennsylvania continuous standard (0.014 µg/L) in seven of the nine elutriates by factors ranging from 3 to 21.

- Two of the 22 tested chlorinated pesticides (4,4'-DDD and 4,4'-DDE) were detected in the total fraction of the modified elutriates at concentrations exceeding Commonwealth of Pennsylvania surface water quality standards.

  — ***4,4'-DDD:*** Concentrations of 4'4'-DDD in the total fraction of the modified elutriates exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in three of the nine samples by factors ranging from 4.2 to 19. Concentrations did not exceed the Pennsylvania maximum standard (1.1 µg/L) in any sample.

  — ***4,4'-DDE:*** Concentrations of 4,4'-DDE in the total fraction of the modified elutriates exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in seven of the nine samples by factors ranging from 2.4 to 24. Concentrations did not exceed the Pennsylvania maximum standard (1.1 µg/L) in any sample.

- Gross alpha was detected in the total fraction of eight of the nine modified elutriates and in the dissolved fraction of three of the nine modified elutriates. Each of the eight total fraction concentrations exceeded the DRBC water quality criteria (3 picocuries per liter [pCi/L]) and seven of the eight total fraction concentrations also exceeded the U.S. Environmental Protection Agency (USEPA) drinking water quality standard (15 pCi/L). Two of the three dissolved fraction concentrations for gross alpha exceeded the DRBC criterion, but did not exceed the USEPA drinking water standard. Gross beta was detected in the total and dissolved fraction of each modified elutriate, but the concentrations did not exceed the USEPA drinking water quality standard (50 pCi/L) or the DRBC water quality criterion (1,000 pCi/L)

- Each of the six tested PFAS was detected in both the total and dissolved fractions for the modified elutriates. Perfluorooctanesulfonic acid (PFOS) concentrations exceeded the Pennsylvania maximum contaminant level of 18 ng/L in both the dissolved and total modified elutriate fractions in two samples (A-0102 and A-0506) by factors ranging from 1.1 to 1.7. Perfluorooctanoic acid (PFOA) concentrations also exceeded the Pennsylvania maximum contaminant level of 14 ng/L in both the dissolved and total fractions of sample A-0102 by factors of 2.3 and 2.4, respectively. None of the detected concentrations of PFAS exceeded the USEPA acute aquatic life criteria.

ES.4 PLACEMENT OPTION ASSESSMENT

Material placement options for the Reserve Basin maintenance dredging sediments include privately owned and operated facilities in the State of New Jersey and Commonwealth of Pennsylvania. Placement/disposal options include Biles Island CDF or Whites Rehandling Basin. Each placement option has facility-specific material acceptance criteria.

Key findings for the proposed placement options are as follows:

<u>Whites Rehandling Basin</u>

A summary of the sediment concentrations that exceed the State of New Jersey non-residential soil criteria is provided in Table ES-2.

- *Arsenic* concentrations in one sample from dredging Area A (A-0708) and each of the three samples from dredging Area B (B-2, B-4, and B-5) exceeded the State of New Jersey non-residential soil standard of 19 mg/kg with concentrations ranging from 22 to 49 mg/kg.

- The *lead* concentration (5,900 mg/kg) in one sample in Area B (B-4) exceeded the New Jersey non-residential soil standard of 800 mg/kg.

<u>Biles Island CDF</u>

A summary of sediment concentrations that exceed the Commonwealth of Pennsylvania non-residential soil criteria and a summary of the modified elutriate concentrations that exceed the Commonwealth of Pennsylvania surface water maximum/continuous water quality standards or other applicable standards (for radioactivity and PFAS) is provided in Table ES-2.

- The *lead* concentration (5,900 mg/kg) in one sediment sample from Area B (B-4) exceeded the Commonwealth of Pennsylvania non-residential soil standard of 1,000 mg/kg (0 to -2 ft).

- *Aluminum* concentrations in the total fraction of each of the nine modified elutriates exceeded the Commonwealth of Pennsylvania maximum surface water quality standard (750 µg/L) by factors ranging from 3.3 to 62.7.

- *Cobalt* concentrations exceeded the Commonwealth of Pennsylvania continuous surface water standard (19 µg/L) in the total fraction in six of the nine modified elutriate samples by factors ranging from 1.2 to 2.9.

- *4,4'-DDD* concentrations exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in the total fraction in three of the nine modified elutriates by factors ranging from 4.2 to 19. Concentrations did not exceed the Commonwealth of Pennsylvania maximum standard (1.1 µg/L) in any sample.

- *4,4'-DDE* concentrations exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in the total fraction of seven of the nine samples by factors ranging from 2.4 to 24. Concentrations did not exceed the Commonwealth of Pennsylvania maximum standard (1.1 µg/L) in any sample.

- Three individual PAHs [acenaphthene, phenanthrene, and benzo(a)anthracene] were detected at concentrations that exceeded the Commonwealth of Pennsylvania continuous standard in the total fractions of the modified elutriate. The a*cenaphthene* concentrations in the total fraction of the modified elutriate at sample location B-5 (28 µg/L) exceeded the Commonwealth of Pennsylvania continuous standard (17 µg/L) but did not exceed the Pennsylvania maximum standard (83 µg/L). The *phenanthrene* concentration (10 µg/L) in the sample location B-5 exceeded both Commonwealth of Pennsylvania continuous (1 µg/L) and maximum (5 µg/L) standards. The *benzo(a)anthracene* concentration exceeded the Commonwealth of Pennsylvania continuous standard (0.1 µg/L) in the total fraction of six of the nine modified elutriates by factors ranging from 1.1 to 3.5.

- *Total PCB Congener* concentrations (ND=0, sum of congeners) in the total fraction of the modified elutriates ranged from 199 to 1,378 ng/L and exceeded the Commonwealth of Pennsylvania continuous standard (14 ng/L) by factors ranging from 14 to 98. Total PCBs calculated based on the sum of homologs had comparable results, ranging from 200 to 1,400 ng/L.

- *Total PCB Aroclor* concentrations (ND=0) ranged from 0 to 0.29 µg/L and exceeded the Commonwealth of Pennsylvania continuous standard (0.014 µg/L) in seven of the nine elutriates by factors ranging from 3 to 21.

- *Gross alpha* was detected in the total fraction of eight of the nine modified elutriates and in the dissolved fraction of three of the nine modified elutriates. Each of the eight total fraction concentrations exceeded the DRBC water quality criteria (3 pCi/L) and seven of the eight total fraction concentrations also exceeded the USEPA drinking water quality standard (15 pCi/L). Two of the three dissolved fraction concentrations for gross alpha exceeded the DRBC criterion but did not exceed the USEPA drinking water standard.

- *PFOS* concentrations exceeded the Pennsylvania MCL of 18 ng/L in both the dissolved and total modified elutriate fractions in two of the nine elutriates samples (A-0102 and A-0506) by factors ranging from 1.1 to 1.7. *PFOA* concentrations also exceeded the Pennsylvania MCL of 14 ng/L in both the dissolved and total fractions of one elutriate (A-0102) by factors of 2.3 and 2.4, respectively. None of the detected concentrations of *PFAS* exceeded the USEPA acute aquatic life criteria.

TABLE ES-1. ANALYTICAL TESTING REQUIREMENTS
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Constituent	Analytical Methods	Whites Rehandling Basin and CDF [(a)]	Biles Island CDF [(b)]
Physical Parameters			
Grain Size	ASTM D422	S	S
pH	EPA 9054D	--	S, W, ME
Water Content	ASTM D2216-90	S	S
Total Solids	SM 2540G	S	S
Total Suspended Solids	SM 2540D	--	W, ME
Inorganic and Organic Parameters			
Chlorinated Pesticides	Low Level SW846 8081B	S	S, W, ME
Volatile Organic Compounds (VOCs)	SW846 8260C	--	S,W, ME
Total Organic Carbon/Dissolved Organic Carbon	Lloyd Kahn / SM 5310C	S	S,W, ME
Metals	SW846 6020A	S	S, W, ME
Mercury	SW846 7471A / SW846 7470A	S	S,W, ME
Trivalent Chromium	SW846 7196A calculation	S	--
Hexavalent Chromium	SW846 7196A	S	--
Dioxin/Furan Congeners[(c)]	EPA 1613B	--	S, W, ME
PFAS Compounds[(d)]	EPA 1633	--	S, W, ME
PCB Congeners (209 congeners)	EPA 1668A	--	S, W, ME
PCB Aroclors	EPA 608 GC/ECD	S	S, W, ME
Total Petroleum Hydrocarbons-DRO (C10 to C20) Total Petroleum Hydrocarbons-GRO (C6 to C10) Total Petroleum Hydrocarbons-ORO (C20 to C34)	SW846 8015D	--	S, W, ME
Semivolatile Organic Compounds (SVOCs) Polycyclic Aromatic Hydrocarbons (PAHs)	Low Level SW846 8270D	S	S, W, ME
Cyanide, Total	SW846 9014	--	S, W, ME
Sulfide	EPA 9030B/9034	--	S, W, ME
Radioactivity (Gross Alpha, Gross Beta, Gamma)	EPA 9310 / 900	--	S, W, ME
TCLP Analysis (Includes Volatiles, Semivolatiles, Pesticides, Herbicides, Metals, Mercury, and TCLP)	SW846 8260C,8270D,8081B, 8151A,6010C,7470A, 1311	--	S

Notes:
(a) As specified by New Jersey Department of Environmental Protection Office of Dredging and Sediment Technology.
(b) As specified by Pennsylvania Department of Environmental Protection Waste Management Program Office. Biles Island Dredging Guidance 20145.
(c) Includes the following 17 dioxin/furan congeners: 2,3,7,8-TCDD, 1,2,3,7,8-PeCDD, 1,2,3,4,7,8-HxCDD, 1,2,3,6,7,8-HxCDD, 1,2,3,7,8,9-HxCDD, 1,2,3,4,6,7,8-HpCDD, OCDD, 2,3,7,8-TCDF, 1,2,3,7,8-PeCDF, 2,3,4,7,8-PeCDF, 1,2,3,4,7,8-HxCDF, 1,2,3,6,7,8-HxCDF, 1,2,3,7,8,9-HxCDF, 2,3,4,6,7,8-HxCDF, 1,2,3,4,6,7,8-HpCDF, 1,2,3,4,7,8,9-HpCDF, OCDF
(d) Includes the following 6 PFAS compounds: Perfluorobutane Sulfonate (PFBS), Perfluorooctane Sulfonate (PFOS), Perfluorooctanoic acid (PFOA), Ammonium salt of hexafluoropropylene oxide dimer acid (HFPO-DA)[GenX], Perfluorohexanesulfonic acid (PFHxS), Perfluorononanoic acid (PFNA)

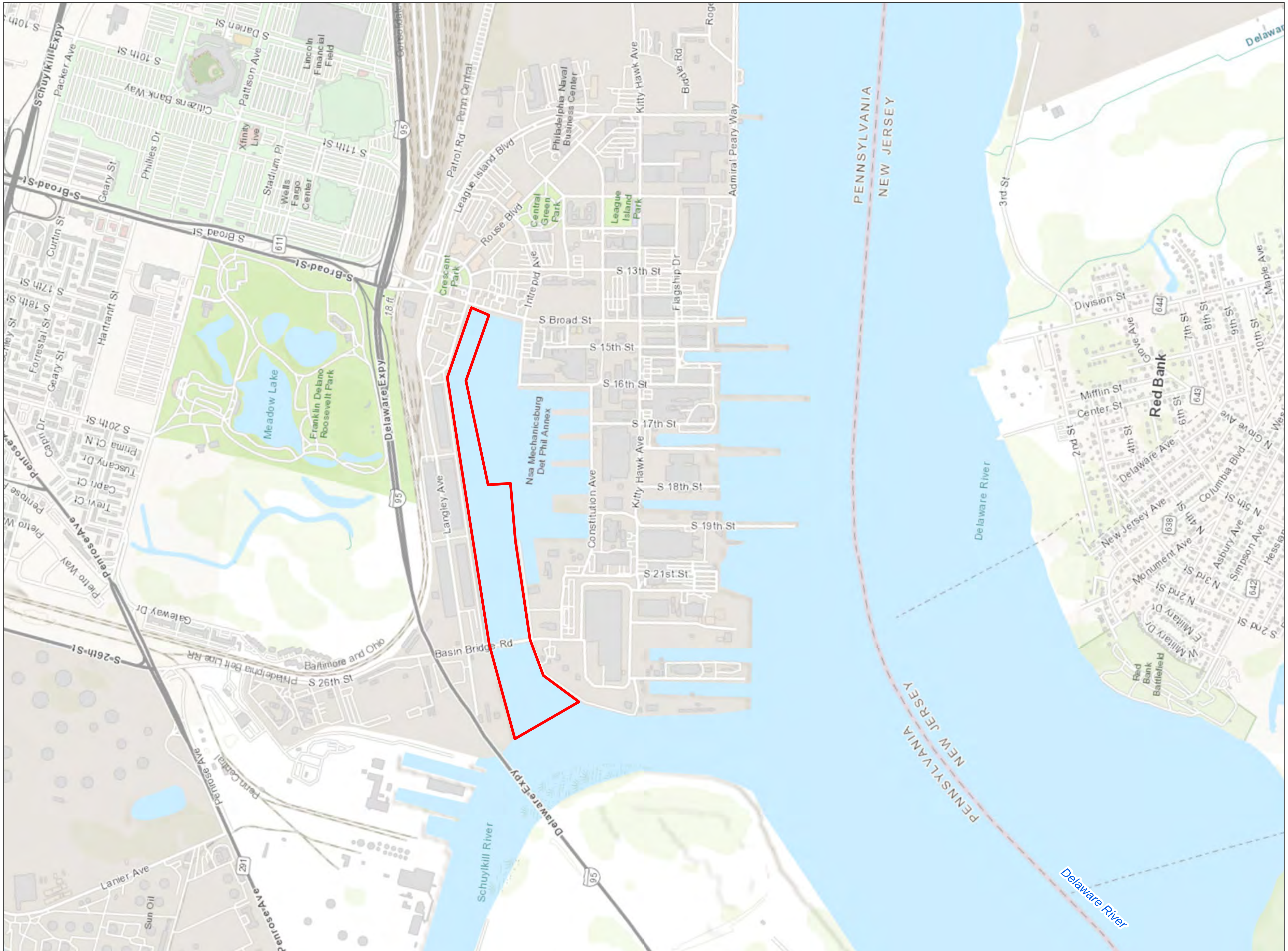
CDF = Confined Disposal Facility
DRO = Diesel Range Organics
GRO = Gasoline Range Organics
ORO = Oil Range Organics
PCB = Polychlorinated Biphenyl
PFAS = Per- and Polyfluoroalkyl Substances
TCLP = Toxicity Characteristic Leaching Procedure
S = sediment; W=site water (total and dissolved fractions); ME = modified elutriates (total and dissolved fractions)

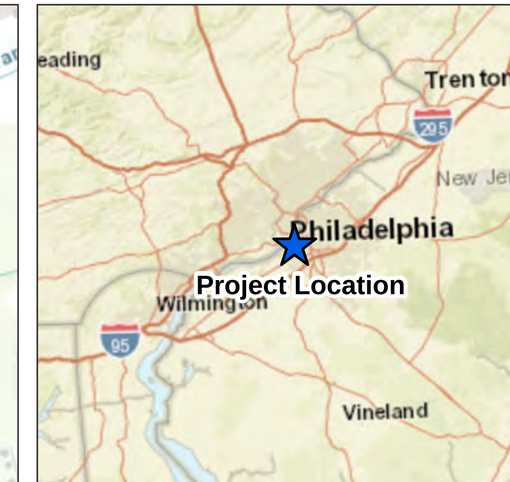
This page left intentionally blank

TABLE ES-2. SUMMARY OF SCREENING CRITERIA EXCEEDANCES BASED ON PLACEMENT OPTION
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

	Whites Rehandling Basin				Biles Island CDF											
	Sediment				Sediment				Elutriate [a]							
Location/Sample ID	NJ Non-Residential Soil Standards (mg/kg)		Detected Concentration (mg/kg)	Magnitude of Exceedance	PA Non-Residential Soil Standards 0 to 2 ft (mg/kg)		Detected Concentration (mg/kg)	Magnitude of Exceedance	PA Maximum WQS - Acute (µg/L)		Detected Concentration (µg/L)	Magnitude of Exceedance	PA Continuous WQS - Chronic (µg/L)		Detected Concentration (µg/L)	Magnitude of Exceedance
PNYRB24-A-0102	--	--	--	--	--	--	--	--	Aluminum (total)	750	47000	62.7	Cobalt (total)	19	39	2.1
													4,4-DDE	0.001	0.0042	4.2
													Total PCBs [b]	0.014	0.405	28.9
PNYRB24-A-0304	--	--	--	--	--	--	--	--	Aluminum (total)	750	24000	32	Cobalt (total)	19	22	1.2
													4,4-DDE	0.001	0.0024	2.4
													Total PCBs [b]	0.014	0.229	16.4
PNYRB24-A-0506	--	--	--	--	--	--	--	--	Aluminum (total)	750	40000	53.3	Cobalt (total)	19	41	2.2
													Benzo(a)anthracene	0.1	0.15	1.5
													4,4-DDE	0.001	0.003	3
													Total PCBs [b]	0.014	0.37	26.4
PNYRB24-A-0708	Arsenic	19	28	1.5	--	--	--	--	Aluminum (total)	750	46000	61.3	Cobalt (total)	19	38	2
													Benzo(a)anthracene	0.1	0.11	1.1
													4,4-DDD	0.001	0.0042	4.2
													4,4-DDE	0.001	0.008	8
													Total PCBs [b]	0.014	0.911	65.1
PNYRB24-A-0910	--	--	--	--	--	--	--	--	Aluminum (total)	750	46000	61.3	Cobalt (total)	19	56	2.9
													Benzo(a)anthracene	0.1	0.11	1.1
													4,4-DDD	0.001	0.0059	5.9
													4,4-DDE	0.001	0.024	24
													Total PCBs [b]	0.014	0.685	48.9
PNYRB24-A-1112	--	--	--	--	--	--	--	--	Aluminum (total)	750	13000	17.3	Benzo(a)anthracene	0.1	0.13	1.3
													4,4-DDE	0.001	0.0028	2.8
													Total PCBs [b]	0.014	0.199	14.2
PNYRB24-B-2	Arsenic	19	49	2.6	--	--	--	--	Aluminum (total)	750	35000	46.7	Cobalt (total)	19	38	2
													Phenanthrene	1	1.1	1.1
													Benzo(a)anthracene	0.1	0.35	3.5
													4,4-DDD	0.001	0.019	19
													4,4-DDE	0.001	0.017	17
													Total PCBs [b]	0.014	1.378	98.4
PNYRB24-B-4	Arsenic	19	44	2.3	Lead	1000	5900	5.9	Aluminum (total)	750	7300	9.7	Benzo(a)anthracene	0.1	0.16	1.6
	Lead	800	5900	7.4									Total PCBs	0.014	0.209	14.9
PNYRB24-B-5	Arsenic	19	22	1.2	--	--	--	--	Aluminum (total)	750	2500	3.3	Acenaphthene	17	28	1.6
									Phenanthrene	5	10	2	Phenanthrene	1	10	10
													Total PCBs [b]	0.014	0.204	14.6

(a) total fraction for organics compared to surface water quality criteria; comparisons for metals based fraction (total or dissolved) specified in surface wtaer quality standards
(b) Total PCB Congeners (ND=0) and total PCB homologs
PA = Pennsylvania
NJ = New Jersey
WQS = water quality standards





Legend

Project Area

0 750 1,500

Feet

Map Date: 7/24/2024
Source: ESRI 2024
Projection: NAD 1983 State Plane Pennsylvania South US Foot

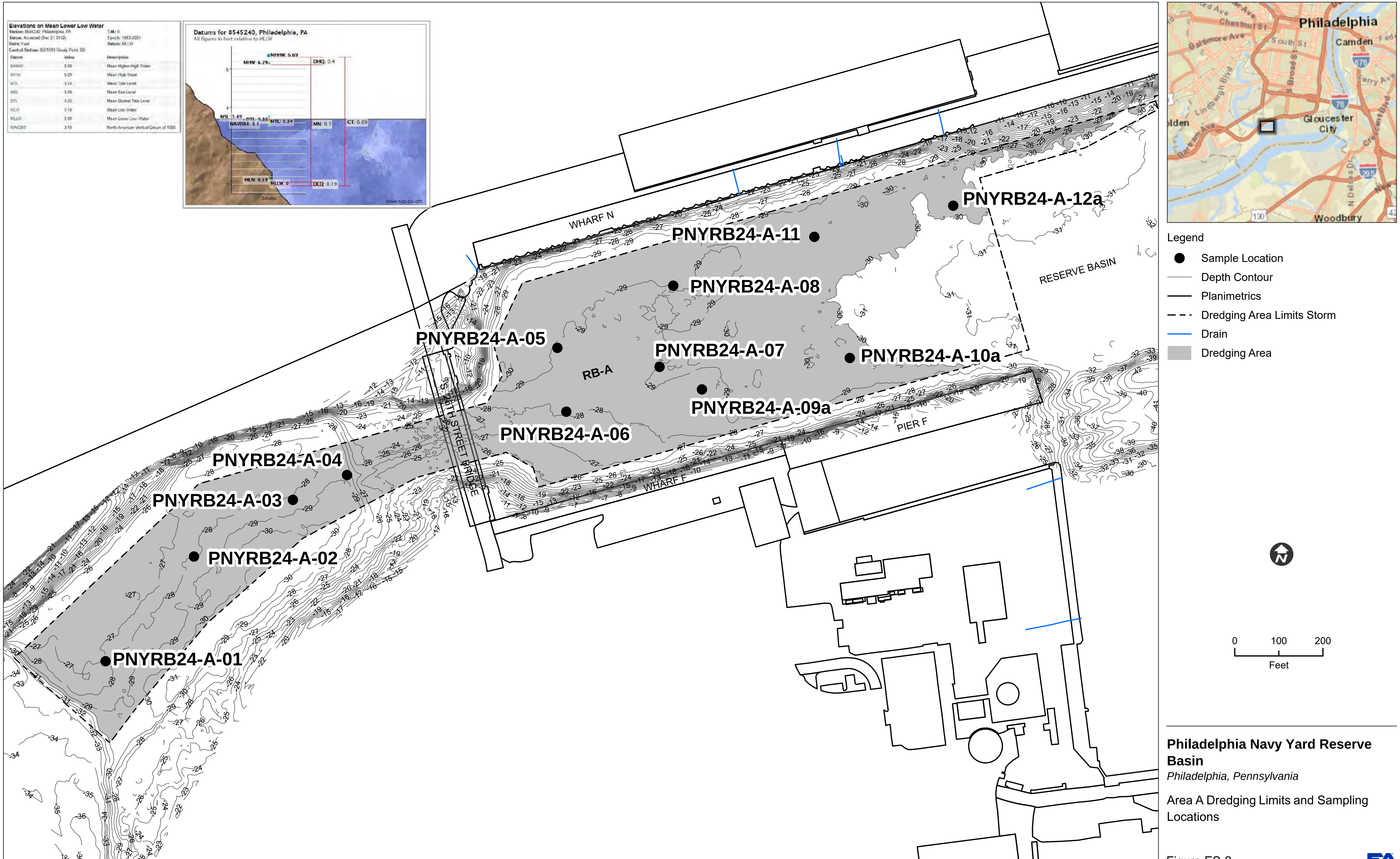
Philadelphia Navy Yard Reserve Basin *Philadelphia, Pennsylvania*

Reserve Basin Maintenance Dredging Project Area

Figure ES-1

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure 1-1 Dredging Location | 7/24/2024 | kburdick

This page left intentionally blank



Philadelphia Navy Yard Reserve Basin

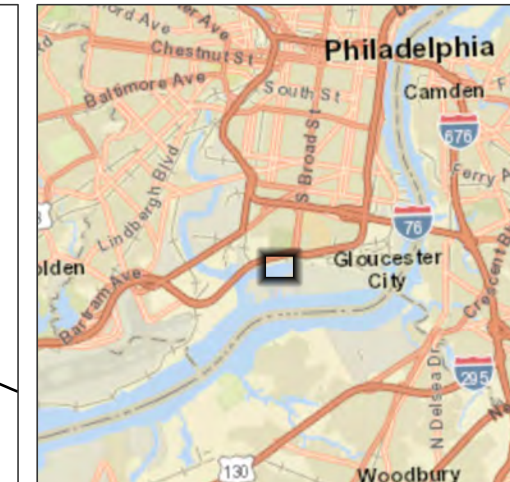
Philadelphia, Pennsylvania

Area A Dredging Limits and Sampling Locations

Figure ES-2

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure X - Proposed Sampling Locations Area A | 3/12/2025 | kburdick

This page left intentionally blank



Legend

- Sample Location
- Depth Contour
- Planimetrics
- Dredging Area Limits
- Storm Drain
- Dredging Area

0 100 200
Feet

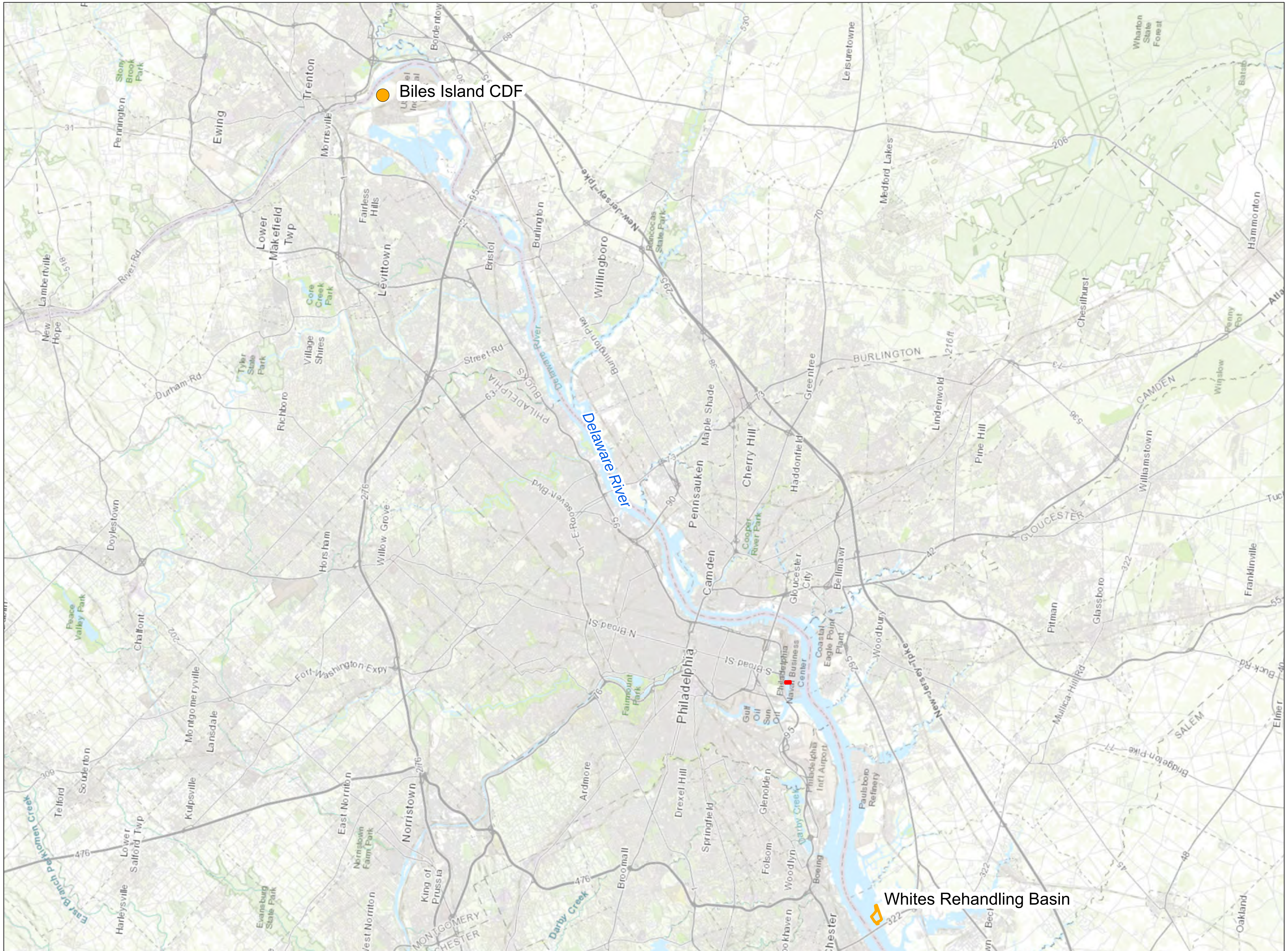
Philadelphia Navy Yard Reserve Basin

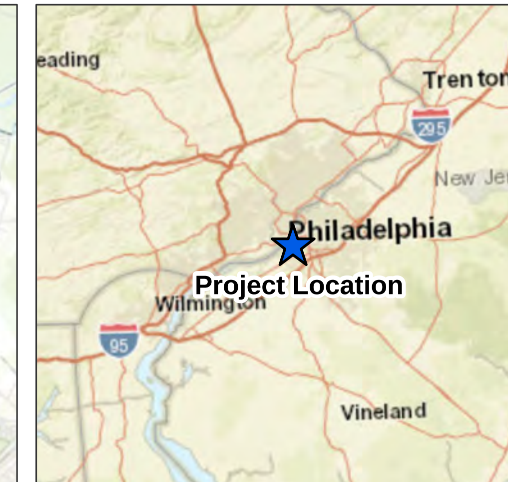
Philadelphia, Pennsylvania

Area B Dredging Limits and Sampling Locations

Figure ES-3

This page left intentionally blank





Legend

- Area of Dredging
- Potential Disposal Locations

0 2.5 5
Miles

Map Date: 7/24/2024
Source: ESRI 2024
Projection: NAD 1983 State Plane Pennsylvania South US Foot

Philadelphia Navy Yard Reserve Basin *Philadelphia, Pennsylvania*

Dredged Material Placement Locations in Pennsylvania and New Jersey

Figure ES-4

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure 1-3 Potential Disposal Locations in PA and NJ | 7/24/2024 | kburdick

This page left intentionally blank

TABLE OF CONTENTS

LIST OF TABLES

LIST OF TABLES (continued)

LIST OF TABLES (continued)

LIST OF TABLES (continued)

LIST OF FIGURES

LIST OF APPENDICES

LIST OF ATTACHMENTS

(included as PDFs in a separate zip file)

LIST OF ACRONYMS AND ABBREVIATIONS

°C	Degrees Celsius
µg/L	Microgram(s) per liter
CAB	Cellulose acetate butyrate
CDF	Confined Disposal Facility
CFR	Code of Federal Regulations
COC	Chain-of-custody
cy	Cubic yard(s)
DRBC	Delaware River Basin Commission
DRO	Diesel range organics
EA	EA Engineering, Science, and Technology, Inc., PBC
ERDC	Engineer Research and Development Center
EET	Eurofins Environment Testing
ft	Foot (feet)
GRO	Gasoline range organics
HFPO-DA	Hexafluoropropylene oxide dimer acid
HPAH	High molecular weight polycyclic aromatic hydrocarbon
ID	Identification
LCS	Laboratory control sample
LPAH	Low molecular weight polycyclic aromatic hydrocarbon
MCL	Maximum contaminant level
MDL	Method detection limit
MET	Modified Elutriate Test
MIDLANT	Mid-Atlantic Division
mg/kg	Milligram(s) per kilogram
mg/L	Milligram(s) per liter
MLLW	Mean lower low water
MS	Matrix spike
MSD	Matrix spike duplicate
NAVFAC	Naval Facilities Engineering Command
ND	Non-detect/not detected
NELAP	National Environmental Laboratory Accreditation Program
ng/L	Nanogram(s) per liter
N.J.A.C.	New Jersey Administrative Code
NJDEP	New Jersey Department of Environmental Protection

LIST OF ACRONYMS AND ABBREVIATIONS (continued)

NOAA	National Oceanic and Atmospheric Administration
ORO	Oil range organics
Pa. Code	Pennsylvania Code
PADEP	Pennsylvania Department of Environmental Protection
PAH	Polycyclic aromatic hydrocarbon
PCB	Polychlorinated biphenyl
pCi/g	Picocurie(s) per gram
pCi/L	Picocurie(s) per liter
PFAS	Per- and polyfluoroalkyl substances
PFBS	Perfluorobutanesulfonic acid
PFNA	Perfluorononanoic acid
PFOA	Perfluorooctanoic acid
PFOS	Perfluorooctanesulfonic acid
pg/L	Picogram(s) per liter
QA	Quality assurance
QAPP	Quality Assurance Project Plan
QC	Quality control
RL	Reporting limit
R/V	Research vessel
SPLP	Synthetic Precipitation Leaching Procedure
SSAP	Sediment Sampling and Analysis Plan
SVOC	Semivolatile organic compound
TCLP	Toxicity Characteristic Leaching Procedure
TEQ	Toxicity equivalency quotient
TPH	Total petroleum hydrocarbons
TOC	Total organic carbon
TSS	Total suspended solids
USACE	U.S. Army Corps of Engineers
USEPA	U.S. Environmental Protection Agency
VOC	Volatile organic compound

This page intentionally left blank

1. PROJECT DESCRIPTION

Naval Support Activity Philadelphia Navy Yard ("Philadelphia Navy Yard") in Philadelphia, Pennsylvania, provides operationally ready, secure shore infrastructure and installation management for military and civil staff and provides a home base for multiple commands, services, and agencies serving warfighter operations. The U.S. Navy proposes to perform maintenance dredging of the entrance to, and portions of, the Reserve Basin at the Philadelphia Navy Yard, located on the Schuylkill River at the confluence with the Delaware River (Figure 1-1). Current usage of the Reserve Basin is for mooring of decommissioned vessels. There is no active dismantling of ships in the area. The goal of this project was to collect the data necessary to characterize the existing physical and chemical attributes of the material proposed for maintenance dredging and to determine the suitability of the material for available placement options at Whites Rehandling Basin and Confined Disposal Facility (CDF) in Logan Township, New Jersey or Biles Island Upland Disposal Facility in Falls Township, Pennsylvania.

The proposed maintenance dredging area is approximately 20 acres and is divided into two subareas: Area A (Figure 1-2) and Area B (Figure 1-3). Both areas require maintenance dredging to a depth of -30 feet (ft) mean lower low water (MLLW) plus 2 ft of overdepth allowance, for a total characterization depth and permitting depth of -32 ft MLLW. Approximately 100,000 cubic yards (cy) of material (95,800 cy from Area A and 4,200 cy from Area B) will require dredging and disposal. The project will be contracted by the U.S. Navy and the dredging is expected to occur in mid to late 2025.

On behalf of Naval Facilities Engineering Systems Command (NAVFAC) Mid-Atlantic Division (MIDLANT), EA Engineering, Science, and Technology, Inc., PBC (EA) was contracted through HDR, Inc. to conduct the dredged material evaluation and permit acquisition activities for the Reserve Basin Maintenance Dredging. The dredged material evaluation consisted of collecting sediment cores to a depth of -32 ft MLLW using vibratory coring methods; collecting site water/elutriate preparation water; conducting analytical testing of bulk sediment, site water, and modified elutriates; and evaluating test results with respect to site-specific criteria for placement at each proposed facility. The data attained have been provided to the New Jersey Department of Environmental Protection (NJDEP), Pennsylvania Department of Environmental Protection (PADEP), and the U.S. Army Corps of Engineers (USACE) Philadelphia District to facilitate approvals and issuance of necessary permits for both dredging and dredged material placement activities.

1.1 PROJECT LOCATION

The Philadelphia Navy Yard Reserve Basin is located in Southeast Pennsylvania on the Schuylkill River, at the confluence with the Delaware River. The proposed maintenance dredging will involve the removal of approximately 100,000 cy of material from two dredging areas (95,800 cy from Area A and 4,200 cy from Area B) to depths of -32 ft MLLW, totaling approximately 20 acres. Previous maintenance dredging activities were performed at the Reserve Basin in 1987 under Permit No. NAPOP-R-87-0122-11. Additional dredging activities occurred in 1990, 1993, and 1995 under the same permit. Following expiration of the 1987 permit, several dredging actions have occurred by USACE in various areas of the basin, with the most recent known dredging in 2015.

1.2 MATERIAL PLACEMENT OPTIONS

Maintenance dredged material from the Reserve Basin dredging areas was evaluated with respect to placement/disposal requirements and criteria for the following facilities:

- White's Rehandling Basin and CDF
- Biles Island CDF

The material placement facilities and requirements are described in the Overview Section 1.4 of the Sediment Sampling and Analysis Plan (SSAP) (EA 2024).

1.2.1 Whites Rehandling Basin and Confined Disposal Facility

Whites Rehandling Basin and CDF is located in Logan Township, New Jersey (Figure 1-4). The facility is owned and operated by Weeks Marine and consists of a partially enclosed open water basin and an adjacent 2,400-acre upland area that includes two upland placement cells (Cell 1 and Cell 2). Dredged material is mechanically dredged, transported, and placed in the rehandling basin using bottom dump scows. The rehandling basin has a capacity of approximately 1.8 million cubic yards. Every 2 to 3 years, material is hydraulically dredged from the rehandling basin and pumped to the upland containment cells. Cells 1 and 2 are approximately 150 and 325 acres in size. As the material dewaters, settles, and consolidates, the effluent is discharged through a permitted outfall from Cell 2 into the rehandling basin.

NJDEP requires testing of bulk sediment to evaluate placement at Whites Rehandling Basin. Bulk sediment data were compared to New Jersey Non-Residential Soil Remediation Standards (New Jersey Administrative Code [N.J.A.C.] 7:26D).

1.2.2 Biles Island Confined Disposal Facility

The Biles Island CDF is a 500-acre mining site along the Delaware River just south of Trenton, New Jersey, and Morrisville, Pennsylvania (Figure 1-4). The facility is divided into five cells and is located approximately 37 miles upstream of the Pier 4 maintenance dredging area. Cell 3 is mainly used for dredging projects in close proximity and for material from downstream that does not pass criteria for Whites Rehandling Basin. Cell 3 is adjacent to the Delaware River and designated for materials from private dredging projects. Material designated for Cell 3 is hydraulically pumped from a scow over the berm into the cell. There is no discharge from this facility. Analytical testing data generated in accordance with the screening and evaluation protocols in the Upland Testing Manual (USACE 2003) also apply to this placement option.

Bulk sediment data were compared to Commonwealth of Pennsylvania non-residential soil standards. Modified elutriate data were compared to Commonwealth of Pennsylvania Water Quality Standards. Additional testing required for this location included Toxicity Characteristic Leaching Procedure (TCLP) and comparison of leachate concentrations to criteria as per 40 Code of Federal Regulations (CFR) Part 261.24. Radioactivity in aqueous media data (modified elutriates) were compared to the Delaware River Basin Commission (DRBC) stream quality objectives as specified in DRBC Water Quality Regulations, 18 CFR Part 410, 4 December 2013 as well as the Federal Drinking Water Standards Radionuclides Rule, 66 Federal Register 76708, 7 December 2000: Volume 65, No. 236. Per- and polyfluoroalkyl substances (PFAS) in aqueous media (modified elutriates) were compared to the Commonwealth of Pennsylvania Maximum Contaminant Levels (MCL) per 25 PA. CODE CH. 109. Safe Drinking Water PFAS MCL Rule 14 January 2023 as well as the US Environmental Protection Agency (USEPA) final recommended acute aquatic life criteria and benchmarks for select PFAS (USEPA 2024a).

1.3 PROJECT OBJECTIVES

The purpose of this project was to collect the data necessary for the characterization of physical and chemical quality of the sediments from the Reserve Basin maintenance dredging area and determine if the material is suitable for placement at White's Rehandling Basin and CDF or Biles Island CDF.

The overall objective of the sampling effort was to obtain and analyze sediment and water samples representative of the area proposed for maintenance. The analytical methods, sample quantities, and sample types are summarized in Table 1-1.

Specific objectives of the dredged material evaluation were to:

- Collect the required volume of sediment and site water for physical and chemical analysis (as appropriate) and modified elutriate preparation.
- Collect sediment cores from specified locations in the Reserve Basin maintenance dredging area within positioning accuracy appropriate for the project objectives.
- Collect sediment cores from 15 locations within the proposed Reserve Basin maintenance dredging area using vibracoring equipment.
- Collect site water for chemical analysis and modified elutriate preparation from one location in the Reserve Basin maintenance dredging area.
- Process cores and create six composite samples for Area A and create three discrete samples for Area B.
- Collect and transfer sediment to appropriate, laboratory-prepared containers and preserve/hold samples for analysis according to protocols that ensure sample integrity.
- Test and characterize sediments for physical characteristics and chemical characteristics and compliance with applicable upland soil standards/criteria.
- Test site water and modified elutriates with regard to chemical characteristics (both total and dissolved fractions) and compliance with applicable Pennsylvania water quality standards.

- Evaluate physical and chemical data with respect to the screening criteria for placement options for the maintenance material: White's Rehandling Basin and Biles Island CDF.

1.4 STUDY DESIGN AND TECHNICAL APPROACH

The technical approach for the sampling/testing program was designed to characterize the maintenance material with respect to placement/disposal requirements for the facilities listed in Overview Section 1.4 of the SSAP. The technical approach was based on the requirements specified in the NJDEP SSAP form (Appendix A) that was approved by NJDEP and PADEP. This form specified the sampling locations and testing requirements for White's Rehandling Basin and for Biles Island CDF. PADEP also reviewed and approved the SSAP (EA 2024).

A total of 15 sampling/coring locations were sampled. Cores from 12 locations in Area A were composited to create six composite samples for the sampling and testing program. Cores from each of three locations in Area B were analyzed as discrete samples. Target coordinates and elevations for each sample location are presented in Table 1-2. Sediment cores were collected using vibratory coring, with a maximum recovery depth approximately 5 ft below the sediment-water interface to target depth of -32 ft MLLW (-30 ft plus 2-ft allowable overdepth).

1.4.1 Field Sampling Program

The field sampling program and sample compositing scheme for the project is summarized in Table 1-2 and included the following tasks:

- Fifteen locations in the Reserve Basin maintenance dredging area were sampled using vibracoring equipment. Fifteen individual samples were analyzed for volatile organic compounds (VOCs); six composite and three discrete core samples were analyzed for physical/chemical constituents and modified elutriate analysis.

- Site water was collected from one location (PNYRB24-A-11) for chemical analysis and modified elutriate preparation.

1.4.2 Analytical Testing of Bulk Sediment, Site Water, and Modified Elutriates

Bulk sediment composites were tested for physical and chemical characteristics. The analytical methods for sediment samples are summarized in Table 1-1. Analytical methods, target analytes, holding times, reporting limits (RLs), and quality assurance (QA)/quality control (QC) protocols are provided in Chapter 3 and detailed in the Analytical Quality Assurance Project Plan (QAPP) (Attachment II of the SSAP, EA 2024). Analytical testing of the bulk sediment was conducted by Eurofins Environment Testing (EET) located in Pittsburgh, Pennsylvania.

The analytical methods for aqueous samples (site water and elutriates) are summarized in Table 1-1. Modified elutriates were prepared using site water and each sediment sample. Analytical methods, target analytes, holding times, RLs, and laboratory QA/QC protocols are provided in Chapter 3 and detailed in the Analytical QAPP (Attachment II of the SSAP, EA 2024). Analytical testing of the site water/elutriate preparation water and modified elutriates (total and dissolved concentrations) was conducted by EET located in Pittsburgh, Pennsylvania.

The analytical testing program (Table 1-1) included the following tasks:

- Physical and chemical analysis of bulk sediment;
- Chemical analysis of site water/elutriate preparation water (total and dissolved concentrations); and
- Preparation and chemical analysis of modified elutriates (total and dissolved concentrations).

A list of the sample identifications (IDs) for the Reserve Basin maintenance dredging area is provided in Table 1-2. In addition to sediment, water, and modified elutriate samples, QC samples were submitted to the laboratory. Field blanks, equipment blanks, and matrix spike (MS)/matrix spike duplicates (MSDs), and field/composite duplicates samples were analyzed. Analytical methods, target analytes, holding times, RLs, and laboratory QA/QC protocols are described in Chapter 3 and detailed in the Analytical QAPP, Attachment II of the SSAP (EA 2024).

1.5 DATA SCREENING AND ANALYSIS

Data screening and analysis included the following tasks:

- Chemical concentrations in bulk sediment composites were compared to the State of New Jersey and Commonwealth of Pennsylvania Non-Residential Soil Remediation Standards.
- Chemical concentrations in site water and modified elutriate samples were compared to the Commonwealth of Pennsylvania Water Quality Standards.
- Concentrations of chemical constituents in the TCLP leachate were compared to maximum concentrations of contaminants for toxicity characteristics (40 CFR 261.24).
- Radioactivity data (modified elutriates) were compared to DRBC stream quality objectives as specified in DRBC Water Quality Regulations, 18 CFR Part 410, 4 December 2013 and Federal Drinking Water Standards Radionuclides Rule, 66 Federal Register 76708, 7 December 2000: Volume 65, No. 236.
- PFAS in aqueous media (modified elutriates) were compared to the Commonwealth of Pennsylvania MCLs per 25 PA. CODE CH. 109. Safe Drinking Water PFAS MCL Rule 14 January 2023 as well as the USEPA final recommended acute aquatic life criteria and benchmarks for select PFAS (USEPA 2024a).
- Evaluation of chemical data in relation to screening criteria for each sample type and disposal location.

The summary of the data screening for each potential placement/disposal location is provided in Table 1-3.

1.6 REPORT ORGANIZATION

This data report contains a comprehensive summary of the sampling program and the results of bulk sediment, site water, and modified elutriate testing for the Reserve Basin maintenance dredging area. An overview of the field sampling and sample processing program is provided in Chapter 2. Chapter 3 presents the analytical testing methods, and Chapter 4 presents the data analysis techniques used in the interpretation of results. Bulk sediment chemistry results are provided in Chapter 5. The modified elutriate chemical data are provided in Chapter 6. A data summary in relation to requirements for various placement/disposal options is presented in Chapter 7. References cited in the report are provided in Chapter 8.

1.7 REPORT APPENDICES AND ATTACHMENTS

The following data are incorporated into the appendices and attachments to the report. The attachments are included only as portable document format (PDF) files in a separate zip file format.

Appendix A – NJDEP SSAP Form. A copy of the NJDEP SSAP form and correspondence with NJDEP are provided in this appendix.

Appendix B – Field Logbooks. A copy of the field logbooks that were maintained during the sediment/water sampling are provided in this appendix.

Appendix C – Core Logs and Photographs. A copy of the core processing logbooks, core logs, and core photographs are provided in this appendix.

Appendix D – Equipment Blank and Field Blank Tables. Data tables with the results of the equipment blanks and field blanks are provided in this appendix.

Appendix E– Synthetic Precipitation Leaching Procedure (SPLP) Data. Data tables and the analytical data report for the SPLP PFAS testing are provided in this appendix.

Attachment I – NELAP Certificates for Eurofins. Copies of the laboratory accreditation certificates are provided in this attachment.

Attachment II – Sediment Analytical Reports. Copies of the laboratory (EET) sediment analytical data reports (including laboratory QA/QC documentation) are provided in this attachment (including sediment field blank).

Attachment III – Site Water Analytical Reports. Copies of the laboratory (EET) analytical data reports (including laboratory QA/QC documentation) are provided in this attachment (including equipment and site water field blank).

Attachment IV – Modified Elutriate Analytical Reports. Copies of the laboratory (EET) analytical data reports (including laboratory QA/QC documentation) are provided in this attachment.

TABLE 1-1. ANALYTICAL TESTING REQUIREMENTS
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Constituent	Analytical Methods	Whites Rehandling Basin and CDF [(a)]	Biles Island CDF [(b)]
Physical Parameters			
Grain Size	ASTM D422	S	S
pH	EPA 9054D	--	S, W, ME
Water Content	ASTM D2216-90	S	S
Total Solids	SM 2540G	S	S
Total Suspended Solids	SM 2540D	--	W, ME
Inorganic and Organic Parameters			
Chlorinated Pesticides	Low Level SW846 8081B	S	S, W, ME
Volatile Organic Compounds (VOCs)	SW846 8260C	--	S,W, ME
Total Organic Carbon/Dissolved Organic Carbon	Lloyd Kahn / SM 5310C	S	S,W, ME
Metals	SW846 6020A	S	S, W, ME
Mercury	SW846 7471A / SW846 7470A	S	S,W, ME
Trivalent Chromium	SW846 7196A calculation	S	--
Hexavalent Chromium	SW846 7196A	S	--
Dioxin/Furan Congeners[(c)]	EPA 1613B	--	S, W, ME
PFAS Compounds[(d)]	EPA 1633	--	S, W, ME
PCB Congeners (209 congeners)	EPA 1668A	--	S, W, ME
PCB Aroclors	EPA 608 GC/ECD	S	S, W, ME
Total Petroleum Hydrocarbons-DRO (C10 to C20) Total Petroleum Hydrocarbons-GRO (C6 to C10) Total Petroleum Hydrocarbons-ORO (C20 to C34)	SW846 8015D	--	S, W, ME
Semivolatile Organic Compounds (SVOCs) Polycyclic Aromatic Hydrocarbons (PAHs)	Low Level SW846 8270D	S	S, W, ME
Cyanide, Total	SW846 9014	--	S, W, ME
Sulfide	EPA 9030B/9034	--	S, W, ME
Radioactivity (Gross Alpha, Gross Beta, Gamma)	EPA 9310 / 900	--	S, W, ME
TCLP Analysis (Includes Volatiles, Semivolatiles, Pesticides, Herbicides, Metals, Mercury, and TCLP)	SW846 8260C,8270D,8081B, 8151A,6010C,7470A, 1311	--	S

Notes:
(a) As specified by New Jersey Department of Environmental Protection Office of Dredging and Sediment Technology.
(b) As specified by Pennsylvania Department of Environmental Protection Waste Management Program Office. Biles Island Dredging Guidance 20145.
(c) Includes the following 17 dioxin/furan congeners: 2,3,7,8-TCDD, 1,2,3,7,8-PeCDD, 1,2,3,4,7,8-HxCDD, 1,2,3,6,7,8-HxCDD, 1,2,3,7,8,9-HxCDD, 1,2,3,4,6,7,8-HpCDD, OCDD, 2,3,7,8-TCDF, 1,2,3,7,8-PeCDF, 2,3,4,7,8-PeCDF, 1,2,3,4,7,8-HxCDF, 1,2,3,6,7,8-HxCDF, 1,2,3,7,8,9-HxCDF, 2,3,4,6,7,8-HxCDF, 1,2,3,4,6,7,8-HpCDF, 1,2,3,4,7,8,9-HpCDF, OCDF
(d) Includes the following 6 PFAS compounds: Perfluorobutane Sulfonate (PFBS), Perfluorooctane Sulfonate (PFOS), Perfluorooctanoic acid (PFOA), Ammonium salt of hexafluoropropylene oxide dimer acid (HFPO-DA)[GenX], Perfluorohexanesulfonic acid (PFHxS), Perfluorononanoic acid (PFNA)

CDF = Confined Disposal Facility
DRO = Diesel Range Organics
GRO = Gasoline Range Organics
ORO = Oil Range Organics
PCB = Polychlorinated Biphenyl
PFAS = Per- and Polyfluoroalkyl Substances
TCLP = Toxicity Characteristic Leaching Procedure
S = sediment; W=site water (total and dissolved fractions); ME = modified elutriates (total and dissolved fractions)

This page left intentionally blank

TABLE 1-2. SAMPLE IDENTIFICATION AND COMPOSITING SCHEME
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA PENNSYLVANIA (2024)

Sampling Location[1]	Sediment Volatile/TPH Sample ID	Sediment Discrete or Composite Sample ID	Site Water Sample ID	Modified Elutriate Sample ID	
				Total	Dissolved
Dredging Area A					
PNYRB24-A-01	PNYRB24-A-01-SED	PNYRB24-A-0102-SED	PNYRB24-WAT-T PNYRB24-WAT-D	PNYRB24-A-0102-MET-T	PNYRB24-A-0102-MET-D
PNYRB24-A-02	PNYRB24-A-02-SED				
PNYRB24-A-03	PNYRB24-A-03-SED	PNYRB24-A-0304-SED		PNYRB24-A-0304-MET-T	PNYRB24-A-0304-MET-D
PNYRB24-A-04	PNYRB24-A-04-SED				
PNYRB24-A-05	PNYRB24-A-05-SED	PNYRB24-A-0506-SED		PNYRB24-A-0506-MET-T	PNYRB24-A-0506-MET-D
PNYRB24-A-06	PNYRB24-A-06-SED				
PNYRB24-A-07	PNYRB24-A-07-SED	PNYRB24-A-0708-SED		PNYRB24-A-0708-MET-T	PNYRB24-A-0708-MET-D
PNYRB24-A-08	PNYRB24-A-08-SED				
PNYRB24-A-09a	PNYRB24-A-09-SED	PNYRB24-A-0910-SED		PNYRB24-A-0910-MET-T	PNYRB24-A-0910-MET-D
PNYRB24-A-10a	PNYRB24-A-10-SED				
PNYRB24-A-11**	PNYRB24-A-11-SED	PNYRB24-A-1112-SED		PNYRB24-A-1112-MET-T	PNYRB24-A-1112-MET-D
PNYRB24-A-12a	PNYRB24-A-12-SED				
Dredging Area B					
PNYRB24-B-2a	PNYRB24-B-2-SED			PNYRB24-B-2-MET-T	PNYRB24-B-2-MET-D
PNYRB24-B-4a	PNYRB24-B-4-SED			PNYRB24-B-4-MET-T	PNYRB24-B-4-MET-D
PNYRB24-B-5a	PNYRB24-B-5-SED			PNYRB24-B-5-MET-T	PNYRB24-B-5-MET-D

Notes:
**Water collection location
Sample IDs with an "a" designation were adjusted from their original proposed locations to be avoid vessels moored within the basin. Adjustments were coordinated with NJDEP.

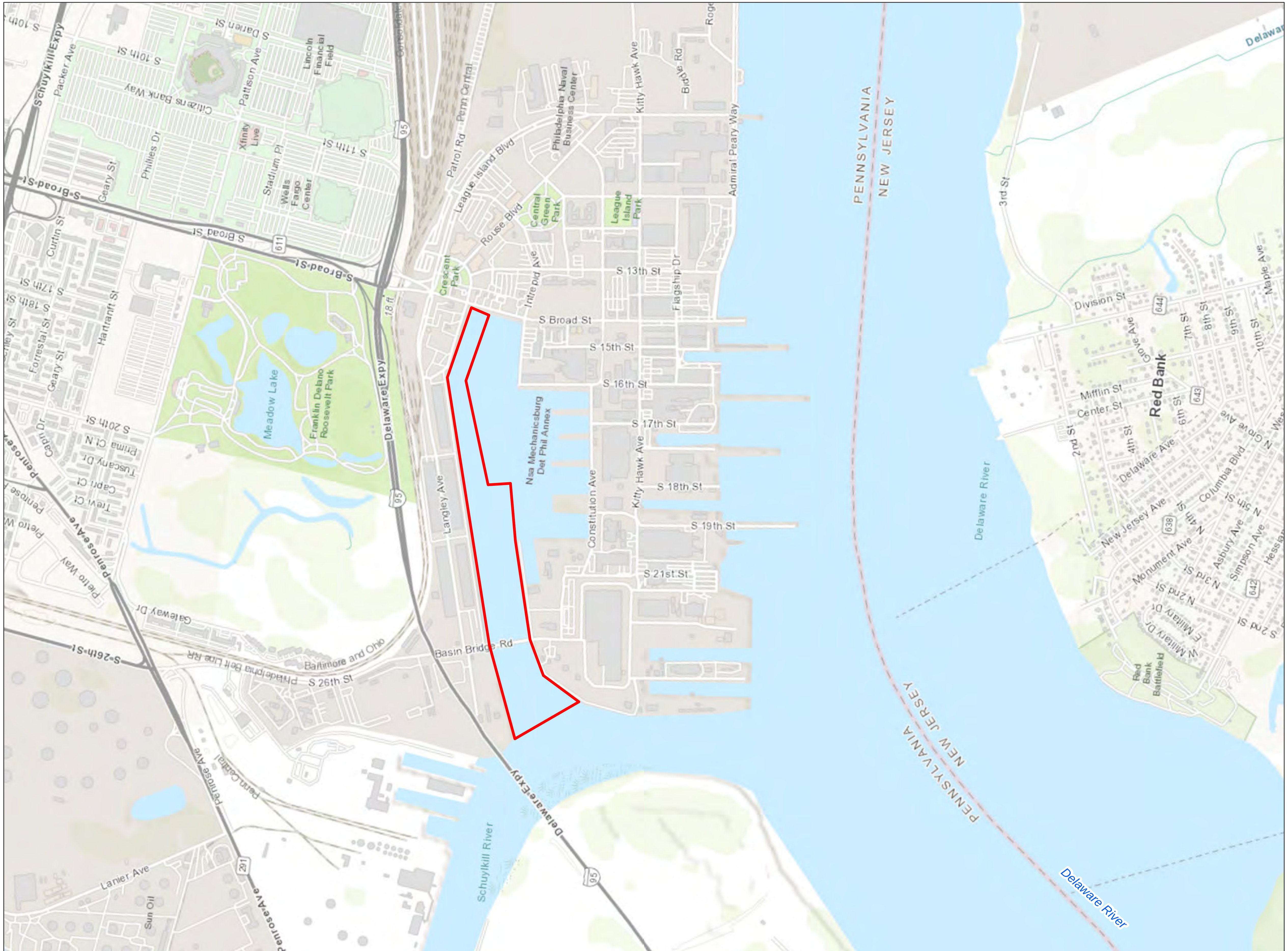
This page left intentionally blank

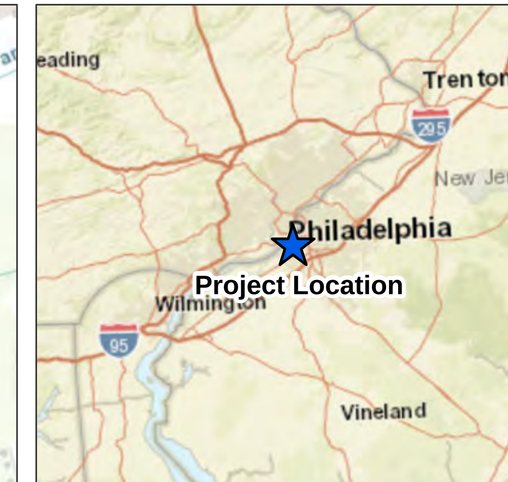
TABLE 1-3. SUMMARY OF DATA SCREENING REQUIREMENTS FOR PLACEMENT OPTIONS
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD
PHILADELPHIA, PENNSYLVANIA (2024)

Screening Criteria / Standards	Source	Whites Rehandling Basin and CDF	Biles Island CDF
NJ Non-Residential Soil	NJAC 7:26D	Sediment	--
PA Non-Residential Soil	25 Pa. Code § 250 Table 3a and 4a	--	Sediment
PA Surface Water	25 Pa. Code § 93.7	--	Modified Elutriates
TCLP	40 CFR 261.24	--	Sediment

Notes:
CDF = Confined Disposal Facility
CFR = Code of Federal Regulations
NJ = New Jersey
NJAC = New Jersey Administrative Code
PA = Pennsylvania
TCLP = Toxicity Characteristic Leaching Procedures

This page left intentionally blank





Legend

Project Area

0 750 1,500
Feet

Map Date: 7/24/2024
Source: ESRI 2024
Projection: NAD 1983 State Plane Pennsylvania South US Foot

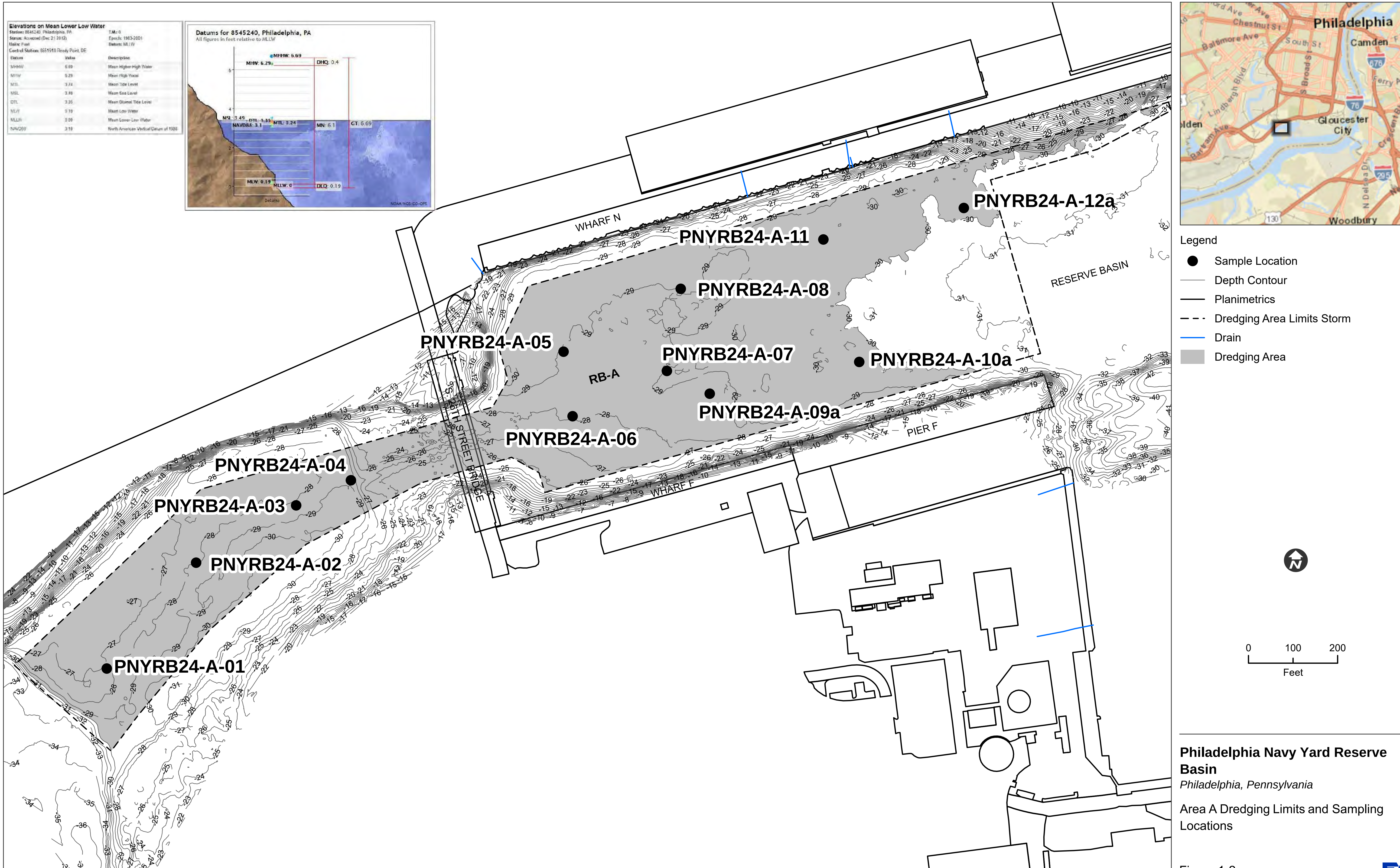
Philadelphia Navy Yard Reserve Basin *Philadelphia, Pennsylvania*

Reserve Basin Maintenance Dredging Project Area

Figure 1-1

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure 1-1 Dredging Location | 7/24/2024 | kburdick

This page left intentionally blank



Legend

- Sample Location
- Depth Contour
- Planimetrics
- Dredging Area Limits Storm
- Drain
- Dredging Area

0 100 200

Feet

Philadelphia Navy Yard Reserve Basin

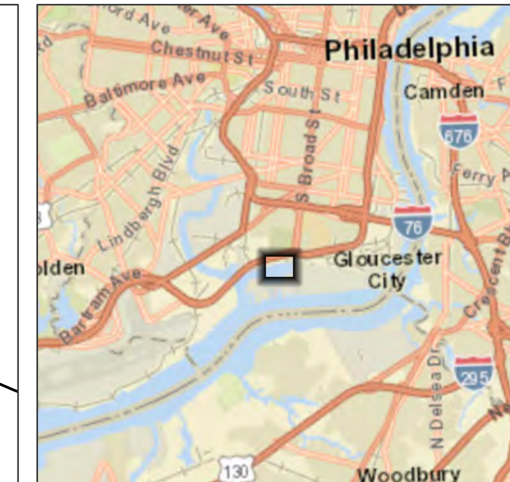
Philadelphia, Pennsylvania

Area A Dredging Limits and Sampling Locations

Figure 1-2

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure X - Proposed Sampling Locations Area A | 3/12/2025 | kburdick

This page left intentionally blank



Legend

- Sample Location
- Depth Contour
- Planimetrics
- Dredging Area Limits
- Storm Drain
- Dredging Area

0 100 200
Feet

Philadelphia Navy Yard Reserve Basin

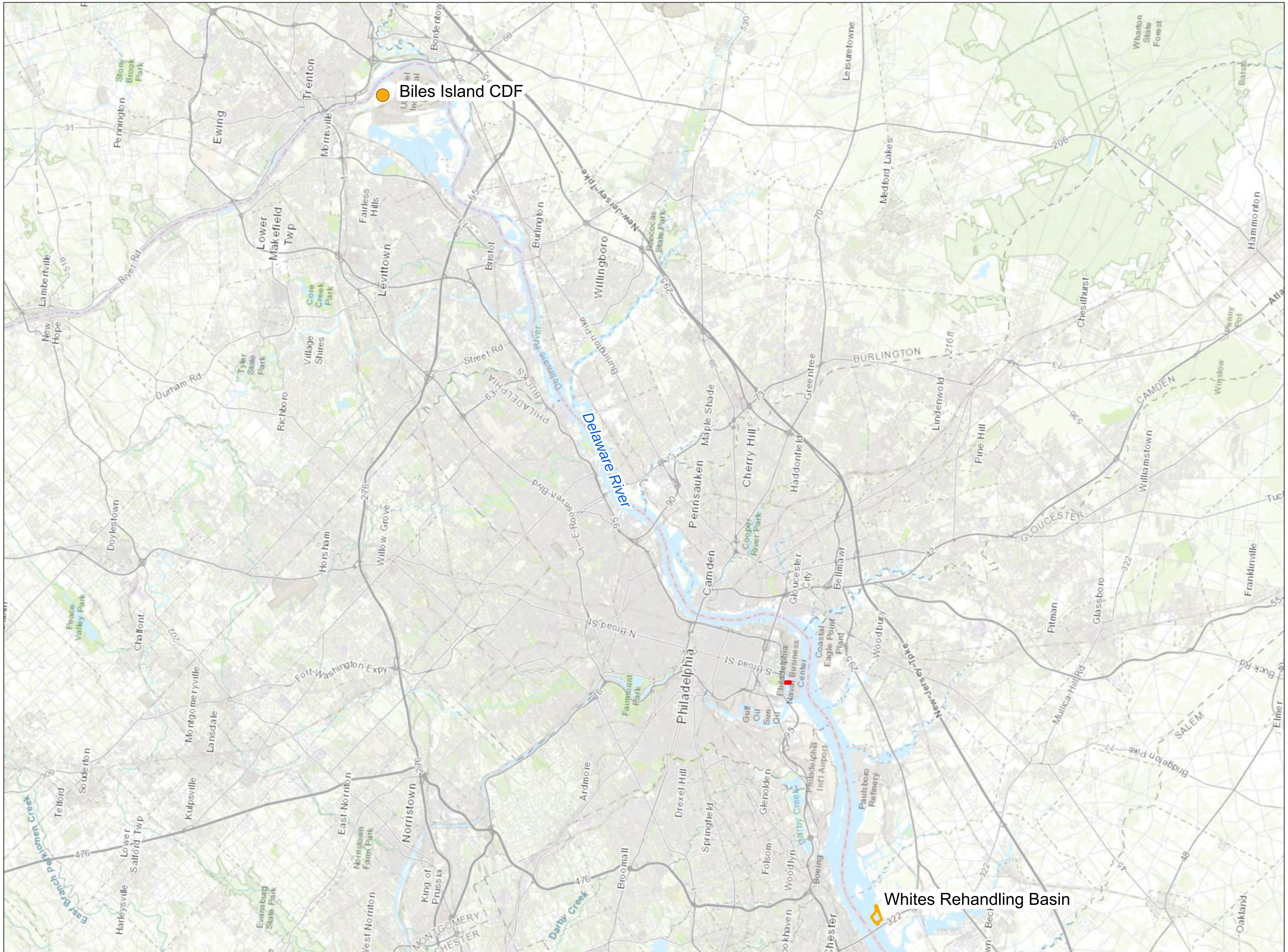
Philadelphia, Pennsylvania

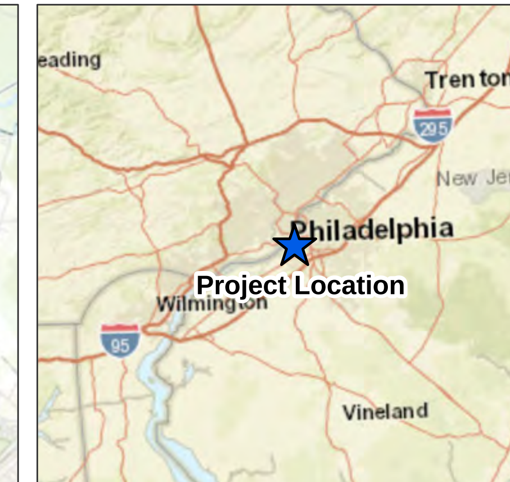
Area B Dredging Limits and Sampling Locations

Figure 1-3

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure X - Proposed Sampling Locations Area B | 3/12/2025 | kburdick

This page left intentionally blank





Legend

- Area of Dredging
- Potential Disposal Locations

0 2.5 5

Miles

Map Date: 7/24/2024
Source: ESRI 2024
Projection: NAD 1983 State Plane Pennsylvania South US Foot

Philadelphia Navy Yard Reserve Basin *Philadelphia, Pennsylvania*

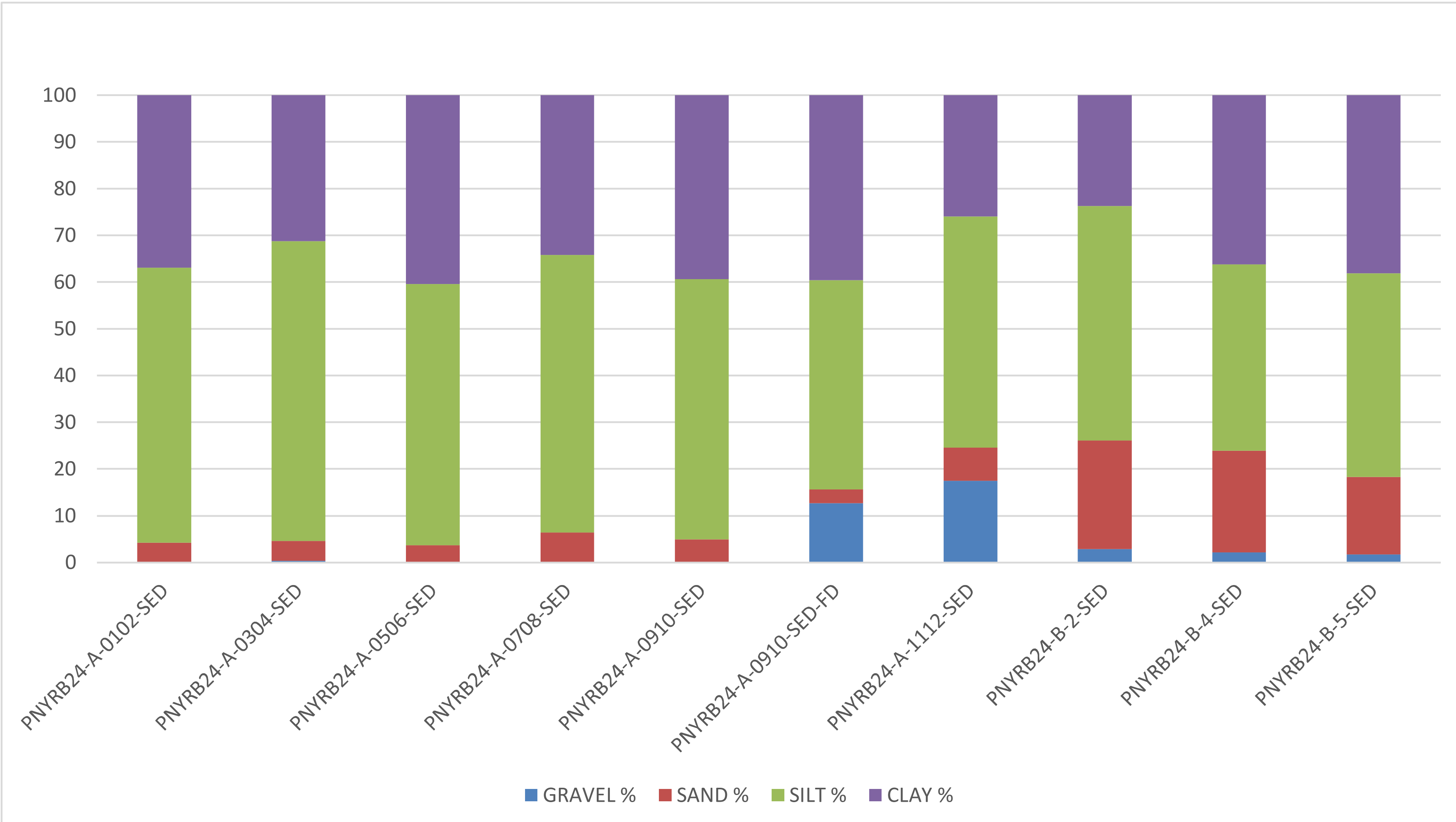
Dredged Material Placement Locations in Pennsylvania and New Jersey

Figure 1-4

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure 1-3 Potential Disposal Locations in PA and NJ | 7/24/2024 | kburdick

This page left intentionally blank

Figure 1-5. Grain Size Distribution for Sediment Samples
Reserve Basin Maintenance Dredging, Philadelphia Navy Yard, Philadelphia, Pennsylvania (December 2024)



2. FIELD SAMPLING AND SAMPLE PROCESSING

The field sampling program was conducted in accordance with the project SSAP (EA 2024). Mobilization for the sampling occurred on 2 December 2024. Sediment collection in the Reserve Basin project area began on 3 December 2024 and continued through 6 December 2024. Cores were processed 5 December through 8 December 2024. Site water/elutriate preparation water samples were collected on 5 December 2024. Coring location coordinates, core penetration depths, and sediment recovery information are provided in Table 2-1.

2.1 SAMPLING OBJECTIVES

The objectives of the field sampling and sample processing efforts included:

- Collection of sediment cores from 15 individual locations utilizing vibracoring equipment to target depths/elevations and within positioning accuracy appropriate for the project objectives;

- Processing and homogenization of sediment from multiple locations according to Table 1-2 to create six composites and three discrete samples that were submitted for bulk sediment physical and chemical analysis and modified elutriate preparation.

- Collection of the required volume of site water from one designated location within the Reserve Basin dredging area for chemical analysis and standard elutriate preparation;

- Measuring and recording water quality information (temperature, salinity, pH, dissolved oxygen, and turbidity) at the water collection location;

- Transfer of water and sediment to appropriate laboratory-prepared containers meeting holding times for analysis according to protocols that ensure sample integrity;

- Completion of appropriate chain-of-custody (COC) documentation.

2.2 SAMPLING LOCATIONS AND ELEVATION DETERMINATION

Sampling locations were specified by NJDEP and PADEP in the approved SSAP form. Within the Reserve Basin, 12 locations were sampled within dredging Area A and 3 locations were sampled within dredging Area B (Figures 1-2 and 1-3, respectively). Northing and easting coordinates are provided in Table 1-2. In the field, Station PNYRB-24-B-2a was shifted approximately 5 ft to the west from the target location due to immediate refusal with the core barrel. Station PNYRB-24-B-4a was blocked by U.S. Navy Barge YC1626. As a result, vibracoring was moved approximately 11 ft southwest for the target location. Station PNYRB-24-A-09 was shifted approximately 28 ft to the south of the target locations in order to collect sufficient recoverable material.

Navigation and positioning were accomplished using a Trimble Differential Global Positioning System (sub-meter accurate) interfaced with HYPACK for horizontal positioning, and a Furuno fathometer (verified by lead line) for determination of water depth. A Trimble R12i Global

Satellite Navigation System interfaced with the Trimble RTX correction network (accurate to ±5 centimeters) was utilized to determine water level and sediment surface elevations. Station locating followed NJDEP (2017) *Data Collection Standards for GIS Data Development.*

Tide corrections for surface elevations, recorded in MLLW, were based off real-time data from the National Oceanic and Atmospheric Administration (NOAA) Center for Oceanographic Products and Services Station 8545240, Philadelphia, Pennsylvania. This tide station is located approximately 5 nautical miles upstream of the work site at the U.S. Coast Guard Station.

2.3 IN SITU WATER QUALITY MEASUREMENTS

Water quality measurements were recorded in situ at the site water collection location using a YSI water quality sonde. Water temperature (degrees Celsius [°C]), salinity (parts per thousand), pH, turbidity (nephelometric turbidity units), and dissolved oxygen (milligrams per liter [mg/L]) measurements were continuously recorded and reported at 5-ft depth increments (Table 2-2).

2.4 SAMPLING VESSEL/PLATFORM

Sediment coring and site water collections were performed aboard the research vessel (R/V) *Recovery* owned and operated by a subcontractor, Athena Technologies, LLC. The R/V *Recovery* is a 32-ft research vessel with a moon pool in the center of the deck to support coring operations. The R/V *Recovery* mobilized to the sampling locations as discussed in the field sampling plan and a triple anchor system was used to stabilize the vessel over the target location.

2.4.1 Sediment Core Collection – Vibracoring

Athena Technologies, LLC provided and operated a concrete-type vibratory coring unit capable of penetrating up to 5 ft below the sediment-water interface. Cores were collected in accordance with the *NJDEP Field Sampling Procedures Manual* (NJDEP 2024). The core barrel contained a single, internal cellulose acetate butyrate (CAB) clear core liner with a 2.875-inch inner diameter. The liner served as a sample container that is held in position by a decontaminated stainless steel core catcher and cutter fixed to the core barrel. At the time of collection, the core barrel and liner were lowered using a tripod and winch system positioned over the moonpool. The vibratory coring was then advanced into the sediment column by delivering mechanical vibration down the pipe to generate high-frequency vibrations that liquefy the sediments at the edge of the cutter and allowed the barrel to cut a continuous cross-section of sediment. The pressure and frequency of vibration was controlled from the R/V *Recovery* to optimize the coring system performance and sample integrity.

Upon reaching the desired penetration depth, the vibrations were discontinued. The recovered sediment sample was retained within the internal CAB liner and held in position during recovery by the core catcher assembly while the coring system was lifted out, through the water column, and lowered to the deck of the R/V *Recovery*. Upon reaching the deck, the CAB liner was removed from the barrel, immediately capped and sealed to maintain physical and chemical properties, labeled, and kept on wet ice (if ambient air temperatures were above 4°C) until it could be transferred to cold storage onshore. A total of 15 stations were sampled.

Cores collected during each workday were stored on wet ice onboard the vessel (if ambient air temperatures are above 4°C), then transferred to and stored in an onshore refrigeration unit (cooled to 4°C) at the staging area. The refrigeration unit at the staging area was secured with a padlock when unattended.

2.4.2 Site Water/Elutriate Preparation Water

Site water and elutriate preparation water was collected from mid-depth at one location (PNYRB24-A-11) in the Reserve Basin maintenance dredging area (Figure 1-2). Site water/elutriate preparation water was collected using monsoon pumps with dedicated Tygon tubing. Water designated for chemical analysis was stored at 4°C in certified cleaned bottles with appropriate preservatives; elutriate preparation water was stored at 4°C in 2.5-gallon certified cleaned, plastic carboys.

Site water designated for chemical analysis was shipped via overnight delivery to EET on the day of collection. Elutriate preparation water was transported to a refrigeration unit at the staging area following collection and hand-delivered to EET for analysis at the completion of the sampling.

Upon receipt at the analytical laboratory, the samples were checked against the COCs, logged, and given a unique accession number. Samples were stored in walk-in refrigeration units (cooled to 4°C) following receipt and prior to analysis. Holding times for the site water samples began when the water samples were collected and placed into the appropriate sample containers. The holding time for the modified elutriate samples was initiated at the completion of the elutriate preparation process. Copies of the COC forms for the site water and modified elutriate samples are provided in the analytical data report files in Attachments III and IV, respectively. The sample containers, preservation techniques, and holding time requirements for site water and modified elutriates are provided in Table 2-3. Methods for the modified elutriate preparation process are described in Section 3.1.9.

2.4.3 Composite and Field Duplicates

A composite duplicate is a separate sample collected during core processing at the same time as a normal sample. Duplicates are utilized to determine the precision of laboratory analytical activities and are indicative of sample homogeneity. For this project, composite duplicates were collected at a frequency of one per 10 samples per matrix and submitted for duplicate analysis.

One field duplicate site water sample was submitted for total and dissolved analyses. A field duplicate is a second sample collected as close as possible to the same point in time and space as the first sample, using identical methods, to assess the precision of the sampling process and the sample's homogeneity.

2.4.4 Matrix Spike (MS)/ Matrix Spike Duplicate (MSD) Samples

A fortified sample (MS) is an aliquot of a field sample that is fortified with the analyte(s) of interest and analyzed to monitor matrix effects associated with a particular sample. Spiked samples were co-located with field duplicate (FD) samples. The final spiked concentration of each analyte in the sample was at least 10 times the calculated method detection limit (MDL). A fortified and duplicate-fortified sample (MSD) was performed for every batch of 20 or fewer samples.

2.4.5 Equipment Blanks

Equipment blanks were collected to determine the extent of contamination, if any, from the sampling equipment used in the project. Three equipment blanks were collected for the Reserve Basin maintenance dredging project: one for the core catcher, one for the core processing equipment, and one for the pump/tubing used to collect the site water/elutriate preparation water. The equipment blanks were collected by pouring deionized water, provided by EA's Ecotoxicology Laboratory, over decontaminated collection equipment and placing the rinsate water in laboratory-prepared containers. The equipment was decontaminated using the procedure outlined in Section 2.7. The equipment blanks were submitted to the analytical laboratory and tested for the same chemical constituents as the sediments and site water. The equipment blanks were sent to EET–Pittsburgh via overnight delivery on the day of collection. COC documentation was submitted with the equipment blanks.

The sample containers, preservation techniques, and holding time requirements for the equipment blanks are provided in Table 2-3. Holding times for the equipment blanks began when the samples were collected and placed into the appropriate sample containers. Analytical results for the equipment blanks are provided in Appendix D and Attachment III.

2.4.6 Trip Blanks

Trip blanks are used to determine contamination for volatile organic samples that may be attributable to field procedures, sample handling, and shipping procedures. Trip blanks consisted of deionized water in 40-milliliter volatile organic analysis vials prepared and provided by EET–Pittsburgh. One trip blank was shipped with each cooler containing water samples designated for volatile organic analysis. Analytical results for the trip blanks are provided in Attachments III and IV.

2.4.7 Field Blanks

A field blank is reagent water or other medium that is placed into a sample container in the field or laboratory, and is treated as a sample in all aspects, including exposure to sampling conditions, sample preservation, storage, and testing. The purpose of the field blank is to determine if the field or sample transport conditions have contaminated the sample. Field blanks were collected at a rate of one field blank sample per 20 parent samples for each media. Field blanks for the water matrix consisted of laboratory grade de-ionized water poured into sample bottles in the field. Field blanks for sediment consisted of play sand purchased from a local retailer, which was then placed into pre-cleaned sample bottles during core processing activities. Analytical results for the field blanks are provided in Appendix D and Attachments II and III.

2.4.8 Temperature Blanks

Temperature blanks consist of analyte-free media (such as de-ionized water) and are used by the analytical laboratory upon sample receipt to determine if the samples maintain the required temperature during shipping to the analytical laboratory. One temperature blank was shipped with each cooler to the analytical laboratory for samples that required preservation to 4°C.

2.5 SAMPLE VOLUME REQUIREMENTS

Required containers and analytical volumes for aqueous and sediment samples are provided in Table 2-3 and Table 2-4, respectively. For each composite sediment sample submitted for analytical testing approximately 4 gallons of sediment was required for bulk sediment analysis and 4 gallons was required for preparation of modified elutriates. Additional sediment volume was required for analysis of composite duplicate samples (analyzed at a rate of 10 percent of total submitted samples), and additional volume was collected from two sampling locations designated for MS/MSD analyses (analyzed at a rate of 5 percent of total submitted samples). Approximately 1 gallon of sediment was archived for each composite for re-analyses if required.

Approximately 8 gallons of site water was collected for the chemical analysis (total and dissolved) of the site water/elutriate preparation water. A total of approximately 10 gallons of site water was required for the preparation of each modified elutriate sample. Additional water was required for QA/QC samples (MS/MSD and field duplicate). A total of approximately 120 gallons of site water was collected from one location (PNYRB24-A-11) within the Reserve Basin maintenance dredging area (Figure 1-2).

2.6 SEDIMENT CORE PROCESSING AND COMPOSITING

Cores were transported to and processed in a protected, designated area at EA’s warehouse facility in Hunt Valley, Maryland, following the completion of daily coring activities. Each core was split lengthwise, visually inspected for grain size stratification, logged showing the depth of sampling (below the sediment surface in ft MLLW), and photographed. Photographs include the core collection location, a length scale, time of collection, and date of collection, and are provided in Appendix C. VOC samples were collected from cores prior to creation of composites and prior to the extraction of sediments from the core liner. Cores were split, and subsamples for VOCs were removed with a modified 6-ounce stainless-steel syringe. Sediments were then extracted/removed from each core, composited by location, and homogenized in cleaned stainless-steel holding containers. Sample composites were created according to Table 1-2, with representative subsamples from each homogenized location combined in equal proportions (by mass) to form the sample composite. Each sample was homogenized until the sediment was thoroughly mixed and of uniform consistency. Sample processing equipment that came into direct contact with the sediment was decontaminated according to the protocols specified in Section 2.7. After the cores were processed, samples were shipped on ice via overnight delivery to EET–Pittsburgh.

The sample containers, preservatives, and holding time requirements for sediment samples are provided in Table 2-4. Holding times for the sediment samples began when the first core of the composite sample was collected.

2.7 EQUIPMENT DECONTAMINATION AND WASTE HANDLING PROCEDURES

Equipment that came into direct contact with sediment during sampling was decontaminated prior to deployment in the field and between each sampling location to minimize cross-contamination. This included the stainless-steel core catcher, and stainless-steel vibracore head. Additionally, core processing equipment was decontaminated prior to the initiation of compositing activities as well as in between each sampling location to minimize cross-contamination. This equipment includes stainless-steel holding containers and stainless-steel processing equipment (spoons, knives, bowls, etc.). While performing the decontamination procedure, phthalate-free nitrile gloves were worn to prevent phthalate contamination of the sampling equipment or the samples.

The decontamination procedure described below was utilized:

- Rinse equipment using site water
- Rinse with 10 percent nitric acid
- Rinse with distilled or de-ionized water
- Rinse with methanol followed by hexane
- Rinse with distilled or de-ionized water
- Air dry (in area not adjacent to the decontamination area)

Decontamination waste liquid generated at the sample processing area will be contained directly in secure drums. The liquid and sediments contained in the drums will be tested, characterized, transported, and disposed of offsite by a subcontractor (See Attachment I of the SSAP [EA 2024]).

2.8 SAMPLE LABELING, CHAIN-OF-CUSTODY, AND DOCUMENTATION

2.8.1 Field Logbook

Sampling activities, sampling locations, water depths, sample IDs, and water quality data were recorded in permanently bound logbooks in indelible ink. Personnel names, local weather conditions, and other information that impacted the field sampling program were also recorded. Each page of the logbook was numbered and dated by the personnel entering information. A full copy of the core collection logbook is provided in Appendix B. A separate logbook containing the core processing information is provided in Appendix C.

2.8.2 Numbering System

The sample numbering system was used to communicate sample location and sample type between the field crew, the core processing crew, and the analytical laboratory(s). Each sample ID that was submitted for physical and/or chemical testing contained information to indicate from which location the sample was collected.

An example of a sample ID is as follows:

PNYRB24-A-01-VOC-SED

where the first field denotes the site designation (PNYRB=Philadelphia Navy Yard Reserve Basin and year [2024]).

The second field indicates the dredging area: either A or B.

The third field indicates the sampling location.

The fourth field indicates either a discrete sample for VOC analysis or a composite sample ID number.

The fifth field (last three letters) of the sample ID indicates the type of sample collected/tested:

- SED – sediment sample submitted for chemical and physical analyses
- WAT – site water submitted for chemical analyses
- MET – water collected to be used in the modified elutriate testing procedure.
- MET-T = modified elutriate total fraction (sample created in laboratory)
- MET-D = modified elutriate dissolved fraction (sample created in laboratory)

QA/QC samples included a fifth field shown below:

- FD – field duplicate (water) or composite duplicate (sediment)
- MS/MSD – matrix spike/matrix spike duplicate
- FB – field blank
- TB – trip blank
- EQB – equipment blank

2.8.3 Core Labels

Upon collection, each core was capped at each end and secured with both electrical and duct tape. The core tube and core caps were labeled with the location ID, core repetition, date/time of collection, top and bottom, and the elevation relative to MLLW below the maintenance material.

An example of a labeled core tube collected from PNYRB24-A-01a, -21 ft to -30 ft MLLW is presented below:

← TOP	PNYRB24-A-01a		BTM→
-21	12/5/24	1440	-30

When multiple core profiles/replication were collected at a location, each set of profiles/replicate was differentiated with an a (core 1), b (core 2), c (core 3), d (core 4), and/or e (core 5).

2.8.4 Sample Labels

Sample containers for the processed sediment and water samples were labeled with the following information:

- Client name
- Project number
- Sample ID
- Sampling location
- Date and time of collection
- Sampler's initials
- Type of analyses required

2.8.5 Sample Packing and Shipping

Immediately following core processing, sediment samples were stored on ice at the sample processing location. Site water, elutriate preparation water, and equipment blanks were stored in ice-filled coolers immediately following collection. Sediments, site water, field blanks, trip blanks, and equipment blanks designated for immediate analytical testing were packaged in bubble wrap, placed in an ice-filled cooler, and shipped via overnight express to EET in Pittsburgh, Pennsylvania. Samples were sent directly to the following address:

Eurofins Environment Testing Laboratory
301 Alpha Drive
Pittsburg, Pennsylvania 15238
(412) 963-7058
Attn: Ms. Carrie Gamber

Sediment samples and elutriate preparation water targeted for modified elutriate analysis were stored in a secure refrigeration unit at the processing area. These samples were hand delivered to EET upon the completion of core/sample processing. Coolers (both shipped and hand delivered) had a copy of the COC form taped to the inside of the top lid.

2.8.6 Chain-of-Custody Records

Sediment and water samples collected in the field were documented on a COC. The COCs were generated using SCRIBE, a U.S. Environmental Protection Agency (USEPA) data management system. COCs accompanied the samples to EET–Pittsburgh. Copies of the COC forms are provided in Attachments II (sediment), III (site water/blanks), and IV (modified elutriates).

TABLE 2-1. CORE COLLECTION AND SEDIMENT RECOVERY INFORMATION
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

Sampling Location	Composite Sample ID	Pennsylvania State Plane Coordinates (South) FIPS-3702 U.S. Survey Feet (NAD 83)		Date	Time	Target Dredging Depth + 2ft Overdepth (ft MLLW)	Target Core Recovery (ft)	Tide Corrected Sediment-Water Interface Elevation (ft MLLW)[a]	Core Replicate	Core Total Linear Recovery (ft)	Actual Core Recovery (ft)
		Easting	Northing								
Dredging Area A											
PNYRB24-A-01	PNYRB24-A-0102-SED	2685654.747	213869.206	12/6/2024	1010	-32	6.8	25.2	A	6.8	6.8
					1015				B	13.2	6.4
PNYRB24-A-02		2685848.339	214114.688	12/6/2024	1035		5.6	26.4	A	5.6	5.6
					1045				B	11.2	5.6
PNYRB24-A-03	PNYRB24-A-0304-SED	2686077.042	214241.003	12/6/2024	1105		4.4	27.6	A	4.4	4.4
					1115				B	8.8	4.4
					1120				C	13.2	4.4
PNYRB24-A-04		2686194.430	214303.256	12/6/2024	1135		5.6	26.4	A	5.6	5.6
					1140				B	11.2	5.6
PNYRB24-A-05	PNYRB24-A-0506-SED	2686671.041	214584.979	12/5/2024	1005		4	28.0	A	4	4
					1010				B	8	4
					1020				C	12	4
PNYRB24-A-06		2686693.570	214438.654	12/5/2024	925		4.6	27.4	A	4.6	4.6
					930				B	8.6	4
					940				C	13.2	4.6
PNYRB24-A-07	PNYRB24-A-0708-SED	2686909.245	214545.690	12/5/2024	840		3.9	28.1	A	3.8	3.8
					850				B	7.7	3.9
					855				C	11.6	3.9
					900				D	15.2	3.6
PNYRB24-A-08		2686937.476	214732.742	12/5/2024	805		4.6	27.4	A	3.5	3.5
					815				B	7.5	4
					820				C	12.1	4.6
PNYRB24-A-09	PNYRB24-A-0910-SED	2687000.056	214466.132	12/4/2024	1505		3.4	28.6	A	3.4	3.4
					1515				B	6.8	3.4
					1520				C	10.2	3.4
					1525				D	13.6	3.4
					1530				E	17	3.4
					1535				F	20.4	3.4
					1540				G	23.4	3
					1545				H	26.8	3.4
					1550				I	30.2	3.4
					1555				J	33.6	3.4
PNYRB24-A-10		2687338.458	214566.803	12/4/2024	1320		3	29.0	A	2.7	2.7
					1330				B	5.7	3
					1340				C	8.7	3
					1350				D	11.7	3
					1400				E	14.4	2.7
					1410				F	17.4	3
					1415				G	20.4	3
					1420				H	23.2	2.8
					1425				I	26	2.8
					1430				J	28.8	2.8
					1440				K	31.8	3
PNYRB24-A-11**	PNYRB24-A-1112-SED	2687258.606	214842.940	12/4/2024	1220		2.1	29.9	A	2.1	2.1
					1225				B	4.2	2.1
					1235				C	6.3	2.1
					1240				D	8.2	1.9
					1250				E	10.2	2
					1255				F	12.3	2.1
PNYRB24-A-12		2687581.749	214914.619	12/4/2024	1115		2.8	29.2	A	1.8	1.8
					1120				B	3.8	2
					1135				C	6.6	2.8
					1145				D	9.4	2.8
					1150				E	11.7	2.3

TABLE 2-1. CORE COLLECTION AND SEDIMENT RECOVERY INFORMATION
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

Sampling Location	Composite Sample ID	Pennsylvania State Plane Coordinates (South) FIPS-3702 U.S. Survey Feet (NAD 83)		Date	Time	Target Dredging Depth + 2ft Overdepth (ft MLLW)	Target Core Recovery (ft)	Tide Corrected Sediment-Water Interface Elevation (ft MLLW)(a)	Core Replicate	Core Total Linear Recovery (ft)	Actual Core Recovery (ft)
		Easting	Northing								
Dredging Area B											
PNYRB24-B-2a	PNYRB24-B-2-SED	2689181.551	215466.453	12/3/2024	1550	-32	7.6	24.4	A	2.5	2.5
PNYRB24-B-2a	PNYRB24-B-2-SED	2689181.551	215466.453	12/3/2024	1625	-32	7.6	24.4	B	6.8	4.3
PNYRB24-B-2a	PNYRB24-B-2-SED	2689168.848	215470.825	12/4/2024	820	-32	6.6	25.4	C	12.7	5.9
PNYRB24-B-2a	PNYRB24-B-2-SED	2689168.848	215470.825	12/4/2024	825	-32	6.6	25.4	D	17.8	5.1
PNYRB24-B-2a	PNYRB24-B-2-SED	2689168.848	215470.825	12/4/2024	835	-32	6.6	25.4	E	24.4	6.6
PNYRB24-B-2a	PNYRB24-B-2-SED	2689168.848	215470.825	12/4/2024	845	-32	6.6	25.4	F	31	6.6
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	920	-32	3.6	28.4	A	2.2	2.2
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	940	-32	3.6	28.4	B	5.8	3.6
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	950	-32	3.6	28.4	C	9.4	3.6
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	1000	-32	3.6	28.4	D	13	3.6
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	1010	-32	3.6	28.4	E	15.9	2.9
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	1020	-32	3.6	28.4	F	19.5	3.6
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	1030	-32	3.6	28.4	G	22.8	3.3
PNYRB24-B-4a	PNYRB24-B-4-SED	2689765.323	215637.938	12/3/2024	1040	-32	3.6	28.4	H	24.3	1.5
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1255	-32	3	29.0	A	2.3	2.3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1325	-32	3	29.0	B	5.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1340	-32	3	29.0	C	8.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1350	-32	3	29.0	D	11.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1405	-32	3	29.0	E	14.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1420	-32	3	29.0	F	17.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1430	-32	3	29.0	G	20.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1445	-32	3	29.0	H	23.3	3
PNYRB24-B-5a	PNYRB24-B-5-SED	2690512.815	215491.748	12/3/2024	1455	-32	3	29.0	I	25.4	2.1

Notes:
(a) Tide corrections for surface elevations were based off real-time data from the NOAA Center for Oceanographic Products and Services (CO-OPS) Station 8545240, Philadelphia, Pennsylvania.
**Water collection location.
ft = Foot (feet)
MLLW = Mean Lower Low Water
NAD83 = North American Datum of 1983

TABLE 2-2. IN SITU WATER QUALITY MEASUREMENTS
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD
PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

Location	Date, Time	Water Depth (ft) Below Surface	Water Temperature (°C)	Salinity (ppt)	pH	Turbidity (NTU)	Dissolved Oxygen (mg/L)
PNY24-WAT	12/5/2024, 1130	Surface	8.5	0.32	7.36	1.3	7.13
		5	8.5	0.32	7.35	1.4	7.11
		10	8.4	0.32	7.39	1.5	7.12
		15	8.4	0.31	7.40	1.4	7.12
		20	8.4	0.31	7.40	1.7	7.14
		25	8.4	0.31	7.42	1.8	7.16
		Bottom	8.5	0.31	7.45	11.5	7.24

Notes:
°C = Degrees Celsius
ft = Feet
mg/L = Milligram(s) per liter
NTU = Nephelometric turbidity units
ppt = Parts per thousand

This page left intentionally blank

TABLE 2-3. REQUIRED CONTAINERS, PRESERVATION TECHNIQUES, AND HOLDING TIMES FOR SEDIMENT SAMPLES[a]
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Parameter	Volume Required [b]	Container [c]	Preservative	Holding Time
Inorganics				
Metals (including Mercury)	32 ounces	G	4º C	6 months (28 days for mercury)
Cyanide, Total	(d)	G	4°C	14 days
Total Solids/Moisture	(d)	G	4°C	NA
Sulfide	(d)	G	4°C	7 days to analysis
Physical Parameters				
Grain Size	32 ounces	P,G	4º C	6 months
Modified Elutriate Test	3.5 x 1 gallon	G	4º C	14 days until elutriate generation
Organics				
Total Organic Carbon	(d)	G	4º C	14 days
Pesticides (Organochlorine), Semivolatile Organics and Polycyclic Aromatic Hydrocarbons, Polychlorinated Biphenyl (PCB) Aroclors	(d)	G	4°C	14 days to extraction/ 40 days to analysis
Dioxins and PCB Congeners	2 x 4 ounces	G	4°C	1 year
Volatile Organic Compounds (VOCs)	4 ounces	G	4º C	14 days to analysis
Total Petroleum Hydrocarbons (TPH) Gasoline Range Organics (GRO)	4 ounces	G	4º C	14 days to analysis
Total Petroleum Hydrocarbons (TPH) Diesel Range Organics (DRO)/ Oil Range Organics (ORO)	4 ounces	G	4º C	14 days until extraction, 14 days after extraction
Radioactivity (Gross Alpha, Gross Beta, Gamma)	4 ounces (2 grams)	G	4º C	180 days from sample collection
PFAS Compounds	4 ounces	P	4º C	90 days
Other Parameters				
pH	Sufficient volume is provided from the 32 ounces noted under TCLP	G	4°C	Immediate

Table 2-3. (Continued)

Parameter	Volume Required [b]	Container [c]	Preservative	Holding Time
Toxicity Characteristic Leaching Procedure (TCLP)				
TCLP (Metals, Mercury)	32 ounces	G	4°C	6 months to leach (28 days Mercury)/ 6 months to analysis (28 days Mercury)
TCLP (SVOC, Pesticides, Herbicides)	Sufficient volume is provided from the 32 oz. noted under TCLP Metals	G	4°C	14 days to leach/7 days to extraction/40 days to analysis
TCLP VOA	4 ounces	G	4°C	14 days to leach/14 days to analysis

Notes:
(a) From time of sample collection.
(b) Additional volume will need to be provided for samples designated as matrix spike/matrix spike duplicates.
(c) P=plastic; G=glass
(d) Can be taken from the 32 oz. noted for Metals.
°C = Degrees Celsius
NA = Not Applicable
SVOC = Semivolatile Organic Compound
TCLP = Toxicity Characteristic Leaching Procedure
VOA = Volatile organic analysis

TABLE 2-4. REQUIRED CONTAINERS, PRESERVATION TECHNIQUES, AND HOLDING TIMES FOR WATER SAMPLES
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Parameter	Volume Required (b)	Container (c)	Preservative	Holding Time
Inorganics				
Metals (including Mercury)	250 mL	P	Nitric Acid, 4°C	6 months (28 days mercury)
Sulfide	250 mL	P	Sodium Hydroxide/Zinc Acetate, 4°C	7 days
Cyanide, Total	250 mL	P	Sodium Hydroxide, 4°C	14 days
Physical Parameters				
Modified Elutriate Test	10 gallon	P	4º C	None specified
Total Suspended Solids	500 mL	P	None	7 days
Organics				
Dioxins, Polychlorinated Biphenyl (PCB) Congeners	4 liters	G (amber)	4°C	1 year to extraction/ 40 days to analysis
Total Organic Carbon/Dissolved Organic Carbon	2 x 40mL	G (amber) VOA Vial	Sulfuric Acid, 4°C	28 days
Volatile Organic Compounds (VOCs)	2 x 40 mL	VOA Vial	Hydrochloric Acid	14 days
Gasoline Range Organics (GRO)	2 x 40 mL	VOA Vial	Hydrochloric Acid	14 days
Pesticides (Organochlorine), PCB Aroclors	4 liters	G	4°C	7 days to extraction/ 40 days to analysis
Semivolatile Organics and Polycyclic Aromatic Hydrocarbons	2 x 250 mL	G (amber)	4°C	7 days to extraction/ 40 days to analysis
Diesel Range Organics (DRO)/ Oil Range Organics (ORO)	2 x 250 mL	G (amber)	4°C	7 days to extraction/ 40 days to analysis
Radioactivity (Gross Alpha, Gross Beta, Gamma)	500 mL	P	Nitric Acid, 4º C	180 days from sample collection
PFAS Compounds	250 mL	P	4°C	28 days

Notes:
(a) From time of sample collection.
(b) Additional volume will need to be provided for samples designated as matrix spike/matrix spike duplicates.
(c) P=plastic; G=glass
°C = Degrees Celsius
mL = Milliliter(s)
PFAS = Per- and Polyfluoroalkyl Substances
VOA = Volatile Organic Analysis

This page left intentionally blank

3. ANALYTICAL TESTING OF SEDIMENT, SITE WATER, AND MODIFIED ELUTRIATES

Analytical testing of sediment, site water, and modified elutriates for the Reserve Basin maintenance dredging project was conducted by EET–Pittsburgh, with support from EET–Burlington (grain size), EET–Lancaster (polychlorinated biphenyl [PCB] congeners, metals/mercury, dioxins/furans, PFAS, gasoline range organics [GRO], diesel range organics [DRO], and oil range organics [ORO]), EET-Cleveland (TCLP), and EET–St. Louis (radioactivity).

The six sediment composite samples from Area A and three individual core samples from Area B were analyzed for the following physical parameters:

- Grain size
- Total solids
- Moisture content

The six sediment composite samples, three individual core samples, site water (total and dissolved fractions), and modified elutriates (total and dissolved fractions) were analyzed for the following (Table 1-1):

- Chlorinated pesticides
- Cyanide, total
- Dioxin/furan congeners
- Metals
- Mercury
- PCB Aroclors
- PCB congeners and homologs
- Polycyclic aromatic hydrocarbons (PAHs)
- PFAS
- Radioactivity
- Semivolatile organic compounds (SVOCs)
- Sulfide
- TCLP (sediment only)
- Total organic carbon (TOC)/dissolved organic carbon
- Total petroleum hydrocarbons (TPH)-DRO (C10-C20)
- TPH-GRO (C6-C10)
- TPH-ORO (C20-C34)
- Total suspended solids (TSS) (total fraction of site water/elutriates only)
- VOCs (discrete sample from individual sediment cores prior to homogenization)

The analytical testing conducted for each sample type is summarized in Table 1-1. The overall testing program for the project is described in detail in the project SSAP (EA 2024). Target analytes, target detection limits, methodologies, modified elutriate preparation procedures, and sample holding times were derived from the following guidance documents:

- USACE Engineer Research and Development Center (ERDC). 2003. *Evaluation of Dredged Material Proposed for Disposal at Island, Nearshore, or Upland Confined Disposal Facilities – Testing Manual.* ERDC/EL TR-03-1.

- USEPA. 2001. *Methods for Collection, Storage, and Manipulation of Sediments for Chemical and Toxicological Analyses: Technical Manual.* EPA-823-B-01-002.

- USEPA/USACE. 1995. *QA/QC Guidance for Sampling and Analysis of Sediments, Water, and Tissues for Dredged Material Evaluations.* EPA-823-B-95-001.

- USEPA/USACE. 1998. *Evaluation of Dredged Material Proposed for Discharge in Waters of the U.S. – Inland Testing Manual.* EPA-823-B-98-004.

- NJDEP. 2024. *Field Sampling Procedures Manual*. March. https://dep.nj.gov/srp/guidance/fspm/

- PADEP. *Information Needed for Disposal of Dredged Material at the Biles Island Confined Disposal Facility in Falls Township, Bucks, County, PA.* Dredging Guidance 20145.

The following sections outline key components of the testing program.

3.1 ANALYTICAL METHODS

Inorganic and organic compounds were determined using the methods listed in Table 3-1, as described in the laboratory's analytical standard operating procedures (EA 2024). Quantitation limits applicable to this project are listed in Table 3-2 (sediment), Table 3-3 (TCLP), and Table 3-4 (aqueous). To meet program-specific regulatory requirements for chemicals of concern, the methods/standard operating procedures were followed as stated with some specific requirements noted below.

3.1.1 Polychlorinated Biphenyl Congeners

PCBs were analyzed and quantified as individual congeners by USEPA Method 1668A. Each of the 209 PCB congeners were determined in the sediment and aqueous matrices. In addition, PCB homolog groups were also reported. Homologs are groups of PCB congeners with the same number of chlorine atoms. Nine homolog groups were reported: dichlorobiphenyls, heptachlorobiphenyls, hexachlorobiphenyls, monochlorobiphenyls, nonachlorobiphenyls, octachlorobiphenyls, pentachlorobiphenyls, tetrachlorobiphenyls, and trichlorobiphenyls.

PCBs were also analyzed as Aroclors using USEPA Method 8082. This method provides the results of seven Aroclors, which consist of mixtures of congeners. The first two digits indicate the number of carbon atoms in the phenyl ring, and the last two digits indicate the percentage of chlorine by mass. Seven Aroclors were evaluated: Aroclor-1016, -1221, -1232, -1242, -1248, -1254, and -1260.

3.1.2 Total Organic Carbon

TOC in sediments was determined using the 1988 USEPA Region 2 combustion procedure (Lloyd Kahn procedure). TOC in aqueous samples was determined using USEPA Method SM 5310C.

3.1.3 Polycyclic Aromatic Hydrocarbons

PAHs were determined utilizing SW846 Method 8270D LL. For those samples where both SVOCs by SW846 Method 8270D LL and PAHs by SW846 Method 8270D LL were requested, both analyses were performed on the same extract. For those samples, the evaluation of method performance was based on the determined recoveries of surrogates and control analytes (laboratory control sample [LCS] and MS/MSDs from the SVOCs by USEPA Method SWA846 8270D LL (full scan gas chromatography/mass spectrometry) analyses because the spiked concentrations exceeded the calibration range for PAHs by gas chromatography/mass spectrometry analyses.

3.1.4 Metals

Because of potential matrix interferences, metals were determined utilizing inductively coupled plasma/mass spectrometry according to the methodology specified, except for mercury. For mercury, samples were analyzed by Cold Vapor Atomic Absorption USEPA Method (SW846 7470A [aqueous] or 7471B [sediment]).

3.1.5 Dioxins/Furan Congeners

Dioxin and furan congeners for the sediment, site water, and modified elutriate samples were reported as 17 individual isomers using USEPA Method 1613B. The results were reported based on a sample-specific estimated detection limit, which considers matrix interferences and provides the most accurate limit of detection for each sample.

3.1.6 Cyanide

Total cyanide was determined by USEPA Method SW846 9014. The laboratory RL using this method is higher than the requested target detection limit; however, this method represents the best available technology for total cyanide determination and, therefore, the lowest possible RL.

3.1.7 PFAS

Six PFAS, at the request of PADEP, were analyzed using USEPA Method 1633: hexafluoropropylene oxide dimer acid (HFPO-DA), perfluorobutanesulfonic acid (PFBS), perfluorohexanesulfonic acid (PFHXS), perfluorononanoic acid (PFNA), perfluorooctanesulfonic acid (PFOS), and perfluorooctanoic acid (PFOA). Following review of the bulk sediment PFAS data, PADEP requested that the Synthetic Precipitation Leaching Procedure (SPLP) be performed and analyzed for PFAS for the Area A composites. Archived material from the initial bulk sediment testing was used for this analysis, and the results of the testing are provided in Appendix E.

3.1.8 Total Petroleum Hydrocarbons

TPH were analyzed as groups of petroleum hydrocarbon compounds with a specific number of carbons in each hydrocarbon chain. TPH-GRO includes compounds with six to 10 carbons in each chain. TPH-DRO includes compounds with 10 to 20 carbons per chain. TPH-ORO includes the heaviest petroleum compounds having 20 to 34 carbons per chain.

3.1.9 Modified Elutriate Test

Modified elutriates simulate the potential release of chemical constituents to effluent during dredged material placement operations and dewatering in a confined disposal facility. The Modified Elutriate Test (MET) is designed to evaluate the settling processes and geochemical changes within the effluent that would be discharged from a confined disposal facility (USACE 2003). Modified elutriate testing includes the determination of the total and dissolved concentrations of target analytes in the full-strength aqueous elutriate sample.

The MET was performed following the procedures in Appendix B of the Inland Testing Manual, June 1998 (USEPA/USACE 1998). The sediment and site water were mixed in a 4-liter cylinder with the sediment concentration equal to the average inflow concentration (based on dry weight), and the mixture was manually mixed and aerated for 1 hour. That mixture was then allowed to settle for a time period equal to the anticipated field mean retention time, up to a maximum period of 24 hours. The supernatant was then siphoned off creating the total fraction, and a portion of the supernatant was centrifuged to remove particulates creating the dissolved fraction. The sample was analyzed for the chemical constituents specified in the Analytical QAPP (Attachment II of the SSAP [EA 2024]). For the MET, EET–Pittsburgh used the method default values of 120 grams per liter for the average field inflow concentration and 24 hours for the field mean retention time, respectively.

The sediment and site water volume requirements needed for the MET were dependent on the number and type of analytical tests to be performed on the elutriate. Quantitation limits for the modified elutriates were the same as those for the aqueous samples (Table 3-4).

3.1.10 Toxicity Characteristic Leaching Procedure (TCLP)

Sediment samples from the Reserve Basin maintenance dredging area were extracted following the TCLP methods specified in SW846 Method 1311. The resultant leachates were analyzed for the parameters specified in Table 3-3. Sample PNYRB24-B-4-SED had an initial lead TCLP result of 10 mg/L, which exceeded the regulatory limit of 5 mg/L. The result was B qualified, indicating lead was detected in the corresponding laboratory method blank. The sample was re-analyzed and the results are presented from the re-analysis.

3.2 DETECTION LIMITS

The detection limit is a statistical concept that corresponds to the minimum concentration of an analyte above which the net analyte signal can be distinguished with a specified probability from the signal because of the noise inherent in the analytical system. The RL is the limit of detection for a target analyte for an individual sample after adjustments to dilutions or percent moisture.

Therefore, the RL of any analyte in the testing program can vary between samples with the same matrix as a result of different percent moistures (sediments) or if a sample is diluted by the analytical laboratory (sediments and waters).

The MDL was developed by USEPA and is defined as “the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero” (40 CFR 136, Appendix B). Quantitation limits applicable to this project are listed in Table 3-2 (sediment), Table 3-3 (TCLP), and Table 3-4 (aqueous). All analytical parameters for water and sediments, except general chemistry parameters and dioxin/furan congeners, were quantified to the MDL and reported to the RL. Detected values greater than or equal to the MDL, but less than the laboratory RL, were qualified by the laboratory as estimated. For sediment analyses, sample weights were adjusted for percent moisture (up to 50 percent moisture), where appropriate, prior to analysis to achieve the lowest possible RLs.

3.3 LABORATORY QUALITY CONTROL SAMPLES

QC samples were analyzed at the frequency stated in the following table. Acceptance criteria are listed in Appendix A of the Analytical QAPP, Attachment II of the SSAP (EA 2024).

QC Sample	Frequency
Method Blanks	1 per analytical batch of 1–20 samples
LCS	1 per analytical batch of 1–20 samples
Surrogates	Spiked into all field and QC samples (Organic Analyses)
Sample Duplicates	1 per analytical batch of 1–20 samples (Inorganic Analyses)
MS/MSD	1 per analytical batch of 1–20 samples

3.3.1 Method Blanks

The method (reagent) blank is used to monitor laboratory contamination. The method blank is usually a sample of laboratory reagent water processed through the same analytical procedure as the sample (i.e., digested, extracted, distilled). One method blank was analyzed at a frequency of one per every analytical preparation batch of 20 or fewer samples.

3.3.2 Laboratory Control Sample

The LCS is a fortified method blank consisting of reagent water or solid fortified with the analytes of interest for single-analyte methods and selected analytes for multi-analyte methods according to the appropriate analytical method. LCSs were prepared and analyzed with each analytical batch, and analyte recoveries were used to monitor analytical accuracy and precision.

3.3.3 Matrix Spike/Matrix Spike Duplicate

A fortified sample (matrix spike) is an aliquot of a field sample that is fortified with the analyte(s) of interest and analyzed to monitor matrix effects associated with a particular sample. Samples to be spiked were chosen at random. The final spiked concentration of each analyte in the sample was at least 10 times the calculated MDL. A duplicate-fortified sample (matrix spike duplicate) was performed for every batch of 20 or fewer samples.

3.3.4 Sample Duplicates

A sample duplicate is a second aliquot of a field sample that is analyzed to monitor analytical precision associated with that particular sample. Sample duplicates were performed for every batch of 20 or fewer samples for those analytes that did not have MS/MSD analyses.

3.3.5 Surrogates

Surrogates are organic compounds that are similar to analytes of interest in chemical composition, extraction, and chromatography, but are not normally found in environmental samples. These compounds were spiked into all blanks, standards, samples, and spiked samples prior to analysis for organic parameters. Generally, surrogates are not used for inorganic analyses. Percent recoveries were calculated for each surrogate. Surrogates were spiked into samples according to the requirements of the reference analytical method (Section 9 of the Analytical QAPP, Attachment II of the SSAP [EA 2024]). Surrogate spike recoveries were evaluated against the limits provided in Appendix A of the QAPP, Attachment II of the SSAP (EA 2024), and were used to assess method performance and sample measurement bias. If sample dilution caused the surrogate concentration to fall below the quantitation limit, surrogate recoveries were not calculated.

TABLE 3-1. ANALYTICAL TESTING METHODS
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

PARAMETER	METHOD	METHOD #	MATRIX[1]	REFERENCE
Organics/Leaching – Extraction and Cleanup				
Sulfuric Acid Cleanup	Liquid-liquid Partitioning	3665A	S	EPA 1997
Sulfur Cleanup	Treatment with Cu or Hg or TBA	3660A/B	S	EPA 1997
Gel-permeation Chromatography	Macromolecule separation by size exclusion	3640A	S	EPA 1997
TCLP Leach/TCLP ZHE	Leaching	1311	S	EPA 1992
MET	Elutriate Generation	ITM	S	EPA 1998
SVOC/PAH prep	Continuous Liquid/Liquid extraction	3520C	W	EPA 1997
PCB/Pesticides prep	Separatory funnel extraction	3510C	W	EPA 1997
PFAS Compounds	Solid phase extraction	1633A	S,W	EPA 2024
PCB/Pest/SVOC/PAH prep	Soxtherm extraction	3541	S	EPA 1997
Organics				
Total Organic Carbon	Combustion Oxidation	Lloyd Kahn	S	EPA 1988
	Chemical Oxidation	5310C	W	APHA 1998
Particulate Organic Carbon	Combustion Oxidation	440.0	S	EPA 1997
Organochlorine Pesticides	Gas Chromatography – ECD	8081B	S,W	EPA 1997
VOCs	Gas Chromatography/Mass Spectrometry	8260C	S,W	EPA 1996
PCB Congeners	High Resolution Mass Spectrometry	1668A	S,W	EPA 1979
SVOCs/PAHs	Gas Chromatography/Mass Spectrometry	8270D	S,W	EPA 1997
PCB Aroclors	Gas Chromatography – ECD	8082	S,W	EPA 1997
Dioxins/Furans	High Resolution Mass Spectrometry	1613B	S,W	EPA 1979
GRO/DRO/ORO	Gas Chromatograph	8015D	S, W	EPA 2003

TABLE 3-1. (continued)

PARAMETER	METHOD	METHOD #	MATRIX[1]	REFERENCE
PFAS Compounds	LC-MS/MS	1633A	S, W	EPA 2024
Metals				
Aluminum	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Antimony	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Arsenic	Atomic Emission – ICP/MS/ Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)	6020A/6010C	S,W	EPA 1997
Barium	Atomic Emission – ICP/MS/ Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)	6020A/6010C	S,W	EPA 1997
Beryllium	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Cadmium	Atomic Emission – ICP/MS/ Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)	6020A/6010C	S,W	EPA 1997
Chromium	Atomic Emission – ICP/MS/ Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)	6020A/6010C	S,W	EPA 1997
Cobalt	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Copper	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Iron	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Lead	Atomic Emission – ICP/MS/ Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)	6020A/6010C	S,W	EPA 1997
Mercury	Atomic Absorption – Cold Vapor	7471B	S	EPA 1997
Mercury	Atomic Absorption – Cold Vapor	7470A	W	EPA 1997
Manganese	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Nickel	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Selenium	Atomic Emission – ICP/MS/ Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)	6020A/6010C	S,W	EPA 1997
Silver	Atomic Emission – ICP/MS/	6020A/6010C	S,W	EPA 1997

TABLE 3-1. (continued)

PARAMETER	METHOD	METHOD #	MATRIX[1]	REFERENCE
	Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP)			
Thallium	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Vanadium	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Zinc	Atomic Emission – ICP/MS	6020A	S,W	EPA 1997
Inorganic Nonmetals				
Cyanide, Total	Colorimetric – Automated	9014	S,W	EPA 1997
TSS	Residue	2540D	W	EPA 1979
pH	Electrometrically	9045D	S	EPA 1997
Sulfide	Ion Chromatograph	9056A	S	EPA 1997
Grain Size (sieve and hydrometer)	------	D422	S	ASTM 2016
Moisture Content	-------	D2216-19	S	ASTM 2019
Total Solids	Gravimetric	2540G	S	APHA 1998
Radioactivity	------	900.0	S,W	EPA 1980

References:

APHA 1998	American Public Health Association. 1998. Standard Methods for the Examination of Water and Wastewater, 20th Edition. APHA, Washington, DC.
ASTM 2016	American Society for Testing and Materials, 2016. Standard Test Method for Particle-Size Analysis of Soils
ASTM 2019	American Society for Testing and Materials, 2019. Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass.
EPA 1979	United States Environmental Protection Agency. 1979. Methods for Chemical analysis of Water and Wastes. EPA-600/4-79-020. U.S. EPA, Cincinnati, Ohio.
EPA 1980	United States Environmental Protection Agency, EMSL. 1980. “Method 900.0: Gross Alpha and Gross Beta Radioactivity in Drinking Water.” Prescribed Procedures for Measurement of Radioactivity in Drinking Water, EPA/600/4/80/032.
EPA 1988	Kahn, Lloyd. 1988. Determination of Total Organic Carbon in Sediment. U.S. EPA Region II. Edison, N.J.
EPA 1992	United States Environmental Protection Agency. 1992. Method 1311: Toxicity Characteristic Leaching Procedure, part of Test Methods for Evaluating Solid Waste, Physical/Chemical Methods. July.

TABLE 3-1. (continued)

PARAMETER	METHOD	METHOD #	MATRIX[1]	REFERENCE
EPA 1997	United States Environmental Protection Agency. June 1997. Test Methods for Evaluating Solid Waste. Physical/Chemical Methods. EPA SW-846, 3rd Edition, including Final Update III. U.S. EPA, Washington, D.C.			
EPA 1998	United States Environmental Protection Agency. 1998. *Evaluation of Dredged Material Proposed for Discharge in Waters of the U.S. – Testing Manual: Inland Testing Manual*. EPA-823-B-98-004. USEPA Office of Water, Office of Science, and Technology, Washington, D.C., and USACE Operations, Construction, and Readiness Division, Washington, D.C. February.			
EPA 2003	United States Environmental Protection Agency. 2003. Method 8015C (SW-846): Nonhalogenated Organics Using GC/FID. Revision 4. Washington, DC.			
EPA 2024	United States Environmental Protection Agency. 2024. Method 1633 Revision A Analysis of Per- and Polyfluoroalkyl Substances (PFAS) in Aqueous, Solid, Biosolids, and Tissue Samples by LC-MS/MS. December.			

Matrix codes: S = sediment; W = water/elutriates

TABLE 3-2. PROJECT LIMITS FOR SEDIMENT SAMPLES
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
Metals - Atomic Emission Inductively Coupled Plasma / Mass Spectrometry (SW846 6020A)			
Aluminum	mg/kg	3	2.12
Antimony	mg/kg	0.1	0.053
Arsenic	mg/kg	0.05	0.029
Barium		0.5	0.306
Beryllium	mg/kg	0.05	0.036
Cadmium	mg/kg	0.05	0.028
Chromium	mg/kg	0.1	0.089
Cobalt	mg/kg	0.025	0.018
Copper	mg/kg	0.15	0.103
Iron	mg/kg	2.5	2.39
Lead	mg/kg	0.05	0.033
Manganese	mg/kg	0.25	0.215
Nickel	mg/kg	0.05	0.047
Selenium	mg/kg	0.25	0.061
Silver	mg/kg	0.05	0.025
Thallium	mg/kg	0.05	0.035
Vanadium	mg/kg	0.05	0.047
Zinc	mg/kg	0.75	0.399
Metals - Cold Vapor (SW846 7471B)			
Mercury	mg/kg	0.0165	0.0106
Dioxins/Furans- High Resolution Mass Spectrometry (HRMS) - (USEPA 1613B) [(b)]			
2,3,7,8-TCDD	ng/kg	1.0	0.41
1,2,3,7,8-PeCDD	ng/kg	5	0.46
1,2,3,4,7,8-HxCDD	ng/kg	5	0.45
1,2,3,6,7,8-HxCDD	ng/kg	5	0.37
1,2,3,7,8,9-HxCDD	ng/kg	5	0.57
1,2,3,4,6,7,8-HpCDD	ng/kg	5	0.49
OCDD	ng/kg	10	1.3
2,3,7,8-TCDF	ng/kg	1	0.35
1,2,3,7,8-PeCDF	ng/kg	5	0.42
2,3,4,7,8-PeCDF	ng/kg	5	0.44
1,2,3,4,7,8-HxCDF	ng/kg	5	0.35
1,2,3,6,7,8-HxCDF	ng/kg	5	0.41
2,3,4,6,7,8-HxCDF	ng/kg	5	0.48
1,2,3,7,8,9-HxCDF	ng/kg	5	0.46
1,2,3,4,6,7,8-HpCDF	ng/kg	5	0.44
1,2,3,4,7,8,9-HpCDF	ng/kg	5	0.48
OCDF	ng/kg	10	0.99
Polycyclic Aromatic Hydrocarbons (PAHs) – Gas Chromatography/Mass Spectrometry – (SW846 8270D LL)			
1-Methylnaphthalene	µg/kg	3.35	0.76
2-Methylnaphthalene	µg/kg	3.35	0.8

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
Acenaphthene	µg/kg	3.35	0.96
Acenaphthylene	µg/kg	3.35	0.73
Anthracene	µg/kg	3.35	0.865
Benzo[a]anthracene	µg/kg	3.35	1.51
Benzo[a]pyrene	µg/kg	3.35	1.45
Benzo[b]fluoranthene	µg/kg	3.35	0.82
Benzo[g,h,i]perylene	µg/kg	3.35	0.72
Benzo[k]fluoranthene	µg/kg	3.35	1
Chrysene	µg/kg	3.35	1.85
Dibenz(a,h)anthracene	µg/kg	3.35	2.14
Fluoranthene	µg/kg	3.35	0.88
Fluorene	µg/kg	3.35	0.655
Indeno[1,2,3-cd]pyrene	µg/kg	3.35	1.66
Naphthalene	µg/kg	3.35	0.65
Phenanthrene	µg/kg	3.35	0.895
Pyrene	µg/kg	3.35	1.17
Organochlorine Pesticides - Gas Chromatography/Electron Capture Detector - (SW846 8081B LL)			
Aldrin	µg/kg	0.0417	0.013
alpha-BHC	µg/kg	0.0417	0.0103
beta-BHC	µg/kg	0.0417	0.0115
delta-BHC	µg/kg	0.0417	0.0132
gamma-BHC (Lindane)	µg/kg	0.0417	0.0108
Chlordane (technical)	µg/kg	0.417	0.179
Chlorobenside	µg/kg	0.1	0.0701
DCPA	µg/kg	0.0417	0.0134
4,4'-DDD	µg/kg	0.0417	0.00875
4,4'-DDE	µg/kg	0.0417	0.0085
4,4'-DDT	µg/kg	0.0417	0.0299
Dieldrin	µg/kg	0.0417	0.0105
Endosulfan I	µg/kg	0.0417	0.0113
Endosulfan II	µg/kg	0.0417	0.0092
Endosulfan sulfate	µg/kg	0.0417	0.0192
Endrin	µg/kg	0.0417	0.0078
Endrin aldehyde	µg/kg	0.0417	0.0149
Heptachlor	µg/kg	0.0417	0.0131
Heptachlor epoxide	µg/kg	0.0417	0.0107
Methoxychlor	µg/kg	0.0417	0.0163
Mirex	µg/kg	0.0417	0.0303
Toxaphene	µg/kg	1.67	1.13
PCB Aroclors - Gas Chromatography (SW846 8082A) [(b)]			
PCB-1016	µg/kg	0.417	0.135
PCB-1221	µg/kg	0.417	0.148
PCB-1232	µg/kg	0.417	0.102
PCB-1242	µg/kg	0.417	0.061
PCB-1248	µg/kg	0.417	0.101
PCB-1254	µg/kg	0.417	0.125
PCB-1260	µg/kg	0.417	0.119

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[a]	Laboratory MDL[a]
PCB Congeners - Gas Chromatography/Electron Capture Detector (SW846 1668A)			
PCB-1	µg/kg	0.01	0.004
PCB-2	µg/kg	0.01	0.0042
PCB-3	µg/kg	0.01	0.004
PCB-4	µg/kg	0.02	0.0034
PCB-5	µg/kg	0.01	0.0035
PCB-6	µg/kg	0.01	0.0032
PCB-7	µg/kg	0.01	0.0034
PCB-8	µg/kg	0.02	0.003
PCB-9	µg/kg	0.01	0.0025
PCB-10	µg/kg	0.01	0.0027
PCB-11	µg/kg	0.02	0.0046
PCB-12	µg/kg	0.02	0.0062
PCB-13	µg/kg	0.02	0.0062
PCB-14	µg/kg	0.01	0.0029
PCB-15	µg/kg	0.01	0.0033
PCB-16	µg/kg	0.01	0.0037
PCB-17	µg/kg	0.01	0.004
PCB-18	µg/kg	0.02	0.0059
PCB-19	µg/kg	0.01	0.0041
PCB-20	µg/kg	0.02	0.0063
PCB-21	µg/kg	0.02	0.0049
PCB-22	µg/kg	0.01	0.0025
PCB-23	µg/kg	0.01	0.0028
PCB-24	µg/kg	0.01	0.0035
PCB-25	µg/kg	0.01	0.0026
PCB-26	µg/kg	0.02	0.0044
PCB-27	µg/kg	0.01	0.003
PCB-28	µg/kg	0.02	0.0063
PCB-29	µg/kg	0.02	0.0044
PCB-30	µg/kg	0.02	0.0059
PCB-31	µg/kg	0.02	0.0046
PCB-32	µg/kg	0.01	0.0032
PCB-33	µg/kg	0.02	0.0049
PCB-34	µg/kg	0.01	0.0021
PCB-35	µg/kg	0.01	0.0022
PCB-36	µg/kg	0.01	0.0027
PCB-37	µg/kg	0.01	0.0023
PCB-38	µg/kg	0.01	0.0027
PCB-39	µg/kg	0.01	0.0023
PCB-40	µg/kg	0.03	0.0074
PCB-41	µg/kg	0.03	0.0074
PCB-42	µg/kg	0.01	0.003
PCB-43	µg/kg	0.02	0.005
PCB-44	µg/kg	0.03	0.015
PCB-45	µg/kg	0.02	0.0046
PCB-46	µg/kg	0.01	0.0043
PCB-47	µg/kg	0.03	0.015
PCB-48	µg/kg	0.01	0.0025
PCB-49	µg/kg	0.02	0.0055

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-50	µg/kg	0.02	0.0065
PCB-51	µg/kg	0.02	0.0046
PCB-52	µg/kg	0.01	0.004
PCB-53	µg/kg	0.02	0.0065
PCB-54	µg/kg	0.01	0.0037
PCB-55	µg/kg	0.01	0.0027
PCB-56	µg/kg	0.01	0.0044
PCB-57	µg/kg	0.01	0.0021
PCB-58	µg/kg	0.01	0.0028
PCB-59	µg/kg	0.03	0.0062
PCB-60	µg/kg	0.01	0.0034
PCB-61	µg/kg	0.04	0.011
PCB-62	µg/kg	0.03	0.0062
PCB-63	µg/kg	0.01	0.0021
PCB-64	µg/kg	0.01	0.0035
PCB-65	µg/kg	0.03	0.015
PCB-66	µg/kg	0.01	0.003
PCB-67	µg/kg	0.01	0.0026
PCB-68	µg/kg	0.01	0.0036
PCB-69	µg/kg	0.02	0.0055
PCB-70	µg/kg	0.04	0.011
PCB-71	µg/kg	0.03	0.0074
PCB-72	µg/kg	0.01	0.0025
PCB-73	µg/kg	0.02	0.005
PCB-74	µg/kg	0.04	0.011
PCB-75	µg/kg	0.03	0.0062
PCB-76	µg/kg	0.04	0.011
PCB-77	µg/kg	0.01	0.0053
PCB-78	µg/kg	0.01	0.0025
PCB-79	µg/kg	0.01	0.0027
PCB-80	µg/kg	0.01	0.0026
PCB-81	µg/kg	0.01	0.0033
PCB-82	µg/kg	0.01	0.0039
PCB-83	µg/kg	0.02	0.0074
PCB-84	µg/kg	0.01	0.0033
PCB-85	µg/kg	0.03	0.0087
PCB-86	µg/kg	0.06	0.014
PCB-87	µg/kg	0.06	0.014
PCB-88	µg/kg	0.02	0.0051
PCB-89	µg/kg	0.01	0.0033
PCB-90	µg/kg	0.03	0.011
PCB-91	µg/kg	0.02	0.0051
PCB-92	µg/kg	0.01	0.0033
PCB-93	µg/kg	0.02	0.0059
PCB-94	µg/kg	0.01	0.0042
PCB-95	µg/kg	0.01	0.0044
PCB-96	µg/kg	0.01	0.0029
PCB-97	µg/kg	0.06	0.014
PCB-98	µg/kg	0.02	0.0061
PCB-99	µg/kg	0.02	0.0074
PCB-100	µg/kg	0.02	0.0059

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-101	µg/kg	0.03	0.011
PCB-102	µg/kg	0.02	0.0061
PCB-103	µg/kg	0.01	0.0034
PCB-104	µg/kg	0.01	0.0026
PCB-105	µg/kg	0.01	0.0029
PCB-106	µg/kg	0.01	0.0032
PCB-107	µg/kg	0.01	0.0036
PCB-108	µg/kg	0.02	0.0054
PCB-109	µg/kg	0.06	0.014
PCB-110	µg/kg	0.02	0.0083
PCB-111	µg/kg	0.01	0.0027
PCB-112	µg/kg	0.01	0.0024
PCB-113	µg/kg	0.03	0.011
PCB-114	µg/kg	0.01	0.0037
PCB-115	µg/kg	0.02	0.0083
PCB-116	µg/kg	0.03	0.0087
PCB-117	µg/kg	0.03	0.0087
PCB-118	µg/kg	0.01	0.004
PCB-119	µg/kg	0.06	0.014
PCB-120	µg/kg	0.01	0.0028
PCB-121	µg/kg	0.01	0.0025
PCB-122	µg/kg	0.01	0.003
PCB-123	µg/kg	0.01	0.0037
PCB-124	µg/kg	0.02	0.0054
PCB-125	µg/kg	0.06	0.014
PCB-126	µg/kg	0.01	0.0026
PCB-127	µg/kg	0.01	0.0032
PCB-128	µg/kg	0.02	0.0062
PCB-129	µg/kg	0.04	0.016
PCB-130	µg/kg	0.01	0.003
PCB-131	µg/kg	0.01	0.0033
PCB-132	µg/kg	0.01	0.0035
PCB-133	µg/kg	0.01	0.0034
PCB-134	µg/kg	0.02	0.0065
PCB-135	µg/kg	0.02	0.0062
PCB-136	µg/kg	0.01	0.0028
PCB-137	µg/kg	0.01	0.0025
PCB-138	µg/kg	0.04	0.016
PCB-139	µg/kg	0.02	0.0041
PCB-140	µg/kg	0.02	0.0041
PCB-141	µg/kg	0.01	0.0036
PCB-142	µg/kg	0.01	0.0034
PCB-143	µg/kg	0.02	0.0065
PCB-144	µg/kg	0.01	0.0025
PCB-145	µg/kg	0.01	0.0029
PCB-146	µg/kg	0.01	0.0033
PCB-147	µg/kg	0.02	0.0074
PCB-148	µg/kg	0.01	0.0033
PCB-149	µg/kg	0.02	0.0074
PCB-150	µg/kg	0.01	0.0024
PCB-151	µg/kg	0.02	0.0062

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[a]	Laboratory MDL[a]
PCB-152	µg/kg	0.01	0.0035
PCB-153	µg/kg	0.02	0.0039
PCB-154	µg/kg	0.01	0.0038
PCB-155	µg/kg	0.01	0.0033
PCB-156	µg/kg	0.02	0.0059
PCB-157	µg/kg	0.02	0.0059
PCB-158	µg/kg	0.01	0.003
PCB-159	µg/kg	0.01	0.003
PCB-160	µg/kg	0.04	0.016
PCB-161	µg/kg	0.01	0.0024
PCB-162	µg/kg	0.01	0.0027
PCB-163	µg/kg	0.04	0.016
PCB-164	µg/kg	0.01	0.0033
PCB-165	µg/kg	0.01	0.0023
PCB-166	µg/kg	0.02	0.0062
PCB-167	µg/kg	0.01	0.0034
PCB-168	µg/kg	0.02	0.0039
PCB-169	µg/kg	0.01	0.0051
PCB-170	µg/kg	0.01	0.0047
PCB-171	µg/kg	0.02	0.0069
PCB-172	µg/kg	0.01	0.0041
PCB-173	µg/kg	0.02	0.0069
PCB-174	µg/kg	0.01	0.0031
PCB-175	µg/kg	0.01	0.0028
PCB-176	µg/kg	0.01	0.0026
PCB-177	µg/kg	0.01	0.0031
PCB-178	µg/kg	0.01	0.0035
PCB-179	µg/kg	0.01	0.0031
PCB-180	µg/kg	0.02	0.01
PCB-181	µg/kg	0.01	0.0032
PCB-182	µg/kg	0.01	0.0029
PCB-183	µg/kg	0.02	0.0066
PCB-184	µg/kg	0.01	0.0024
PCB-185	µg/kg	0.02	0.0066
PCB-186	µg/kg	0.01	0.0021
PCB-187	µg/kg	0.01	0.003
PCB-188	µg/kg	0.01	0.0026
PCB-189	µg/kg	0.01	0.0051
PCB-190	µg/kg	0.01	0.0036
PCB-191	µg/kg	0.01	0.0033
PCB-192	µg/kg	0.01	0.0032
PCB-193	µg/kg	0.02	0.01
PCB-194	µg/kg	0.01	0.0032
PCB-195	µg/kg	0.01	0.004
PCB-196	µg/kg	0.01	0.003
PCB-197	µg/kg	0.01	0.0022
PCB-198	µg/kg	0.02	0.0064
PCB-199	µg/kg	0.02	0.0064
PCB-200	µg/kg	0.01	0.0028
PCB-201	µg/kg	0.01	0.0023
PCB-202	µg/kg	0.01	0.0026

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[a]	Laboratory MDL[a]
PCB-203	µg/kg	0.01	0.0041
PCB-204	µg/kg	0.01	0.0029
PCB-205	µg/kg	0.01	0.0021
PCB-206	µg/kg	0.01	0.0043
PCB-207	µg/kg	0.01	0.0028
PCB-208	µg/kg	0.01	0.003
PCB-209	µg/kg	0.01	0.0029
PCB-12/13	µg/kg	0.02	0.012
PCB-18/30	µg/kg	0.02	0.014
PCB-20/28	µg/kg	0.02	0.0056
PCB-21/33	µg/kg	0.02	0.0066
PCB-26/29	µg/kg	0.02	0.0046
PCB-40/41/71	µg/kg	0.03	0.013
PCB-43/73	µg/kg	0.02	0.0048
PCB-44/47/65	µg/kg	0.03	0.015
PCB-45/51	µg/kg	0.02	0.0096
PCB-49/69	µg/kg	0.02	0.0054
PCB-50/53	µg/kg	0.02	0.006
PCB-59/62/75	µg/kg	0.03	0.016
PCB-61/70/74/76	µg/kg	0.04	0.011
PCB-83/99	µg/kg	0.02	0.0054
PCB-85/116/117	µg/kg	0.03	0.012
PCB-86/87/97/109/119/125	µg/kg	0.06	0.046
PCB-88/91	µg/kg	0.02	0.0064
PCB-90/101/113	µg/kg	0.03	0.0141
PCB-93/100	µg/kg	0.02	0.007
PCB-98/102	µg/kg	0.02	0.0064
PCB-108/124	µg/kg	0.02	0.004
PCB-110/115	µg/kg	0.02	0.0068
PCB-128/166	µg/kg	0.02	0.0098
PCB-129/138/160/163	µg/kg	0.04	0.023
PCB-134/143	µg/kg	0.02	0.0086
PCB-135/151	µg/kg	0.02	0.0074
PCB-139/140	µg/kg	0.02	0.006
PCB-147/149	µg/kg	0.02	0.0048
PCB-153/168	µg/kg	0.02	0.0039
PCB-156/157	µg/kg	0.02	0.0048
PCB-171/173	µg/kg	0.02	0.0086
PCB-180/193	µg/kg	0.02	0.0074
PCB-183/185	µg/kg	0.02	0.0052
PCB-198/199	µg/kg	0.02	0.0074
Total Monochlorobiphenyls	µg/kg	0.01	--
Total Dichlorobiphenyls	µg/kg	0.02	--
Total Trichlorobiphenyls	µg/kg	0.02	--
Total Tetrachlorobiphenyls	µg/kg	0.04	--
Total Pentachlorobiphenyls	µg/kg	0.06	--
Total Hexachlorobiphenyls	µg/kg	0.04	--
Total Heptachlorobiphenyls	µg/kg	0.02	--
Total Octachlorobiphenyls	µg/kg	0.02	--
Total Nonachlorobiphenyls	µg/kg	0.01	--
Polychlorinated biphenyls, Total	µg/kg	0.06	--

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
Semivolatile Organic Compounds (SVOCs) – Gas Chromatography/Mass Spectrometry – (SW846 8270D LL)			
1,2,4-Trichlorobenzene	µg/kg	16.5	5.06
1,2-Dichlorobenzene	µg/kg	16.5	5.33
1,2-Diphenylhydrazine (as Azobenzene)	µg/kg	16.5	7.42
1,3-Dichlorobenzene	µg/kg	16.5	5.53
1,4-Dichlorobenzene	µg/kg	16.5	6.09
2,2'-oxybis[1-chloropropane]	µg/kg	3.35	1.24
2,4,6-Trichlorophenol	µg/kg	16.5	5.52
2,4-Dichlorophenol	µg/kg	3.35	1.29
2,4-Dimethylphenol	µg/kg	16.5	5.6
2,4-Dinitrophenol	µg/kg	165	104
2,4-Dinitrotoluene	µg/kg	16.5	9.93
2,6-Dinitrotoluene	µg/kg	16.5	6.46
2-Chloronaphthalene	µg/kg	3.35	0.765
2-Chlorophenol	µg/kg	16.5	6.11
2-Methylphenol	µg/kg	16.5	4.78
2-Nitrophenol	µg/kg	16.5	6.14
3,3'-Dichlorobenzidine	µg/kg	16.5	15.6
4,6-Dinitro-2-methylphenol	µg/kg	85	28.8
4-Bromophenyl phenyl ether	µg/kg	16.5	7.07
4-Chloro-3-methylphenol	µg/kg	16.5	5.85
4-Chlorophenyl phenyl ether	µg/kg	16.5	5.54
4-Nitrophenol	µg/kg	85	11.7
Benzidine	µg/kg	335	126
Benzoic acid	µg/kg	85	37.1
Benzyl alcohol	µg/kg	83.5	22.6
Bis(2-chloroethoxy)methane	µg/kg	16.5	6.1
Bis(2-chloroethyl)ether	µg/kg	3.35	1.03
Bis(2-ethylhexyl) phthalate	µg/kg	165	17.8
Butyl benzyl phthalate	µg/kg	83.5	11.5
Dibenzofuran	µg/kg	16.5	6.12
Diethyl phthalate	µg/kg	16.5	5.85
Dimethyl phthalate	µg/kg	33.5	18.5
Di-n-butyl phthalate	µg/kg	165	34.7
Di-n-octyl phthalate	µg/kg	16.5	9.7
Hexachlorobenzene	µg/kg	3.35	1.2
Hexachlorobutadiene	µg/kg	3.35	0.975
Hexachlorocyclopentadiene	µg/kg	16.5	4.51
Hexachloroethane	µg/kg	16.5	5.88
Isophorone	µg/kg	16.5	6.24
Methylphenol, 3 & 4	µg/kg	16.5	4.9
Nitrobenzene	µg/kg	33.4	6.1
N-Nitrosodimethylamine	µg/kg	16.5	6.32
N-Nitrosodi-n-propylamine	µg/kg	3.35	1.13

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
N-Nitrosodiphenylamine	µg/kg	16.5	5.55
Pentachlorophenol	µg/kg	85	26.8
Phenol	µg/kg	16.5	5.05
Volatile Organic Compounds – Gas Chromatography/Mass Spectrometry (SW846 8260C) [(b)]			
Benzene	µg/kg	5	2.88
Toluene	µg/kg	5	3.8
Ethylbenzene	µg/kg	5	3.8
Xylenes, Total	µg/kg	10.0	7.21
m-Xylene & p-Xylene	µg/kg	5.00	3.92
o-Xylene	µg/kg	5.00	3.46
1,1,1-Trichloroethane	µg/kg	250	83.5
1,1,2,2-Tetrachloroethane	µg/kg	250	190
1,1,2-Trichloroethane	µg/kg	250	131
1,1-Dichloroethane	µg/kg	250	128
1,1-Dichloroethene	µg/kg	250	194
1,2,4-Trichlorobenzene	µg/kg	250	127
1,2-Dichlorobenzene	µg/kg	250	143
1,2-Dichloroethane	µg/kg	250	150
1,2-Dichloropropane	µg/kg	250	120
1,3-Dichlorobenzene	µg/kg	250	154
1,4-Dichlorobenzene	µg/kg	250	139
2-Butanone (MEK)	µg/kg	1250	491
2-Chloroethyl vinyl ether	µg/kg	500	336
Acrolein	µg/kg	5000	1710
Acrylonitrile	µg/kg	5000	1350
Bromoform	µg/kg	250	126
Bromomethane	µg/kg	250	114
Carbon tetrachloride	µg/kg	250	102
Chlorodibromomethane	µg/kg	250	124
Chloroethane	µg/kg	250	146
Chloroform	µg/kg	250	168
Chloromethane	µg/kg	250	198
cis-1,3-Dichloropropene	µg/kg	250	111
Dichlorobromomethane	µg/kg	250	117
Dichlorodifluoromethane	µg/kg	250	203
Methylene Chloride	µg/kg	250	226
Tetrachloroethene	µg/kg	250	189
trans-1,2-Dichloroethene	µg/kg	250	154
trans-1,3-Dichloropropene	µg/kg	250	113
Trichloroethene	µg/kg	250	136
Trichlorofluoromethane	µg/kg	250	208
Vinyl chloride	µg/kg	250	180
Total Petroleum Hydrocarbons (TPH) – Gas Chromatography (8015D) [(b)]			
Gasoline Range Organics (C6-C10)	mg/kg	1	0.41
Diesel Range Organics (C10-C20)	mg/kg	12	5
Oil Range Organics (C20-C34)	mg/kg	12	5

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-2 (continued)

Parameter	Units	Laboratory RL[a]	Laboratory MDL[a]
Wet Chemistry Parameters			
Total Organic Carbon (Lloyd Kahn)	mg/kg	1000	971
Cyanide, Total (EPA 9010C/9014)	mg/kg	0.25	0.19
pH (EPA 9045)	SU	0.1	--
Sulfide (EPA 9030B/9034)	mg/kg	30	10.0
PFAS (EPA 1633)			
Perfluorooctanoic acid (PFOA)	µg/kg	0.2	0.06
Perfluorobutanesulfonic acid (PFBS)	µg/kg	0.2	0.07
Perfluorooctanesulfonic acid (PFOS)	µg/kg	0.2	0.05
Hexafluoropropylene Oxide Dimer Acid (HFPO-DA)	µg/kg	0.2	0.11
Perfluorononanoic acid (PFNA)	µg/kg	0.2	0.06
Perfluorohexanesulfonic acid (PFHxS)	µg/kg	0.2	0.05
Radioactivity (EPA 900.0)			
Gross alpha	pCi/g	10.0	--
Gross beta	pCi/g	10.0	--

mg/kg – milligrams per kilogram; µg/kg – micrograms per kilogram; ng/kg – nanograms per kilogram

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-3. PROJECT LIMITS FOR TCLP SAMPLES
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]	Recommended TDL[(b)]
Metals - Atomic Emission Inductively Coupled Plasma / Mass Spectrometry (SW846 1311/6010C)				
Arsenic	mg/L	0.01	0.0083	5.0
Barium	mg/L	0.2	0.0273	100
Cadmium	mg/L	0.005	0.0003	1.0
Chromium	mg/L	0.005	0.00309	5.0
Lead	mg/L	0.01	0.0044	5.0
Selenium	mg/L	0.01	0.00373	1.0
Silver	mg/L	0.005	0.00091	5.0
Metals - Cold Vapor (SW846 1311/7470A)				
Mercury	mg/L	0.0002	0.00013	0.20
Volatile Organics - Gas Chromatography/Mass Spectrometry - (SW846 1311/8260C)				
1,1-Dichloroethene	mg/L	0.2	0.0189	0.7
1,2-Dichloroethane	mg/L	0.2	0.0115	0.5
2-Butanone (Methyl Ethyl Ketone)	mg/L	1	0.0514	200
Benzene	mg/L	0.2	0.0119	0.5
Carbon tetrachloride	mg/L	0.2	0.0176	0.5
Chlorobenzene	mg/L	0.2	0.01	100
Chloroform	mg/L	0.2	0.012	6.0
Tetrachloroethene	mg/L	0.2	0.00934	0.7
Trichloroethene	mg/L	0.2	0.0138	0.5
Vinyl Chloride	mg/L	0.2	0.00814	0.2
Semivolatile Organics - Gas Chromatography/Mass Spectrometry - (SW846 1311/8270D)				
1,4-Dichlorobenzene	mg/L	0.05	0.00451	7.5
2,4-Dinitrotoluene	mg/L	0.05	0.0079	0.13
2,4,5-Trichlorophenol	mg/L	0.05	0.00792	400
2,4,6-Trichlorophenol	mg/L	0.05	0.00945	2.0
2-Methylphenol	mg/L	0.05	0.00403	200
3 & 4-Methylphenol	mg/L	0.05	0.00787	200
Hexachlorobenzene	mg/L	0.05	0.00547	0.13
Hexachlorobutadiene	mg/L	0.05	0.00837	0.5
Hexachloroethane	mg/L	0.05	0.00402	3.0
Nitrobenzene	mg/L	0.05	0.0122	2.0
Pentachlorophenol	mg/L	0.25	0.00754	100
Pyridine	mg/L	0.1	0.00815	5.0
Organochlorine Pesticides - Gas Chromatography/ Electron Capture Detector - (SW846 1311/8081B)				
Chlordane (technical)	mg/L	0.005	0.0029	0.03
Endrin	mg/L	0.0005	9.1E-05	0.02
Gamma-BHC (Lindane)	mg/L	0.0005	0.00012	0.4
Heptachlor	mg/L	0.0005	0.00018	0.008
Heptachlor epoxide	mg/L	0.0005	0.00014	0.008
Methoxychlor	mg/L	0.0005	0.00031	10
Toxaphene	mg/L	0.04	0.0196	0.5

mg/L – milligrams per liter
(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change.
(b) Target Detection Limit (TDL) for TCLP parameters are the Toxicity Characteristic Rule's Regulatory Level (40 CFR 261.24, June 2018).

TABLE 3-3 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]	Recommended TDL[(b)]
Chlorophenoxy Acid Herbicides - Gas Chromatography/ Electron Capture Detector - (SW846 1311/8151A)				
2,4-D	mg/L	0.04	0.0197	10
2,4,5-TP (Silvex)	mg/L	0.01	0.00304	1.0

mg/L – milligrams per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change.

(b) Target Detection Limit (TDL) for TCLP parameters are the Toxicity Characteristic Rule's Regulatory Level (40 CFR 261.24, June 2018).

TABLE 3-4. PROJECT LIMITS FOR AQUEOUS SAMPLES
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (2024)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
Metals - Atomic Emission Inductively Coupled Plasma / Mass Spectrometry (SW846 6020A)			
Aluminum	µg/L	30	21.8
Antimony	µg/L	2	0.967
Arsenic	µg/L	1	0.563
Beryllium	µg/L	1	0.274
Cadmium	µg/L	1	0.217
Chromium	µg/L	2	1.53
Cobalt	µg/L	0.5	0.261
Copper	µg/L	2	1.14
Iron	µg/L	50	27.7
Lead	µg/L	1	0.376
Manganese	µg/L	5	1.96
Nickel	µg/L	1	0.517
Selenium	µg/L	5	1.53
Silver	µg/L	1	0.223
Thallium	µg/L	1	0.472
Vanadium	µg/L	1	0.776
Zinc	µg/L	15	6
Metals - Cold Vapor (SW846 7470A)			
Mercury	µg/L	0.200	0.13
Polycyclic Aromatic Hydrocarbons (PAHs) – Gas Chromatography/Mass Spectrometry/Selected Ion Monitoring - (SW846 8270D LL) [(b)]			
1-Methylnaphthalene	µg/L	0.19	0.056
2-Methylnaphthalene	µg/L	0.19	0.062
Acenaphthene	µg/L	0.19	0.065
Acenaphthylene	µg/L	0.19	0.065
Anthracene	µg/L	0.19	0.049
Benzo[a]anthracene	µg/L	0.19	0.075
Benzo[a]pyrene	µg/L	0.19	0.053
Benzo[b]fluoranthene	µg/L	0.19	0.097
Benzo[g,h,i]perylene	µg/L	0.19	0.069
Benzo[k]fluoranthene	µg/L	0.19	0.088
Chrysene	µg/L	0.19	0.081
Dibenz(a,h)anthracene	µg/L	0.19	0.072
Fluoranthene	µg/L	0.19	0.06
Fluorene	µg/L	0.19	0.069
Indeno[1,2,3-cd]pyrene	µg/L	0.19	0.085
Naphthalene	µg/L	0.19	0.059
Phenanthrene	µg/L	0.19	0.157
Pyrene	µg/L	0.19	0.054
Organochlorine Pesticides - Gas Chromatography/Electron Capture Detector - (SW846 8081B LL)			
Aldrin	µg/L	0.0013	0.00036
alpha-BHC	µg/L	0.0013	0.00024
beta-BHC	µg/L	0.0013	0.00037

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
delta-BHC	µg/L	0.0013	0.00064
gamma-BHC (Lindane)	µg/L	0.0013	0.00029
Chlordane (technical)	µg/L	0.0125	0.00726
Chlorobenside	µg/L	0.0032	0.00176
Dachtal (DCPA)	µg/L	0.0025	0.00045
4,4'-DDD	µg/L	0.0013	0.00053
4,4'-DDE	µg/L	0.0013	0.00072
4,4'-DDT	µg/L	0.0013	0.00069
Dieldrin	µg/L	0.0013	0.00027
Endosulfan I	µg/L	0.0013	0.00069
Endosulfan II	µg/L	0.0013	0.00094
Endosulfan sulfate	µg/L	0.0013	0.00064
Endrin	µg/L	0.0013	0.00023
Endrin aldehyde	µg/L	0.0013	0.00052
Heptachlor	µg/L	0.0013	0.00045
Heptachlor epoxide	µg/L	0.0013	0.00034
Methoxychlor	µg/L	0.0013	0.00078
Mirex	µg/L	0.0013	0.00047
Toxaphene	µg/L	0.1	0.0491
PCB Aroclors - Gas Chromatography (SW846 8082A) [(b)]			
PCB-1016	µg/L	0.01	0.00476
PCB-1221	µg/L	0.01	0.00572
PCB-1232	µg/L	0.01	0.00521
PCB-1242	µg/L	0.01	0.00913
PCB-1248	µg/L	0.01	0.00299
PCB-1254	µg/L	0.01	0.00952
PCB-1260	µg/L	0.01	0.00392
Dioxins/Furans- High Resolution Mass Spectrometry (HRMS) - (USEPA 1613B) [(b)]			
2,3,7,8-TCDD	pg/L	10	2.2
1,2,3,7,8-PeCDD	pg/L	50	2.6
1,2,3,4,7,8-HxCDD	pg/L	50	3.1
1,2,3,6,7,8-HxCDD	pg/L	50	2.9
1,2,3,7,8,9-HxCDD	pg/L	50	3
1,2,3,4,6,7,8-HpCDD	pg/L	50	3.9
OCDD	pg/L	100	11
2,3,7,8-TCDF	pg/L	10	3.8
1,2,3,7,8-PeCDF	pg/L	50	2.2
2,3,4,7,8-PeCDF	pg/L	50	1.8
1,2,3,4,7,8-HxCDF	pg/L	50	2.6
1,2,3,6,7,8-HxCDF	pg/L	50	3
2,3,4,6,7,8-HxCDF	pg/L	50	2.8
1,2,3,7,8,9-HxCDF	pg/L	50	2.6
1,2,3,4,6,7,8-HpCDF	pg/L	50	3.1
1,2,3,4,7,8,9-HpCDF	pg/L	50	3.2
OCDF	pg/L	100	7.6
PCB Congeners - Gas Chromatography/Electron Capture Detector (SW846 1668A) [(b)]			
PCB-1	ng/L	0.04	0.0091
PCB-2	ng/L	0.04	0.0097

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-3	ng/L	0.04	0.0097
PCB-4	ng/L	0.06	0.012
PCB-5	ng/L	0.04	0.0082
PCB-6	ng/L	0.04	0.0068
PCB-7	ng/L	0.04	0.0086
PCB-8	ng/L	0.06	0.0092
PCB-9	ng/L	0.04	0.0096
PCB-10	ng/L	0.04	0.009
PCB-11	ng/L	0.06	0.025
PCB-12	ng/L	0.08	0.016
PCB-13	ng/L	0.08	0.016
PCB-14	ng/L	0.04	0.0081
PCB-15	ng/L	0.04	0.0061
PCB-16	ng/L	0.04	0.0091
PCB-17	ng/L	0.04	0.0098
PCB-18	ng/L	0.08	0.016
PCB-19	ng/L	0.04	0.0093
PCB-20	ng/L	0.08	0.012
PCB-21	ng/L	0.08	0.015
PCB-22	ng/L	0.04	0.0078
PCB-23	ng/L	0.04	0.0075
PCB-24	ng/L	0.04	0.012
PCB-25	ng/L	0.04	0.0069
PCB-26	ng/L	0.08	0.013
PCB-27	ng/L	0.04	0.0079
PCB-28	ng/L	0.08	0.012
PCB-29	ng/L	0.08	0.013
PCB-30	ng/L	0.08	0.016
PCB-31	ng/L	0.04	0.0088
PCB-32	ng/L	0.04	0.0095
PCB-33	ng/L	0.08	0.015
PCB-34	ng/L	0.04	0.0061
PCB-35	ng/L	0.04	0.007
PCB-36	ng/L	0.04	0.0074
PCB-37	ng/L	0.04	0.0075
PCB-38	ng/L	0.04	0.0081
PCB-39	ng/L	0.04	0.0079
PCB-40	ng/L	0.12	0.018
PCB-41	ng/L	0.12	0.018
PCB-42	ng/L	0.04	0.0065
PCB-43	ng/L	0.08	0.012
PCB-44	ng/L	0.12	0.026
PCB-45	ng/L	0.08	0.016
PCB-46	ng/L	0.04	0.0077
PCB-47	ng/L	0.12	0.026
PCB-48	ng/L	0.04	0.0076
PCB-49	ng/L	0.08	0.012
PCB-50	ng/L	0.08	0.012
PCB-51	ng/L	0.08	0.016
PCB-52	ng/L	0.04	0.0098
PCB-53	ng/L	0.08	0.012
PCB-54	ng/L	0.04	0.0096

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-55	ng/L	0.04	0.0072
PCB-56	ng/L	0.04	0.01
PCB-57	ng/L	0.04	0.0074
PCB-58	ng/L	0.04	0.0078
PCB-59	ng/L	0.12	0.017
PCB-60	ng/L	0.04	0.0071
PCB-61	ng/L	0.16	0.022
PCB-62	ng/L	0.12	0.017
PCB-63	ng/L	0.04	0.0054
PCB-64	ng/L	0.04	0.007
PCB-65	ng/L	0.12	0.026
PCB-66	ng/L	0.04	0.0097
PCB-67	ng/L	0.04	0.0078
PCB-68	ng/L	0.04	0.0046
PCB-69	ng/L	0.08	0.012
PCB-70	ng/L	0.16	0.022
PCB-71	ng/L	0.12	0.018
PCB-72	ng/L	0.04	0.006
PCB-73	ng/L	0.08	0.012
PCB-74	ng/L	0.16	0.022
PCB-75	ng/L	0.12	0.017
PCB-76	ng/L	0.16	0.022
PCB-77	ng/L	0.04	0.007
PCB-78	ng/L	0.04	0.0071
PCB-79	ng/L	0.04	0.008
PCB-80	ng/L	0.04	0.006
PCB-81	ng/L	0.04	0.0081
PCB-82	ng/L	0.04	0.011
PCB-83	ng/L	0.08	0.015
PCB-84	ng/L	0.04	0.012
PCB-85	ng/L	0.12	0.029
PCB-86	ng/L	0.24	0.061
PCB-87	ng/L	0.24	0.061
PCB-88	ng/L	0.08	0.02
PCB-89	ng/L	0.04	0.01
PCB-90	ng/L	0.12	0.024
PCB-91	ng/L	0.08	0.02
PCB-92	ng/L	0.04	0.0077
PCB-93	ng/L	0.08	0.023
PCB-94	ng/L	0.04	0.0091
PCB-95	ng/L	0.04	0.0095
PCB-96	ng/L	0.04	0.01
PCB-97	ng/L	0.24	0.061
PCB-98	ng/L	0.08	0.018
PCB-99	ng/L	0.08	0.015
PCB-100	ng/L	0.08	0.023
PCB-101	ng/L	0.12	0.024
PCB-102	ng/L	0.08	0.018
PCB-103	ng/L	0.04	0.0091
PCB-104	ng/L	0.04	0.009
PCB-105	ng/L	0.04	0.0093
PCB-106	ng/L	0.04	0.008

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-107	ng/L	0.04	0.0091
PCB-108	ng/L	0.08	0.013
PCB-109	ng/L	0.24	0.061
PCB-110	ng/L	0.08	0.02
PCB-111	ng/L	0.04	0.0091
PCB-112	ng/L	0.04	0.012
PCB-113	ng/L	0.12	0.024
PCB-114	ng/L	0.04	0.0073
PCB-115	ng/L	0.08	0.02
PCB-116	ng/L	0.12	0.029
PCB-117	ng/L	0.12	0.029
PCB-118	ng/L	0.04	0.011
PCB-119	ng/L	0.24	0.061
PCB-120	ng/L	0.04	0.012
PCB-121	ng/L	0.04	0.0096
PCB-122	ng/L	0.04	0.0079
PCB-123	ng/L	0.04	0.01
PCB-124	ng/L	0.08	0.013
PCB-125	ng/L	0.24	0.061
PCB-126	ng/L	0.04	0.0086
PCB-127	ng/L	0.04	0.0065
PCB-128	ng/L	0.08	0.017
PCB-129	ng/L	0.16	0.03
PCB-130	ng/L	0.04	0.0072
PCB-131	ng/L	0.04	0.0092
PCB-132	ng/L	0.04	0.0073
PCB-133	ng/L	0.04	0.0089
PCB-134	ng/L	0.08	0.011
PCB-135	ng/L	0.08	0.024
PCB-136	ng/L	0.04	0.0096
PCB-137	ng/L	0.04	0.009
PCB-138	ng/L	0.16	0.03
PCB-139	ng/L	0.08	0.016
PCB-140	ng/L	0.08	0.016
PCB-141	ng/L	0.04	0.01
PCB-142	ng/L	0.04	0.011
PCB-143	ng/L	0.08	0.011
PCB-144	ng/L	0.04	0.012
PCB-145	ng/L	0.04	0.011
PCB-146	ng/L	0.04	0.0076
PCB-147	ng/L	0.08	0.011
PCB-148	ng/L	0.04	0.0095
PCB-149	ng/L	0.08	0.011
PCB-150	ng/L	0.04	0.009
PCB-151	ng/L	0.08	0.024
PCB-152	ng/L	0.04	0.0065
PCB-153	ng/L	0.08	0.011
PCB-154	ng/L	0.04	0.012
PCB-155	ng/L	0.04	0.007
PCB-156	ng/L	0.08	0.018
PCB-157	ng/L	0.08	0.018
PCB-158	ng/L	0.04	0.0084

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-159	ng/L	0.04	0.0084
PCB-160	ng/L	0.16	0.03
PCB-161	ng/L	0.04	0.0073
PCB-162	ng/L	0.04	0.0094
PCB-163	ng/L	0.16	0.03
PCB-164	ng/L	0.04	0.0096
PCB-165	ng/L	0.04	0.0073
PCB-166	ng/L	0.08	0.017
PCB-167	ng/L	0.04	0.0077
PCB-168	ng/L	0.08	0.011
PCB-169	ng/L	0.04	0.0085
PCB-170	ng/L	0.04	0.0082
PCB-171	ng/L	0.08	0.018
PCB-172	ng/L	0.04	0.0083
PCB-173	ng/L	0.08	0.018
PCB-174	ng/L	0.04	0.0087
PCB-175	ng/L	0.04	0.0079
PCB-176	ng/L	0.04	0.008
PCB-177	ng/L	0.04	0.0092
PCB-178	ng/L	0.04	0.011
PCB-179	ng/L	0.04	0.0071
PCB-180	ng/L	0.08	0.017
PCB-181	ng/L	0.04	0.0086
PCB-182	ng/L	0.04	0.008
PCB-183	ng/L	0.08	0.016
PCB-184	ng/L	0.04	0.0079
PCB-185	ng/L	0.08	0.016
PCB-186	ng/L	0.04	0.008
PCB-187	ng/L	0.04	0.0086
PCB-188	ng/L	0.04	0.01
PCB-189	ng/L	0.04	0.008
PCB-190	ng/L	0.04	0.0077
PCB-191	ng/L	0.04	0.0078
PCB-192	ng/L	0.04	0.0078
PCB-193	ng/L	0.08	0.017
PCB-194	ng/L	0.04	0.011
PCB-195	ng/L	0.04	0.009
PCB-196	ng/L	0.04	0.0095
PCB-197	ng/L	0.04	0.009
PCB-198	ng/L	0.08	0.018
PCB-199	ng/L	0.08	0.018
PCB-200	ng/L	0.04	0.008
PCB-201	ng/L	0.04	0.0091
PCB-202	ng/L	0.04	0.0096
PCB-203	ng/L	0.04	0.0089
PCB-204	ng/L	0.04	0.0086
PCB-205	ng/L	0.04	0.011
PCB-206	ng/L	0.04	0.01
PCB-207	ng/L	0.04	0.009
PCB-208	ng/L	0.04	0.0097
PCB-209	ng/L	0.04	0.013
PCB-12/13	ng/L	0.08	--

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
PCB-18/30	ng/L	0.08	--
PCB-20/28	ng/L	0.08	--
PCB-21/33	ng/L	0.08	--
PCB-26/29	ng/L	0.08	--
PCB-40/41/71	ng/L	0.12	--
PCB-43/73	ng/L	0.08	--
PCB-44/47/65	ng/L	0.12	--
PCB-45/51	ng/L	0.08	--
PCB-49/69	ng/L	0.08	--
PCB-50/53	ng/L	0.08	--
PCB-59/62/75	ng/L	0.12	--
PCB-61/70/74/76	ng/L	0.16	--
PCB-83/99	ng/L	0.08	--
PCB-85/116/117	ng/L	0.12	--
PCB-86/87/97/109/119/125	ng/L	0.24	--
PCB-88/91	ng/L	0.08	--
PCB-90/101/113	ng/L	0.12	--
PCB-93/100	ng/L	0.08	--
PCB-98/102	ng/L	0.08	--
PCB-108/124	ng/L	0.08	--
PCB-110/115	ng/L	0.08	--
PCB-128/166	ng/L	0.08	--
PCB-129/138/160/163	ng/L	0.16	--
PCB-134/143	ng/L	0.08	--
PCB-135/151	ng/L	0.08	--
PCB-139/140	ng/L	0.08	--
PCB-147/149	ng/L	0.08	--
PCB-153/168	ng/L	0.08	--
PCB-156/157	ng/L	0.08	--
PCB-171/173	ng/L	0.08	--
PCB-180/193	ng/L	0.08	--
PCB-183/185	ng/L	0.08	--
PCB-198/199	ng/L	0.08	--
Total Monochlorobiphenyls	ng/L	0.04	--
Total Dichlorobiphenyls	ng/L	0.08	--
Total Trichlorobiphenyls	ng/L	0.08	--
Total Tetrachlorobiphenyls	ng/L	0.16	--
Total Pentachlorobiphenyls	ng/L	0.24	--
Total Hexachlorobiphenyls	ng/L	0.16	--
Total Heptachlorobiphenyls	ng/L	0.08	--
Total Octachlorobiphenyls	ng/L	0.08	--
Total Nonachlorobiphenyls	ng/L	0.04	--
Polychlorinated biphenyls, Total	ng/L	0.24	--
Semivolatile Organic Compounds (SVOCs) – Gas Chromatography/Mass Spectrometry - (SW846 8270D LL)			
1,2,4-Trichlorobenzene	µg/L	1	0.244
1,2-Dichlorobenzene	µg/L	1	0.21
1,2-Diphenylhydrazine(as Azobenzene)	µg/L	1	0.196
1,3-Dichlorobenzene	µg/L	1	0.215
1,4-Dichlorobenzene	µg/L	1	0.237
2,2'-oxybis[1-chloropropane]	µg/L	0.19	0.058

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
2,4,6-Trichlorophenol	µg/L	1	0.224
2,4-Dichlorophenol	µg/L	0.19	0.051
2,4-Dimethylphenol	µg/L	1	0.585
2,4-Dinitrophenol	µg/L	10	3.25
2,4-Dinitrotoluene	µg/L	1	0.353
2,6-Dinitrotoluene	µg/L	1	0.173
2-Chloronaphthalene	µg/L	0.19	0.059
2-Chlorophenol	µg/L	1	0.225
2-Methylphenol	µg/L	1	0.557
2-Nitrophenol	µg/L	1	0.193
3,3'-Dichlorobenzidine	µg/L	1	0.583
4,6-Dinitro-2-methylphenol	µg/L	5	1.47
4-Bromophenyl phenyl ether	µg/L	1	0.319
4-Chloro-3-methylphenol	µg/L	1	0.435
4-Chlorophenyl phenyl ether	µg/L	1	0.221
4-Nitrophenol	µg/L	5	0.94
Benzidine	µg/L	20	9.13
Benzoic acid	µg/L	5	2.53
Benzyl alcohol	µg/L	5	1.68
Bis(2-chloroethoxy)methane	µg/L	1	0.152
Bis(2-chloroethyl)ether	µg/L	0.19	0.04
Bis(2-ethylhexyl) phthalate	µg/L	10	6.23
Butyl benzyl phthalate	µg/L	2	0.936
Dibenzofuran	µg/L	1	0.19
Diethyl phthalate	µg/L	1	0.567
Dimethyl phthalate	µg/L	2	0.2
Di-n-butyl phthalate	µg/L	10	4.91
Di-n-octyl phthalate	µg/L	1	0.685
Hexachlorobenzene	µg/L	0.19	0.056
Hexachlorobutadiene	µg/L	0.19	0.069
Hexachlorocyclopentadiene	µg/L	1	0.497
Hexachloroethane	µg/L	1	0.133
Isophorone	µg/L	1	0.188
Methylphenol, 3 & 4	µg/L	1	0.372
Nitrobenzene	µg/L	2	0.5
N-Nitrosodimethylamine	µg/L	1	0.259
N-Nitrosodi-n-propylamine	µg/L	0.19	0.071
N-Nitrosodiphenylamine	µg/L	1	0.119
Pentachlorophenol	µg/L	5	0.847
Phenol	µg/L	1	0.487
Volatile Organic Compounds (VOCs) – Gas Chromatography/Mass Spectrometry - (SW846 8260C LL)			
1,1,1-Trichloroethane	µg/L	1	0.602
1,1,2,2-Tetrachloroethane	µg/L	1	0.602
1,1,2-Trichloroethane	µg/L	1	0.453
1,1-Dichloroethane	µg/L	1	0.556
1,1-Dichloroethene	µg/L	1	0.947
1,2,4-Trichlorobenzene	µg/L	1	0.773
1,2-Dichlorobenzene	µg/L	1	0.695
1,2-Dichloroethane	µg/L	1	0.574
1,2-Dichloropropane	µg/L	1	0.659

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

TABLE 3-4 (continued)

Parameter	Units	Laboratory RL[(a)]	Laboratory MDL[(a)]
1,3-Dichlorobenzene	µg/L	1	0.504
1,4-Dichlorobenzene	µg/L	1	0.544
2-Butanone (MEK)	µg/L	5	2.58
2-Chloroethyl vinyl ether	µg/L	2	1.72
Acrolein	µg/L	20	16
Acrylonitrile	µg/L	20	16.9
Benzene	µg/L	1	0.596
Bromoform	µg/L	1	0.978
Bromomethane	µg/L	1	0.887
Carbon tetrachloride	µg/L	1	0.878
Chlorodibromomethane	µg/L	1	0.84
Chloroethane	µg/L	1	0.899
Chloroform	µg/L	1	0.598
Chloromethane	µg/L	1	0.899
cis-1,3-Dichloropropene	µg/L	1	0.591
Dichlorobromomethane	µg/L	1	0.641
Dichlorodifluoromethane	µg/L	1	0.83
Ethylbenzene	µg/L	1	0.507
Methylene Chloride	µg/L	1	0.886
Tetrachloroethene	µg/L	1	0.467
Toluene	µg/L	1	0.457
trans-1,2-Dichloroethene	µg/L	1	0.67
trans-1,3-Dichloropropene	µg/L	1	0.583
Trichloroethene	µg/L	1	0.688
Trichlorofluoromethane	µg/L	1	0.87
Vinyl chloride	µg/L	1	0.407
Wet Chemistry Parameters			
Total Organic Carbon (SM 5310C)	mg/L	1.0	0.508
Cyanide, Total (SW846 9014)	µg/L	10	8
Sulfide (SW846 9030B/9034)	mg/L	3.0	2.09
Total Suspended Solids (EPA 2540D)	mg/L	0.5	0.5
Total Petroleum Hydrocarbons (8015D)			
C10-C20 DRO	mg/L	0.2	0.074
C20-C34 ORO	mg/L	0.2	0.074
C6-C10 GRO	µg/L	50	23
Radioactivity (9310)			
Gross alpha	pCi/L	3	--
Gross beta	pCi/L	4	--
PFAS (1633)			
Perfluorooctanoic acid (PFOA)	ng/L	2	0.9
Perfluorononanoic acid (PFNA)	ng/L	2	0.5
Perfluorobutanesulfonic acid (PFBS)	ng/L	2	0.5
Perfluorohexanesulfonic acid (PFHxS)	ng/L	2	0.8
Perfluorooctanesulfonic acid (PFOS)	ng/L	2	0.5
Hexafluoropropylene Oxide Dimer Acid (HFPO-DA)	ng/L	3	1.2

mg/L – milligrams per liter; µg/L – micrograms per liter; pg/L – picograms per liter; ng/L – nanogram per liter

(a) RL=Reporting Limit, MDL = Method Detection Limit. Values ≥ MDL and < RL will be qualified as estimated. MDLs are required to be updated periodically, and are subject to change; dioxin and furan congeners are quantified to a sample-specific EDL, calculated based on the instrument noise for each sample at the time it is analyzed

This page left intentionally blank

4. DATA ANALYSIS AND SCREENING

Results of the physical and chemical testing were evaluated using applicable screening criteria and statistical methods. Details of the data analysis methods used in this evaluation are provided in the following sections.

4.1 DATA ANALYSIS

4.1.1 Total PCB Aroclors and Congeners

The total PCB Aroclor concentrations were determined by summing the concentrations of the seven Aroclors. Three values were reported for total PCB Aroclors, representing the following methods for treating concentrations below the analytical detection limit:

- Non-detects (NDs) = zero (ND=0)
- ND=½RL
- ND=RL

Substituting the RL (ND=RL) for each non-detected (ND) analyte provides the most conservative estimate of the concentration. This method, however, tends to produce results that are biased high, especially in data sets where the majority of samples are NDs. This overestimation is important to consider when comparing the calculated total values to criteria values.

Total PCBs were calculated 1) by summing the concentrations of the nine homolog groups and 2) a separate calculation was conducted by summing the individual 209 congeners.

4.1.2 Calculation of Total Polycyclic Aromatic Hydrocarbons

PAHs were also summed because PAHs are usually found in mixtures containing two or more compounds (Agency for Toxic Substances and Disease Registry 1995). Total PAH concentrations were calculated for each sample by summing the concentrations of the individual PAHs. In addition, total PAHs were calculated as total low molecular weight PAHs (LPAHs) (two or three carbon rings) and total high molecular weight PAHs (HPAHs) (four, five, or six carbon rings). HPAHs and LPAHs have different sources and behave differently in marine environments. LPAHs are often associated with petroleum, while HPAHs are associated with combustion products (NOAA 1989).

- LPAHs included in the total LPAH (as per USEPA/USACE 2008): 1-methylnaphthalene, 2-methylnaphthalene, anthracene, acenaphthene, fluorene, naphthalene, and phenanthrene.

- HPAHs included in the total HPAH (as per USEPA/USACE 2008): benzo(a)anthracene, benzo(a)pyrene, chrysene, dibenzo(a,h)anthracene, fluoranthene, and pyrene.

Three values were reported for total PAHs, representing the following method for treating concentrations below the analytical detection limit:

- ND=0

- ND=½RL
- ND=RL

Substituting the reporting limit (ND=RL) for each non-detect provides a conservative estimate of the concentration. This method, however, tends to produce results that are biased high, especially in data sets where the majority of samples are non-detects. This overestimation is important to consider when comparing the calculated total values to criteria.

4.1.3 Dioxin/Furans - Calculation of Toxic Equivalency Quotients

The toxic equivalency quotients (TEQs) for dioxin and furan congeners in sediment were calculated following the approach recommended by the World Health Organization (WHO) (Van den Berg et al. 2006). Each congener was multiplied by a WHO-recommended toxicity equivalency factor for human health (Van den Berg et al. 2006) and then the congener concentrations were summed. Concentrations that were flagged "I" (estimated maximum possible concentration) were not included in the TEQ calculation (USEPA 2020). Concentrations that were flagged with a "B" (detected in blank) were only included in the TEQ calculation for 2,3,7,8-TCDD and 2,3,7,8-TCDF. The dioxin TEQs were calculated using ND=RL, ND=1/2RL, and ND=RL. Substituting the reporting limit (ND=RL) for each value below the RL provides a conservative estimate of the concentration. This method, however, tends to produce results that are biased high, especially in data sets where the majority of samples are below the RL.

4.2 DATA SCREENING – SEDIMENT

4.2.1 Comparison to New Jersey and Pennsylvania Non-Residential Soil Standards

Concentrations of detected chemical constituents were compared to the State of New Jersey non-residential soil remediation standards and Commonwealth of Pennsylvania non-residential soil standards (N.J.A.C. 7:26D and 25 Pennsylvania Code (Pa. Code) § 250 Table 3a and 4a, respectively). The New Jersey and Pennsylvania non-residential soil standards used to screen the sediment data for this project are provided in Table 4-1 and Table 4-2, respectively.

4.2.2 Evaluation of TCLP Data

Concentrations of chemical constituents in the TCLP leachate were compared to maximum concentrations of contaminants for toxicity characteristics (Table 4-3) (40 CFR 261.24). The toxicity characteristics are used to identify the potential for toxicity and chemical characteristics of potential leachate from upland sites. The TCLP test is routinely required for dredged material placement at landfills and upland locations. If analyte concentrations in TCLP leachate exceed the regulatory criteria, the materials may require management as a hazardous waste or additional delineation of source material may be warranted to evaluate placement alternatives.

4.2.3 Evaluation of PFAS

Six PFAS were analyzed in bulk sediment at the request of PADEP. The PFAS data were compared to non-residential soil standards available for New Jersey and Pennsylvania (N.J.A.C. 7:26D and 25 Pennsylvania Code (Pa. Code) § 250 Table 3a and 4a, respectively). Both the Commonwealth of Pennsylvania and the State of New Jersey have non-residential soil standards for PFOS and PFOA. The Commonwealth of Pennsylvania has non-residential soil standards for PFBS. The State of New Jersey has non-residential soil standards for PFNA.

4.3 DATA SCREENING – SITE WATER AND MODIFIED ELUTRIATES

Analyte concentrations detected in the total and dissolved fractions of the site water and modified elutriates were screened against the applicable Commonwealth of Pennsylvania Surface Water Maximum/Continuous Water Quality Standards (25 Pa. Code § 93.7) for disposal in the Biles Island CDF. PADEP defines a criterion maximum concentration for acute protection and a criterion continuous concentration for chronic protection. Surface water quality for the metals cadmium, copper, lead, nickel, silver, and zinc are expressed as a function of water hardness. The surface water quality criterion for pentachlorophenol was calculated as a function of pH. The Commonwealth of Pennsylvania surface water standards for chemical constituents detected in site water and modified elutriate samples are provided in Table 4-4 and are provided on the site water and elutriate data tables. Designations for which standards apply to dissolved and total fractions of the site water and elutriates are provided on the site water and elutriate data tables.

4.3.1 Evaluation of Radioactivity

The radioactivity data were compared to the Drinking Water Standards-Radionuclides Rule, 66 Federal Register 76708, 7 December 2000; Volume 65, No. 236.and to the DRBC stream quality objectives as specified in DRBC Water Quality Regulations, 18 CFR Part 410, 4 December 2013.

4.3.2 Evaluation of PFAS

Six PFAS were analyzed in the modified elutriates and site water sample. In addition, the same six PFAS were analyzed in SPLP samples for Area A following PADEP review of the bulk sediment data (Appendix E). The Commonwealth of Pennsylvania has maximum contaminant levels (MCLs) established for two PFAS in drinking water: PFOS and PFOA. In addition, acute surface water quality criteria for freshwater aquatic life from USEPA (2024a) were also used for comparison.

4.3.3 Polychlorinated Biphenyls

The Pennsylvania surface water quality continuous criterion for Total PCBs (0.014 microgram per liter [µg/L]) is based on the USEPA Federal aquatic water quality criterion and, as stated by USEPA (2024b), applies to the sum of all PCB Aroclors or congeners, or homologs. As a result, the criterion was applied to the total form of PCB Aroclors, congeners, and homologs.

This page intentionally left blank

TABLE 4-1. STATE OF NEW JERSEY SOIL REMEDIATION STANDARDS (mg/kg)

Chemical Constituents	CAS No.	Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway	Non-Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway
General Chemistry			
Cyanide, Total	57-12-5	47	780
Metals			
Aluminum	7429-90-5	78,000	NA(1)
Antimony	7440-36-0	31	520
Arsenic	7440-38-2	19(2)	19(2)
Barium	7440-39-3	16,000	260,000
Beryllium	7440-41-7	160	2,600
Cadmium	7440-43-9	71	1,100
Cobalt	7440-48-4	23	390
Copper	7440-50-8	3,100	52,000
Lead	7439-92-1	200**(5)	800(6)
Manganese	7439-96-5	1,900	31,000
Mercury	7439-97-6	23	390
Nickel	7440-02-0	1,600	26,000
Selenium	7782-49-2	390	6,500
Silver	7440-22-4	390	6,500
Vanadium	7440-62-2	390	6,500
Zinc	7440-66-6	23,000	390,000
PCB Aroclors			
Polychlorinated biphenyls, Total	1336-36-3	0.25	1.1
PFAS Compounds			
Perfluorononanoic acid (PFNA)	375-95-1	0.047*	0.67*
Perfluorooctanoic acid (PFOA)	335-67-1	0.13*	1.8*
Perfluorooctane sulfonate (PFOS)	1763-23-1	0.11*	1.6*
Dioxin/Furan Congeners			
2,3,7,8-TCDD	1746-01-6	0.000051(8)	0.00081(8)
Pesticides			
4,4'-DDD	72-54-8	2.30	11
4,4'-DDE	72-55-9	2.00	11
4,4'-DDT	50-29-3	1.90	10
Aldrin	309-00-2	0.041	0.21
alpha-BHC	319-84-6	0.09	0.41
beta-BHC	319-85-7	0.3	1.40
Dieldrin	60-57-1	0.03	0.16
Endosulfan sulfate	1031-07-8	NR	NR
Endrin	72-20-8	19	270
gamma-BHC (Lindane)	58-89-9	0.57	2.80
Heptachlor	76-44-8	0.15	0.81
Heptachlor epoxide	1024-57-3	0.08	0.40
Methoxychlor	72-43-5	320	4,600
Toxaphene	8001-35-2	0.49	2.3

TABLE 4-1. STATE OF NEW JERSEY SOIL REMEDIATION STANDARDS (mg/kg)

Chemical Constituents	CAS No.	Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway	Non-Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway
Semivolatile Organic Compounds			
1,2,4-Trichlorobenzene	120-82-1	780	13,000
1,2-Dichlorobenzene	95-50-1	6,700	110,000
1,2-Diphenylhydrazine(as Azobenzene)	122-66-7	NR	NR
1,3-Dichlorobenzene	541-73-1	6,700	110,000
1,4-Dichlorobenzene	106-46-7	780	13,000
2,2'-oxybis[1-chloropropane]	108-60-1	NR	NR
2,4,5-Trichlorophenol	95-95-4	6,300	91,000
2,4,6-Trichlorophenol	88-06-2	49	230
2,4-Dichlorophenol	120-83-2	190	2,700
2,4-Dimethylphenol	105-67-9	1,300	18,000
2,4-Dinitrophenol	51-28-5	130	1,800
2,4-Dinitrotoluene	121-14-2	NR	NR
2,6-Dinitrotoluene	606-20-2	NR	NR
2-Chlorophenol	95-57-8	390	6,500
2-Methylnaphthalene	91-57-6	240	3,300
2-Methylphenol	95-48-7	320	4,600
3,3'-Dichlorobenzidine	91-94-1	1.2	5.7
4,6-Dinitro-2-methylphenol	534-52-1	NR	NR
Acenaphthene	83-32-9	3,600	50,000
Acenaphthylene	208-96-8	NR	NR
Anthracene	120-12-7	18,000	250,000
Benzidine	92-87-5	NR	NR
Benzo[a]anthracene	56-55-3	5.1	23
Benzo[a]pyrene	50-32-8	0.51	2.3
Benzo[b]fluoranthene	205-99-2	5.1	23
Benzo[g,h,i]perylene	191-24-2	NR	NR
Benzo[k]fluoranthene	207-08-9	51	230
Bis(2-chloroethyl)ether	111-44-4	0.63	3.3
Bis(2-ethylhexyl) phthalate	117-81-7	39	180
Butyl benzyl phthalate	85-68-7	290	1,300
Chrysene	218-01-9	510	2,300
Dibenz(a,h)anthracene	53-70-3	0.5	2.3
Diethyl phthalate	84-66-2	51,000	730,000
Di-n-butyl phthalate	84-74-2	6,300	91,000
Di-n-octyl phthalate	117-84-0	630	9,100
Fluoranthene	206-44-0	2,400	33,000
Fluorene	86-73-7	2,400	33,000
Hexachlorobenzene	118-74-1	0.43	2.3
Hexachlorobutadiene	87-68-3	8.9	47.0
Hexachlorocyclopentadiene	77-47-4	470	7,800
Hexachloroethane	67-72-1	17	91
Indeno[1,2,3-cd]pyrene	193-39-5	5.1	23
Isophorone	78-59-1	570	2,700
Methylphenol, 3 & 4	106-44-5	630	9,100

TABLE 4-1. STATE OF NEW JERSEY SOIL REMEDIATION STANDARDS (mg/kg)

Chemical Constituents	CAS No.	Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway	Non-Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway
Naphthalene	91-20-3	2,500	34,000
Nitrobenzene	98-95-3	160	2,600
N-Nitrosodimethylamine	62-75-9	NR	NR
N-Nitrosodi-n-propylamine	621-64-7	0.17(7)	0.36
N-Nitrosodiphenylamine	86-30-6	110	520
Pentachlorophenol	87-86-5	1.0	4.4
Phenanthrene	85-01-8	NR	NR
Phenol	108-95-2	19,000	270,000
Pyrene	129-00-0	1,800	25,000
Volatile Organic Compounds			
1,1,1-Trichloroethane	71-55-6	160,000	NA(1)
1,1,2,2-Tetrachloroethane	79-34-5	3.5	18
1,1,2-Trichloroethane	79-00-5	12	64
1,1-Dichloroethane	75-34-3	120	640
1,1-Dichloroethene	75-35-4	11	180
1,2-Dibromo-3-Chloropropane	96-12-8	0.87	4.5
1,2-Dibromoethane	106-93-4	0.35	1.8
1,2-Dichloroethane	107-06-2	5.8	30.0
1,2-Dichloropropane	78-87-5	19	98
2-Butanone (MEK)	78-93-3	47,000	780,000
Acetone	67-64-1	70,000	NA(1)
Acrolein	107-02-8	NR	NR
Acrylonitrile	107-13-1	NR	NR
Benzene	71-43-2	3.0	16
Bromoform	75-25-2	88	460
Bromomethane	74-83-9	110	1,800
Carbon disulfide	75-15-0	NA	NA
Carbon tetrachloride	56-23-5	7.6	40
Chlorobenzene	108-90-7	510	8,400
Chlorodibromomethane	124-48-1	8.3	43
Chloroethane	75-00-3	NA	NA
Chloroform	67-66-3	780	13,000
Chloromethane	74-87-3	NA	NA
cis-1,2-Dichloroethene	156-59-2	780	13,000
Dichlorobromomethane	75-27-4	11	59
Dichlorodifluoromethane	75-71-8	16,000	260,000
Ethylbenzene	100-41-4	7,800	130,000
Methyl acetate	79-20-9	78,000	NA
Methylene Chloride	75-09-2	50	260
Styrene	100-42-5	16,000	260,000
Tetrachloroethene	127-18-4	330	1,700
Toluene	108-88-3	6,300	100,000
trans-1,2-Dichloroethene	156-60-5	1,300	22,000
Trichloroethene	79-01-6	15	79
Trichlorofluoromethane	75-69-4	23,000	390,000

TABLE 4-1. STATE OF NEW JERSEY SOIL REMEDIATION STANDARDS (mg/kg)

Chemical Constituents	CAS No.	Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway	Non-Residential Soil Remediation Standard for the Ingestion-Dermal Exposure Pathway
Vinyl chloride	75-01-4	0.97	5.0
Xylenes, Total	1330-20-7	12,000	190,000

Notes:

Source: NJ Code 7:26D; last updated 06 May 2024

mg/kg = Milligram(s) per kilogram

NA = Standard not available

NR = Compound Not Regulated

PCB = Polychlorinated Biphenyl

PFAS = Per- and Polyfluoroalkyl Substances

1 – Standard not applicable because health-based criterion exceeds one million mg/kg

2 – The direct contact standard for arsenic is based on natural background

5 – Standard based on the Integrated Exposure Uptake Biokinetic (IEUBK) model for lead in children

6 – Standard based on the Adult Lead Methodology (ALM)

7 – Standard set at soil reporting limit

8 – This standard is used for comparison to site soil data that have been converted to sample-specific TCDD-TEQ values through application of the Toxicity Equivalence Factor Methodology (USEPA 2010)

* Interim standard effective 10/17/2022

TABLE 4-2. COMMONWEALTH OF PENNSYLVANIA NON-RESIDENTIAL SOIL STANDARDS (mg/kg)

Chemical Constituent	CAS No.	Non-Residential			
		Surface (0-2 ft)		Subsurface (2-15 ft)	
General Chemistry					
Ammonia, distilled	7664-41-7	10000	C	10000	C
Cyanide, Total	57-12-5	1900	G	190000	C
Metals					
Aluminum	7429-90-5	190000	C	190000	C
Antimony	7440-36-0	1300	G	190000	C
Arsenic	7440-38-2	61	G	190000	C
Barium	7440-39-3	190000	C	190000	C
Beryllium	7440-41-7	6400	G	190000	C
Cadmium	7440-43-9	1600	G	190000	C
Chromium (III)	16065-83-1	190000	C	190000	C
Chromium (VI)	18540-29-9	180	G	140000	N
Cobalt	7440-48-4	960	G	190000	N
Copper	7440-50-8	100000	G	190000	C
Iron	7439-89-6	190000	C	190000	C
Lead	7439-92-1	1000	S	190000	C
Manganese	7439-96-5	190000	C	190000	C
Mercury	7439-97-6	510	G	190000	C
Nickel	7440-02-0	64000	G	190000	C
Selenium	7782-49-2	16000	G	190000	C
Silver	7440-22-4	16000	G	190000	C
Thallium	7440-28-0	32	G	190000	C
Vanadium	7440-62-2	16000	G	190000	C
Zinc	7440-66-6	190000	C	190000	C
PCB Aroclors					
PCB-1016	12674-11-2	220	G	10000	C
PCB-1221	11104-28-2	23	N	27	N
PCB-1232	11141-16-5	46	G	10000	C
PCB-1242	53469-21-9	46	G	10000	C
PCB-1248	12672-29-6	46	G	10000	C
PCB-1254	11097-69-1	64	G	10000	C
PCB-1260	11096-82-5	46	G	190000	C
Total PCBs (Aroclors)	1336-36-3	46	G	190000	C
PFAS Compounds					
Perfluorobutane Sulfonate (PFBS)	375-73-5	960	G	10000	C
Perfluorooctane Sulfonate (PFOS)	1763-23-1	64	G	190000	C
Perfluorooctanoic Acid (PFOA)	335-67-1	64	G	190000	C
Dioxin/Furan Congeners					
2,3,7,8-TCDD	1746-01-6	0.0007	G	190000	C
Pesticides					
4,4'-DDD	72-54-8	380	G	190000	C
4,4'-DDE	72-55-9	270	G	190000	C
4,4'-DDT	50-29-3	270	G	190000	C
Aldrin	309-00-2	5.4	G	190000	C

TABLE 4-2. COMMONWEALTH OF PENNSYLVANIA
NON-RESIDENTIAL SOIL STANDARDS (mg/kg)

Chemical Constituent	CAS No.	Non-Residential			
		Surface (0-2 ft)		Subsurface (2-15 ft)	
alpha-BHC	319-84-6	14	G	190000	C
beta-BHC	319-85-7	51	G	190000	C
DCPA	1861-32-1	32000	G	190000	C
Dieldrin	60-57-1	5.7	G	190000	C
Endosulfan I	959-98-8	19000	G	190000	C
Endosulfan II	33213-65-9	19000	G	190000	C
Endosulfan sulfate	1031-07-8	19000	G	190000	C
Endrin	72-20-8	960	G	190000	C
gamma-BHC (Lindane)	58-89-9	83	G	190000	C
Heptachlor	76-44-8	20	G	190000	C
Heptachlor epoxide	1024-57-3	10	G	190000	C
Methoxychlor	72-43-5	16000	G	190000	C
Toxaphene	8001-35-2	83	G	190000	C
Semivolatile Organic Compounds					
1,2,4-Trichlorobenzene	120-82-1	160	N	190	N
1,2-Dichlorobenzene	95-50-1	10000	C	10000	C
1,2-Diphenylhydrazine(as Azobenzene)	122-66-7	10	N	12	N
1,3-Dichlorobenzene	541-73-1	10000	C	10000	C
1,4-Dichlorobenzene	106-46-7	200	N	230	N
2,2'-oxybis[1-chloropropane]	108-60-1	220	N	250	N
2,4,5-Trichlorophenol	95-95-4	190000	C	190000	C
2,4,6-Trichlorophenol	88-06-2	3200	G	190000	C
2,4-Dichlorophenol	120-83-2	9600	G	190000	C
2,4-Dimethylphenol	105-67-9	10000	C	10000	C
2,4-Dinitrophenol	51-28-5	6400	G	190000	C
2,4-Dinitrotoluene	121-14-2	290	G	190000	C
2,6-Dinitrotoluene	606-20-2	61	G	190000	C
2-Chloronaphthalene	91-58-7	190000	C	190000	C
2-Chlorophenol	95-57-8	10000	C	10000	C
2-Methylnaphthalene	91-57-6	240	N	270	N
2-Methylphenol	95-48-7	160000	G	190000	C
2-Nitrophenol	88-75-5	26000	G	190000	C
3,3'-Dichlorobenzidine	91-94-1	200	G	190000	C
4,6-Dinitro-2-methylphenol	534-52-1	260	G	190000	C
4-Chloro-3-methylphenol	59-50-7	190000	C	190000	C
4-Nitrophenol	100-02-7	26000	G	190000	C
Acenaphthene	83-32-9	190000	C	190000	C
Acenaphthylene	208-96-8	190000	C	190000	C
Anthracene	120-12-7	190000	C	190000	C
Benzidine	92-87-5	0.4	G	190000	C
Benzo[a]anthracene	56-55-3	130	G	190000	C
Benzo[a]pyrene	50-32-8	91	G	190000	C
Benzo[b]fluoranthene	205-99-2	76	G	190000	C

TABLE 4-2. COMMONWEALTH OF PENNSYLVANIA NON-RESIDENTIAL SOIL STANDARDS (mg/kg)

Chemical Constituent	CAS No.	Non-Residential			
		Surface (0-2 ft)		Subsurface (2-15 ft)	
Benzo[g,h,i]perylene	191-24-2	190000	C	190000	C
Benzo[k]fluoranthene	207-08-9	76	G	190000	C
Benzoic acid	65-85-0	190000	C	190000	C
Benzyl alcohol	100-51-6	10000	C	10000	C
Bis(2-chloroethoxy)methane	111-91-1	9600	G	10000	C
Bis(2-chloroethyl)ether	111-44-4	6.7	N	7.6	N
Bis(2-ethylhexyl) phthalate	117-81-7	6500	G	10000	C
Butyl benzyl phthalate	85-68-7	10000	C	10000	C
Chrysene	218-01-9	760	G	190000	C
Dibenz(a,h)anthracene	53-70-3	22	G	190000	C
Dibenzofuran	132-64-9	3200	G	190000	C
Diethyl phthalate	84-66-2	10000	C	10000	C
Di-n-butyl phthalate	84-74-2	10000	C	10000	C
Di-n-octyl phthalate	117-84-0	10000	C	10000	C
Fluoranthene	206-44-0	130000	G	190000	C
Fluorene	86-73-7	130000	G	190000	C
Hexachlorobenzene	118-74-1	57	G	190000	C
Hexachlorobutadiene	87-68-3	1200	G	10000	C
Hexachlorocyclopentadiene	77-47-4	10000	C	10000	C
Hexachloroethane	67-72-1	230	N	270	N
Indeno[1,2,3-cd]pyrene	193-39-5	76	G	190000	C
Isophorone	78-59-1	10000	C	10000	C
Methylphenol, 3 & 4	106-44-5	16000	G	190000	C
Naphthalene	91-20-3	66	N	77	N
Nitrobenzene	98-95-3	55	N	63	N
N-Nitrosodimethylamine	62-75-9	0.16	N	0.18	N
N-Nitrosodi-n-propylamine	621-64-7	1.1	N	1.3	N
N-Nitrosodiphenylamine	86-30-6	860	N	990	N
Pentachlorophenol	87-86-5	230	G	190000	C
Phenanthrene	85-01-8	190000	C	190000	C
Phenol	108-95-2	16000	N	18000	N
Pyrene	129-00-0	96000	G	190000	C
Pyridine	110-86-1	3200	G	10000	C
Volatile Organic Compounds					
1,1,1-Trichloroethane	71-55-6	10000	C	10000	C
1,1,2,2-Tetrachloroethane	79-34-5	38	N	44	N
1,1,2-Trichloro-1,2,2-trifluoroethane	76-13-1	10000	C	10000	C
1,1,2-Trichloroethane	79-00-5	16	N	18	N
1,1-Dichloroethane	75-34-3	1400	N	1600	N
1,1-Dichloroethene	75-35-4	10000	C	10000	C
1,2-Dibromo-3-Chloropropane	96-12-8	0.37	N	0.42	N
1,2-Dibromoethane	106-93-4	3.7	N	4.2	N
1,2-Dichloroethane	107-06-2	85	N	98	N

TABLE 4-2. COMMONWEALTH OF PENNSYLVANIA
NON-RESIDENTIAL SOIL STANDARDS (mg/kg)

Chemical Constituent	CAS No.	Non-Residential			
		Surface (0-2 ft)		Subsurface (2-15 ft)	
1,2-Dichloropropane	78-87-5	0.6	N	0.69	N
2-Butanone (MEK)	78-93-3	10000	C	10000	C
2-Hexanone	591-78-6	2400	N	2700	N
4-Methyl-2-pentanone (MIBK)	108-10-1	10000	C	10000	C
Acetone	67-64-1	10000	C	10000	C
Acrolein	107-02-8	1.6	N	1.8	N
Acrylonitrile	107-13-1	33	N	37	N
Benzene	71-43-2	280	N	330	N
Bromoform	75-25-2	2000	N	2300	N
Bromomethane	74-83-9	400	N	460	N
Carbon disulfide	75-15-0	10000	C	10000	C
Carbon tetrachloride	56-23-5	370	N	430	N
Chlorobenzene	108-90-7	3900	N	4500	N
Chlorodibromomethane	124-48-1	1100	G	10000	C
Chloroethane	75-00-3	10000	C	10000	C
Chloroform	67-66-3	96	N	110	N
Chloromethane	74-87-3	1200	N	1400	N
cis-1,2-Dichloroethene	156-59-2	6400	G	10000	C
Cyclohexane	110-82-7	10000	C	10000	C
Dichlorobromomethane	75-27-4	60	N	69	N
Dichlorodifluoromethane	75-71-8	8000	N	9100	N
Ethylbenzene	100-41-4	880	N	1000	N
Isopropylbenzene	98-82-8	10000	C	10000	C
Methyl acetate	79-20-9	10000	C	10000	C
Methyl tert-butyl ether	1634-04-4	8500	N	9800	N
Methylene Chloride	75-09-2	10000	C	10000	C
Styrene	100-42-5	10000	C	10000	C
Tetrachloroethene	127-18-4	3200	N	3600	N
Toluene	108-88-3	10000	C	10000	C
trans-1,2-Dichloroethene	156-60-5	10000	C	10000	C
Trichloroethene	79-01-6	160	N	180	N
Trichlorofluoromethane	75-69-4	10000	C	10000	C
Vinyl chloride	75-01-4	61	G	290	N
Xylenes, Total	1330-20-7	7900	N	9100	N

Notes:
ft = Foot (feet)
mg/kg = Milligram(s) per kilogram
PCB = Polychlorinated Biphenyl
PFAS = Per- and Polyfluoroalkyl Substances
G = Ingestion
N = Inhalation
C = Cap
Source: Pennsylvania Code. Title 25 Environmental Protection, Chapter 250 Administration of Land Recycling Program, Subchapter G Demonstration of Attainment, updated 11 November 2023.

TABLE 4-2. COMMONWEALTH OF PENNSYLVANIA NON-RESIDENTIAL SOIL STANDARDS (mg/kg)

Chemical Constituent	CAS No.	Non-Residential	
		Surface (0-2 ft)	Subsurface (2-15 ft)

Table 3 - Medium-Specific Concentrations (MSCs) for Organic Regulated Substances in Soil - A. Direct Contact Numeric Values

Table 4 - Medium-Specific Concentrations (MSCs) for Inorganic Regulated Substances in Soil - A. Direct Contact Numeric Values

This page left intentionally blank

TABLE 4-3. TOXICITY CHARACTERISTIC LEACHING PROCEDURE (TCLP) REGULATORY GUIDELINES

(Maximum Concentration of Contaminants for Toxicity Characteristics)

Chemical Name	**Regulatory Level (milligrams per liter)**
Metals	
Arsenic	5
Barium	100
Cadmium	1
Chromium	5
Lead	5
Mercury	0.2
Selenium	1
Silver	5
Pesticides and Herbicides	
2, 4, 5-TP (Silvex)	1
2, 4-D	10
Chlordane	0.03
Endrin	0.02
Heptachlor (and its Epoxide)	0.008
Gamma-BHC	0.4
Methoxychlor	10
Toxaphene	0.5
Semivolatile Organic Compounds (SVOCs)	
O-Cresol*	200
M-Cresol*	200
P-Cresol*	200
Cresol	200
1, 4 Dichlorobenzene	7.5
2,4 Dinitrotoluene	0.13
Hexachlorobenzene	0.13
Hexachlorobutadiene	0.5
Hexachloroethane	3
Nitrobenzene	2
Pentachlorophenol	100
2, 4, 5-Trichorophenol	400
2, 4, 6-Trichorophenol	2
Pyridine	5
Volatile Organic Compounds (VOCs)	
Benzene	0.5
Carbon Tetrachloride	0.5
Chlorobenzene	100
Chloroform	6
1, 2 Dichoroethane	0.5
1, 1 Dichloroethylene	0.7
2-Butanone	200
Tetrachloroethylene	0.7
Trichloroethylene	0.5
Vinyl Chloride	0.2

Note:
*If o-, m-, p-Cresol concentration cannot be differentiated, the total cresol concentration is used. The regulatory level of total cresol is 200 milligrams per liter.
Source: 40 CFR 261.24.

TABLE 4-4. COMMONWEALTH OF PENNSYLVANIA SURFACE WATER QUALITY STANDARDS (µg/L)

Chemical Constituent	Criteria Continuous Concentration (Chronic)	Criteria Maximum Concentration (Acute)
General Chemistry		
Cyanide, Total	5.2	22
Metals		
Aluminum	NA	750
Antimony	220	1100
Arsenic	150	340
Barium	4100	21000
Boron	1600	8100
Cadmium	0.31*	2.79*
Chromium (III)	*	*
Chromium (VI)	11	16
Cobalt	19	95
Copper	11.9*	18.5*
Lead	3.6*	93*
Mercury	0.77	1.4
Nickel	69*	623*
Selenium	4.6	NA
Silver	NA	5.74*
Thallium	13	65
Vanadium	100	510
Zinc	157*	156*
PCB Congeners		
Polychlorinated biphenyls, Total	0.014	NA
Pesticides		
4,4'-DDD	0.001	1.1
4,4'-DDE	0.001	1.1
4,4'-DDT	0.001	1.1
Aldrin	0.1	3
gamma-BHC	NA	0.95
Chlordane (technical)	0.0043	2.4
Dieldrin	0.056	0.24
Endosulfan I	0.056	0.22
Endosulfan II	0.056	0.22
Endrin	0.036	0.086
Heptachlor	0.0038	0.52
Heptachlor epoxide	0.0038	0.5
Toxaphene	0.0002	0.73

TABLE 4-4. COMMONWEALTH OF PENNSYLVANIA SURFACE WATER QUALITY STANDARDS (µg/L)

Chemical Constituent	Criteria Continuous Concentration (Chronic)	Criteria Maximum Concentration (Acute)
Semivolatile Organic Compounds		
1,2,4-Trichlorobenzene	26	130
1,2-Dichlorobenzene	160	820
1,2-Diphenylhydrazine(as Azobenzene)	3	15
1,3-Dichlorobenzene	69	350
1,4-Dichlorobenzene	150	730
2,4,6-Trichlorophenol	91	460
2,4-Dichlorophenol	340	1700
2,4-Dimethylphenol	130	660
2,4-Dinitrophenol	130	660
2,4-Dinitrotoluene	320	1600
2,6-Dinitrotoluene	200	990
2-Chlorophenol	110	560
2-Nitrophenol	1600	8000
4,6-Dinitro-2-methylphenol	16	80
4-Bromophenyl phenyl ether	54	270
4-Chloro-3-methylphenol	30	160
4-Nitrophenol	470	2300
Acenaphthene	17	83
Benzidine	59	300
Benzo[a]anthracene	0.1	0.5
Bis(2-chloroethyl)ether	6000	30000
Bis(2-ethylhexyl) phthalate	910	4500
Butyl benzyl phthalate	35	140
Diethyl phthalate	800	4000
Dimethyl phthalate	500	2500
Di-n-butyl phthalate	21	110
Fluoranthene	40	200
Hexachlorobutadiene	2	10
Hexachlorocyclopentadiene	1	5
Hexachloroethane	12	60
Isophorone	2100	10000
Methylphenol, 3 & 4	160	800
Naphthalene	43	140
Nitrobenzene	810	4000
N-Nitrosodimethylamine	3400	17000
N-Nitrosodiphenylamine	59	300
Pentachlorophenol	12.2**	16**
Phenanthrene	1	5

TABLE 4-4. COMMONWEALTH OF PENNSYLVANIA SURFACE WATER QUALITY STANDARDS (µg/L)

Chemical Constituent	Criteria Continuous Concentration (Chronic)	Criteria Maximum Concentration (Acute)
Volatile Organic Compounds		
1,1,1-Trichloroethane	610	3000
1,1,2,2-Tetrachloroethane	210	1000
1,1,2-Trichloroethane	680	3400
1,1-Dichloroethene	1500	7500
1,2-Dichloroethane	3100	15000
1,2-Dichloropropane	2200	11000
2-Butanone (MEK)	32000	230000
2-Chloroethyl vinyl ether	3500	18000
2-Hexanone	4300	21000
4-Methyl-2-pentanone (MIBK)	5000	26000
Acetone	86000	450000
Acrolein	3	3
Acrylonitrile	130	650
Benzene	130	640
Bromoform	370	1800
Bromomethane	110	550
Carbon tetrachloride	560	2800
Chlorobenzene	240	1200
Chloroform	390	1900
Ethylbenzene	580	2900
Methyl Bromide	110	550
Methyl Chloride	5500	28000
Methylene Chloride	2400	12000
Tetrachloroethene	140	700
Toluene	330	1700
trans-1,2-Dichloroethene	1400	6800
Trichloroethene	450	2300
Xylenes, Total	210	1100

Notes:

Source: Pennsylvania Code. 25 Chapter 93.7. Commonwealth of Pennsylvania Water Quality Standards (Table 5). Effective 09 November 2023.

* = site specific hardness of 140 mg/L used to calculate water quality criteria

** = site specific pH of 7.6 used to calculate criterion

µg/L = micrograms per liter

NA = Not Applicable

PCB = Polychlorinated Biphenyl

This page left intentionally blank

5. BULK SEDIMENT CHEMISTRY

The physical and chemical characteristics of six composite samples (plus one duplicate composite), 15 individual VOC samples (plus two duplicates), and three discrete cores were analyzed from the Reserve Basin maintenance dredging project area to assess the sediment quality of the maintenance material proposed for dredging and disposal. This chapter presents the results of the bulk sediment chemical analyses, and comparisons of detected chemical constituents to State of New Jersey and Commonwealth of Pennsylvania non-residential soil standards, and TCLP regulatory criteria.

Bulk sediments were evaluated for target analytes identified in the approved project SSAP (EA 2024). Project-specific analytical methods and detection limits for sediment samples are provided in Tables 3-1 and 3-2, respectively. Sample weights were adjusted for percent moisture (up to 50 percent moisture) prior to analysis to achieve the lowest possible detection limits. Analytical results are reported on a dry weight basis. Detected concentrations of target analytes are bolded in each data table. Applicable screening criteria are presented on each data table and exceedances of criteria are flagged and footnoted.

Analytical narratives that include an evaluation of laboratory QA/QC results and copies of final raw data sheets (Form I's) are provided in Attachment II. EET–Pittsburgh will retain and archive the results of these analyses for seven years from the date of issuance of the final results.

5.1 PHYSICAL ANALYSES

Physical properties (grain size, moisture content, total solids) for each composite sample are provided in Table 5-1. Grain size composition for each sample is depicted in Figure 5-1.

Grain size analyses indicated that the sediments were predominantly fine grained throughout most of dredging Area A (ranging from 75.4 to 96.4 percent silt+clay) and were a mix and fines (ranging from 73.9 to 81.7 silt+clay) in dredging Area B along the edge of Wharf N. Percent solids in the sediments ranged from 32 to 44.4 percent (Table 5-1).

5.2 GENERAL CHEMISTRY AND TOTAL ORGANIC CARBON

Results of general chemistry (pH, total cyanide, sulfide) and TOC analyses for each sediment sample are provided Table 5-2. Total cyanide was detected in five of the six composites from Area A (ranging from 0.35 to 0.8 milligram per kilogram [mg/kg]) and in each of the three composites from Area B (ranging from 1.4 to 8 mg/kg); none of the detected concentrations exceeded the State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards. Sulfide concentrations ranged from 210 to 1,200 mg/kg in Area A and ranged from 2,100 to 3,700 mg/kg in Area B. TOC ranged from 3.8 to 4.8 percent in Area A and from 6.2 to 9.5 percent in Area B samples (Table 5-2). pH ranged from 6.5 to 7 in Area A and ranged from 6.9 to 7.2 in Area B.

5.3 METALS

The results of the metals analyses for the sediment samples are presented in Table 5-3. Each of the 25 tested metals, with the exception of hexavalent chromium, was detected in the Reserve Bason sediment samples. Arsenic concentrations exceeded New Jersey non-residential soil standard (19 mg/kg) in one composite for dredging Area A (A-07/08; 28 mg/kg) and in each of the three samples in dredging Area B (ranging from 22 to 49 mg/kg). Arsenic concentrations did not exceed Commonwealth of Pennsylvania non-residential soil standards. The lead concentration for one location in Area B (B-4) exceeded both Commonwealth of Pennsylvania and the State of New Jersey non-residential soil standards (1,000 and 800 mg/kg, respectively), with a concentration of 5,900 mg/kg. No other metals were detected at concentrations that exceeded State of New Jersey or Commonwealth of Pennsylvania soil criteria (Table 5-3).

5.4 POLYCYCLIC AROMATIC HYDROCARBONS

Results of the PAH analysis for the sediment samples are presented in Table 5-4. Each of the 18 tested PAHs were detected in the Area A and Area B sediment samples. None of the individual PAH concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards. Total PAH concentrations (ND=RL) ranged from 2.14 to 8.65 mg/kg in dredging Area A from 13.3 to 26.9 mg/kg in dredging Area B (Table 5-4). There are no State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards for total PAHs.

5.5 PCB CONGENERS AND HOMOLOGS

Results of the PCB congener and homolog analyses for the sediments composites are presented in Table 5-5. There are no Commonwealth of Pennsylvania non-residential soil standards for individual or total PCB congeners. A total of 185 of the 209 tested PCB congeners were detected in the sediment samples from the Reserve Basin. Total PCB congeners and total PCB homologs (ND=0) ranged from 0.0097 to 0.19 mg/kg in Area A and ranged from 0.092 to 0.3 mg/kg in Area B (Table 5-5). None of the total PCB congener or total PCB homolog concentrations exceeded the State of New Jersey non-residential soil standard of 1.1 mg/kg. PCB AROCLORS

Results of the PCB Aroclors analyses for the sediment composites are presented in Table 5-6. There are no State of New Jersey non-residential soil standards for individual PCB Aroclors; however, there are non-residential standard total PCB Aroclors. There are Commonwealth of Pennsylvania non-residential soil standards for individual PCB Aroclors and total PCB Aroclors.

Of the seven tested PCB Aroclors, two Aroclors (PCB-1248 and PCB-1260) were detected in the Reserve Basin sediment samples; none of the detected concentrations exceeded the Commonwealth of Pennsylvania non-residential soil standards (46 mg/kg for both PCB-1248 and PCB-1260). Total PCB Aroclor concentrations (ND=RL) ranged from 0.024 to 0.365 mg/kg in Area A and from 0.24 to 0.79 mg/kg in Area B. None of the total PCB Aroclor concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards (1.1 and 46 mg/kg, respectively) (Table 5-6).

5.6 DIOXIN/FURAN CONGENERS

Results of the dioxin/furan analysis for the sediment composites are presented in Table 5-7. The concentration of 2,3,7,8-TCDD was compared to the Commonwealth of Pennsylvania and State of New Jersey non-residential soil standards. The State of New Jersey states that the soil standard for 2,3,7,8-TCDD is used for comparison to site soil data that have been converted to sample-specific TEQ values through application of the Toxicity Equivalency Factor methodology (Van den Berg et al. 2006). 2,3,7,8-TCDD was detected in five sediment samples in Area A, including one duplicate composite (Table 5-7); 2,3,7,8-TCDD was not detected in the samples from Area B. The TEQ (ND=RL) ranged from 0.000021 to 0.000242 mg/kg in Area A. Estimated concentrations of 2,3,7,8-TCDD and the calculated TEQ did not exceed either the Commonwealth of Pennsylvania or State of New Jersey soil standards for 2,3,7,8-TCDD.

5.7 SEMIVOLATILE ORGANIC COMPOUNDS

Results of the SVOC analysis for the sediment composites are presented in Table 5-8. Five of the 46 tested SVOCs (benzoic acid, bis(2-ethylhexyl)phthalate, dibenzofuran, methylphenol 3&4, and phenol) were detected in the sediment composites, with the majority of concentrations estimated below the laboratory RL. None of the detected SVOCs exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

5.8 VOLATILE ORGANIC COMPOUNDS

Results of the VOC analysis for the individual sampling locations are presented in Table 5-9. Four VOCs (2-butanone, chlorodibromomethane, chloroform, and dichlorobromomethane) were detected in the Reserve Basin sediment samples. Many of the detected concentrations were estimated below the laboratory RL. Three detected VOCs were also detected in the laboratory method blank. None of the detected VOCs exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards (Table 5-9).

5.9 CHLORINATED PESTICIDES

Results of the pesticide analysis for the sediment composites are presented in Table 5-10. Eleven of the 22 tested chlorinated pesticides were detected in the Reserve Basin sediment samples. None of the tested chlorinated pesticides were detected at concentrations that exceeded the State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

5.10 TOTAL PETROLEUM HYDROCARBONS

TPH results are presented in Table 5-11. TPH-GRO (C6-C10) was detected in samples from four of the 12 locations in Area A (ranging from 1.4 to 76 mg/kg) and in samples from each of the three locations in Area B (ranging from 12 to 15 mg/kg). TPH-DRO (C10-C20) was detected in samples from four of the 12 locations in Area A (ranging from 14 to 900 mg/kg) and in samples from each of the three locations in Area B (ranging from 210 to 1,000 mg/kg). TPH-ORO (C20-C34) was detected in samples from nine of the 12 locations in Area A (ranging from 15 to 1,200 mg/kg) and in samples from each of the three locations in Area B (ranging from 370 to 2,400 mg/kg). Overall, the highest concentrations of TPH were detected in samples from A-07, B-2, and B-4. There are no Commonwealth of Pennsylvania or State of New Jersey soil standards for comparison.

5.11 RADIOACTIVITY

Radioactivity results are presented in Table 5-12. Gross alpha was detected in five of the six composite samples from Area A (ranging from 7.2 to 25.6 picocuries per gram [pCi/g]) and in one sample in Area B (B-4) at a concentration of 8.9 pCi/g. Gross beta was detected in in five of the six composite samples from Area A [ranging from 7.2 to 25.6 pCi/g] and in one sample in Area B (B-4) at a concentration of 8.9 pCi/g. each of sediment samples in Area A(ranging from 8.05 to 22.4 pCi/g) and in two of the samples in Area B (ranging from 5.37 to 5.57 pCi/g. There are no Commonwealth of Pennsylvania or State of New Jersey soil standards for comparison.

5.12 PFAS

Six PFAS were analyzed in bulk sediment samples. Results are presented in Table 5-13. Four of the six test tested compounds were detected in the Reserve Basin sediments. Two PFAS, HFPO-DA and PFBS, were not detected in the samples. None of the concentrations of the four detected PFAS exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards. In addition, the same six PFAS were analyzed in SPLP samples for Area A following PADEP review of the bulk sediment data (Appendix E).

5.13 TCLP LEACHATE

The results of the TCLP leachate analyses for the composite samples are presented in Table 5-14. Eight constituents (arsenic, barium, cadmium, chromium, lead, selenium, silver, and chloroform) were detected in at least one of the leachates for Reserve Basin sediment samples. Results presented for lead from sample location B-4 were re-analyzed due to an anomalous concentration elevated above the TCLP lead criterion (5 mg/L). The sample was re-analyzed and the concentration reported in the re-analysis was below the TCLP lead criterion. None of the detected concentrations of TCLP leachate exceeded the TCLP criteria.

5.14 SUMMARY OF BULK SEDIMENT CHARACTERISTICS

The physical and chemical characteristics of six composite samples (plus one field duplicate), 15 individual VOC samples (plus two field duplicates), and three discrete core samples were analyzed from the Reserve Basin maintenance dredging project area to assess the sediment quality of the maintenance material proposed for dredging and disposal. Concentrations of detected analytes in sediment samples were compared to State of New Jersey non-residential soil standards (Table 4-1), Commonwealth of Pennsylvania non-residential soil standards (Table 4-2), and TCLP regulatory criteria (40 CFR 261.24; Table 4-3).

A summary of the bulk sediment evaluation is provided below.

Physical Characteristics and TOC. Grain size analyses indicated that the sediments were predominantly fine-grained throughout dredging Area A (ranging from 75.4 to 96.4 percent silt+clay) and were a mix and fines (ranging from 73.9 to 81.7 silt+clay) in dredging Area B along the edge of Wharf N. Percent solids in the sediments ranged from 32 to 44.4 percent. TOC ranged from 3.8 to 4.8 percent in Area A and from 6.2 to 9.5 percent in the Area B samples.

Metals. Arsenic concentrations exceeded the State of New Jersey non-residential soil standard (19 mg/kg) in one composite for dredging Area A (A-07/08; 28 mg/kg) and in each of the three samples in dredging Area B (ranging from 22 to 49 mg/kg). Arsenic concentrations did not exceed Commonwealth of Pennsylvania non-residential soil standards. The lead concentration for one location in Area B (B-4) exceeded both Commonwealth of Pennsylvania and State of New Jersey non-residential soil standards (1,000 and 800 mg/kg, respectively), with a concentration of 5,900 mg/kg.

PAHs. Each of the 18 tested PAHs were detected in the Area A and Area B sediment samples. None of the individual PAH concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards. Total PAH concentrations (ND=RL) ranged from 2.14 to 8.65 mg/kg in dredging Area A from 13.3 to 26.9 mg/kg in dredging Area B.

PCB Congeners and Homologs. Total PCB congeners (ND=0) and total PCB homologs ranged from 0.0097 to 0.19 mg/kg in Area A and ranged from 0.092 to 0.3 mg/kg in Area B. None of the total PCB congener or total PCB homolog concentrations exceeded the State of New Jersey non-residential soil standard of 1.1 mg/kg.

PCB Aroclors. Two PCB Aroclors (PCB-1248 and PCB-1260) were detected in the Reserve Basin sediment samples. None of the detected Aroclor concentrations exceeded the Commonwealth of Pennsylvania non-residential soil standards (46 mg/kg for both PCB-1248 and PCB-1260). Total PCB Aroclor concentrations (ND=RL) ranged from 0.024 to 0.365 mg/kg in Area A and from 0.24 to 0.79 mg/kg in Area B. None of the total PCB Aroclor concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards (1.1 and 46 mg/kg, respectively).

Dioxins/Furan Congeners: 2,3,7,8-TCDD was detected in four composite samples from Area A and in one duplicate composite sample; each concentration was estimated below the laboratory RL. Estimated 2,3,7,8-TCDD concentrations did not exceed Commonwealth of Pennsylvania non-residential soil standards. The dioxin TEQ (ND=RL) ranged from 0.000021 to 0.000242 mg/kg.

SVOCs. Five of the 46 tested SVOCs (benzoic acid, bis(2-ethylhexyl)phthalate, dibenzofuran, methylphenol 3&4, and phenol) were detected in the sediment composites, with the majority of concentrations estimated below the laboratory RL. None of the detected SVOCs exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

VOCs. Four VOCs (2-butanone, chlorodibromomethane, chloroform, and dichlorobromomethane) were detected in the sediment samples. Many of the detected concentrations were estimated below the laboratory RL. None of the detected VOCs exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

Chlorinated Pesticides. Eleven of the 22 tested chlorinated pesticides were detected in the Reserve Basin sediment samples. None of the detected concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

TPH. TPH-GRO (C6-C10) was detected in samples from four of the 12 locations in Area A (ranging from 1.4 to 76 mg/kg) and in samples from each of the three locations in Area B (ranging from 12 to 15 mg/kg). TPH-DRO (C10-C20) was detected in samples from four of the 12 locations in Area A (ranging from 14 to 900 mg/kg) and in samples from each of the three locations in Area B (ranging from 210 to 1,000 mg/kg). TPH-ORO (C20-C34) was detected in samples from nine of the 12 locations in Area A (ranging from 15 to 1,200 mg/kg) and in samples from each of the three locations in Area B (ranging from 370 to 2,400 mg/kg). Overall, the highest concentrations of TPH were detected in samples from A-07, B-2, and B-4.

Radioactivity. Gross alpha was detected in five of the six composite samples from Area A (ranging from 7.2 to 25.6 pCi/g) and in one sample in Area B (B-4) at a concentration of 8.9 pCi/g. Gross beta was detected in in five of the six composite samples from Area A (ranging from 7.2 to 25.6 pCi/g) and in one sample in Area B (B-4) at a concentration of 8.9 pCi/g. each of sediment samples in Area A(ranging from 8.05 to 22.4 pCi/g) and in two of the samples in Area B (ranging from 5.37 to 5.57 pCi/g).

PFAS. Four of the six tests tested PFAS were detected in the Reserve Basin sediments. None of the detected concentrations exceeded State of New Jersey or Commonwealth of Pennsylvania non-residential soil standards.

TCLP. Eight constituents (arsenic, barium, cadmium, chromium, lead, selenium, silver, and chloroform) were detected in at least one of the leachates for Reserve Basin sediment samples. None of the detected concentrations exceeded TCLP criteria.

TABLE 5-1. PHYSICAL CHARACTERISTICS OF SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

		Dredging Area A							Dredging Area B		
ANALYTE	UNITS	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PHYSICAL PROPERTIES											
PERCENT MOISTURE	%	**56.1**	**55.6**	**60.6**	**60.2**	**61**	**59.9**	**59.5**	**59.4**	**60.6**	**68**
PERCENT SOLIDS	%	**43.9**	**44.4**	**39.4**	**39.8**	**39**	**40.1**	**40.5**	**40.6**	**39.4**	**32**
MOISTURE CONTENT	%	**133.2**	**129.1**	**155.5**	**123.3**	**147.5**	**155.7**	**195.7**	**147.7**	**97.4**	**197.9**
GRAIN SIZE											
GRAVEL	%	0	**0.4**	0	0	0	**12.7**	**17.5**	**2.9**	**2.2**	**1.7**
SAND	%	**4.2**	**4.2**	**3.7**	**6.4**	**4.9**	**2.9**	**7.1**	**23.2**	**21.7**	**16.6**
COARSE SAND	%	0	**0.2**	0	**1.7**	**1.2**	0	**1.6**	**0.6**	**1.8**	**0.8**
MEDIUM SAND	%	**0.6**	**1**	**0.7**	**3**	**1.2**	**0.9**	**3.7**	**5.9**	**7.4**	**4.1**
FINE SAND	%	**3.6**	**3**	**3**	**1.7**	**2.5**	**2**	**1.8**	**16.7**	**12.5**	**11.7**
SILT	%	**58.9**	**64.1**	**55.9**	**59.4**	**55.7**	**44.8**	**49.4**	**50.2**	**39.9**	**43.6**
CLAY	%	**36.9**	**31.3**	**40.4**	**34.2**	**39.4**	**39.6**	**26**	**23.7**	**36.2**	**38.1**
SILT+CLAY	%	**95.8**	**95.4**	**96.3**	**93.6**	**95.1**	**84.4**	**75.4**	**73.9**	**76.1**	**81.7**

Notes:
Bold values represent detected concentrations.
FD = Field duplicate.
% = Percent.

This page left intentionally blank

TABLE 5-2. GENERAL CHEMISTRY CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	**UNITS**	**NJ Non-Residential Standards[a]**	**PA Non-Residential Standards[b] (-0 to -2 ft)**	**PA Non-Residential Standards[b] (-2 to -15 ft)**	**PNYRB24-A-0102-SED**	**PNYRB24-A-0304-SED**	**PNYRB24-A-0506-SED**	**PNYRB24-A-0708-SED**	**PNYRB24-A-0910-SED**	**PNYRB24-A-0910-SED-FD**	**PNYRB24-A-1112-SED**	**PNYRB24-B-2-SED**	**PNYRB24-B-4-SED**	**PNYRB24-B-5-SED**
CYANIDE, TOTAL	mg/kg	780	1,900	190,000	**0.35**	**0.43**	**0.5**	**0.73**	**0.62**	**0.8**	0.45 U	**8**	**1.8**	**1.4**
pH	s.u.	--	--	--	**7.5**	**7.4**	**6.5**	**6.8**	**6.8**	**6.5**	**6.7**	**7.2**	**6.9**	**7.2**
SULFIDE	mg/kg	--	--	--	**210**	**380**	**240**	**540**	**490**	**730**	**1200**	**3700**	**2100**	**3100**
TOTAL ORGANIC CARBON	%	--	--	--	**4.7**	**4.7**	**4.5**	**4.6**	**4.8**	**4.6**	**3.8**	**9.5**	**6.2**	**8.2**

Notes:
Sources: (a) N.J.A.C. 7:26D.
(b) 25 Pa. Code § 250 Table 3a and 4a.
Bold values represent detected concentrations. RL is reported for non-detected constituents.
-- = No value available.
FD = Field duplicate.
ft = Foot (feet).
mg/kg = Milligram(s) per kilogram.
RL = Laboratory reporting limit.
s.u. = standard units
% = percent

This page left intentionally blank

TABLE 5-3. METALS CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

ANALYTE	UNITS	NJ Non-Residential Standards[a]	PA Non-Residential Standards[b] (-0 to -2 ft)	PA Non-Residential Standards[b] (-2 to -15 ft)	Dredging Area A							Dredging Area B		
					PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
ALUMINUM	mg/kg	--	190,000	190,000	30000 B	27000	28000	32000	32000	31000	27000	34000	18000	20000
ANTIMONY	mg/kg	520	1,300	190,000	0.84	0.48 B	0.52 B	0.89 B	0.84 B	0.91 B	0.63 B	5.9	6.6 B	2.1 B
ARSENIC	mg/kg	19	61	190,000	12	11	12	28	16	17	13	49	44	22
BARIUM	mg/kg	260,000	190,000	190,000	240	230	260	310	290	280	240	610	810	320
BERYLLIUM	mg/kg	2,600	6,400	190,000	1.8	2	2.1	3.3	2.5	2.5	2	2.8	3	2
CADMIUM	mg/kg	1,100	1,600	190,000	1.9	1.9	2.6	4.2	3.4	3.6	2.3	5.5	6.6	3
CALCIUM	mg/kg	--	--	--	7100 B	6600	5500	5300	5800	5400	4700	4200	3500	35000
CHROMIUM	mg/kg	--	--	--	69	66 B	71 B	140 B	120 B	120 B	70 B	240	140 B	91 B
CHROMIUM, HEXAVALENT	mg/kg	--	180	140,000	4.5 U	4.5 U	5.1 U	5.1 U	5.2 U	5.0 U	5.0 U	5.0 U	5.1 U	1.3 U
CHROMIUM, TRIVALENT	mg/kg	--	190,000	190,000	69	66	71	140	120	120	70	240	140	91
COBALT	mg/kg	390	960	190,000	27	28	29	36	33	32	28	37	31	31
COPPER	mg/kg	52,000	100,000	190,000	91	73	90	140	120	120	91	450	190	210
IRON	mg/kg	--	190,000	190,000	43000 B	41000	42000	53000	46000	44000	39000	51000	47000	61000
LEAD	mg/kg	800	1,000	190,000	100 B	87	100	200	160	150	96	790 B	5900	280
MAGNESIUM	mg/kg	--	--	--	6500	6900	6600	6000	7200	6800	6100	5200	5200	7500
MANGANESE	mg/kg	31,000	190,000	190,000	1600	1600	1500	1700	1600	1500	950	770	640	1700
MERCURY	mg/kg	390	510	190,000	0.17	0.29 H	0.38 H	1.1 H	0.34 H	0.43 H	0.26 H	2.7	1.3 H	1.9 H
NICKEL	mg/kg	26,000	64,000	190,000	46	46	48	61	57	55	46	85	61	52
POTASSIUM	mg/kg	--	--	--	2900	2700	2600	2700	3100	3000	2800	3300	2600	2500
SELENIUM	mg/kg	6,500	16,000	190,000	1.5	1.3	1.4	4.4	2.3 J	1.8	1.4	4.3	4.4	2.2 J
SILVER	mg/kg	6,500	16,000	190,000	1.8	0.76 B	1.2 B	2.2 B	1.7 B	1.7 B	0.97 B	8	2.4 B	2.0 B
SODIUM	mg/kg	--	--	--	300	270	270	260	320	310	340	330	300	390
THALLIUM	mg/kg	--	32	190,000	0.24	0.24	0.26	0.34	0.31	0.3	0.24	0.44	0.39	0.35
VANADIUM	mg/kg	6,500	16,000	190,000	61	54 B	52 B	77 B	70 B	70 B	56 B	93	59 B	67 B
ZINC	mg/kg	390,000	190,000	190,000	510	460	550	850	740	730	510	2100	2100	980

Notes:

Sources: (a) N.J.A.C. 7:26D.
(b) 25 Pa. Code § 250 Table 3a and 4a.

Bold values represent detected concentrations. Shaded concentrations screening criteria.
RL is reported for non-detected constituents.

-- = No value available.

mg/kg = Milligram(s) per kilogram.

RL = Laboratory reporting limit.

B = Compound was detected in the laboratory method blank.

H = analyte analyzed outside of holding time

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

Exceeds New Jersey Non-Residential Standards.

Exceeds New Jersey and Pennsylvania Non-Residential Standards (0-2').

This page left intentionally blank

TABLE 5-4. POLYCYCLIC AROMATIC HYDROCARBON CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards(a)	PA Non-Residential Standards(b) (-0 to -2 ft)	PA Non-Residential Standards(b) (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
LOW MOLECULAR WEIGHT PAHs (LPAH)														
1-METHYLNAPHTHALENE	mg/kg	--	--	--	0.037 J	0.011 J	0.038 J	0.053	0.058 J	0.084 J	0.04 J	0.82	0.49	0.37
2-METHYLNAPHTHALENE	mg/kg	3,300	240	270	0.064	0.026 J	0.081 J	0.14	0.13	0.21	0.093	1.8	1.1	0.88
ACENAPHTHENE	mg/kg	50,000	190,000	190,000	0.076	0.038	0.059 J	0.091	0.096 J	0.13	0.061 J	0.77	0.3	0.85
ACENAPHTHYLENE	mg/kg	--	190,000	190,000	0.07	0.041	0.082 J	0.09	0.093 J	0.17	0.08 J	0.4	0.19 J	0.28 J
ANTHRACENE	mg/kg	250,000	190,000	190,000	0.16	0.056	0.14	0.15	0.16	0.34	0.13	0.99	0.46	0.85
FLUORENE	mg/kg	33,000	130,000	190,000	0.11	0.045	0.075 J	0.1	0.11 J	0.17	0.075 J	1.1	0.49	0.75
NAPHTHALENE	mg/kg	34,000	66	77	0.095	0.033 J	0.1	0.15	0.17	0.27	0.14	1.7	0.93	0.84
PHENANTHRENE	mg/kg	--	190,000	190,000	0.44	0.18	0.36	0.51	0.45	0.77	0.27	3.9	1.7	3.7
TOTAL LPAHs (ND=0)	mg/kg	--	--	--	1.05	0.43	0.935	1.28	1.27	2.14	0.889	11.5	5.66	8.52
TOTAL LPAHs (ND=1/2RL)	mg/kg	--	--	--	1.05	0.43	0.935	1.28	1.27	2.14	0.889	11.5	5.66	8.52
TOTAL LPAHs (ND=RL)	mg/kg	--	--	--	1.05	0.43	0.935	1.28	1.27	2.14	0.889	11.5	5.66	8.52
HIGH MOLECULAR WEIGHT PAHs (HPAH)														
BENZO(A)ANTHRACENE	mg/kg	23	130	190,000	0.31	0.16	0.36	0.41	0.34	0.59	0.32	1.3	0.69	1.6
BENZO(A)PYRENE	mg/kg	2.3	91	190,000	0.26	0.17	0.36	0.39	0.37	0.58	0.34	0.85	0.47	1.5
BENZO(B)FLUORANTHENE	mg/kg	23	76	190,000	0.34	0.23	0.48	0.52	0.45	0.67	0.45	1.1	0.65	2
BENZO(GHI)PERYLENE	mg/kg	--	190,000	190,000	0.2	0.14	0.27	0.31	0.31	0.52	0.31	0.68	0.46	1.3
BENZO(K)FLUORANTHENE	mg/kg	230	76	190,000	0.11	0.073	0.15	0.17	0.18	0.36	0.17	0.3	0.26	0.8
CHRYSENE	mg/kg	2,300	760	190,000	0.33	0.19	0.43	0.53	0.51	0.81	0.45	1.9	1.2	2.3
DIBENZO(A,H)ANTHRACENE	mg/kg	2.3	22	190,000	0.055	0.032 J	0.1	0.069	0.13 U	0.11 J	0.073 J	0.2 J	0.25 U	0.43
FLUORANTHENE	mg/kg	33,000	130,000	190,000	0.63	0.34	0.63	0.77	0.68	1.1	0.52	3.1	1.3	3.7
INDENO(1,2,3-CD)PYRENE	mg/kg	23	76	190,000	0.17	0.096	0.22	0.24	0.23	0.37	0.22	0.46	0.33	0.92
PYRENE	mg/kg	25,000	96,000	190,000	0.64	0.28	0.71	0.92	0.81	1.4	0.62	4	2	3.8
TOTAL HPAHs (ND=0)	mg/kg	--	--	--	3.05	1.71	3.71	4.33	3.88	6.51	3.47	13.9	7.36	18.4
TOTAL HPAHs (ND=1/2RL)	mg/kg	--	--	--	3.05	1.71	3.71	4.33	3.95	6.51	3.47	13.9	7.49	18.4
TOTAL HPAHs (ND=RL)	mg/kg	--	--	--	3.05	1.71	3.71	4.33	4.01	6.51	3.47	13.9	7.61	18.4
TOTAL PAHs														
TOTAL PAHs (ND=0)	mg/kg	--	--	--	4.1	2.14	4.65	5.61	5.15	8.65	4.36	25.4	13.0	26.9
TOTAL PAHs (ND=1/2RL)	mg/kg	--	--	--	4.1	2.14	4.65	5.61	5.21	8.65	4.36	25.4	13.1	26.9
TOTAL PAHs (ND=RL)	mg/kg	--	--	--	4.1	2.14	4.65	5.61	5.28	8.65	4.36	25.4	13.3	26.9

Notes:

Sources: (a) N.J.A.C. 7:26D.

(b) 25 Pa. Code § 250 Table 3a and 4a.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

-- = No value available.

FD = Field duplicate.

ft = Foot (feet).

mg/kg = Milligram(s) per kilogram.

RL = Laboratory reporting limit.

PAH = Polycyclic aromatic hydrocarbon.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 5-5. PCB CONGENER AND HOMOLOG CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PCB-1	mg/kg	--	0.000031	0.000054	0.000063	0.00025	0.00019	0.00021	0.000032	0.00015 U	0.000093 J	0.00015 U
PCB-2	mg/kg	--	0.000012 J	0.000022	0.000024	0.000083 JI	0.000053	0.00005	0.000015	0.00015 U	0.00015 U	0.00015 U
PCB-3	mg/kg	--	0.00003	0.000056	0.000064	0.00024	0.00021	0.00019	0.000039	0.00021 U	0.00021 U	0.00021 U
PCB-4	mg/kg	--	0.00014	0.0002	0.00018	0.0011	0.00074	0.00056	0.000065	0.00072	0.0006 I	0.00017 J
PCB-5	mg/kg	--	0.000019 U	0.000019 U	0.000019 U	0.00019 U	0.000022 I	0.000017 JI	0.000019 U	0.00019 U	0.00019 U	0.00019 U
PCB-6	mg/kg	--	0.000034	0.000073	0.000069	0.00035	0.00022	0.00018	0.000034	0.0002 I	0.0001 I	0.00005 J
PCB-7	mg/kg	--	0.0000059 JI	0.000012 I	0.000015 I	0.0001 JI	0.000061	0.000046	0.0000077 JI	0.00011 U	0.00011 U	0.00011 U
PCB-8	mg/kg	--	0.000072	0.00017	0.00016	0.001	0.00063	0.00054	0.000081	0.00057	0.00027	0.00017
PCB-9	mg/kg	--	0.0000071 JI	0.000016	0.000016 I	0.00011	0.000065	0.000056	0.0000069 JI	0.000046 J	0.00009 U	0.00009 U
PCB-10	mg/kg	--	0.000008 J	0.000014 JI	0.000014 JI	0.0001 J	0.000067	0.000055	0.000015 U	0.000076 JI	0.00015 U	0.00015 U
PCB-11	mg/kg	--	0.000033 J	0.000052 J	0.000051 JI	0.00067 U	0.000061 JI	0.000049 J	0.000039 J	0.00067 U	0.00067 U	0.00067 U
PCB-12/13	mg/kg	--	0.000026	0.000057	0.000056	0.00031 I	0.0002	0.00015	0.000031	0.00016 U	0.000095 J	0.00016 U
PCB-14	mg/kg	--	0.000009 U	0.000009 U	0.000009 U	0.00009 U	0.000009 U	0.000009 U	0.000009 U	0.00009 U	0.00009 U	0.00009 U
PCB-15	mg/kg	--	0.000076	0.00018	0.00021	0.0012	0.00074	0.00056	0.00012	0.00055	0.00035	0.00021
PCB-16	mg/kg	--	0.000051	0.00013	0.00012	0.0011	0.00056	0.00048	0.000055	0.00098	0.00068	0.00032
PCB-17	mg/kg	--	0.000081	0.00019	0.00019	0.0013	0.00065	0.00049	0.000087	0.0011	0.00078	0.00033
PCB-18/30	mg/kg	--	0.00012	0.00028	0.00028	0.0025	0.0013	0.00097	0.00012	0.0024	0.0016	0.00073
PCB-19	mg/kg	--	0.00011	0.00015	0.00015	0.0003	0.00018	0.00018	0.00005	0.00027	0.00022	0.00011
PCB-20/28	mg/kg	--	0.00022	0.00055	0.00056	0.0036	0.0019	0.0014	0.00027	0.003	0.002	0.0011
PCB-21/33	mg/kg	--	0.000051	0.00013	0.00014	0.001	0.00049	0.00036	0.000062	0.00064	0.00038	0.00028
PCB-22	mg/kg	--	0.000047	0.00011	0.00012	0.00087	0.00044	0.00034	0.000055	0.00065	0.0004	0.00013 U
PCB-23	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-24	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.000034 J	0.00002 I	0.000008 U	0.000002 JI	0.00008 U	0.00008 U	0.00008 U
PCB-25	mg/kg	--	0.000085	0.00016	0.00016	0.00049	0.00033	0.00022	0.000074	0.00008 U	0.00033	0.00013
PCB-26/29	mg/kg	--	0.000058	0.00013	0.00015	0.00072	0.00045	0.00029	0.000074	0.00059	0.00046	0.00021 J
PCB-27	mg/kg	--	0.000033	0.000062	0.000075	0.00024	0.00013	0.00011	0.000029	0.00021	0.00021	0.000069 J
PCB-31	mg/kg	--	0.00014	0.00035	0.00037	0.0023	0.0013	0.0009	0.00017	0.0022	0.0015	0.00011 U
PCB-32	mg/kg	--	0.000066	0.00013	0.00013	0.00087	0.00043	0.0003	0.00006	0.00071	0.00067	0.00024
PCB-34	mg/kg	--	0.000008 U	0.0000052 J	0.0000057 J	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-35	mg/kg	--	0.0000045 J	0.0000098	0.0000094 I	0.000057 J	0.000021 I	0.000017	0.0000044 J	0.00009 U	0.00009 U	0.00009 U
PCB-36	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-37	mg/kg	--	0.000036	0.000097	0.00011	0.00067	0.00032	0.00022	0.000054	0.00055	0.00034	0.00024
PCB-38	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-39	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-40/71	mg/kg	--	0.00012	0.00033	0.00036	0.0018	0.0011	0.00062	0.00014	0.0022	0.0012	0.00072
PCB-41	mg/kg	--	0.000008 U	0.000026 I	0.000024	0.00024	0.00013	0.000092 I	0.0000071 JI	0.00028 I	0.00017 I	0.000089
PCB-42	mg/kg	--	0.000059	0.00018	0.0002	0.001	0.00056	0.00038	0.000085	0.0012	0.00065	0.0004
PCB-43	mg/kg	--	0.000008	0.000024 I	0.000026 I	0.00016	0.000094	0.000062 I	0.00001	0.00008 U	0.00008 U	0.00008 U
PCB-44/47/65	mg/kg	--	0.00031	0.00092	0.00091	0.0042	0.0022	0.0014	0.00035	0.0063	0.0028	0.0018
PCB-45	mg/kg	--	0.00003 I	0.000067	0.000062	0.00057	0.00031	0.00024	0.000038	0.00088	0.00041 I	0.00022
PCB-46	mg/kg	--	0.000023	0.000046	0.000048	0.00028	0.00015	0.00011	0.000022	0.00008 U	0.00025 I	0.00008 U

TABLE 5-5. PCB CONGENER AND HOMOLOG CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PCB-48	mg/kg	--	**0.000028**	**0.00009**	**0.000091**	**0.0007**	**0.00037**	**0.00023**	**0.000037**	**0.00087**	**0.00044**	**0.00025**
PCB-49/69	mg/kg	--	**0.00021**	**0.00061**	**0.00065**	**0.0027**	**0.0015**	**0.00091**	**0.00026**	**0.0036**	**0.0018**	**0.0011**
PCB-50/53	mg/kg	--	**0.000096**	**0.00023**	**0.00023**	**0.0007**	**0.00039**	**0.00028**	**0.000079**	**0.00078**	**0.00057**	**0.00029**
PCB-51	mg/kg	--	**0.000098**	**0.00024**	**0.00024**	**0.00028**	**0.00016**	**0.00011**	**0.000067**	0.00009 U	**0.00023 I**	**0.00013 I**
PCB-52	mg/kg	--	**0.00033**	**0.00098**	**0.0011**	**0.0062**	**0.003**	**0.0019**	**0.0004**	**0.0055 I**	**0.0042**	**0.0028**
PCB-54	mg/kg	--	**0.000024**	**0.000042**	**0.00004**	**0.000032 J**	**0.00002**	**0.000015**	**0.000015**	0.00008 U	**0.000027 J**	0.00008 U
PCB-55	mg/kg	--	**0.0000032 J**	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-56	mg/kg	--	**0.000063**	**0.00019**	**0.0002**	**0.0013**	**0.00065**	**0.00038**	**0.00009**	**0.0017**	**0.0008**	**0.00052**
PCB-57	mg/kg	--	**0.0000026 J**	**0.0000076 J**	0.000008 U	0.00008 U	**0.000019**	0.000008 U	**0.0000042 J**	0.00008 U	0.00008 U	0.00008 U
PCB-58	mg/kg	--	0.000008 U	**0.00002**	0.000008 U	**0.00013**	0.000008 U	**0.000026 I**	**0.000003 J**	**0.00032**	0.00008 U	**0.000063 J**
PCB-59/62/75	mg/kg	--	**0.000021 J**	**0.000066**	**0.000075**	**0.00034**	**0.00019**	**0.00012**	**0.000031**	**0.00037**	**0.00024**	**0.00013 J**
PCB-60	mg/kg	--	**0.000017 J**	**0.000051**	**0.000053**	0.00027 U	**0.00021**	**0.00011**	**0.000019 J**	**0.00065**	**0.0003**	**0.00018 J**
PCB-61/70/74/76	mg/kg	--	**0.00022**	**0.00068**	**0.00071**	**0.0053**	**0.0023**	**0.0014**	**0.00029**	**0.0095**	**0.0033**	**0.0024**
PCB-63	mg/kg	--	**0.0000077 J**	**0.000024**	**0.000026**	0.00008 U	**0.000069**	**0.000038**	**0.0000095**	0.00008 U	**0.000082**	0.00008 U
PCB-64	mg/kg	--	**0.000086**	**0.00026**	**0.00028**	**0.0017**	**0.00086**	**0.00053**	**0.00012**	**0.0025**	**0.0011**	**0.00071**
PCB-66	mg/kg	--	**0.00012**	**0.0004**	**0.00045**	**0.0028**	**0.0013**	**0.0008**	**0.0002**	**0.0038**	**0.0017**	**0.0012**
PCB-67	mg/kg	--	**0.00002**	**0.000044**	**0.000054**	0.00008 U	**0.000099**	**0.000056**	**0.000022**	0.00008 U	**0.00012**	0.00008 U
PCB-68	mg/kg	--	**0.0000038 J**	**0.0000094**	**0.000013**	**0.000025 JI**	**0.000016**	0.000008 U	**0.0000048 J**	0.00008 U	0.00008 U	0.00008 U
PCB-72	mg/kg	--	**0.0000057 J**	**0.000018**	**0.000019**	0.00008 U	**0.000037**	**0.000018**	**0.0000068 J**	0.00008 U	0.00008 U	0.00008 U
PCB-73	mg/kg	--	**0.0000049 JI**	**0.000013**	**0.000012**	0.00008 U	0.000008 U	0.000008 U	**0.0000039 J**	0.00008 U	0.00008 U	0.00008 U
PCB-77	mg/kg	--	**0.000014**	**0.000048**	**0.000052**	**0.00027**	**0.00013**	**0.00008**	**0.000023**	**0.00031**	**0.00017**	**0.00014**
PCB-78	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-79	mg/kg	--	0.000008 U	**0.000011**	0.000008 U	0.00008 U	**0.0000089 I**	0.000008 U	**0.0000031 JI**	0.00008 U	0.00008 U	0.00008 U
PCB-80	mg/kg	--	0.000008 U	**0.0000036 JI**	**0.0000048 J**	0.00008 U	**0.000011**	**0.0000047 JI**	**0.0000022 J**	**0.000048 JI**	0.00008 U	0.00008 U
PCB-81	mg/kg	--	0.000009 U	0.000009 U	0.000009 U	**0.000047 J**	0.000009 U	0.000009 U	0.000009 U	**0.00011**	0.00009 U	0.00009 U
PCB-82	mg/kg	--	**0.000037**	**0.00012**	**0.00012**	0.00017 U	**0.00034**	**0.00021**	**0.000045**	**0.0022**	**0.00063**	**0.0005**
PCB-83	mg/kg	--	**0.00002**	**0.000068**	**0.000073**	**0.00043**	**0.00018**	0.000008 U	**0.000029**	**0.00099**	**0.0003**	**0.00024**
PCB-84	mg/kg	--	**0.00007**	**0.00024**	**0.00026**	0.00008 U	**0.00063**	**0.00038**	**0.00011**	**0.0049**	**0.0013**	**0.0011**
PCB-85/116/117	mg/kg	--	**0.00005**	**0.00016 I**	**0.00018**	**0.001**	**0.00043**	**0.00026**	**0.000069**	**0.0025**	**0.00079**	**0.00062**
PCB-86/87/97/109/119/125	mg/kg	--	**0.00019**	**0.00065**	**0.00066**	**0.0042**	**0.0016**	**0.00097**	**0.00028**	**0.012**	**0.003**	**0.0026**
PCB-88	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-89	mg/kg	--	0.000008 U	**0.0000086**	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-90/101/113	mg/kg	--	**0.00033**	**0.0012**	**0.0012**	**0.0085**	**0.0028**	**0.0015**	**0.00051**	**0.018**	**0.0054**	**0.0044**
PCB-91	mg/kg	--	**0.000053**	**0.00019**	**0.00021**	**0.00095**	**0.0004**	**0.00022**	**0.000087**	**0.0022**	**0.00071**	**0.00058**
PCB-92	mg/kg	--	**0.000094**	**0.00031**	**0.00032**	**0.0016**	**0.00068**	**0.00035**	**0.00013**	**0.0034**	**0.001**	**0.00085**
PCB-93/100	mg/kg	--	**0.000029**	**0.0001**	**0.00009**	0.00016 U	**0.00007**	**0.000037**	**0.000025**	0.00016 U	0.00016 U	**0.000064 JI**
PCB-94	mg/kg	--	**0.00001 I**	**0.000034**	**0.000034**	0.00008 U	**0.000025**	**0.000014 I**	**0.0000091**	0.00008 U	0.00008 U	0.00008 U
PCB-95	mg/kg	--	**0.00026**	**0.00081**	**0.00079**	**0.007**	**0.0022**	**0.0013**	**0.00039**	**0.016**	**0.0044**	**0.0032**
PCB-96	mg/kg	--	**0.0000049 J**	**0.000018**	**0.000015 I**	**0.000048 J**	**0.000028**	**0.000018**	**0.0000065 J**	**0.0001**	**0.000047 J**	**0.000031 J**
PCB-98/102	mg/kg	--	**0.000022**	**0.000078**	**0.000076**	**0.0003 I**	**0.00013**	**0.000059**	**0.000023**	**0.00048**	**0.000082 J**	0.00016 U
PCB-99	mg/kg	--	**0.00015**	**0.00061**	**0.0006**	**0.0032**	**0.0012**	**0.00007**	**0.00026**	**0.007**	**0.002**	**0.0018**

TABLE 5-5. PCB CONGENER AND HOMOLOG CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PCB-103	mg/kg	--	**0.000015**	**0.00005**	**0.000051**	**0.00014**	**0.000066**	**0.000034**	**0.000017**	0.00008 U	0.00008 U	0.00008 U
PCB-104	mg/kg	--	**0.0000048 J**	**0.000013**	**0.000012**	0.00008 U	**0.000005 JI**	**0.0000041 JI**	**0.000004 J**	0.00008 U	0.00008 U	0.00008 U
PCB-105	mg/kg	--	**0.000067**	**0.00024**	**0.00024**	**0.0018**	**0.00062**	**0.00038**	**0.000091**	**0.0053**	**0.0014**	**0.0011**
PCB-106	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	**0.000047 J**
PCB-107	mg/kg	--	**0.000018**	**0.000066**	**0.000067**	**0.00038**	**0.00014**	**0.000087**	**0.000028**	**0.00083**	**0.00022**	0.00008 U
PCB-108/124	mg/kg	--	**0.0000063 J**	**0.000023**	**0.000023**	**0.00018**	**0.000062**	**0.000034**	**0.0000097 J**	**0.00058**	**0.00013 J**	**0.00013 J**
PCB-110/115	mg/kg	--	**0.00035**	**0.0012**	**0.0012**	**0.0076**	**0.0028**	**0.0016**	**0.00055**	**0.019**	**0.0052**	**0.0046**
PCB-111	mg/kg	--	0.000009 U	0.000009 U	0.000009 U	0.00009 U	0.000009 U	0.000009 U	0.000009 U	0.00009 U	0.00009 U	0.00009 U
PCB-112	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	**0.0006**	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-114	mg/kg	--	**0.000004 J**	**0.000014**	**0.000013 I**	**0.00011**	**0.000045**	**0.000023**	**0.0000055 J**	**0.00033**	0.00008 U	0.00008 U
PCB-118	mg/kg	--	**0.00018**	**0.00066**	**0.00064**	**0.0044**	**0.0016**	**0.00098**	**0.00028**	**0.012**	**0.0032**	**0.0026**
PCB-120	mg/kg	--	0.000009 U	**0.000011**	**0.0000089 J**	0.00009 U	**0.000012 I**	**0.0000074 J**	**0.0000044 JI**	0.00009 U	0.00009 U	0.00009 U
PCB-121	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-122	mg/kg	--	0.000008 U	**0.0000074 J**	**0.0000072 JI**	**0.000056 JI**	**0.000024**	**0.000013**	**0.0000039 J**	0.00008 U	0.00008 U	0.00008 U
PCB-123	mg/kg	--	**0.0000031 J**	**0.000011**	**0.000014**	**0.000085**	**0.00003**	**0.000017**	**0.0000051 J**	0.00008 U	**0.000054 J**	**0.00026**
PCB-126	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	**0.0000042 J**	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-127	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-128/166	mg/kg	--	**0.000053**	**0.0002**	**0.00019**	**0.0012**	**0.0004**	**0.00024**	**0.000089**	**0.0026**	**0.0007**	**0.00071**
PCB-129/138/163	mg/kg	--	**0.00048**	**0.0019**	**0.0017**	**0.012**	**0.0036**	**0.0021**	**0.00074**	**0.019**	**0.0063**	**0.0062**
PCB-130	mg/kg	--	**0.00003**	**0.00012**	**0.00012**	**0.00062**	**0.00022**	**0.00015**	**0.000055 I**	**0.0012**	**0.00037**	**0.00036**
PCB-131	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-132	mg/kg	--	**0.00013**	**0.00051**	**0.0005**	**0.0035**	**0.0011**	**0.00063**	**0.00021**	**0.0066**	**0.0022**	**0.002**
PCB-133	mg/kg	--	**0.000019**	**0.000051**	**0.000044**	**0.00028**	**0.00011**	**0.000042**	**0.000018**	0.00008 U	0.00008 U	0.00008 U
PCB-134	mg/kg	--	**0.000021 I**	**0.000082 I**	**0.00008**	**0.00052**	**0.00018**	**0.0001**	**0.000036**	**0.0011**	**0.00032**	**0.00033**
PCB-135/151	mg/kg	--	**0.0002**	**0.00079**	**0.00075**	**0.0046**	**0.0016**	**0.00089**	**0.00031**	**0.006**	**0.0026**	**0.0022**
PCB-136	mg/kg	--	**0.000057**	**0.00024**	**0.00023**	**0.0016**	**0.00051**	**0.00028**	**0.000093**	**0.0025**	**0.00096**	**0.00078**
PCB-137	mg/kg	--	**0.000012**	**0.000046**	0.000008 U	**0.00027**	**0.000082**	**0.00007**	**0.000019**	**0.00092**	**0.00022**	**0.00019**
PCB-139/140	mg/kg	--	**0.0000083 J**	**0.000032**	**0.000031**	**0.00016**	**0.000066**	**0.000031**	**0.000013 J**	**0.00033**	**0.000083 JI**	**0.000085 J**
PCB-141	mg/kg	--	**0.00006**	**0.00025**	**0.00022**	**0.0024**	**0.0005**	**0.00031**	**0.000097**	**0.0031**	**0.00098**	**0.00096**
PCB-142	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-143	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-144	mg/kg	--	**0.000019**	**0.000072**	**0.000074**	0.00008 U	**0.00018**	**0.000096**	**0.000031**	**0.00078**	**0.00029**	**0.00028**
PCB-145	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-146	mg/kg	--	**0.00013**	**0.00043**	**0.0004**	**0.0022**	**0.00082**	**0.00042**	**0.00017**	**0.0027**	**0.0011**	**0.00095**
PCB-147/149	mg/kg	--	**0.00041**	**0.0017**	**0.0016**	**0.01**	**0.0033**	**0.0018**	**0.00067**	**0.015**	**0.006**	**0.0051**
PCB-148	mg/kg	--	**0.0000058 JI**	**0.000017**	0.000008 U	0.00008 U	**0.000033**	**0.000012**	**0.0000045 J**	0.00008 U	0.00008 U	0.00008 U
PCB-150	mg/kg	--	**0.0000034 JI**	**0.000011**	0.000008 U	0.00008 U	**0.000017**	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-152	mg/kg	--	0.000008 U	**0.0000045 JI**	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-153/168	mg/kg	--	**0.00039**	**0.0016**	**0.0015**	**0.01**	**0.0031**	**0.0018**	**0.00063**	**0.014**	**0.0056**	**0.0052**
PCB-154	mg/kg	--	**0.000026 J**	**0.000087**	**0.000075**	**0.00028 J**	**0.00014**	**0.000063**	**0.000032 J**	**0.00021 JI**	0.00041 U	0.00041 U
PCB-155	mg/kg	--	0.000009 U	0.000009 U	0.000009 U	0.00009 U	0.000009 U	0.000009 U	0.000009 U	0.00009 U	0.00009 U	0.00009 U

TABLE 5-5. PCB CONGENER AND HOMOLOG CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PCB-156/157	mg/kg	--	**0.000034**	**0.00014**	**0.00014**	**0.00092**	**0.00029**	**0.00017**	**0.000056**	**0.0021**	**0.00058**	**0.0006**
PCB-158	mg/kg	--	**0.000037**	**0.00015**	**0.00013**	**0.001**	**0.00031**	**0.00018**	**0.00006**	**0.0019**	**0.00055**	**0.00057**
PCB-159	mg/kg	--	**0.0000042 J**	**0.000021**	**0.000016**	**0.00012**	**0.000038**	**0.00002**	**0.000008 I**	**0.000098**	**0.000057 J**	**0.000052 J**
PCB-160	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-161	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-162	mg/kg	--	0.000008 U	**0.0000036 JI**	**0.0000038 J**	0.00008 U	**0.0000061 J**	**0.0000048 JI**	0.000008 U	**0.000044 J**	0.00008 U	0.00008 U
PCB-164	mg/kg	--	**0.00003**	**0.00013**	**0.00015**	**0.00078**	**0.00024**	**0.00015**	**0.000055**	**0.0012**	**0.00041**	**0.00042**
PCB-165	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-167	mg/kg	--	**0.000013**	**0.000053**	**0.000044**	**0.00037**	**0.000099**	**0.000058**	**0.000022**	**0.00068**	**0.00021**	**0.00021**
PCB-169	mg/kg	--	0.000008 U	**0.0000021 JI**	**0.0000025 J**	0.00008 U	**0.000004 JI**	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-170	mg/kg	--	**0.00013**	**0.00061**	**0.00044**	**0.0034**	**0.00092**	**0.00057**	**0.00023**	**0.0033**	**0.0013**	**0.0016**
PCB-171/173	mg/kg	--	**0.000046**	**0.00021**	**0.00015**	**0.0012**	**0.00034**	**0.00021**	**0.000084**	**0.0012**	**0.0005**	**0.00054**
PCB-172	mg/kg	--	**0.000026**	**0.00012**	**0.000084**	**0.00061**	**0.00019**	**0.00011**	**0.000048**	**0.00059**	**0.00028**	**0.0003**
PCB-174	mg/kg	--	**0.00016**	**0.00075**	**0.00056**	**0.0041**	**0.0013**	**0.00083**	**0.0003**	**0.0041**	**0.002**	**0.0019**
PCB-175	mg/kg	--	**0.0000077 J**	**0.000037 I**	**0.000028**	**0.00018**	**0.000064 I**	**0.000041**	**0.000015 I**	**0.00018**	**0.000075 JI**	**0.000094**
PCB-176	mg/kg	--	**0.000022**	**0.0001**	**0.000078**	**0.00058**	**0.00018**	**0.00011**	**0.000037 I**	**0.00059**	**0.00028**	**0.00029**
PCB-177	mg/kg	--	**0.00013**	**0.00048**	**0.00036**	**0.0025**	**0.00083**	**0.00048**	**0.00019**	**0.0022**	**0.00099**	**0.0011**
PCB-178	mg/kg	--	**0.00006**	**0.0002**	**0.00014**	**0.00083**	**0.00033**	**0.00017**	**0.000068**	**0.00079**	**0.00041**	**0.00037**
PCB-179	mg/kg	--	**0.000091**	**0.00035**	**0.00028**	**0.0019**	**0.00064**	**0.00036**	**0.00014**	**0.0019**	**0.00098**	**0.00092**
PCB-180/193	mg/kg	--	**0.00031**	**0.0014**	**0.001**	**0.0077**	**0.0023**	**0.0014**	**0.00054**	**0.0073**	**0.0034**	**0.0037**
PCB-181	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-182	mg/kg	--	0.000008 U	**0.0000098**	**0.000006 JI**	0.00008 U	0.000008 U	0.000008 U	**0.0000047 JI**	0.00008 U	0.00008 U	0.00008 U
PCB-183/185	mg/kg	--	**0.00011**	**0.00051**	**0.00034**	**0.0028**	**0.00086**	**0.00046**	**0.00019**	**0.0027**	**0.0012**	**0.0013**
PCB-184	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-186	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-187	mg/kg	--	**0.0003**	**0.0011**	**0.00082**	**0.0051**	**0.0018**	**0.0011**	**0.00045**	**0.0049**	**0.0026**	**0.0024**
PCB-188	mg/kg	--	0.000049 U	0.000049 U	0.000049 U	0.00049 U	0.000049 U	0.000049 U	0.000049 U	0.00049 U	0.00049 U	0.00049 U
PCB-189	mg/kg	--	**0.0000052 J**	**0.000024**	**0.000016**	**0.00012**	**0.000037**	**0.000022**	**0.00001**	**0.00014**	**0.00006 JI**	**0.000077 J**
PCB-190	mg/kg	--	**0.000027**	**0.00013**	**0.000091**	**0.00067**	**0.00019**	**0.00012**	**0.000047**	**0.00063**	**0.00027**	**0.0003**
PCB-191	mg/kg	--	**0.0000055 J**	**0.000026**	**0.000016 I**	**0.00015**	**0.000042**	**0.000025**	**0.000011 I**	**0.00013**	**0.000061 J**	**0.00007 J**
PCB-192	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-194	mg/kg	--	**0.000083**	**0.00045**	**0.00027**	**0.002**	**0.00055**	**0.00032**	**0.00013**	**0.0019**	**0.001**	**0.00098**
PCB-195	mg/kg	--	**0.000034**	**0.00019**	**0.00011**	**0.00084**	**0.00024**	**0.00014**	**0.000056**	**0.00075**	**0.00036**	**0.00039**
PCB-196	mg/kg	--	**0.000044**	**0.00026**	**0.00017**	**0.0011**	**0.00031**	**0.0002**	**0.000073**	**0.0011**	**0.00059**	**0.00058**
PCB-197/200	mg/kg	--	**0.000014 J**	**0.000074**	**0.000047**	**0.00031**	**0.0001**	**0.000068**	**0.000024**	**0.00038**	**0.00019**	**0.00017**
PCB-198/199	mg/kg	--	**0.00011**	**0.00057**	**0.0004**	**0.0024**	**0.00077**	**0.00053**	**0.00018**	**0.003**	**0.0017**	**0.0013**
PCB-201	mg/kg	--	0.000045 U	**0.00007**	**0.000046**	**0.00031 J**	**0.0001**	**0.000064**	**0.000022 J**	**0.00037 J**	**0.00024 J**	0.00045 U
PCB-202	mg/kg	--	**0.000024**	**0.00011**	**0.000083**	**0.00058**	**0.00016**	**0.0001**	**0.000036**	**0.00077**	**0.00055**	**0.0003**
PCB-203	mg/kg	--	**0.000061**	**0.00033**	**0.00022**	**0.0015**	**0.00043**	**0.00027**	**0.0001**	**0.0017**	**0.001**	**0.0008**
PCB-204	mg/kg	--	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.000008 U	0.000008 U	0.000008 U	0.00008 U	0.00008 U	0.00008 U
PCB-205	mg/kg	--	**0.0000041 J**	**0.000022**	0.000008 U	**0.000088**	**0.000025**	**0.000016**	**0.0000072 J**	0.00008 U	**0.000041 J**	0.00008 U

TABLE 5-5. PCB CONGENER AND HOMOLOG CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PCB-206	mg/kg	--	**0.000083**	**0.00043**	**0.00037**	**0.0016**	**0.00043**	**0.00048**	**0.00016**	**0.003**	**0.0019**	**0.0013**
PCB-207	mg/kg	--	**0.0000077 J**	**0.000037**	**0.000031**	**0.00017**	**0.000046**	**0.000052**	**0.000014**	**0.00025**	**0.00017**	**0.00013**
PCB-208	mg/kg	--	**0.000033**	**0.00015**	**0.00016**	**0.00074**	**0.00016**	**0.00021**	**0.000063**	**0.0013**	**0.00085**	**0.00046**
PCB-209	mg/kg	--	**0.00009**	**0.00043**	**0.00037**	**0.0028**	**0.00042**	**0.0012**	**0.00018**	**0.004**	**0.0019**	**0.0012**
TOTAL PCBs (ND=0)	mg/kg	1.1	**0.0097**	**0.0337**	**0.0310**	**0.1927**	**0.0745**	**0.0473**	**0.0138**	**0.2954**	**0.1159**	**0.0920**
TOTAL PCBs (ND=1/2RL)	mg/kg	1.1	**0.0099**	**0.0339**	**0.0312**	**0.1956**	**0.0747**	**0.0475**	**0.0140**	**0.2988**	**0.1194**	**0.0961**
TOTAL PCBs (ND=RL)	mg/kg	1.1	**0.0101**	**0.0340**	**0.0314**	**0.1985**	**0.0748**	**0.0477**	**0.0141**	**0.3023**	**0.1229**	**0.1002**
TOTAL DICHLOROBIPHENYLS	mg/kg	--	**0.0004 I**	**0.00077 I**	**0.00077 I**	**0.0043 I**	**0.0028 I**	**0.0022 I**	**0.00038 I**	**0.0022 I**	**0.0014 I**	**0.0006 I**
TOTAL HEPTACHLOROBIPHENYLS	mg/kg	--	**0.0014**	**0.0061 I**	**0.0044 I**	**0.032**	**0.01 I**	**0.006**	**0.0024 I**	**0.031**	**0.014 I**	**0.015**
TOTAL HEXACHLOROBIPHENYLS	mg/kg	--	**0.0022 I**	**0.0086 I**	**0.008**	**0.053**	**0.017 I**	**0.0096 I**	**0.0034 I**	**0.082 I**	**0.03 I**	**0.027**
TOTAL MONOCHLOROBIPHENYLS	mg/kg	--	**0.000073**	**0.00013**	**0.00015**	**0.00057 I**	**0.00045**	**0.00045**	**0.000086**	0.00015 U	**0.000093 JI**	0.00015 U
TOTAL NONACHLOROBIPHENYLS	mg/kg	--	**0.00012**	**0.00062**	**0.00056**	**0.0025**	**0.00064**	**0.00074**	**0.00024**	**0.0046**	**0.0029**	**0.0019**
TOTAL OCTACHLOROBIPHENYLS	mg/kg	--	**0.00037**	**0.0021**	**0.0013**	**0.0091**	**0.0027**	**0.0017**	**0.00063**	**0.01**	**0.0057**	**0.0045**
TOTAL PENTACHLOROBIPHENYLS	mg/kg	--	**0.002 I**	**0.0069 I**	**0.0069 I**	**0.042 I**	**0.016 I**	**0.0092 I**	**0.003 I**	**0.11**	**0.03**	**0.025 I**
TOTAL TETRACHLOROBIPHENYLS	mg/kg	--	**0.0019 I**	**0.0056 I**	**0.0059 I**	**0.031 I**	**0.016 I**	**0.0099 I**	**0.0023 I**	**0.041 I**	**0.021 I**	**0.013 I**
TOTAL TRICHLOROBIPHENYLS	mg/kg	--	**0.0011**	**0.0025**	**0.0026 I**	**0.016**	**0.0085 I**	**0.0063**	**0.0012 I**	**0.013**	**0.0096 I**	**0.0038 I**
TOTAL PCBs - HOMOLOGS	mg/kg	1.1	**0.0097 I**	**0.034 I**	**0.031 I**	**0.19 I**	**0.075 I**	**0.047 I**	**0.014 I**	**0.3 I**	**0.12 I**	**0.092 I**

Notes:
Bold values represent detected concentrations.
RL is reported for non-detected constituents.
-- = No value available.
FD = Field duplicate.
mg/kg = Milligram(s) per kilogram.
PCB = Polychlorinated biphenyl.
RL = Laboratory reporting limit.
J = Compound was detected, but below the reporting limit (value is estimated).
I = Value is EMPC (estimated maximum possible concentration).
U = Compound was analyzed, but not detected.
B = Compound was found in the blank and sample.

This page left intentionally blank

TABLE 5-6. PCB AROCLOR CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards[(a)]	PA Non-Residential Standards[(b)] (-0 to -2 ft)	PA Non-Residential Standards[(b)] (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
PCB-1016	mg/kg	--	220	10,000	0.00095 U	0.00093 U	0.0011 U	0.0010 U	0.0011 U	0.0010 U	0.0010 U	0.01 U	0.01 U	0.013 U
PCB-1221	mg/kg	--	23	27	0.00095 U	0.00093 U	0.0011 U	0.0010 U	0.0011 U	0.0010 U	0.0010 U	0.01 U	0.01 U	0.013 U
PCB-1232	mg/kg	--	46	10,000	0.00095 U	0.00093 U	0.0011 U	0.0010 U	0.0011 U	0.0010 U	0.0010 U	0.01 U	0.01 U	0.013 U
PCB-1242	mg/kg	--	46	10,000	0.00095 U	0.00093 U	0.0011 U	0.0010 U	0.0011 U	0.0010 U	0.0010 U	0.01 U	0.01 U	0.013 U
PCB-1248	mg/kg	--	46	10,000	0.00095 U	0.00093 U	**0.024**	**0.07**	**0.075**	**0.2**	**0.046**	**0.42**	**0.094**	**0.17**
PCB-1254	mg/kg	--	64	10,000	0.00095 U	0.00093 U	0.0011 U	0.0010 U	0.0011 U	0.0010 U	0.0010 U	0.01 U	0.01 U	0.013 U
PCB-1260	mg/kg	--	46	190,000	**0.018**	**0.053**	**0.033**	**0.092**	**0.074**	**0.16**	**0.062**	**0.32**	**0.096**	**0.23**
TOTAL PCBs (ND=0)	mg/kg	1.1	46	190,000	**0.018**	**0.053**	**0.057**	**0.162**	**0.149**	**0.36**	**0.108**	**0.74**	**0.19**	**0.4**
TOTAL PCBs (ND=1/2RL)	mg/kg	1.1	46	190,000	**0.021**	**0.056**	**0.06**	**0.165**	**0.152**	**0.363**	**0.111**	**0.765**	**0.215**	**0.433**
TOTAL PCBs (ND=RL)	mg/kg	1.1	46	190,000	**0.024**	**0.059**	**0.063**	**0.167**	**0.155**	**0.365**	**0.113**	**0.79**	**0.24**	**0.465**

Notes:
Sources: (a) N.J.A.C. 7:26D.
(b) 25 Pa. Code § 250 Table 3a and 4a.
Bold values represent detected concentrations. RL is reported for non-detected constituents.
RL is reported for non-detected constituents.
-- = No value available.
mg/kg = Milligram(s) per kilogram.
RL = Laboratory reporting limit.
PCB = Polychlorinated biphenyl.
J = compound was detected, but below the reporting limit (value is estimated)
U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 5-7. DIOXIN AND FURAN CONGENER CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

						Dredging Area A							Dredging Area B		
ANALYTE	UNITS	TEF[a]	NJ Non-Residential Standards[a]	PA Non-Residential Standards[b] (-0 to -2 ft)	PA Non-Residential Standards[b] (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
2,3,7,8-TCDD	mg/kg	1	0.00081	0.0007	190,000	**0.00000089 JI**	**0.00000094 J**	**0.0000012**	0.00002 U	0.00002 U	**0.0000025 J**	**0.0000013 JI**	0.00002 U	0.00002 U	0.00002 U
1,2,3,7,8-PECDD	mg/kg	1	--	--	--	**0.0000051 I**	**0.0000032 J**	**0.0000052 I**	0.0001 U	0.0001 U	0.000025 U	0.000025 U	0.0001 U	0.0001 U	0.0001 U
1,2,3,4,7,8-HXCDD	mg/kg	0.1	--	--	--	**0.0000052**	**0.0000048 J**	**0.0000052**	0.0001 U	0.0001 U	0.000025 U	0.000025 U	0.0001 U	0.0001 U	0.0001 U
1,2,3,6,7,8-HXCDD	mg/kg	0.1	--	--	--	**0.000013**	**0.000013**	**0.000011**	0.0001 U	0.0001 U	**0.000022 J**	**0.000014 J**	0.0001 U	0.0001 U	0.0001 U
1,2,3,7,8,9-HXCDD	mg/kg	0.1	--	--	--	**0.000011**	**0.00001**	**0.0000085**	0.0001 U	0.0001 U	**0.000014 J**	**0.00001 J**	0.0001 U	0.0001 U	0.0001 U
1,2,3,4,6,7,8-HPCDD	mg/kg	0.01	--	--	--	**0.00035**	**0.00042**	**0.00024**	**0.00087**	**0.00057**	**0.00052**	**0.00037**	**0.00071**	**0.00046**	**0.00073**
OCDD	mg/kg	0.0003	--	--	--	**0.0086**	**0.0094**	**0.0056**	**0.017**	**0.012**	**0.0099**	**0.0081**	**0.013**	**0.0094**	**0.013**
2,3,7,8-TCDF	mg/kg	0.1	--	--	--	**0.0000066**	**0.00001**	**0.0000066**	**0.000012 JI**	**0.000013 J**	0.0000050 U	**0.0000096**	**0.000015 JI**	**0.00001 J**	**0.000012 J**
1,2,3,7,8-PECDF	mg/kg	0.03	--	--	--	**0.0000037 J**	**0.0000037 J**	**0.0000043 JI**	0.0001 U	0.0001 U	0.000025 U	0.000025 U	0.0001 U	0.0001 U	0.0001 U
2,3,4,7,8-PECDF	mg/kg	0.3	--	--	--	**0.0000091**	**0.0000096**	**0.0000084**	0.0001 U	0.0001 U	**0.000017 J**	**0.000011 J**	0.0001 U	0.0001 U	0.0001 U
1,2,3,4,7,8-HXCDF	mg/kg	0.1	--	--	--	**0.000014**	**0.000014**	**0.000019**	0.0001 U	0.0001 U	**0.000019 J**	**0.000012 J**	0.0001 U	0.0001 U	0.0001 U
1,2,3,6,7,8-HXCDF	mg/kg	0.1	--	--	--	**0.0000087**	**0.0000088**	**0.0000079**	0.0001 U	0.0001 U	**0.000014 J**	0.000025 U	0.0001 U	0.0001 U	0.0001 U
2,3,4,6,7,8-HXCDF	mg/kg	0.1	--	--	--	**0.0000084**	**0.0000094**	**0.0000073**	0.0001 U	0.0001 U	**0.000014 J**	0.000025 U	0.0001 U	0.0001 U	0.0001 U
1,2,3,7,8,9-HXCDF	mg/kg	0.1	--	--	--	**0.0000020 J**	**0.0000023 J**	**0.0000031 JI**	0.0001 U	0.0001 U	0.000025 U	0.000025 U	0.0001 U	0.0001 U	0.0001 U
1,2,3,4,6,7,8-HPCDF	mg/kg	0.01	--	--	--	**0.00011**	**0.00011**	**0.000069**	**0.00026**	**0.00018**	**0.00017**	**0.000092**	**0.00045**	**0.00032**	**0.00026**
1,2,3,4,7,8,9-HPCDF	mg/kg	0.01	--	--	--	**0.0000056 I**	**0.0000065**	**0.0000058 I**	0.0001 U	0.0001 U	**0.00001 J**	0.000025 U	0.0001 U	0.0001 U	0.0001 U
OCDF	mg/kg	0.0003	--	--	--	**0.00026**	**0.00024**	**0.00015**	**0.00052**	**0.00037**	**0.00031**	**0.0002**	**0.00058**	**0.00041**	**0.00054**
WHO TEQ (ND=0)	mg/kg	--	0.00081	0.0007	190000	**0.000023**	**0.000023**	**0.000021**	**0.000018**	**0.000013**	**0.000026**	**0.000016**	**0.000017**	**0.000012**	**0.000015**
WHO TEQ (ND=1/2RL)	mg/kg	--	0.00081	0.0007	190000	**0.000023**	**0.000023**	**0.000021**	**0.000130**	**0.000125**	**0.000042**	**0.000034**	**0.000129**	**0.000124**	**0.000127**
WHO TEQ (ND=RL)	mg/kg	--	0.00081	0.0007	190000	**0.000023**	**0.000023**	**0.000021**	**0.000242**	**0.000237**	**0.000057**	**0.000052**	**0.000241**	**0.000236**	**0.000239**

Notes:

Sources: (a) Van den Berg, M, et al. 2006. The 2005 World Health Organization Re-evaluation of Human and Mammalian Toxic Equivalency Factors for Dioxins and Dioxin-Like Compounds. *Toxicological Sciences 93(2):223-241.*

(b) 25 Pa. Code § 250 Table 3a and 4a.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

-- = No value available.

FD = Field duplicate.

ft = Foot (feet).

mg/kg = Milligram(s) per kilogram.

RL = Laboratory reporting limit.

TEF = Toxicity equivalency factor.

I = Value is EMPC (estimated maximum possible concentration).

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 5-8. SEMIVOLATILE ORGANIC COMPOUND CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards[a]	PA Non-Residential Standards[b] (-0 to -2 ft)	PA Non-Residential Standards[b] (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
1,2,4-TRICHLOROBENZENE	mg/kg	13,000	160	190	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
1,2-DICHLOROBENZENE	mg/kg	110,000	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
1,2-DIPHENYLHYDRAZINE(AS AZOBENZENE	mg/kg	--	10	12	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
1,3-DICHLOROBENZENE	mg/kg	110,000	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
1,4-DICHLOROBENZENE	mg/kg	13,000	200	230	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2,2'-OXYBIS[1-CHLOROPROPANE]	mg/kg	--	220	250	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
2,4,6-TRICHLOROPHENOL	mg/kg	230	3,200	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2,4-DICHLOROPHENOL	mg/kg	2,700	9,600	190,000	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
2,4-DIMETHYLPHENOL	mg/kg	18,000	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2,4-DINITROPHENOL	mg/kg	1,800	6,400	190,000	1.9 U	1.8 U	4.2 U	2.1 U	6.3 U	6.1 U	4.1 U	12 U	13 U	15 U
2,4-DINITROTOLUENE	mg/kg	--	290	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2,6-DINITROTOLUENE	mg/kg	--	61	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2-CHLORONAPHTHALENE	mg/kg	--	190,000	190,000	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
2-CHLOROPHENOL	mg/kg	6,500	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2-METHYLPHENOL	mg/kg	4,600	160,000	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
2-NITROPHENOL	mg/kg	--	26,000	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
3,3'-DICHLOROBENZIDINE	mg/kg	5.7	200	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
4,6-DINITRO-2-METHYLPHENOL	mg/kg	--	260	190,000	0.97 U	0.95 U	2.1 U	1.1 U	3.2 U	3.1 U	2.1 U	6.2 U	6.5 U	7.9 U
4-BROMOPHENYL PHENYL ETHER	mg/kg	--	--	--	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
4-CHLORO-3-METHYLPHENOL	mg/kg	--	190,000	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
4-CHLOROPHENYL PHENYL ETHER	mg/kg	--	--	--	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
4-NITROPHENOL	mg/kg	--	26,000	190,000	0.97 U	0.95 U	2.1 U	1.1 U	3.2 U	3.1 U	2.1 U	6.2 U	6.5 U	7.9 U
BENZIDINE	mg/kg	--	0.4	190,000	3.8 U	3.7 U	8.4 U	4.2 U	13 U	12 U	8.3 U	25 U	25 U	31 U
BENZOIC ACID	mg/kg	--	190,000	190,000	0.97 U	0.95 U	2.1 U	**0.63 J**	3.2 U	3.1 U	2.1 U	6.2 U	6.5 U	7.9 U
BENZYL ALCOHOL	mg/kg	--	10,000	10,000	0.95 U	0.93 U	2.1 U	1 U	3.2 U	3.1 U	2.1 U	6.1 U	6.4 U	7.7 U
BIS(2-CHLOROETHOXY)METHANE	mg/kg	--	9,600	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
BIS(2-CHLOROETHYL)ETHER	mg/kg	3.3	6.7	7.6	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
BIS(2-ETHYLHEXYL) PHTHALATE	mg/kg	180	6,500	10,000	**0.46 J**	1.8 U	**0.48 J**	**0.68 J**	**0.96 J**	**4.2 J**	**0.66 J**	**3.2 J**	**1.6 J**	15 U
BUTYL BENZYL PHTHALATE	mg/kg	1,300	10,000	10,000	0.95 U	0.93 U	2.1 U	1 U	3.2 U	3.1 U	2.1 U	6.1 U	6.4 U	7.7 U
DIBENZOFURAN	mg/kg	--	3,200	190,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	**0.51 J**	1.3 U	1.5 U
DIETHYL PHTHALATE	mg/kg	730,000	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U

TABLE 5-8. SEMIVOLATILE ORGANIC COMPOUND CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards[a]	PA Non-Residential Standards[b] (-0 to -2 ft)	PA Non-Residential Standards[b] (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
DIMETHYL PHTHALATE	mg/kg	--	--	--	0.38 U	0.37 U	0.84 U	0.42 U	1.3 U	1.2 U	0.83 U	2.5 U	2.5 U	3.1 U
DI-N-BUTYL PHTHALATE	mg/kg	91,000	10,000	10,000	1.9 U	1.8 U	4.2 U	2.1 U	6.3 U	6.1 U	4.1 U	12 U	13 U	15 U
DI-N-OCTYL PHTHALATE	mg/kg	9,100	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
HEXACHLOROBENZENE	mg/kg	2.3	57	190,000	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
HEXACHLOROBUTADIENE	mg/kg	47	1,200	10,000	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
HEXACHLOROCYCLOPENTADIENE	mg/kg	7,800	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
HEXACHLOROETHANE	mg/kg	91	230	270	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
ISOPHORONE	mg/kg	2,700	10,000	10,000	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
METHYLPHENOL, 3 & 4	mg/kg	9,100	16,000	190,000	**0.11 J**	0.18 U	**0.14 J**	**0.22**	0.63 U	**0.34 J**	**0.14 J**	**1.8**	**0.86 J**	**0.66 J**
NITROBENZENE	mg/kg	2,600	55	63	0.38 U	0.37 U	0.84 U	0.42 U	1.3 U	1.2 U	0.82 U	2.4 U	2.5 U	3.1 U
N-NITROSODIMETHYLAMINE	mg/kg	--	0.16	0.18	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
N-NITROSODI-N-PROPYLAMINE	mg/kg	0.36	1.1	1.3	0.038 U	0.037 U	0.084 U	0.042 U	0.13 U	0.12 U	0.083 U	0.25 U	0.25 U	0.31 U
N-NITROSODIPHENYLAMINE	mg/kg	520	860	990	0.19 U	0.18 U	0.42 U	0.21 U	0.63 U	0.61 U	0.41 U	1.2 U	1.3 U	1.5 U
PENTACHLOROPHENOL	mg/kg	4.4	230	190,000	0.97 U	0.95 U	2.1 U	1.1 U	3.2 U	3.1 U	2.1 U	6.2 U	6.5 U	7.9 U
PHENOL	mg/kg	270,000	16,000	18,000	**0.091 J**	0.18 U	0.42 U	**0.09 J**	0.63 U	**0.2 J**	0.41 U	1.2 U	1.3 U	1.5 U

Notes:
Sources: (a) N.J.A.C. 7:26D.
(b) 25 Pa. Code § 250 Table 3a and 4a.
Bold values represent detected concentrations. RL is reported for non-detected constituents.
-- = No value available.
FD = Field duplicate.
ft = Foot (feet).
mg/kg = Milligram(s) per kilogram.
RL = Laboratory reporting limit.
J = Compound was detected, but below the reporting limit (value is estimated).
U = Compound was analyzed, but not detected.

TABLE 5-9. VOLATILE ORGANIC COMPOUND CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A													Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards[a]	PA Non-Residential Standards[b] (-0 to -2 ft)	PA Non-Residential Standards[b] (-2 to -15 ft)	PNYRB24-A-01-SED	PNYRB24-A-02-SED	PNYRB24-A-03-SED	PNYRB24-A-04-SED	PNYRB24-A-05-SED	PNYRB24-A-06-SED	PNYRB24-A-07-SED	PNYRB24-A-07-SED-FD	PNYRB24-A-08-SED	PNYRB24-A-09-SED	PNYRB24-A-10-SED	PNYRB24-A-11-SED	PNYRB24-A-12-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
1,1,1-TRICHLOROETHANE	mg/kg	--	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,1,2,2-TETRACHLOROETHANE	mg/kg	18	38	44	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,1,2-TRICHLOROETHANE	mg/kg	64	16	18	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,1-DICHLOROETHANE	mg/kg	640	1,400	1,600	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,1-DICHLOROETHENE	mg/kg	180	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,2,4-TRICHLOROBENZENE	mg/kg	13,000	160	190	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,2-DICHLOROBENZENE	mg/kg	110,000	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,2-DICHLOROETHANE	mg/kg	30	85	98	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,2-DICHLOROPROPANE	mg/kg	98	0.6	0.69	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,3-DICHLOROBENZENE	mg/kg	110,000	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
1,4-DICHLOROBENZENE	mg/kg	13,000	200	230	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
2-BUTANONE (MEK)	mg/kg	780,000	10,000	10,000	0.052 U	0.054 U	0.051 U	0.052 U	0.055 U	**0.032 J**	**0.044 J**	0.052 U	**0.025 J**	**0.03 J**	**0.057 J**	**0.044 J**	**0.075**	**0.19**	**0.16**	**0.12**
2-CHLOROETHYL VINYL ETHER	mg/kg	--	--	--	0.021 U	0.022 U	0.02 U	0.021 U	0.022 U	0.021 U	0.021 U	0.021 U	0.021 U	0.02 U	0.027 U	0.027 U	0.025 U	0.025 U	0.025 U	0.031 U
ACROLEIN	mg/kg	--	1.6	1.8	0.21 U	0.22 U	0.2 U	0.21 U	0.22 U	0.21 U	0.21 U	0.21 U	0.21 U	0.2 U	0.27 U	0.27 U	0.25 U	0.25 U	0.25 U	0.31 U
ACRYLONITRILE	mg/kg	--	33	37	0.21 U	0.22 U	0.2 U	0.21 U	0.22 U	0.21 U	0.21 U	0.21 U	0.21 U	0.2 U	0.27 U	0.27 U	0.25 U	0.25 U	0.25 U	0.31 U
BENZENE	mg/kg	16	280	330	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
BROMOFORM	mg/kg	460	2,000	2,300	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
BROMOMETHANE	mg/kg	1,800	400	460	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
CARBON TETRACHLORIDE	mg/kg	40	370	430	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
CHLORODIBROMOMETHANE	mg/kg	43	1,100	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	**0.0056 JB**	0.01 U	0.01 U	**0.0058 JB**	0.01 U	**0.0082 JB**	**0.0068 JB**	**0.0064 JB**	0.012 U	0.013 U	0.015 U
CHLOROETHANE	mg/kg	--	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
CHLOROFORM	mg/kg	13,000	96	110	0.01 U	0.011 U	0.01 U	0.01 U	**0.02 B**	**0.019 B**	0.01 U	0.01 U	**0.019 B**	**0.018 B**	**0.026 B**	**0.021 B**	**0.023 B**	**0.026 B**	**0.024 B**	**0.032 B**
CHLOROMETHANE	mg/kg	--	1,200	1,400	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
CIS-1,3-DICHLOROPROPENE	mg/kg	--	--	--	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
DICHLOROBROMOMETHANE	mg/kg	59	60	69	0.01 U	0.011 U	0.01 U	0.01 U	**0.0067 JB**	**0.0059 JB**	0.01 U	0.01 U	**0.0055 JB**	**0.0069 JB**	**0.0084 JB**	**0.0064 JB**	**0.0064 JB**	0.012 U	0.013 U	0.015 U
DICHLORODIFLUOROMETHANE	mg/kg	260,000	8,000	9,100	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
ETHYLBENZENE	mg/kg	130,000	880	1,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
METHYLENE CHLORIDE	mg/kg	260	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
TETRACHLOROETHENE	mg/kg	1,700	3,200	3,600	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
TOLUENE	mg/kg	100,000	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
TRANS-1,2-DICHLOROETHENE	mg/kg	22,000	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
TRANS-1,3-DICHLOROPROPENE	mg/kg	--	--	--	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
TRICHLOROETHENE	mg/kg	79	160	180	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
TRICHLOROFLUOROMETHANE	mg/kg	390,000	10,000	10,000	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U
VINYL CHLORIDE	mg/kg	5	61	290	0.01 U	0.011 U	0.01 U	0.01 U	0.011 U	0.011 U	0.01 U	0.01 U	0.01 U	0.01 U	0.014 U	0.013 U	0.013 U	0.012 U	0.013 U	0.015 U

Notes:

Sources: (a) N.J.A.C. 7:26D.

(b) 25 Pa. Code § 250 Table 3a and 4a.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

-- = No value available.

FD = Field duplicate.

ft = Foot (feet).

mg/kg = Milligram(s) per kilogram.

RL = Laboratory reporting limit.

B = Compound was detected in the laboratory method blank.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 5-10. CHLORINATED PESTICIDE CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards[(a)]	PA Non-Residential Standards[(b)] (-0 to -2 ft)	PA Non-Residential Standards[(b)] (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
4,4'-DDD	mg/kg	11	380	190,000	**0.0021**	**0.0017**	**0.0013**	**0.003**	**0.0044**	**0.032**	**0.1**	**0.11**	**0.0093**	**0.011**
4,4'-DDE	mg/kg	11	270	190,000	**0.015**	**0.0082**	**0.0072**	**0.016**	**0.043**	**0.097**	**0.018**	**0.063**	**0.02**	**0.042**
4,4'-DDT	mg/kg	10	270	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	**0.1**	**0.62**	**0.24**	**0.011 P**	0.0013 U
CHLORINATED PESTICIDES														
ALDRIN	mg/kg	0.21	5.4	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
ALPHA-BHC	mg/kg	0.41	14	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	**0.02**	**0.0061 J**	0.0051 U	0.0010 U	0.0013 U
BETA-BHC	mg/kg	1.4	51	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	**0.0035**	**0.0036 J**	0.0051 U	0.0010 U	0.0013 U
CHLORDANE (TECHNICAL)	mg/kg	--	--	--	0.0047 U	0.0047 U	0.01 U	0.01 U	0.011 U	0.01 U	0.1 U	0.051 U	0.01 U	0.013 U
CHLOROBENSIDE	mg/kg	--	--	--	0.0011 U	0.0011 U	0.0025 U	0.0025 U	0.0025 U	0.0025 U	0.024 U	0.012 U	0.0025 U	0.0031 U
DACHTAL (DCPA)	mg/kg	--	32,000	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
DELTA-BHC	mg/kg	--	--	--	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
DIELDRIN	mg/kg	0.16	5.7	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	**0.11**	**0.034**	0.0010 U	0.0013 U
ENDOSULFAN I	mg/kg	--	19,000	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	**0.0065 J**	**0.0054**	0.0010 U	0.0013 U
ENDOSULFAN II	mg/kg	--	19,000	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
ENDOSULFAN SULFATE	mg/kg	--	19,000	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
ENDRIN	mg/kg	270	960	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
ENDRIN ALDEHYDE	mg/kg	--	--	--	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	**0.0033 JP**	0.0010 U	0.0013 U
GAMMA-BHC	mg/kg	2.8	83	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	**0.0072**	0.01 U	0.0051 U	0.0010 U	0.0013 U
HEPTACHLOR	mg/kg	0.81	20	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	**0.0019 J**	0.0010 U	0.0013 U
HEPTACHLOR EPOXIDE	mg/kg	0.4	10	190,000	**0.0024**	**0.0014**	**0.0013**	**0.0026**	0.0011 U	0.0010 U	**0.0033 J**	**0.01**	**0.003**	**0.0056**
METHOXYCHLOR	mg/kg	4,600	16,000	190,000	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
MIREX	mg/kg	--	--	--	0.00047 U	0.00047 U	0.0010 U	0.0010 U	0.0011 U	0.0010 U	0.01 U	0.0051 U	0.0010 U	0.0013 U
TOXAPHENE	mg/kg	2.3	83	190,000	0.019 U	0.019 U	0.042 U	0.042 U	0.042 U	0.042 U	0.41 U	0.21 U	0.042 U	0.052 U

Notes:

Sources: (a) N.J.A.C. 7:26D.

(b) 25 Pa. Code § 250 Table 3a and 4a.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

-- = No value available**.**

FD = Field duplicate.

ft = Foot (feet).

mg/kg = Milligram(s) per kilogram.

RL = Laboratory reporting limit.

J = Compound was detected, but below the reporting limit (value is estimated).

P = The percent difference between the original and confirmation analysis is greater than 40%.

U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 5-11. TOTAL PETROLEUM HYDROCARBON CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

		Dredging Area A													Dredging Area B		
ANALYTE	UNITS	PNYRB24-A-01-SED	PNYRB24-A-02-SED	PNYRB24-A-03-SED	PNYRB24-A-04-SED	PNYRB24-A-05-SED	PNYRB24-A-06-SED	PNYRB24-A-07-SED	PNYRB24-A-07-SED-FD	PNYRB24-A-08-SED	PNYRB24-A-09-SED	PNYRB24-A-10-SED	PNYRB24-A-11-SED	PNYRB24-A-12-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
TPH-GRO (C6-C10)	mg/kg	3.5 U	3.4 U	**1.4 J**	**6.5**	14 U	13 U	**26**	**76**	32 U	**6.2**	18 U	17 U	16 U	**15 J**	**14 J**	**12 J**
TPH-DRO (C10-C20)	mg/kg	27 U	26 U	27 U	25 U	**16 J**	25 U	**900**	**120**	**15 J**	**14 J**	33 U	31 U	31 U	**1000**	**1000**	**210**
TPH-ORO (C20-C34)	mg/kg	**29**	**110**	**15 J**	**27**	**99**	**38**	**1200**	**180**	**92**	**33**	33 U	31 U	31 U	**1500**	**2400**	**370**

Notes:

Bold values represent detected concentrations; RL is reported for non-detected constituents.

FD = Field duplicate

mg/kg = milligram(s) per kilogram

RL = laboratory reporting limit

DRO = Diesel range organics

GRO = Gasoline range organics

ORO = Oil range organics

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected

This page left intentionally blank

TABLE 5-12. RADIOACTIVITY CONCENTRATIONS (pCi/g) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

		Dredging Area A							Dredging Area B		
ANALYTE	UNITS	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
GROSS ALPHA	pCi/g	**25.6**	**17.3**	**7.2**	**12.3**	8.17 U	**15.4**	10.5 U	8.7 U	**8.9**	6.84 U
GROSS BETA	pCi/g	**21.1**	**22.4**	**7.06**	**8.05**	**8.11**	**14.5**	**10.8**	4.42 U	**5.37**	**5.57**

Notes:

Bold values represent detected concentrations. RL is reported for non-detected constituents.

FD = Field duplicate

pCi/g = picocuries per gram

RL = laboratory reporting limit

U = Compound was analyzed, but not detected

This page left intentionally blank

TABLE 5-13. PER- AND POLYFLUOROALKYL SUBSTANCES (PFAS) CONCENTRATIONS (MG/KG) IN SEDIMENT

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

					Dredging Area A							Dredging Area B		
ANALYTE	UNITS	NJ Non-Residential Standards[(a)]	PA Non-Residential Standards[(b)] (-0 to -2 ft)	PA Non-Residential Standards[(b)] (-2 to -15 ft)	PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
HEXAFLUOROPROPYLENE OXIDE DIMER ACID (HFPO-DA)	mg/kg	--	--	--	0.00022 U	0.00022 U	0.00024 U	0.00021 U	0.00026 U	0.00027 U	0.00028 U	0.00036 U	0.00031 U	0.00029 U
PERFLUOROBUTANESULFONIC ACID (PFBS)	mg/kg	--	960	10,000	0.00022 U	0.00022 U	0.00024 U	0.00021 U	0.00026 U	0.00027 U	0.00028 U	0.00036 U	0.00031 U	0.00029 U
PERFLUOROHEXANESULFONIC ACID (PFHXS)	mg/kg	--	--	--	0.00022 U	0.00022 U	**0.0014**	**0.000074 J**	**0.000097 J**	**0.00018 J**	**0.00010 J**	0.00036 U	0.00031 U	**0.00011 J**
PERFLUORONONANOIC ACID (PFNA)	mg/kg	0.67	--	--	**0.00069**	**0.00016 J**	**0.00024**	**0.00014 J**	**0.00015 J**	**0.00013 J**	**0.00011 J**	**0.00048**	**0.00011 J**	**0.00015 J**
PERFLUOROOCTANESULFONIC ACID (PFOS)	mg/kg	1.6	64	190,000	**0.0014**	**0.0007**	**0.017**	**0.0008**	**0.0017**	**0.00098**	**0.00092**	**0.0006**	**0.00093**	**0.0017**
PERFLUOROOCTANOIC ACID (PFOA)	mg/kg	1.8	64	190,000	**0.00031**	**0.00012 J**	**0.00047**	**0.00013 JI**	**0.00015 J**	**0.00020 JI**	**0.00015 J**	**0.00020 J**	0.00031 U	**0.00020 J**

Notes:
Sources: (a) N.J.A.C. 7:26D.
(b) 25 Pa. Code § 250 Table 3a and 4a.
Bold values represent detected concentrations. RL is reported for non-detected constituents.
RL is reported for non-detected constituents.
-- = No value available.
mg/kg = Milligram(s) per kilogram.
RL = Laboratory reporting limit.
I = Value is EMPC (estimated maximum possible concentration).
J = compound was detected, but below the reporting limit (value is estimated)
U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 5-14. TOXICITY CHARACTERISTIC LEACHING PROCEDURE (TCLP) RESULTS (MG/L)

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

ANALYTE	UNITS	RL	TCLP Screening Value*	Dredging Area A							Dredging Area B		
				PNYRB24-A-0102-SED	PNYRB24-A-0304-SED	PNYRB24-A-0506-SED	PNYRB24-A-0708-SED	PNYRB24-A-0910-SED	PNYRB24-A-0910-SED-FD	PNYRB24-A-1112-SED	PNYRB24-B-2-SED	PNYRB24-B-4-SED	PNYRB24-B-5-SED
METALS													
ARSENIC	mg/L	0.05	5	**0.045 J**	**0.035 J**	**0.044 JB**	**0.046 JB**	**0.045 JB**	**0.038 JB**	**0.015 JB**	**0.19 B**	**0.029 JB**	**0.034 JB**
BARIUM	mg/L	0.5	100	**0.46 JB**	**0.59 B**	**0.43 JB**	**0.44 JB**	**0.53 B**	**0.48 JB**	**0.61 B**	**1.5 B**	**0.83 B**	**0.67 B**
CADMIUM	mg/L	0.05	1	**0.016 J**	**0.015 J**	**0.0062 J**	0.05 U	**0.013 J**	**0.0062 J**	0.05 U	**0.011 J**	**0.034 J**	**0.002 J**
CHROMIUM	mg/L	0.05	5	**0.0034 JB**	**0.0033 JB**	**0.0065 JB**	**0.01 JB**	**0.01 JB**	**0.015 JB**	**0.0092 JB**	**0.016 JB**	**0.0083 JB**	**0.0037 JB**
LEAD	mg/L	0.05	5	**0.026 JB**	**0.033 JB**	**0.016 JB**	**0.0072 JB**	**0.025 JB**	**0.014 JB**	**0.0038 JB**	**0.77 B**	**0.75**	**0.022 JB**
MERCURY	mg/L	0.002	0.2	0.002 U	0.002 U	0.002 U	0.002 U	0.002 U	0.002 U	0.002 U	0.002 U	0.002 U	0.002 U
SELENIUM	mg/L	0.05	1	0.05 U	**0.0077 J**	0.25 U	0.25 U	0.25 U	0.25 U	0.05 U	0.05 U	0.05 U	0.05 U
SILVER	mg/L	0.05	5	0.05 U	0.05 U	**0.0018 J**	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
CHLORINATED PESTICIDES													
CHLORDANE (TECHNICAL)	mg/L	0.005	--	0.005 U	0.005 U	0.005 U	0.005 U	0.005 U	0.005 U	0.005 U	0.005 U	0.005 U	0.005 U
ENDRIN	mg/L	0.0005	0.02	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U
GAMMA-BHC (LINDANE)	mg/L	0.0005	0.4	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U
HEPTACHLOR	mg/L	0.0005	0.008	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U
HEPTACHLOR EPOXIDE	mg/L	0.0005	0.008	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U
METHOXYCHLOR	mg/L	0.0005	10	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U	0.0005 U
TOXAPHENE	mg/L	0.04	0.5	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U
HERBICIDES													
2,4-D	mg/L	0.004	10	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U	0.04 U
2,4,5-TP (SILVEX)	mg/L	0.001	1	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U
SEMIVOLATILE ORGANIC COMPOUNDS (SVOCs)													
1,4-DICHLOROBENZENE	mg/L	0.05	7.5	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
2,4,5-TRICHLOROPHENOL	mg/L	0.05	400	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
2,4,6-TRICHLOROPHENOL	mg/L	0.05	2	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
2,4-DINITROTOLUENE	mg/L	0.05	0.13	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
2-METHYLPHENOL	mg/L	0.05	200	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
HEXACHLOROBENZENE	mg/L	0.05	0.13	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
HEXACHLOROBUTADIENE	mg/L	0.05	0.5	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
HEXACHLOROETHANE	mg/L	0.05	3	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
METHYLPHENOL, 3 & 4	mg/L	0.05	200	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
NITROBENZENE	mg/L	0.05	2	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
PENTACHLOROPHENOL	mg/L	0.25	100	0.25 U	0.25 U	0.25 U	0.25 U	0.25 U	0.25 U	0.25 U	0.25 U	0.25 U	0.25 U
PYRIDINE	mg/L	0.1	5	0.1 U	0.1 U	0.1 U	0.1 U	0.1 U	0.1 U	0.1 U	0.1 U	0.1 U	0.1 U
VOLATILE ORGANIC COMPOUNDS (VOCs)													
1,1-DICHLOROETHENE	mg/L	0.2	0.7	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
1,2-DICHLOROETHANE	mg/L	0.2	0.5	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
2-BUTANONE (MEK)	mg/L	1	200	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
BENZENE	mg/L	0.2	0.5	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
CARBON TETRACHLORIDE	mg/L	0.2	0.5	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
CHLOROBENZENE	mg/L	0.2	100	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
CHLOROFORM	mg/L	0.2	6	**0.042 JB**	**0.054 JB**	**0.077 JB**	**0.066 JB**	**0.071 JB**	**0.04 JB**	**0.038 JB**	**0.045 JB**	**0.036 JB**	**0.032 JB**
TETRACHLOROETHENE	mg/L	0.2	0.7	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
TRICHLOROETHENE	mg/L	0.2	0.5	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U
VINYL CHLORIDE	mg/L	0.2	0.2	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U

Notes:

**Source*: 40 CFR 261.24

Bold values represent detected concentrations. RL is reported for non-detected constituents.

mg/L = Milligram(s) per liter

RL = Average reporting limit

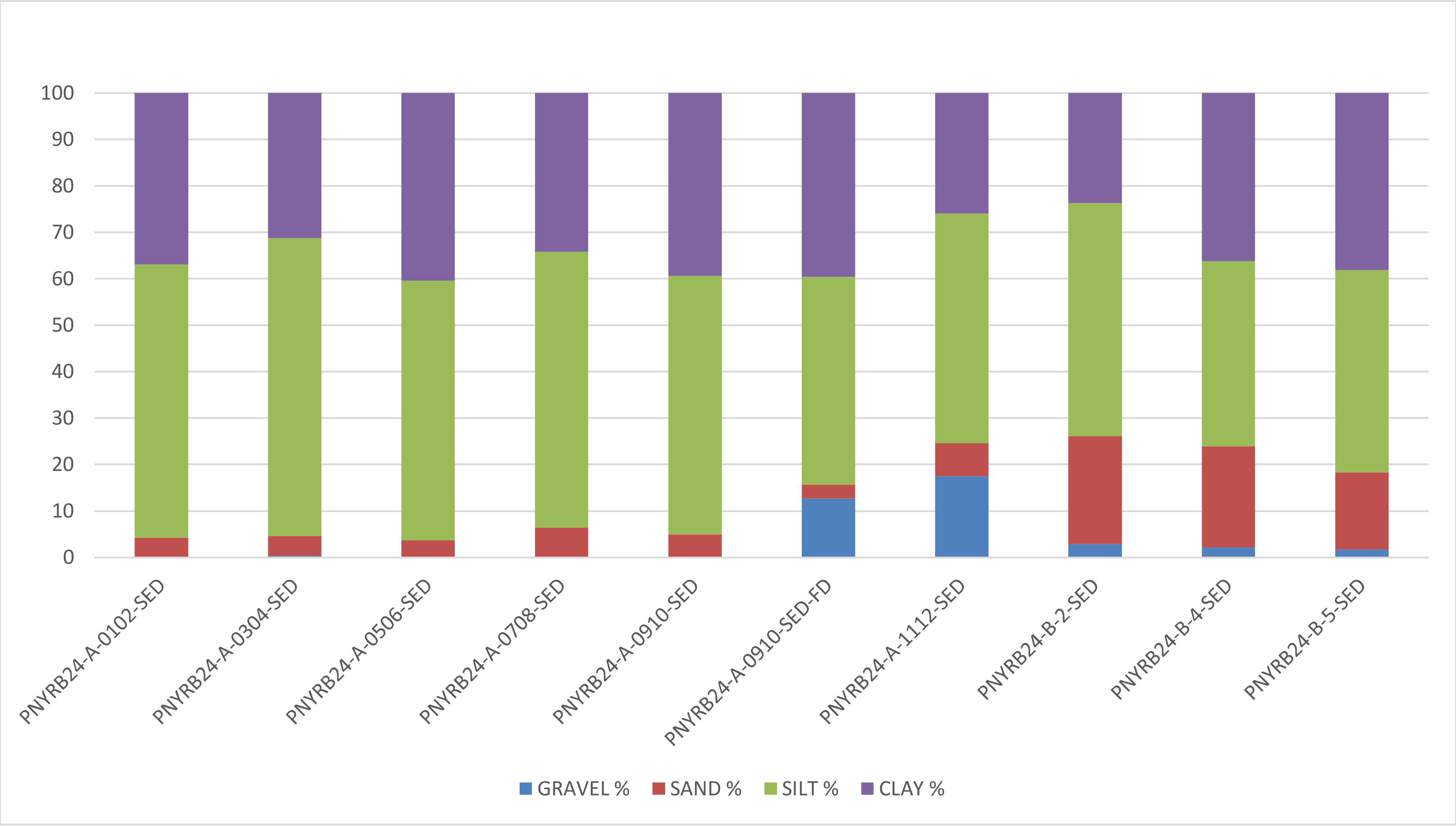
B = Compound was detected in the laboratory method blank

J = Compound was detected, but below the reporting limit (value is estimated)

U = Compound was analyzed, but not detected

This page left intentionally blank

Figure 5-1. Grain Size Distribution for Sediment Samples
Reserve Basin Maintenance Dredging, Philadelphia Navy Yard, Philadelphia, Pennsylvania (December 2024)



This page left intentionally blank

6. SITE WATER AND MODIFIED ELUTRIATE TESTING

Modified elutriates were prepared using sediment composites and site water from the Reserve Basin maintenance dredging area. Concentrations of metals and organic constituents in the full strength modified elutriates were compared to the Commonwealth of Pennsylvania maximum/continuous surface water standards (Sections 4.3.1).

6.1 MODIFIED ELUTRIATES – COMPARISON TO SURFACE WATER STANDARDS

The composite sediment samples from the Reserve Basin maintenance dredging were used in the modified elutriate testing. One site water sample was collected from PNYRB24-A-11 for chemical analysis and combined with the sediment samples to create the modified elutriates. Site water collection methods, holding times, and preservation techniques are described in Chapter 2. Modified elutriate samples that were created and tested for the Reserve Basin maintenance dredging are summarized in Table 1-2. Methodology for creating the modified elutriates is provided in Chapter 3.

Results of the modified elutriate analyses include analyte concentrations in both the total and dissolved fractions. Total concentrations of organics were compared to applicable water quality standards. Either dissolved or total concentrations of metals were compared to applicable water quality standards based on the fraction specified for each metal in the Commonwealth of Pennsylvania standards.

Results of the site water and modified elutriate chemical analyses are presented in Tables 6-1 through 6-11. Definitions of inorganic, organic, and dioxin data qualifiers are presented in the data tables. The TSS concentrations for each sample are provided on the data tables; TSS in the site water was 2.5 mg/L and TSS concentrations for the modified elutriate samples ranged from 34 to 1,400 mg/L.

Copies of final summary data sheets (Form I's) and analytical narratives that include an evaluation of laboratory QA/QC results are provided in Attachment III (site water) and Attachment IV (modified elutriates).

6.1.1 General Chemistry

Results of the general chemistry analyses for site water and modified elutriates for the Reserve Basin are presented in Table 6-1. Total cyanide was the only constituent that has Commonwealth of Pennsylvania surface water standards.

6.1.1.1 Site Water

Total cyanide and sulfide were not detected in the total fraction of the site water sample. The concentration of TOC in the site water was 2.9 mg/L) and the pH was 7.6.

6.1.1.2 Modified Elutriates

Sulfide was not detected in any of the modified elutriate samples. Two samples (A-0708 and B-2) had total cyanide concentrations (8.9 and 14 µg/L, respectively) that exceeded the Pennsylvania continuous criterion (5.2 µg/L) but did not exceed the Commonwealth of Pennsylvania maximum criterion (22 µg/L). TOC in the elutriates ranged from 5.5 to 15 mg/L and pH ranged from 6.7 to 7.5.

6.1.2 Metals

Results of the metals analyses for the site water and modified elutriates are presented in Table 6-2. The concentrations of metals detected in the dissolved and total fractions were compared to the Commonwealth of Pennsylvania maximum/continuous surface water standards. The following metals have water quality criteria that are based on dissolved fractions: cadmium, copper, lead, mercury, nickel, selenium, silver, and zinc.

6.1.2.1 Site Water

Dissolved Fraction

Fourteen of the tested metals (aluminum, antimony, arsenic, barium, chromium, cobalt, copper, iron, lead, manganese, nickel, selenium, vanadium, and zinc) were detected in the dissolved fraction of either the parent or field duplicate site water sample (Table 6-2). None of the detected or estimated concentrations exceeded Commonwealth of Pennsylvania maximum/continuous surface water standards.

Total Fraction

Thirteen of the tested metals (aluminum, antimony, arsenic, barium, cobalt, copper, iron, lead, manganese, nickel, thallium, vanadium, and zinc) were detected in the total fraction of either the parent or field duplicate site water sample (Table 6-2). None of the detected or estimated concentrations exceeded Commonwealth of Pennsylvania maximum/continuous surface water standards.

6.1.2.2 Modified Elutriates

Dissolved Fraction

No metals were detected in the dissolved fraction of the elutriates at a concentration that exceeded applicable Commonwealth of Pennsylvania maximum/continuous surface water standards (Table 6-2).

Total Fraction

Two metals (aluminum and cobalt) were detected in the total fraction of the elutriate at a concentration that exceeded applicable water quality criteria. Aluminum concentrations exceeded the Commonwealth of Pennsylvania maximum standard (750 µg/L) in each of the total fraction elutriates by a factor ranging from 3.3 to 62.7. Cobalt exceeded the Commonwealth of Pennsylvania continuous criterion of 19 µg/L in six of the nine total fraction elutriates but did not exceed the maximum criterion of 95 µg/L. Cobalt in the total fraction exceeded the continuous criterion by factors ranging from 1.2 to 2.9 (Table 6-2).

6.1.3 Polycyclic Aromatic Hydrocarbons (PAHs)

The results of the PAH analysis for the site water and modified elutriates are presented in Table 6-3. Concentrations of individual PAHs detected in the total fraction of the modified elutriates were compared to the Commonwealth of Pennsylvania continuous/maximum surface water standards.

6.1.3.1 Site Water

Two individual PAHs (acenaphthene and naphthalene) were detected in the total fractions of the site water, at concentrations below Commonwealth of Pennsylvania continuous/maximum surface water criteria (Table 6-3). The total PAH concentration (ND=RL) was 3.2 µg/L in the total fraction of site water and 3.96 µg/L in the field duplicate sample (Table 6-3).

6.1.3.2 Modified Elutriates

Total PAH concentrations (ND=RL) in the modified elutriates ranged from 3.07 to 118 µg/L for the total fraction (Tables 6-3).

Three individual PAHs [acenaphthene, phenanthrene, and benzo(a)anthracene] were detected at concentrations that exceeded the Commonwealth of Pennsylvania continuous standard in the total fractions of the modified elutriates. Acenaphthene concentrations in the total fraction of the modified elutriate at sample location B-5 (28 µg/L) exceeded the Pennsylvania continuous standard (17 µg/L) but did not exceed the Commonwealth of Pennsylvania maximum standard (83 µg/L) (Table 6-3). The phenanthrene concentration (10 µg/L) in the same sample (B-5) exceeded both Commonwealth of Pennsylvania continuous (1 µg/L) and maximum (5 µg/L) standards (Table 6-3). Benzo(a)anthracene exceeded the Commonwealth of Pennsylvania continuous standard (0.1 µg/L) in six samples of the total fraction of the modified elutriate by factors ranging from 1.1 to 3.5 (Table 6-3).

6.1.4 PCB Congeners

The results of the PCB congener and homolog analyses for the site water and modified elutriates are presented in Table 6-4. Concentrations of total PCB congeners and total PCB homologs for the total fraction of the modified elutriates were compared to the Commonwealth of Pennsylvania continuous surface water standard.

6.1.4.1 Site Water

Seventeen individual PCB congeners were detected in the total fractions of either the site water and/or the field duplicate sample, each at concentrations estimated below the laboratory RL (Table 6-4). Total PCB concentrations (ND=0, sum of congeners) in the total fraction of the site water ranged from 0.5 to 0.63 nanograms per liter (ng/L) and did not exceed the Commonwealth of Pennsylvania continuous standard of 14 ng/L. Total PCBs calculated based on the sum of homologs had equivalent results, also ranging from 0.5 to 0.63 ng/L.

6.1.4.2 Modified Elutriates

The total PCB concentrations (ND=0, sum of congeners) in the total fractions of the modified elutriates ranged from 199 to 1,378 ng/L and exceeded the Commonwealth of Pennsylvania continuous standard (14 ng/L) by factors ranging from 14 to 98 (Table 6-4). Total PCBs calculated based on the sum of homologs had comparable results, ranging from 200 to 1,400 ng/L.

6.1.5 PCB Aroclors

The results of PCB Aroclors in site water and modified elutriates are presented in Table 6-5.

6.1.5.1 Site Water

PCB Aroclors were not detected in the site water sample (Table 6-5).

6.1.5.2 Modified Elutriates

PCB -1254) was detected in the total fraction of seven of the nine modified elutriates, and PCB-1248 and PCB-1260 were both detected in the total fraction of one modified elutriate sample (A-0910). Total PCB Aroclor concentrations (ND=0) ranged from 0 to 0.29 µg/L (Table 6-5) and exceeded the Commonwealth of Pennsylvania continuous standard (0.014 µg/L) in seven of the nine elutriates by factors ranging from 3 to 21.

6.1.6 Chlorinated Pesticides

The results of the chlorinated pesticide analyses for the site water and modified elutriates are presented in Table 6-6. Concentrations of chlorinated pesticides detected in the total fraction of the modified elutriates were compared to the Commonwealth of Pennsylvania maximum/continuous surface water standards.

6.1.6.1 Site Water

Two chlorinated pesticides (beta-BHC and endosulfan I) were detected in the total faction of the modified elutriates; the concentrations of endosulfan I was below the Commonwealth of Pennsylvania maximum/continuous surface water standard. There are no Commonwealth of Pennsylvania surface water standards for beta-BHC (Table 6-6).

6.1.6.2 Modified Elutriates

Three of the 22 tested chlorinated pesticides (4,4’-DDD, 4,4’-DDE, and heptachlor epoxide) were detected in the total fractions of the modified elutriates (Table 6-6).

- ***4,4’-DDD:*** In three of the nine elutriates, concentrations of 4’4’-DDD in the total fraction of the modified elutriates exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) by factors ranging from 4.2 to 19 (Table 6-6). Concentrations did not exceed the Pennsylvania maximum standard (1.1 µg/L) in any sample.

- ***4,4’-DDE:*** Concentrations of 4,4’-DDE in the total fraction of the modified elutriates exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in seven of the nine elutriates by factors ranging from 2.4 to 24 (Table 6-6).

- ***Heptachlor Epoxide:*** Concentrations of heptachlor epoxide were detected in the total fraction of the modified in four of nine samples; none of the detected concentrations in the elutriates exceeded the Commonwealth of Pennsylvania surface water quality standards.

6.1.7 Semivolatile Organic Compounds (SVOCs)

The results of the SVOC analyses for the site water and modified elutriates are presented in Table 6-7. Concentrations of SVOCs detected in the total fraction of the modified elutriates were compared to Commonwealth of Pennsylvania maximum/continuous surface water standards.

6.1.7.1 Site Water

No SVOCs were detected in the total fraction of the site water. Diethyl phthalate was detected in the site water field duplicate in the dissolved fraction of the sample.

6.1.7.2 Modified Elutriates

Ten of the 46 tested SVOCs (2-nitrophenol, 4-nitrophenol, bis[2-ethylhexyl]phthalate, dibenzofuran, diethyl phthalate, dimethyl phthalate, di-n-octyl phthalate, hexachloroethane, 3&4 methylphenol, and phenol) were detected in either the dissolved or total fractions of the modified elutriates, some of which were estimated at concentrations below the laboratory RL. None of the detected SVOC concentrations in the total fraction exceeded Commonwealth of Pennsylvania maximum/continuous surface water standards.

6.1.8 Volatile Organic Compounds (VOCs)

The results of the VOC analyses for the site water and modified elutriates are presented in Table 6-8. Concentrations of VOCs detected in the total fraction of the modified elutriates were compared to Commonwealth of Pennsylvania maximum/continuous surface water standards.

6.1.8.1 Site Water

No VOCs were detected in the total fraction of the site water. One VOC (methylene chloride) was detected in the dissolved fraction of the site water.

6.1.8.2 Modified Elutriates

Two of the 35 tested VOCs (2-butanone and methylene chloride) were detected in both the dissolved and total fractions of the modified elutriates. None of the detected concentrations for the total fraction exceeded Commonwealth of Pennsylvania maximum/continuous surface water standards.

6.1.9 Radioactivity

The results of the radioactivity analyses for the site water and modified elutriates are presented in Table 6-9. Concentrations of radioactivity detected in the dissolved and total fraction of the modified elutriates were compared to the USEPA Drinking Water Standards-Radionuclides Rule, 66 Federal Register 76708, 7 December 2000; Volume 65, No. 236 and the DRBC stream quality objectives as specified in DRBC Water Quality Regulations, 18 CFR Part 410, 4 December 2013.

6.1.9.1 Site Water

Gross beta was detected in the total fractions of the site water; however, the detected concentration did not exceed USEPA radioactivity drinking water standards (50 picocuries per liter [pCi/L]) or the DRBC water quality criterion (1,000 pCi/L). Gross alpha was not detected in the site water sample (Table 6-9).

6.1.9.2 Modified Elutriates

Gross alpha was detected in the total fraction of eight of the nine modified elutriates and in the dissolved fraction of three of the nine elutriates the modified elutriates. Each of the eight total fraction concentrations exceeded the DRBC water quality criteria (3 pCi/L) and seven of the eight total fraction concentrations also exceeded the USEPA drinking water quality standard (15 pCi/L). Two of the three dissolved fraction concentrations for gross alpha exceeded the DRBC criterion, but did not exceed the USEPA drinking water standard. Gross beta was detected in the total and dissolved fraction of each modified elutriate, but the concentrations did not exceed the USEPA drinking water quality standard (50 pCi/L) or the DRBC water quality criterion (1,000 pCi/L) (Table 6-9).

6.1.10 Total Petroleum Hydrocarbons

The results of the TPH analyses for the site water and modified elutriates are presented in Table 6-9. TPH was only analyzed in the total fraction of the elutriates. There are no drinking water standards for TPH.

6.1.10.1 Site Water

TPH compounds were not detected in the total fraction of the site water (Table 6-9).

6.1.10.2 Modified Elutriates

TPH-DRO (C10-C20) was detected in each of the total fractions of the modified elutriates at concentrations ranging from an estimated value of 0.072 to 2.3 mg/L. TPH-ORO (C20-C34) was detected in each of the total fractions of the modified elutriates at concentrations ranging from 0.43 to 3.1 mg/L. TPH-GRO was detected in one sample (B-4) with a concentration estimated below the laboratory RL (Table 6-9).

6.1.11 Dioxins/Furans

Results of the dioxin and furan compounds in site water and modified elutriates are provided in Table 6-10. There are no Commonwealth of Pennsylvania surface water quality standards for comparison.

6.1.11.1 Site Water

None of the individual dioxin or furan congeners were detected in site water (Table 6-10).

6.1.11.2 Modified Elutriates

Sixteen of the 17 dioxin/furan congeners were detected in either the dissolved or total fractions of the modified elutriates. The dioxin TEQ (ND = 0) ranged from 0.479 to 279 picograms per liter (pg/L) in the dissolved fraction and from 1.72 to 44.9 pg/L in the total fraction for the modified elutriates (Table 6-10).

6.1.12 PFAS

Analytical results of six PFAS are provided in Table 6-11. Concentrations of PFAS were compared to the Commonwealth of Pennsylvania MCLs for PFOA and PFOS in drinking water and to the USEPA (2024a) aquatic life criterion (acute values).

6.1.12.1 Site Water

Five of the six PFAS were detected in both the total and dissolved fractions of the site water sample. The detected concentrations of PFOS and PFOA in site water did not exceed the Pennsylvania MCL for drinking water (Table 6-11). In addition, none of the concentrations for the five detected PFAS exceeded the USEPA aquatic life criteria.

6.1.12.2 Modified Elutriates

Each of the six tested PFAS was detected in both the total and dissolved fractions for the modified elutriates. PFOS concentrations exceeded the Pennsylvania MCL of 18 ng/L in both the dissolved and total modified elutriate fractions in two samples (A-0102 and A-0506) by factors ranging from 1.1 to 1.7. PFOA concentrations also exceeded the Pennsylvania MCL of 14 ng/L in both the dissolved and total fractions of sample A-0102 by factors of 2.3 and 2.4, respectively (Table 6-11). None of the detected concentrations of PFAS exceeded the USEPA acute aquatic life criteria.

6.2 MODIFIED ELUTRIATE SUMMARY

Modified elutriates were prepared using sediment composites and site water from the Reserve Basin maintenance dredging area. Concentrations of metals and organic constituents in the full strength modified elutriates were compared to the Commonwealth of Pennsylvania maximum/continuous surface water quality standards.

Results of the modified elutriate analyses include analyte concentrations in both the total and dissolved fractions. Total concentrations of organics were compared to applicable water quality standards. Either dissolved or total concentrations of metals were compared to applicable water quality standards based on the fraction specified for each metal in the Commonwealth of Pennsylvania standards.

Results of the modified elutriate testing indicated:

- Aluminum exceeded the Commonwealth of Pennsylvania maximum standard (750 µg/L) in each of the total fraction elutriates. The Commonwealth of Pennsylvania maximum standard was exceeded by factors ranging from 3.3 to 62.7.

- Cobalt exceeded the Commonwealth of Pennsylvania continuous criterion (19 µg/L) in six of the nine total fraction elutriates but did not exceed the maximum criterion of 95 µg/L. Cobalt in the total fraction exceeded the continuous criterion by factors ranging from 1.2 to 2.9

- Three individual PAHs [acenaphthene, phenanthrene, and benzo(a)anthracene] were detected at concentrations that exceeded the Commonwealth of Pennsylvania continuous standard in the total fractions of the modified elutriate. Acenaphthene concentrations in the total fraction of the modified elutriate at sample location B-5 (28 µg/L) exceeded the Commonwealth of Pennsylvania continuous standard (17 µg/L) but did not exceed the Pennsylvania maximum standard (83 µg/L). The phenanthrene concentration (10 µg/L) in the same sample (B-5) exceeded both Commonwealth of Pennsylvania continuous (1 µg/L) and maximum (5 µg/L) standards. Benzo(a)anthracene exceeded the Commonwealth of Pennsylvania continuous standard (0.1 µg/L) in six samples of the total fraction of the modified elutriate by factors ranging from 1.1 to 3.5.

- Total PCB concentrations (ND=0, sum of congeners) in the total fractions of the modified elutriates ranged from 199 to 1,378 ng/L and exceeded the Commonwealth of Pennsylvania continuous standard (14 ng/L) by factors ranging from 14 to 98. Total PCBs calculated based on the sum of homologs had comparable results, ranging from 200 to 1,400 ng/L.

- Total PCB Aroclor concentrations (ND=0) ranged from 0 to 0.29 µg/L and exceeded the Commonwealth of Pennsylvania continuous standard (0.014 µg/L) in seven of the nine elutriates by factors ranging from 3 to 21.

- Two of the 22 tested chlorinated pesticides (4,4'-DDD and 4,4'-DDE) were detected in the total fraction of the modified elutriates at concentrations exceeding Commonwealth of Pennsylvania surface water quality standards.

    — ***4,4'-DDD:*** Concentrations of 4'4'-DDD in the total fraction of the modified elutriates exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in three of the nine samples by factors ranging from 4.2 to 19. Concentrations did not exceed the Pennsylvania maximum standard (1.1 µg/L) in any sample.

— ***4,4'-DDE:*** Concentrations of 4,4'-DDE in the total fraction of the modified elutriates exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in seven of the nine samples by factors ranging from 2.4 to 24. Concentrations did not exceed the Pennsylvania maximum standard (1.1 µg/L) in any sample.

- Gross alpha was detected in the total fraction of eight of the nine modified elutriates and in the dissolved fraction of three of the nine modified elutriates. Each of the eight total fraction concentrations exceeded the DRBC water quality criteria (3 pCi/L) and seven of the eight total fraction concentrations also exceeded the USEPA drinking water quality standard (15 pCi/L). Two of the three dissolved fraction concentrations for gross alpha exceeded the DRBC criterion, but did not exceed the USEPA drinking water standard. Gross beta was detected in the total and dissolved fraction of each modified elutriate, but the concentrations did not exceed the USEPA drinking water quality standard (50 pCi/L) or the DRBC water quality criterion (1,000 pCi/L)

- Each of the six tested PFAS was detected in both the total and dissolved fractions for the modified elutriates. PFOS concentrations exceeded the Pennsylvania MCL of 18 ng/L in both the dissolved and total modified elutriate fractions in two samples (A-0102 and A-0506) by factors ranging from 1.1 to 1.7. PFOA concentrations also exceeded the Pennsylvania MCL of 14 ng/L in both the dissolved and total fractions of sample A-0102 by factors of 2.3 and 2.4, respectively. None of the detected concentrations of PFAS exceeded the USEPA acute aquatic life criteria.

This page intentionally left blank

TABLE 6-1. GENERAL CHEMISTRY CONCENTRATIONS IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

				Site Water		Modified Elutriates								
		PA SW	PA SW	PNYRB24-WAT	PNYRB24-WAT-FD	PNYRB24-A-0102-MET	PNYRB24-A-0304-MET	PNYRB24-A-0506-MET	PNYRB24-A-0708-MET	PNYRB24-A-0910-MET	PNYRB24-A-1112-MET	PNYRB24-B-2-MET	PNYRB24-B-4-MET	PNYRB24-B-5-MET
ANALYTE	**UNITS**	**Maximum[b]**	**Continuous[b]**	**Total**	**Total**	**Total**	**Total**	**Total**	**Total**	**Total**	**Total**	**Total**	**Total**	**Total**
CYANIDE, TOTAL	µg/L	22	5.2	10 U	10 U	10 U	10 U	10 UH	**8.9 J**	10 U	10 UH	**14 H**	10 U	10 U
SULFIDE	mg/L	--	--	3 U	3 U	3 U	3 U	3 U	3 U	3 U	3 U	3 U	3 U	3 U
pH	SU	--	--	**7.6**	**7.6**	**7.4**	**7.2**	**7.1**	**7.2**	**7.2**	**7.3**	**6.7**	**7.4**	**7.5**
TOTAL ORGANIC CARBON	mg/L	--	--	**2.9**	**2.9**	**15**	**11**	**10**	**10**	**8.7**	**5.5**	**12**	**6.3**	**8.2**

Notes:

Sources: (a) 25 Pa. Code § 93.7.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

RL is reported for non-detected constituents.

Pennsylvania maximum standards are comparable to USEPA acute criteria and Pennsylvania continuous standards are comparable to USEPA chronic criteria.

-- = No value available.

RL = Laboratory reporting limit.

H = analyte analyzed outside of holding time.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

Analyte concentration exceeds Pennsylvania Surface Water Continuous Concentration Standards.

TABLE 6-2. METALS CONCENTRATIONS IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

				Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Maximum[a]	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
ALUMINUM	µg/L	750	--	**48**	**70**	**31**	**73**	**45**	**47000**	**31**	**24000**	**25**	**40000**	**52**	**46000**	**27**	**46000**	**42**	**13000**	**26**	**35000**	25 U	**7300**	25 U	**2500**
ANTIMONY	µg/L	1,100[2]	220[2]	**0.64 J**	**0.77 J**	**0.57 J**	**0.61 J**	**0.78 J**	**1.7**	**0.68 J**	**1.2**	**0.49 J**	**1.3**	**0.62 J**	**1.7**	**0.63 J**	**3**	**0.62 J**	**0.91 J**	**1.5**	**5.6**	**2.8**	**7.5**	**0.44 J**	**0.94 J**
ARSENIC	µg/L	340[2]	150[2]	**0.94 J**	**0.79 J**	**0.91 J**	**0.69 J**	**3.3**	**22**	**3.7**	**14**	**2.1**	**21**	**3.1**	**28**	**3**	**29**	**1.8 J**	**8.3**	**3.7**	**68**	**2.7**	**18**	**1.9 J**	**8.8**
BARIUM	µg/L	21,000[2]	4,100[2]	**36**	**37**	**36**	**36**	**110**	**500**	**93**	**310**	**150**	**600**	**110**	**580**	**120**	**630**	**110**	**260**	**53**	**460**	**97**	**200**	**170**	**210**
BERYLLIUM	µg/L	--	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	**3.3**	0.5 U	**1.7**	0.5 U	**3.1**	0.5 U	**3.7**	**0.26 J**	**4.2**	0.5 U	**1.1**	0.5 U	**3.2**	0.5 U	**0.84**	0.5 U	**0.18 J**
CADMIUM	µg/L	2.79[1]	0.31[1]	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	**3**	0.5 U	**1.4**	0.5 U	**3**	0.5 U	**3.7**	0.5 U	**4.7**	0.5 U	**0.82**	0.5 U	**7.2**	0.5 U	**1.8**	0.5 U	**0.4 J**
CHROMIUM	µg/L	--	--	**3.3**	2 U	**6.1**	2 U	2 U	**110**	2 U	**55**	2 U	**110**	**0.58 J**	**170**	2 U	**160**	2 U	**34**	**0.82 J**	**250**	2 U	**32**	2 U	**11**
COBALT	µg/L	95[2]	19[2]	**0.57**	**0.26 J**	**0.31 J**	0.5 U	**2.5**	**39**	**2.1**	**22**	**2.9**	**41**	**2.8**	**38**	**2.9**	**56**	**1.2**	**12**	**2.4**	**38**	**1.9**	**10**	**2.9**	**7.1**
COPPER	µg/L	18.5[1]	11.9[1]	**4.2**	**3.8**	**3.8**	**3.9**	**1.1**	**150**	**1.2**	**81**	**0.8 J**	**160**	**0.96 J**	**180**	**0.88 J**	**210**	**0.41 J**	**45**	**0.54 J**	**340**	**0.55 J**	**73**	1 U	**25**
IRON	µg/L	--	--	**77**	**130**	**65**	**130**	**56**	**49000**	**37 J**	**26000**	**270**	**54000**	**220**	**54000**	**1900**	**75000**	**32 J**	**16000**	**540**	**53000**	**870**	**16000**	**430**	**15000**
LEAD	µg/L	93[1]	3.6[1]	**0.23 J**	**0.4 J**	**0.2 J**	**0.42 J**	**0.67**	**210**	**0.5**	**110**	**0.31 J**	**210**	**0.56**	**240**	**0.22 J**	**270**	**0.21 J**	**68**	**0.41 J**	**540**	**1.2**	**1400**	0.5 U	**40**
MANGANESE	µg/L	--	--	**14**	**23**	**11**	**25**	**960**	**1500**	**1100**	**1600**	**2200**	**3500**	**2000**	**3200**	**2800**	**5900 B**	**2100**	**2500**	**1000**	**1600**	**2000**	**2400 B**	**1300**	**1400 B**
MERCURY	µg/L	1.4[1]	0.77[1]	0.2 U	0.2 U	0.2 U	0.2 U	0.2 U	**0.17 J**	0.2 U	**0.14 J**	0.2 U	**0.23**	0.2 U	**0.57**	0.2 U	**0.6**	0.2 U	**0.1 J**	0.2 U	**3.5**	0.2 U	**1**	0.2 U	**0.24**
NICKEL	µg/L	623[1]	69[1]	**15**	**1.4**	**5.6**	**1.6**	**4.1**	**68**	**3.1**	**35**	**2.7**	**62**	**2.5**	**67**	**2.6**	**84**	**1.6**	**18**	**3.5**	**80**	**2.5**	**16**	**3.4**	**9.6**
SELENIUM	µg/L	--	4.6[1]	**0.29 J**	1 U	1 U	1 U	**0.42 J**	**1.8**	**0.4 J**	**1.1**	1 U	**1.5**	1 U	**2**	**0.35 J**	**2.1**	1 U	**0.53 J**	1 U	**3.7**	1 U	**0.96 J**	1 U	**0.32 J**
SILVER	µg/L	5.74[1]	--	0.5 U	0.5 U	0.5 U	0.5 U	0.5 U	**3.1**	0.5 U	**1.1**	0.5 U	**2.4**	0.5 U	**4**	0.5 U	**2.2**	0.5 U	**0.59**	0.5 U	**8.4**	0.5 U	**0.75**	0.5 U	**0.36 J**
THALLIUM	µg/L	65[2]	13[2]	0.5 U	**0.18 J**	0.5 U	0.5 U	0.5 U	**0.34 J**	0.5 U	**0.19 J**	0.5 U	**0.33 J**	0.5 U	**0.36 J**	**0.3 J**	**0.48 J**	0.5 U	0.5 U	0.5 U	**0.55**	0.5 U	**0.17 J**	0.5 U	0.5 U
VANADIUM	µg/L	510[2]	100[2]	**0.83 J**	4 U	**0.83 J**	**0.86 J**	**1.1 J**	**75**	**1.1 J**	**36**	4 U	**64**	4 U	**83**	**0.89 J**	**100**	4 U	**23**	4 U	**89**	4 U	**19**	4 U	**8.2**
ZINC	µg/L	156[1]	157[1]	**4.5 J**	**6.6 J**	**4.5 J**	**6.3 J**	10 U	**710**	10 U	**360**	10 U	**710**	10 U	**810**	10 U	**1100**	10 U	**200**	10 U	**1700**	10 U	**400**	10 U	**110**

Notes:

Sources: (a) 25 Pa. Code § 93.7. (1) = Dissolved metal screening criterion.
(2) = Total recoverable metal screening criterion.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

RL is reported for non-detected constituents.

Pennsylvania maximum standards are comparable to USEPA acute criteria and PA continuous standards are comparable to USEPA chronic criteria.

Pennsylvania surface water screening standards for cadmium, copper, lead, mercury, nickel, silver, and zinc calculated using hardness-based equation. Site water hardness of 140 mg/L from PNYRB24-WAT was used in calculation.

-- = No value available. RL = Laboratory reporting limit

µg/L = Microgram(s) per liter. B = Compound was detected in the laboratory method blank.

mg/L = Milligram(s) per liter. J = Compound was detected, but below the reporting limit (value is estimated.

TSS = Total suspended solids. U = Compound was analyzed, but not detected.

Analyte concentration exceeds Pennsylvania Surface Water Continuous Concentration Standards.

Analyte concentration exceeds Pennsylvania Surface Water Maximum Concentration Standards.

This page left intentionally blank

TABLE 6-3. POLYCYCLIC AROMATIC HYDROCARBON CONCENTRATIONS (µg/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

				Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Maximum[a]	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
LOW MOLECULAR WEIGHT PAHs (LPAH)																									
1-METHYLNAPHTHALENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	**0.22**	**0.18 J**	**0.11 J**	**0.13 J**	0.21 U	**0.12 J**	0.22 U	**0.15 J**	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	**0.33**	0.22 U	**0.25**	**0.16 J**	4.3 U
2-METHYLNAPHTHALENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	**0.1 J**	**0.33**	**0.11 J**	**0.25**	0.21 U	**0.29**	0.22 U	**0.4**	0.41 U	**0.19 J**	0.19 U	**0.1 J**	0.21 U	**0.77**	0.22 U	**0.57**	**0.25**	**1.9 J**
ACENAPHTHENE	µg/L	83	17	0.22 U	**0.098 J**	0.22 U	0.22 U	**0.62**	**0.62**	**1.4**	**1.2**	**0.18 J**	**0.24**	**0.18 J**	**0.28**	**0.18 J**	**0.15 J**	**0.066 J**	0.22 U	**0.55**	**0.64**	**0.2 J**	**0.29**	**1.1**	**28**
ACENAPHTHYLENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	**0.081 J**	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	**0.18 J**	0.22 U	0.22 U	**0.09 J**	4.3 U
ANTHRACENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	**0.11 J**	**0.12 J**	**0.15 J**	**0.2**	0.21 U	**0.17 J**	0.22 U	**0.13 J**	0.41 U	**0.11 J**	0.19 U	**0.075 J**	0.21 U	**0.36**	0.22 U	**0.23**	**0.13 J**	**2.6 J**
FLUORENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	**0.36**	**0.26**	**0.51**	**0.37**	**0.089 J**	**0.16 J**	**0.099 J**	**0.25**	0.41 U	**0.13 J**	0.19 U	**0.087 J**	**0.32**	**0.61**	0.22 U	**0.26**	**0.95**	**15**
NAPHTHALENE	µg/L	140	43	0.22 U	**0.062 J**	**0.12 J**	0.22 U	0.2 U	**0.47**	**0.22**	**0.41**	0.21 U	**0.34**	0.22 U	**0.52**	0.41 U	**0.25**	0.19 U	**0.12 J**	**0.088 J**	**0.77**	0.22 U	**0.62**	**0.29**	**2.4 J**
PHENANTHRENE	µg/L	5	1	0.22 U	0.19 U	0.22 U	0.22 U	**0.33**	**0.44**	**0.37**	**0.42**	**0.26**	**0.59**	**0.32**	**0.57**	0.41 U	**0.46**	0.19 U	**0.27**	**0.76**	**1.1**	**0.19 J**	**0.68**	**1.4**	**10**
TOTAL LPAHs (ND=0)	µg/L	--	--	0	**0.16**	**0.12**	0	**1.74**	**2.5**	**2.87**	**2.98**	**0.529**	**1.91**	**0.599**	**2.3**	**0.18**	**1.29**	**0.066**	**0.652**	**1.72**	**4.76**	**0.39**	**2.9**	**4.37**	**59.9**
TOTAL LPAHs (ND=1/2RL)	µg/L	--	--	**0.88**	**0.73**	**0.89**	**0.88**	**1.94**	**2.5**	**2.97**	**3.08**	**1.05**	**2.02**	**1.15**	**2.4**	**1.62**	**1.49**	**0.731**	**0.982**	**2.14**	**4.76**	**1.05**	**3.01**	**4.37**	**64.2**
TOTAL LPAHs (ND=RL)	µg/L	--	--	**1.76**	**1.3**	**1.66**	**1.76**	**2.14**	**2.5**	**3.07**	**3.17**	**1.58**	**2.13**	**1.7**	**2.5**	**3.05**	**1.69**	**1.4**	**1.31**	**2.56**	**4.76**	**1.71**	**3.12**	**4.37**	**68.5**
HIGH MOLECULAR WEIGHT PAHs (HPAH)																									
BENZO(A)ANTHRACENE	µg/L	0.5	0.1	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	**0.15 J**	0.22 U	**0.11 J**	0.41 U	**0.11 J**	0.19 U	**0.13 J**	**0.1 J**	**0.35**	0.22 U	**0.16 J**	**0.19 J**	4.3 U
BENZO(A)PYRENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	**0.12 J**	**0.077 J**	**0.19 J**	0.22 U	**0.11 J**	**0.082 J**	4.3 U
BENZO(B)FLUORANTHENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	**0.2 J**	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	**0.22**	0.21 U	**0.27**	0.22 U	**0.17 J**	0.21 U	4.3 U
BENZO(GHI)PERYLENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	**0.13 J**	**0.1 J**	**0.2 J**	0.22 U	0.22 U	0.21 U	4.3 U
BENZO(K)FLUORANTHENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	**0.12 J**	0.22 U	0.22 U	0.21 U	4.3 U
CHRYSENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	**0.09 J**	**0.23**	0.22 U	**0.11 J**	0.41 U	**0.12 J**	0.19 U	**0.18 J**	**0.16 J**	**0.57**	0.22 U	**0.3**	**0.18 J**	**2.9 J**
DIBENZO(A,H)ANTHRACENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
FLUORANTHENE	µg/L	200	40	0.22 U	0.19 U	0.22 U	0.22 U	**0.13 J**	**0.24**	**0.28**	**0.29**	**0.13 J**	**0.44**	**0.099 J**	**0.29**	**0.15 J**	**0.29**	0.19 U	**0.22**	**0.3**	**1**	**0.099 J**	**0.52**	**0.24**	**8.8**
INDENO(1,2,3-CD)PYRENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	**0.1 J**	0.21 U	**0.13 J**	0.22 U	0.22 U	0.21 U	4.3 U
PYRENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	**0.11 J**	**0.21**	**0.24**	**0.23**	**0.13 J**	**0.4**	**0.11 J**	**0.3**	**0.16 J**	**0.26**	0.19 U	**0.22**	**0.37**	**1.3**	**0.17 J**	**0.71**	**0.19 J**	**7.9**
TOTAL HPAHs (ND=0)	µg/L	--	--	0	0	0	0	**0.24**	**0.45**	**0.52**	**0.52**	**0.35**	**1.42**	**0.209**	**0.81**	**0.31**	**0.78**	0	**1.32**	**1.11**	**4.13**	**0.269**	**1.97**	**0.882**	**19.6**
TOTAL HPAHs (ND=1/2RL)	µg/L	--	--	**1.1**	**0.95**	**1.1**	**1.1**	**1.04**	**1.25**	**1.32**	**1.28**	**1.09**	**1.97**	**1.09**	**1.41**	**1.95**	**1.38**	**0.95**	**1.54**	**1.53**	**4.24**	**1.15**	**2.41**	**1.41**	**34.7**
TOTAL HPAHs (ND=RL)	µg/L	--	--	**2.2**	**1.9**	**2.2**	**2.2**	**1.84**	**2.05**	**2.12**	**2.04**	**1.82**	**2.52**	**1.97**	**2.01**	**3.59**	**1.98**	**1.9**	**1.76**	**1.95**	**4.35**	**2.03**	**2.85**	**1.93**	**49.7**
TOTAL PAHs																									
TOTAL PAHs (ND=0)	µg/L	--	--	0	**0.16**	**0.12**	0	**1.98**	**2.95**	**3.39**	**3.5**	**0.879**	**3.33**	**0.808**	**3.11**	**0.49**	**2.07**	**0.066**	**1.97**	**2.83**	**8.89**	**0.659**	**4.87**	**5.25**	**79.5**
TOTAL PAHs (ND=1/2RL)	µg/L	--	--	**1.98**	**1.68**	**1.99**	**1.98**	**2.98**	**3.75**	**4.29**	**4.36**	**2.14**	**3.99**	**2.24**	**3.81**	**3.57**	**2.87**	**1.68**	**2.52**	**3.67**	**9.0**	**2.2**	**5.42**	**5.78**	**98.9**
TOTAL PAHs (ND=RL)	µg/L	--	--	**3.96**	**3.2**	**3.86**	**3.96**	**3.98**	**4.55**	**5.19**	**5.21**	**3.4**	**4.65**	**3.67**	**4.51**	**6.64**	**3.67**	**3.3**	**3.07**	**4.51**	**9.11**	**3.74**	**5.97**	**6.3**	**118**

Notes:

Source: (a) 25 Pa. Code § 93.7.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

RL is reported for non-detected constituents.

Pennsylvania maximum standards are comparable to USEPA acute criteria and Pennsylvania continuous standards are comparable to USEPA chronic criteria.

-- = No value available.

µg/L = Microgram(s) per liter. J = Compound was detected, but below the reporting limit (value is estimated).

mg/L = Milligram(s) per liter. U = Compound was analyzed, but not detected.

RL = Laboratory reporting limit.

PAH = Polycyclic aromatic hydrocarbon.

TSS = Total suspended solids.

Analyte concentration exceeds Pennsylvania Surface Water Continuous Concentration Standards.

Analyte concentration exceeds Pennsylvania Surface Water Continuous and Maximum Concentration Standards.

This page left intentionally blank

TABLE 6-4. PCB CONGENER CONCENTRATIONS (ng/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Site Water				Modified Elutriates																	
			PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
PCB-1	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	0.16 U	1.6 U	0.16 U	1.6 U	0.16 U	1.6 U	0.16 U	**0.44 J**	0.16 U	**0.46 J**	0.16 U	1.6 U	0.16 U	**0.4 J**	1.6 U	1.6 U	1.6 U	0.16 U
PCB-2	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	**0.26 JB**	**0.02 JI**	0.8 U	0.079 U	0.8 U	0.08 U	**0.3 J**	**0.036 JB**	**0.25 J**	**0.032 JB**	0.8 U	**0.036 JB**	**0.29 JI**	**0.26 JB**	0.8 U	0.8 U	**0.044 JB**
PCB-3	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.02 J**	**0.36 JB**	0.081 U	**0.23 JB**	**0.034 JB**	**0.26 JB**	**0.022 JI**	**0.63 J**	**0.033 JB**	**0.7 J**	**0.023 JB**	0.8 U	**0.025 JB**	**0.7 J**	0.8 U	0.8 U	0.8 U	**0.065 JB**
PCB-4	ng/L	--	**0.037 JI**	0.078 U	0.079 U	0.077 U	**0.08 JI**	**2.4 I**	**0.35 I**	**1.8 I**	**0.46**	**1.8 I**	**0.78**	**0.9 I**	**0.94**	**0.9 I**	**0.18**	0.8 U	**0.66**	**2.1**	**0.77 JI**	0.8 U	0.8 U	**0.19**
PCB-5	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-6	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.022 JI**	**0.36 J**	0.081 U	**0.25 JI**	0.079 U	**0.46 J**	**0.028 J**	**0.67 J**	**0.03 JI**	**0.57 J**	**0.032 J**	0.8 U	0.081 U	**0.6 J**	0.8 U	0.8 U	0.8 U	**0.04 J**
PCB-7	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-8	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.063 JI**	**0.78 J**	**0.042 JI**	**0.53 JI**	**0.045 JI**	**1.1**	**0.056 J**	**1.8 I**	**0.081 I**	**1.6**	**0.074 JI**	**0.5 JI**	**0.047 J**	**1.8**	**0.39 JI**	0.8 U	0.8 U	**0.12**
PCB-9	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-10	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.05 JI**	0.8 U	**0.052 JI**	0.8 U	**0.031 J**	0.8 U	**0.059 J**	0.8 U	**0.12**	**0.41 JI**	0.08 U	0.8 U	**0.066 J**	**0.52 JI**	0.8 U	0.8 U	0.8 U	**0.049 JI**
PCB-11	ng/L	--	0.23 U	**0.089 JB**	**0.089 JB**	0.23 U	**0.14 JIB**	**1.1 JB**	**0.13 JB**	**0.68 JB**	**0.089 JIB**	**1.1 JB**	**0.088 JB**	**1.2 JIB**	**0.33 B**	**0.86 JB**	**0.29 B**	**0.78 JIB**	**0.32 B**	**0.9 JB**	**2.3 JB**	**0.69 JIB**	2.4 U	**0.29 B**
PCB-12/13	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.034 JI**	**0.62 JI**	**0.021 JI**	**0.3 J**	**0.025 J**	**0.42 J**	**0.029 J**	**0.95 I**	**0.036 JI**	**0.82**	**0.02 J**	**0.24 JI**	0.081 U	**0.85**	0.8 U	0.8 U	0.8 U	**0.044 J**
PCB-14	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-15	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.085**	**1.6**	**0.083**	**0.95**	**0.13**	**1.5**	**0.3**	**4.2**	**0.34**	**4.4**	**0.05 J**	**1**	**0.19**	**6.2**	0.8 U	**0.23 JI**	**8.7**	0.079 U
PCB-16	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.03 J**	**1.4**	**0.093**	**0.7 JI**	**0.3**	**1.5**	**0.71**	**1.8**	**0.66**	**1.1**	**0.12**	0.8 U	**0.5**	**4.1**	**0.25 J**	0.8 U	**27**	**0.36**
PCB-17	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.046 J**	**2.1**	**0.17**	**1.4**	**0.5**	**2.2**	**0.76**	**2.4**	**0.82 B**	**1.4**	**0.2 B**	**0.27 JI**	**0.44 B**	**4.4**	**0.32 JB**	0.8 U	**33**	**0.39 B**
PCB-18/30	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.12 J**	**2.9**	**0.31**	**1.8**	**0.75**	**3.5**	**1.6**	**5.1**	**1.8**	**3.2**	**0.3**	1.6 U	**1.1**	**11**	**0.61 J**	1.6 U	**70**	**1.1**
PCB-19	ng/L	--	**0.029 J**	**0.03 J**	**0.031 J**	**0.029 JI**	**0.58**	**2.1**	**0.73**	**1.7**	**0.29**	**1.3**	**0.33**	**1.7 I**	**0.38**	**1.7**	**0.12**	**0.34 JI**	**0.19**	**2.8**	0.8 U	**0.25 J**	**9.7**	**0.27**
PCB-20/28	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.22**	**6.8 B**	**0.37**	**3.9 B**	**1 B**	**6.7 B**	**2.2**	**14**	**1.5 B**	**11**	**0.43 B**	**2.3**	**0.78 B**	**21**	**0.6 JB**	**0.72 J**	**110**	**1.6 B**
PCB-21/33	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	0.16 U	**1.3 J**	0.16 U	**0.93 J**	**0.12 J**	**1.6**	**0.08 JI**	**1.9**	**0.077 J**	**1.4 J**	**0.1 J**	**0.46 J**	0.16 U	**3.1**	1.6 U	1.6 U	**11**	**0.16**
PCB-22	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.026 J**	**1.2**	**0.027 J**	**0.78 J**	**0.11**	**1.2**	**0.099**	**2.1**	**0.059 J**	**1.7**	**0.075 J**	**0.42 J**	**0.063 J**	**4**	0.8 U	0.8 U	**15**	**0.16**
PCB-23	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-24	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	**0.027 JI**	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-25	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.071 J**	**2.4**	**0.052 J**	**0.86**	**0.37**	**1.8**	**0.3**	**1.8**	**0.15**	**1.3**	**0.14**	**0.46 J**	0.081 U	**2.1**	0.8 U	0.8 U	**7.9**	0.079 U
PCB-26/29	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.046 J**	**1.8**	**0.073 J**	**0.94 J**	**0.36**	**1.8**	**0.51**	**2.4**	**0.34**	**1.6**	**0.14 J**	**0.42 J**	**0.041 J**	**3.2**	1.6 U	1.6 U	**18**	**0.16**
PCB-27	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.17**	**0.86**	**0.17**	**0.6 J**	**0.18**	**0.73 J**	**0.16**	**1.1**	**0.24**	**1**	**0.064 J**	0.8 U	**0.1**	**1.7**	0.8 U	0.8 U	**7.2**	**0.18**
PCB-31	ng/L	--	0.077 U	0.078 U	0.079 U	**0.019 JI**	**0.099**	**6.2 B**	**0.17**	**3.5 B**	**0.95 B**	**6.5 B**	**1.4**	**7.4**	**0.79 B**	**5**	**0.27 B**	**1.2**	**0.16 B**	**12**	**0.45 JB**	**0.43 J**	**86**	**0.75 B**
PCB-32	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.25**	**1.5**	**0.24**	**0.87**	**0.28**	**1.4**	**0.56**	**3.4**	**0.62**	**3**	**0.13**	**0.28 JI**	**0.32**	**5.8**	**0.23 J**	0.8 U	**23**	**0.56**
PCB-34	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-35	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	**0.27 JI**	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-36	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-37	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.061 J**	**1.2**	**0.078 J**	**0.71 J**	**0.14**	**1.1**	**0.39**	**3.9**	**0.3**	**2.9**	**0.052 J**	**0.77 J**	**0.15**	**5**	0.8 U	**0.26 J**	**18**	**0.41**
PCB-38	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-39	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-40/71	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.78**	**4.2**	**0.67**	**2.4**	**0.88**	**3.8**	**1.4**	**9.6**	**1.5**	**7.9**	**0.37**	**1.5 J**	**0.66**	**14**	1.6 U	1.6 U	**69**	**1.5**
PCB-41	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	**0.033 J**	0.8 U	**0.055 J**	**0.29 J**	**0.12**	**0.51 J**	**0.15**	**0.52 J**	0.08 U	0.8 U	**0.094**	**1.4**	0.8 U	0.8 U	0.8 U	**0.097 I**
PCB-42	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.35**	**2.2**	**0.32**	**1.2**	**0.48**	**2.1**	**0.8**	**5.1**	**0.84**	**4.2**	**0.21**	**0.68 J**	**0.34**	**7.1**	0.8 U	0.8 U	**34**	**0.8**
PCB-43	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.039 JI**	**0.3 J**	**0.052 J**	0.8 U	**0.078 JI**	**0.34 J**	**0.12**	**0.63 J**	**0.15**	**0.59 J**	**0.034 J**	0.8 U	**0.065 J**	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-44/47/65	ng/L	--	0.23 U	0.24 U	0.24 U	**0.06 J**	**1.6**	**11**	**1.8**	**6.9**	**2.2**	**9.8**	**3.4**	**20**	**3.2**	**15**	**0.88**	**2.4**	**1.4**	**26**	**0.85 J**	2.4 U	**160**	**3.4**
PCB-45	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.16**	**0.72 J**	**0.11**	0.8 U	**0.2**	**0.74 J**	**0.37**	**2.8 I**	**0.43**	**2.4**	**0.082**	**0.44 J**	**0.25**	**4.7**	0.8 U	**0.2 J**	0.8 U	**0.5**
PCB-46	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.12**	**0.55 J**	**0.1**	**0.31 J**	**0.11 I**	**0.47 J**	**0.19**	**1.4**	**0.21**	**1.1**	**0.054 J**	**0.28 J**	**0.11**	**2**	0.8 U	0.8 U	0.8 U	0.079 U
PCB-48	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.072 JI**	**1.1**	**0.14**	**0.66 J**	**0.25**	**1.1**	**0.48**	**2.4**	**0.52**	**1.7**	**0.11**	0.8 U	**0.24**	**4**	0.8 U	0.8 U	**24**	**0.44**
PCB-49/69	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**1.1**	**7.8**	**1.2**	**4.5**	**1.6**	**6.9**	**2.2**	**14**	**2.3**	**10**	**0.68**	**1.7**	**0.82**	**16**	**0.45 J**	**0.4 J**	**100**	**2.2**
PCB-50/53	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.72**	**3**	**0.68**	**2**	**0.48**	**2.2**	**0.56**	**4**	**0.6**	**3.3**	**0.19**	**0.98 J**	**0.24**	**5**	1.6 U	**0.69 J**	**23**	**0.61**
PCB-51	ng/L	--	0.077 U	**0.02 J**	0.079 U	0.077 U	**0.77**	**3.2**	**0.8**	**2.7**	**0.45**	**2.1**	**0.34**	**2.1**	**0.39**	**1.7**	**0.17**	**0.85**	**0.075 J**	**1.3**	0.8 U	**0.38 JI**	**24**	**0.26**
PCB-52	ng/L	--	**0.049 J**	**0.063 J**	**0.054 J**	**0.071 JI**	**2.3**	**12 B**	**2**	**6.8 B**	**2.6 B**	**11 B**	**4.6**	**29**	**4.5 B**	**22**	**1.1 B**	**3.7**	**2.2 B**	**44**	**1.2 B**	**1**	**330**	**5.8 B**

TABLE 6-4. PCB CONGENER CONCENTRATIONS (ng/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Site Water				Modified Elutriates																	
			PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
PCB-54	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.16	0.62 J	0.18	0.49 J	0.073 J	0.39 J	0.047 J	0.27 J	0.05 J	0.23 J	0.024 J	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	1.7 I	0.031 JI
PCB-55	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	8 U	0.079 U
PCB-56	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.23	2.7	0.23	1.6	0.47	2.6	1.1	7.3	0.9	5.9	0.2	1	0.39	10	0.8 U	0.25 J	62	0.85
PCB-57	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.02 JI	0.8 U	0.026 J	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	1.2 I	0.079 U
PCB-58	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.02 JI	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.2
PCB-59/62/75	ng/L	--	0.23 U	0.24 U	0.24 U	0.23 U	0.18 J	0.87 J	0.14 J	2.4 U	0.18 JI	0.81 J	0.27	1.9 J	0.33	1.7 J	0.087 J	2.4 U	0.13 J	2.6	2.4 U	2.4 U	12 I	0.28
PCB-60	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.039 J	0.73 J	0.065 J	0.47 J	0.087	0.61 J	0.27	1.9	0.2	1.2	0.027 J	0.25 J	0.13	3.3	0.8 U	0.8 U	22	0.31
PCB-61/70/74/76	ng/L	--	0.31 U	0.31 U	0.31 U	0.31 U	0.66	9.9	1	5.8	1.8	9.3	4.3	25	3.3	18	0.69	3.1 J	1.3	37	3.2 U	1 J	310	3.8
PCB-63	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.026 JI	0.3 J	0.035 J	0.8 U	0.073 J	0.31 J	0.12	0.67 J	0.1	0.51 J	0.026 J	0.8 U	0.036 JI	0.95	0.8 U	0.8 U	0.8 U	0.095
PCB-64	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.58	3.2	0.47	1.8	0.72	3	1.2	8.6	1.4	7	0.3	0.96	0.59	13	0.26 J	0.28 J	67	1.4
PCB-66	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.44	5.4	0.65	3.6	1	5.5	2.4	16	2	12	0.45	2.3	0.8	20	0.35 J	0.63 J	0.8 U	2.2
PCB-67	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081	0.98	0.057 J	0.27 JI	0.19	0.76 J	0.23	1.2	0.16	0.84	0.065 J	0.8 U	0.045 J	1.2	0.8 U	0.8 U	0.8 U	0.091
PCB-68	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.034 J	0.8 U	0.03 J	0.8 U	0.031 J	0.8 U	0.041 J	0.25 J	0.039 JI	0.21 J	0.021 J	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	1.9	0.079 U
PCB-72	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.044 J	0.25 J	0.036 J	0.8 U	0.06 J	0.23 J	0.071 J	0.45 J	0.06 J	0.35 J	0.02 J	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-73	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.041 JI	0.8 U	0.028 JI	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-77	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.079 J	0.79 J	0.067 J	0.39 J	0.1	0.6 J	0.22	2.1	0.18	1.7	0.034 J	0.43 J	0.08 J	2.7	0.8 U	0.8 U	15	0.27
PCB-78	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-79	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.02 JI	0.8 U	0.036 JI	0.8 U	0.029 J	0.3 J	0.037 J	0.25 JI	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-80	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-81	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	1.4 I	0.079 U
PCB-82	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.32	1.6	0.23	0.72 J	0.35	1.4	0.59	3.9	0.63	3.2	0.12	0.87	0.32	6.7	0.8 U	0.25 JI	62	1.1
PCB-83	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.2	1 I	0.16	0.55 JI	0.26 I	0.82	0.31	2	0.35	1.4	0.086	0.53 J	0.15	2.6 I	0.8 U	0.26 JI	22 I	0.51
PCB-84	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.71	3.2	0.51	1.7	0.83	3	1.3	9.4	1.4	6.1	0.32	1.7	0.67	13	0.27 J	0.93	77	2.2
PCB-85/116/117	ng/L	--	0.23 U	0.24 U	0.24 U	0.23 U	0.45	2.3 J	0.34	1.2 JI	0.53	2 J	0.73	4.9	0.8	4	0.18 J	1 J	0.33	8.1	2.4 U	2.4 U	73	1.3
PCB-86/87/97/109/119/125	ng/L	--	0.46 U	0.47 U	0.47 U	0.46 U	1.7	8.6	1.3	4.4 J	2	7.8	3.1	20	3.2	15	0.72	3.6 J	1.4	32	4.8 U	1.2 J	340	5.6
PCB-88	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-89	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.024 J	0.8 U	0.081 U	0.8 U	0.033 J	0.8 U	0.034 J	0.24 JI	0.053 J	0.8 U	0.08 U	0.8 U	0.03 J	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-90/101/113	ng/L	--	0.23 U	0.24 U	0.24 U	0.058 J	3.6	17	2.7	8.8	4.1	15	6	39	5.9	26	1.5	7.6	2.5	56	0.95 J	2.1 J	690	10
PCB-91	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.6	2.6	0.42	1.4	0.65	2.4	0.83	5.1	0.82	3.8	0.26	1.3	0.28	6.8	0.8 U	0.54 J	0.8 U	1.2
PCB-92	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	1	4.8	0.81	2.6	1.1	4.2	1.3	8	1.4	6	0.38	2.1 I	0.47	11	0.8 U	0.48 J	130	1.9
PCB-93/100	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	0.33	1.5 J	0.34	1 J	0.25	1 J	0.15 J	1.1 J	0.2	0.72 J	0.072 J	1.6 U	0.16 U	1.6 U	1.6 U	1.6 U	1.6 U	0.16 U
PCB-94	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.11	0.45 J	0.11	0.34 JI	0.078 J	0.35 J	0.063 J	0.39 JI	0.07 J	0.28 J	0.026 J	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.13
PCB-95	ng/L	--	0.039 J	0.062 J	0.056 J	0.057 J	2.1	10 B	1.6	5.7 B	2.4 B	9.3 B	4.9	29	3.7 B	21	0.96 B	6.1	1.9 B	46	0.82 B	3.5	470	6.2 B
PCB-96	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.045 J	0.24 J	0.042 J	0.8 U	0.045 J	0.8 U	0.039 JI	0.28 J	0.056 J	0.22 J	0.08 U	0.8 U	0.026 J	0.37 JI	0.8 U	0.8 U	2.6	0.058 JI
PCB-98/102	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	0.25	0.97 J	0.22	0.63 JI	0.21	0.89 J	0.22	1.5 JI	0.28	1.1 J	0.08 J	1.6 U	0.09 J	1.6	1.6 U	1.6 U	1.6 U	0.29
PCB-99	ng/L	--	0.077 U	0.022 J	0.079 U	0.026 J	1.6	7.5	1.3	4.3	2	7.4	2.5	17	2.8	12	0.74	3.8	0.89	22	0.36 J	0.63 JI	270	4
PCB-103	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.17	0.72 JI	0.16	0.46 J	0.14	0.56 J	0.13	0.85	0.16	0.57 J	0.054 J	0.27 J	0.028 J	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-104	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.045 J	0.2 JI	0.044 J	0.8 U	0.022 J	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-105	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.49	3.3	0.41	1.8	0.59	2.8	1.4	10	1	7	0.21	1.5	0.49	15	0.8 U	0.54 J	150	2.2
PCB-106	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-107	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.18	0.9	0.12	0.43 J	0.19	0.83	0.35	2.4	0.27	1.8	0.072 J	0.5 J	0.094	3.2	0.8 U	0.8 U	25	0.43
PCB-108/124	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	0.053 JI	1.6 U	0.042 J	1.6 U	0.051 J	1.6 U	0.14 J	1 J	0.11 J	0.69 J	0.16 U	1.6 U	0.051 J	1.5 J	1.6 U	1.6 U	20	0.24
PCB-110/115	ng/L	--	0.15 U	0.061 J	0.05 J	0.067 J	3.3	16	2.4	8	3.8	15	5.7	38	5.7	27	1.4	8.3	2.4	56	0.91 J	3.6	640	9.7
PCB-111	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-112	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-114	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.03 J	0.23 J	0.03 J	0.8 U	0.034 J	0.8 U	0.076 J	0.49 JI	0.07 J	0.42 J	0.08 U	0.8 U	0.027 J	0.94	0.8 U	0.8 U	0.8 U	0.12 I

TABLE 6-4. PCB CONGENER CONCENTRATIONS (ng/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Site Water				Modified Elutriates																	
			PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
PCB-118	ng/L	--	0.077 U	0.078 U	0.079 U	**0.03 J**	**1.3**	**8.5**	**1.2**	**5**	**1.7**	**7.7**	**3.8**	**27**	**2.8**	**19**	**0.67**	**4.7**	**1.1**	**38**	**0.42 J**	**1.5**	**420**	**5.6**
PCB-120	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.033 J**	0.8 U	**0.023 J**	0.8 U	**0.048 J**	0.8 U	**0.021 JI**	**0.23 J**	**0.033 J**	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-121	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-122	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	**0.024 J**	0.8 U	**0.045 J**	0.8 U	**0.042 J**	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-123	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	**0.21 J**	**0.021 J**	0.8 U	0.079 U	0.8 U	**0.061 JI**	0.8 U	**0.055 J**	**0.39 J**	0.08 U	0.8 U	0.081 U	**0.77 J**	0.8 U	0.8 U	**19 I**	**0.094 I**
PCB-126	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.021 J**	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-127	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-128/166	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.51**	**2.4**	**0.37**	**1.5 J**	**0.62**	**2.2**	**0.91**	**6**	**0.67**	**4.3**	**0.21 I**	**1.4 J**	**0.26**	**7.7**	1.6 U	**1 JI**	**83**	**1.5**
PCB-129/138/163	ng/L	--	0.23 U	0.24 U	0.24 U	0.23 U	**4.7**	**21**	**3.8**	**12**	**5.4**	**21**	**8**	**50**	**7.5**	**35**	**1.9**	**12**	**2.9**	**72**	**1.1 J**	**7.8**	**850**	**13**
PCB-130	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.33**	**1.3 I**	**0.24**	**0.76 J**	**0.39**	**1.4**	**0.49**	**2.8**	**0.52**	**2.2**	**0.15**	**0.88**	**0.17**	**4.2**	0.8 U	**0.29 JI**	**48**	**0.76**
PCB-131	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	**0.14**
PCB-132	ng/L	--	0.077 U	**0.02 J**	0.079 U	0.077 U	**1.4**	**6.1**	**1.1**	**3.2**	**1.6**	**6.2**	**2.5**	**15**	**2.4**	**11**	**0.62**	**3.6**	**1.1**	**24**	**0.34 J**	**2.2**	**240**	**4**
PCB-133	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.13**	**0.59 J**	**0.12**	**0.39 J**	**0.12**	**0.46 J**	0.08 U	**1.1**	**0.13**	**0.8**	**0.04 J**	**0.26 J**	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-134	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.26**	**0.98**	**0.18 I**	**0.57 J**	**0.29**	**1**	**0.4**	**2.2 I**	**0.41**	**1.7**	**0.11**	**0.49 JI**	**0.19**	**3.9**	0.8 U	**0.26 JI**	**40**	**0.66**
PCB-135/151	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**2.4**	**8.9**	**1.9**	**5**	**2.6**	**9.3**	**3.2**	**19**	**3.9**	**14**	**0.95**	**4.6**	**1.4**	**27**	**0.48 J**	**2.9**	**280**	**5.1**
PCB-136	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.73**	**2.6**	**0.6**	**1.7**	**0.86**	**2.9**	**1**	**6.5**	**1.4**	**4.5**	**0.32**	**1.4**	**0.63**	**10**	**0.23 JI**	**0.88**	**90**	**1.9**
PCB-137	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.11 I**	**0.54 J**	**0.097**	**0.33 J**	**0.14 I**	**0.57 J**	**0.23**	**1.2 I**	**0.2**	**0.89**	**0.049 J**	0.8 U	**0.076 J**	**1.7 I**	0.8 U	0.8 U	**50 I**	**0.37**
PCB-139/140	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.1 JI**	**0.43 JI**	**0.077 J**	1.6 U	**0.098 J**	**0.49 JI**	**0.13 J**	**0.81 JI**	**0.14 J**	**0.53 J**	0.16 U	1.6 U	**0.044 J**	1.6 U	1.6 U	1.6 U	1.6 U	**0.18**
PCB-141	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.59**	**2.8**	**0.49**	**1.8**	**0.62**	**2.5**	**1.4**	**7.9**	**0.93**	**5.6**	**0.25**	**1.8**	**0.45**	**13**	0.8 U	**2.1**	**170**	**2**
PCB-142	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-143	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-144	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.23**	**0.81**	**0.18**	**0.47 J**	0.079 U	**0.72 J**	**0.36**	**2**	**0.38**	**1.4**	**0.091**	**0.41 J**	**0.19**	**3.1**	0.8 U	**0.32 JI**	0.8 U	0.079 U
PCB-145	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-146	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**1.3**	**4.7**	**0.97**	**2.7**	**1.3**	**4.8**	**1.7**	**9.5**	**1.7**	**7.2**	**0.47**	**2.6**	**0.5**	**13**	**0.21 J**	**1.4**	0.8 U	**2.1**
PCB-147/149	ng/L	--	0.15 U	**0.067 J**	**0.045 J**	**0.058 J**	**4.9**	**19**	**3.9**	**11**	**5.5**	**20**	**7.6 B**	**44 B**	**7.5**	**32 B**	**2.1**	**11 B**	**3.1**	**67 B**	**1.1 J**	**6.7 B**	1.6 U	**11**
PCB-148	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.056 JI**	0.8 U	**0.054 J**	0.8 U	**0.046 JI**	0.8 U	**0.037 JI**	0.8 U	**0.06 J**	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-150	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.04 JI**	0.8 U	**0.031 J**	0.8 U	**0.036 JI**	0.8 U	**0.027 J**	0.8 U	**0.044 J**	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-152	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-153/168	ng/L	--	0.15 U	**0.06 J**	0.16 U	0.15 U	**4.2**	**17**	**3.5**	**10**	**4.8**	**18**	**7.2 B**	**41 B**	**6.8**	**30 B**	**1.8**	**10 B**	**2.5**	**63 B**	**0.94 J**	**8.4 B**	**720 B**	**11**
PCB-154	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.24**	**0.98**	**0.23**	**0.58 JI**	**0.27**	**0.96**	**0.22**	**1.4**	**0.33**	**0.98**	**0.095**	**0.42 JI**	**0.058 JI**	**1.1**	0.8 U	0.8 U	**12**	**0.2**
PCB-155	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-156/157	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.31**	**1.8**	**0.25**	**1 J**	**0.38**	**1.5 J**	**0.73**	**4.7**	**0.5**	**3.3**	**0.13 J**	**1 J**	**0.22**	**7.1**	1.6 U	**0.59 J**	**93**	**1.2**
PCB-158	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.36**	**1.7**	**0.28**	**0.96**	**0.41**	**1.6**	**0.63**	**4.1**	**0.57**	**2.9**	**0.15**	**0.96**	**0.23**	**6**	0.8 U	**0.76 J**	**70**	**1.1**
PCB-159	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.054 J**	**0.29 J**	**0.047 J**	0.8 U	**0.072 J**	**0.24 JI**	**0.13**	**0.84 I**	**0.091**	**0.61 J**	**0.024 JI**	**0.2 J**	**0.033 J**	**0.87**	0.8 U	**0.36 JI**	**9.9**	**0.14**
PCB-160	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-161	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	8 U	0.079 U
PCB-162	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.032 JI**	**0.21 J**	**0.034 J**	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	**0.059 J**	0.8 U	**0.021 J**	0.8 U	**0.025 J**	0.8 U	0.8 U	0.8 U	**7.4**	**0.099**
PCB-164	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.31**	**1.3**	**0.26**	**0.81 I**	**0.38**	**1.4**	**0.54**	**3.5 I**	**0.54**	**2.6**	**0.14**	**1.1 I**	**0.22**	**5**	0.8 U	**0.71 JI**	**35 I**	**0.88**
PCB-165	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-167	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.12**	**0.65 J**	**0.095**	**0.35 J**	**0.13**	**0.58 J**	**0.26**	**1.6**	**0.18**	**1.2**	**0.05 J**	**0.46 J**	**0.072 J**	**2.5**	0.8 U	**0.26 JI**	**34**	**0.43**
PCB-169	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-170	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**1.1**	**6.1**	**0.95**	**3.7**	**1.3**	**4.9**	**2.7**	**15**	**1.5**	**11**	**0.51**	**4.2**	**0.65**	**21**	1.6 U	**4.9**	**240**	**3.4**
PCB-171/173	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**0.44**	**2.3**	**0.38**	**1.3 J**	**0.52**	**1.9**	**0.95**	**5.4**	**0.58**	**4**	**0.2**	**1.4 J**	**0.26**	**6.7**	1.6 U	**1.8**	**70**	**1.2**
PCB-172	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.23**	**1.2**	**0.19**	**0.71 J**	**0.26**	**1**	**0.54**	**3**	**0.29**	**2.3**	**0.099**	**0.85**	**0.12**	**3.8**	0.8 U	**1.4**	**42**	**0.67**
PCB-174	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	**1.6**	**8.5**	**1.5**	**4.8**	**1.9**	**6.9**	**4 B**	**21 B**	**2.4**	**17 B**	**0.78**	**5.3 B**	**0.97**	**27 B**	1.6 U	**11 B**	**300 B**	**4.7**
PCB-175	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	**0.1 I**	**0.39 JI**	**0.078 J**	**0.24 J**	**0.095**	**0.34 J**	**0.15**	**0.86**	**0.13**	**0.65 J**	**0.04 J**	**0.22 J**	**0.047 J**	**1.1**	0.8 U	**0.4 J**	**13**	**0.2**

TABLE 6-4. PCB CONGENER CONCENTRATIONS (ng/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Site Water				Modified Elutriates																	
			PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
PCB-176	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.28	1.1	0.24	0.61 J	0.33	1.1	0.48	2.5	0.45	1.9	0.12	0.66 J	0.19	3.9	0.8 U	0.85	41	0.7
PCB-177	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	1	5.3	0.92	3.1	1.2	4.3	2.3 B	13 B	1.3	9.8 B	0.46	3.2 B	0.56	15 B	0.8 U	5.4 B	180 B	2.6
PCB-178	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.49	2	0.42	1.3	0.54	1.9	0.8	4.3	0.67	3.2	0.19	1.1	0.22	4.9	0.8 U	1.9	50	0.95
PCB-179	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	1	3.9	0.86	2.3	1.1	3.9	1.7 B	8.5 B	1.7	6.6 B	0.43	2.2 B	0.64	12 B	0.23 J	3.1 B	120 B	2.4
PCB-180/193	ng/L	--	0.15 U	0.056 J	0.16 U	0.15 U	2.8	14	2.3	8.5	3.2	12	6.9 B	37 B	3.6	29 B	1.2	10 B	1.5	50 B	0.65 J	19 B	600 B	8.2
PCB-181	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-182	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.024 J	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.028 J	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-183/185	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	1.1	5.4	0.92	3.2	1.3	4.4	2.4 B	13 B	1.5	9.7 B	0.5	3.4 B	0.6	17 B	1.6 U	6.5 B	180 B	3.2
PCB-184	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-186	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-187	ng/L	--	0.15 U	0.038 J	0.16 U	0.15 U	2.6	12	2.3	7.5	3	11	5.1 B	26 B	3.8	21 B	1.2	6.9 B	1.2	31 B	0.57 J	15 B	390 B	6.1
PCB-188	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-189	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.049 J	0.22 J	0.043 J	0.8 U	0.06 J	0.23 J	0.1	0.6 J	0.071 J	0.48 J	0.022 J	0.8 U	0.03 J	0.88	0.8 U	0.8 U	12	0.14
PCB-190	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.23	1.3	0.21	0.82	0.28	1.1	0.57	3.3	0.28	2.4	0.11	0.9	0.12	4.3	0.8 U	0.96 I	46	0.68
PCB-191	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.057 J	0.27 J	0.039 J	0.8 U	0.064 J	0.22 J	0.12	0.47 JI	0.069 J	0.48 J	0.023 J	0.8 U	0.027 JI	0.82	0.8 U	0.28 JI	0.8 U	0.14
PCB-192	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-194	ng/L	--	0.12 U	0.12 U	0.12 U	0.12 U	0.8	3.7	0.6	2.3	0.96	3.2	2.3 B	11 B	1.2	8.9 B	0.34	3.8 B	0.47	16 B	1.2 U	6.2 B	150 B	2
PCB-195	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.34	1.7	0.27	1	0.41	1.4	0.95	4.5	0.46	4.1	0.15	1.6	0.2	6.3	0.8 U	3	64	0.85
PCB-196	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.45	2.1	0.38	1.3	0.57	1.9	1.2 B	5.9 B	0.77	5.2 B	0.2	1.8 B	0.29	8.7 B	0.8 U	4.4 B	84 B	1.3
PCB-197/200	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	0.17	0.77 J	0.12 J	0.4 JI	0.18	0.68 J	0.46	2.1	0.25	1.8	0.078 J	1.6 U	0.11 J	2.7	1.6 U	2.1	27	0.44
PCB-198/199	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	1.2	5.9	0.99	3.3	1.4	4.9	3.4 B	16 B	1.8	15 B	0.5	5.1 B	0.8	26 B	0.44 J	16 B	210 B	3.1
PCB-201	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.17	0.66 J	0.14	0.39 J	0.19	0.65 J	0.42	2	0.25	1.8	0.068 J	0.5 J	0.09	2.7	0.8 U	2	25	0.43
PCB-202	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.29	1.3	0.22	0.73 J	0.32	1.1	0.77 B	3.7 B	0.36	3.3 B	0.12	1 B	0.2	6.6 B	0.8 U	4.1 B	38 B	0.66
PCB-203	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.67	3.4	0.54	1.9	0.81	2.8	1.9 B	9 B	0.9	7.9 B	0.29	3 B	0.44	15 B	0.24 JI	8.5 B	130 B	1.9
PCB-204	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.081 U	0.8 U	0.081 U	0.8 U	0.079 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.08 U	0.8 U	0.081 U	0.8 U	0.8 U	0.8 U	0.8 U	0.079 U
PCB-205	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.041 J	0.8 U	0.036 J	0.8 U	0.049 J	0.8 U	0.099	0.43 J	0.054 J	0.41 J	0.08 U	0.8 U	0.081 U	0.71 JI	0.8 U	0.25 JI	7.2 I	0.11
PCB-206	ng/L	--	0.077 U	0.041 J	0.024 J	0.024 J	1.4	6.7	0.98	3.3	1.6	4.7	2.4 B	13 B	1.4	9.6 B	0.49	5.3 B	1.3	44 B	1.2 I	12 B	190 B	5
PCB-207	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U	0.11	0.65 J	0.094	0.35 JI	0.15	0.45 J	0.24	1.4	0.19	1.1	0.044 J	0.46 J	0.12	3.1	0.8 U	1.4	16	0.35
PCB-208	ng/L	--	0.12 U	0.12 U	0.12 U	0.12 U	0.57	2.8	0.39	1.2	0.64	2	1	5.7	0.65	4.2	0.2	1.9	0.53	18	0.44 J	5.9	65	2
PCB-209	ng/L	--	0.15 U	0.16 U	0.16 U	0.15 U	1.7	12	1.2	5.2	1.9	7.9	2.4	18	1.8	14	0.6	5.6	3.1	64	1.3 J	11	190	7.8
TOTAL PCBs (ND=0)*	ng/L	14	0.154	0.629	0.349	0.499	75.9	405	64.8	229	92.2	370	153	911	131	685	35.2	199	57.6	1,378	22.5	209	11,426	204
TOTAL PCBs (ND=1/2RL)*	ng/L	14	8.26	8.34	8.48	8.11	77.8	427	66.7	259	94.1	394	155	931	132	704	37.6	235	60.2	1,402	86.7	254	11,466	206
TOTAL PCBs (ND=RL)*	ng/L	14	16.4	16.0	16.6	15.7	79.8	449	68.7	288	96.0	419	157	950	134	724	40.0	271	62.8	1,427	151	299	11,507	209

TABLE 6-4. PCB CONGENER CONCENTRATIONS (ng/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

ANALYTE	UNITS	PA SW Continuous[(a)]	Site Water			
			PNYRB24-WAT		PNYRB24-WAT-FD	
			Dissolved	Total	Dissolved	Total
			TSS = 2.5 mg/L		TSS = 2.6 mg/L	
TOTAL PCB HOMOLOGS						
TOTAL DICHLOROBIPHENYLS	ng/L	--	**0.037 JI**	**0.089 IB**	**0.089 IB**	0.077 U
TOTAL HEPTACHLOROBIPHENYLS	ng/L	--	0.077 U	**0.094 I**	0.079 U	0.077 U
TOTAL HEXACHLOROBIPHENYLS	ng/L	--	0.077 U	**0.15 I**	**0.045 JI**	**0.058 JI**
TOTAL MONOCHLOROBIPHENYLS	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U
TOTAL NONACHLOROBIPHENYLS	ng/L	--	0.077 U	**0.041 J**	**0.024 J**	**0.024 JI**
TOTAL OCTACHLOROBIPHENYLS	ng/L	--	0.077 U	0.078 U	0.079 U	0.077 U
TOTAL PENTACHLOROBIPHENYLS	ng/L	--	**0.039 JI**	**0.15 I**	**0.11 I**	**0.24 I**
TOTAL TETRACHLOROBIPHENYLS	ng/L	--	**0.049 JI**	**0.083 I**	**0.054 JI**	**0.13 I**
TOTAL TRICHLOROBIPHENYLS	ng/L	--	**0.029 JI**	**0.03 JI**	**0.031 JI**	**0.048 JI**
TOTAL PCBs - HOMOLOGS**	ng/L	14	**0.15 I**	**0.63 IB**	**0.35 IB**	**0.5 IB**

ANALYTE	Modified Elutriates									
	PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET	
	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
	TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L	
TOTAL PCB HOMOLOGS										
TOTAL DICHLOROBIPHENYLS	**0.47 IB**	**6.9 IB**	**0.68 IB**	**4.5 IB**	**0.78 IB**	**6.4 IB**	**1.3 B**	**9.7 IB**	**1.9 IB**	**9.6 IB**
TOTAL HEPTACHLOROBIPHENYLS	**13 I**	**64 I**	**11**	**38 I**	**15**	**55**	**29 B**	**150 IB**	**18**	**120 B**
TOTAL HEXACHLOROBIPHENYLS	**23 I**	**96 I**	**19 I**	**55 I**	**26 I**	**98 I**	**38 IB**	**230 IB**	**37**	**160 IB**
TOTAL MONOCHLOROBIPHENYLS	**0.02 JI**	**0.62 JIB**	**0.02 JI**	**0.23 JB**	**0.034 JIB**	**0.26 JIB**	**0.022 JI**	**1.4**	**0.069 JB**	**1.4**
TOTAL NONACHLOROBIPHENYLS	**2.1**	**10**	**1.5**	**4.9 I**	**2.4**	**7.2**	**3.6 B**	**20 B**	**2.2**	**15 B**
TOTAL OCTACHLOROBIPHENYLS	**4.1**	**20**	**3.3**	**11 I**	**4.9**	**17**	**11 B**	**55 B**	**6**	**48 B**
TOTAL PENTACHLOROBIPHENYLS	**19 I**	**92 IB**	**15**	**49 IB**	**21 IB**	**82 IB**	**34 I**	**220 I**	**32 B**	**160 I**
TOTAL TETRACHLOROBIPHENYLS	**11 I**	**72 IB**	**11 I**	**42 IB**	**14 IB**	**65 IB**	**25**	**160 I**	**24 IB**	**120 I**
TOTAL TRICHLOROBIPHENYLS	**1.7 I**	**32 B**	**2.5 I**	**19 IB**	**5.4 B**	**31 IB**	**9.1 I**	**49 I**	**7.8 IB**	**36 I**
TOTAL PCBs - HOMOLOGS**	**76 IB**	**400 IB**	**65 IB**	**230 IB**	**92 IB**	**370 IB**	**150 IB**	**910 IB**	**130 IB**	**680 IB**

ANALYTE	Modified Elutriates							
	PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
	TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
TOTAL PCB HOMOLOGS								
TOTAL DICHLOROBIPHENYLS	**0.65 IB**	**2.5 IB**	**1.3 IB**	**13 IB**	**3.5 IB**	**0.92 IB**	**8.7 B**	**0.73 IB**
TOTAL HEPTACHLOROBIPHENYLS	**5.9 I**	**40 IB**	**7.1 I**	**200 B**	**1.5 I**	**72 IB**	**2100 B**	**35**
TOTAL HEXACHLOROBIPHENYLS	**9.7 I**	**55 IB**	**14 I**	**330 IB**	**4.4 I**	**37 IB**	**4200 IB**	**58**
TOTAL MONOCHLOROBIPHENYLS	**0.055 JB**	0.8 U	**0.061 JB**	**1.4 I**	**0.26 JIB**	0.8 U	0.8 U	**0.11 B**
TOTAL NONACHLOROBIPHENYLS	**0.73**	**7.7 B**	**2**	**65 B**	**1.6 I**	**19 B**	**280 B**	**7.4**
TOTAL OCTACHLOROBIPHENYLS	**1.7**	**17 IB**	**2.6**	**85 IB**	**0.68 JI**	**47 IB**	**740 IB**	**11**
TOTAL PENTACHLOROBIPHENYLS	**7.9 IB**	**44 I**	**13 B**	**320 I**	**3.7 IB**	**16 I**	**3200 I**	**53 IB**
TOTAL TETRACHLOROBIPHENYLS	**5.8 IB**	**21 I**	**10 IB**	**220**	**3.1 IB**	**4.8 I**	**1300 I**	**25 IB**
TOTAL TRICHLOROBIPHENYLS	**2.1 IB**	**6.9 I**	**3.8 IB**	**80 I**	**2.5 IB**	**1.7 I**	**440**	**6.1 B**
TOTAL PCBs - HOMOLOGS**	**35 IB**	**200 IB**	**58 IB**	**1,400 IB**	**22 IB**	**210 IB**	**12,000 IB**	**200 IB**

Notes:

Source: (a) 25 Pa. Code § 93.7.

*Total PCBs based on summation of individual congeners. Total PCB criterion compared to total fraction of the modified elutriates.

**Total PCBs based on summation of homologs. Total PCB criterion compared to total fraction of the modified elutriates.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

RL is reported for non-detected constituents.

Pennsylvania maximum standards are comparable to USEPA acute criteria and Pennsylvania continuous standards are comparable to USEPA chronic criteria.

-- = No value available.

mg/L = Milligram(s) per liter. B = Compound was detected in the laboratory method blan.

ng/L = Nanogram(s) per liter. J = Compound was detected, but below the reporting limit (value is estimatec.

RL = Laboratory reporting limit. I or Q = Result is the estimated maximum possible concentration of this analyte.

TSS = Total suspended solids. U = Compound was analyzed, but not detected.

Analyte concentration exceeds Pennsylvania Surface Water Continuous Concentration Standards.

This page left intentionally blank

TABLE 6-5. PCB AROCLOR CONCENTRATIONS (µg/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Site Water				Modified Elutriates																	
			PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
	PA SW		Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	Continuous[a]	UNITS	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
PCB-1016	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.0097 U	0.0098 U
PCB-1221	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.0097 U	0.0098 U
PCB-1232	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.0097 U	0.0098 U
PCB-1242	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.0097 U	0.0098 U
PCB-1248	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	**0.023**	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.0097 U	0.0098 U
PCB-1254	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	**0.017**	**0.045**	**0.015**	**0.046**	**0.016**	**0.048**	**0.029**	**0.072**	**0.0096 J**	0.0097 U	**0.013**	**0.039**	0.01 U	**0.29**	0.0097 U	0.0097 U	0.0097 U	**0.012**
PCB-1260	--	µg/L	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	**0.015**	0.01 U	0.01 U	0.01 U	0.01 U	0.0097 U	0.0097 U	0.0097 U	0.0098 U
TOTAL PCBs (ND=0)	0.014	µg/L	0	0	0	0	**0.017**	**0.045**	**0.015**	**0.046**	**0.016**	**0.048**	**0.029**	**0.072**	**0.0096**	**0.038**	**0.013**	**0.039**	0	**0.29**	0	0	0	**0.012**
TOTAL PCBs (ND=1/2RL)	0.014	µg/L	**0.035**	**0.035**	**0.035**	**0.035**	**0.047**	**0.075**	**0.045**	**0.076**	**0.046**	**0.078**	**0.059**	**0.102**	**0.039**	**0.062**	**0.043**	**0.069**	**0.035**	**0.32**	**0.034**	**0.034**	**0.034**	**0.041**
TOTAL PCBs (ND=RL)	0.014	µg/L	**0.07**	**0.07**	**0.07**	**0.07**	**0.077**	**0.105**	**0.075**	**0.106**	**0.076**	**0.108**	**0.089**	**0.132**	**0.068**	**0.087**	**0.073**	**0.099**	**0.07**	**0.35**	**0.068**	**0.068**	**0.068**	**0.071**

Notes:

Source: (a) 25 Pa. Code § 93.7.

*Total PCBs based on summation of individual Aroclors. Total PCB criterion compared to total fraction of the modified elutriates.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

RL is reported for non-detected constituents.

-- = No value available.

µg/L = Microgram(s) per liter.

RL = Laboratory reporting limit.

TSS = Total suspended solids.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

Analyte concentration exceeds Pennsylvania Surface Water Continuous Concentration Standards.

This page left intentionally blank

TABLE 6-6. CHLORINATED PESTICIDE CONCENTRATIONS (µg/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

				Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Maximum[(a)]	Continuous[(a)]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
4,4'-DDD	µg/L	1.1	0.001	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	**0.00098 J**	**0.0042**	0.014 U	**0.0059 J**	0.0013 U	0.0013 U	**0.0014**	**0.019**	0.0013 U	0.013 U	0.0013 U	0.013 U
4,4'-DDE	µg/L	1.1	0.001	0.0013 U	0.0013 U	0.0013 U	0.0013 U	**0.0013**	**0.0042**	**0.0011 J**	**0.0024**	**0.0017**	**0.003**	**0.0021**	**0.008**	0.014 U	**0.024**	0.0013 U	**0.0028**	0.0013 U	**0.017**	0.0013 U	0.013 U	0.0013 U	0.013 U
4,4'-DDT	µg/L	1.1	0.001	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
CHLORINATED PESTICIDES																									
ALDRIN	µg/L	3	0.1	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
ALPHA-BHC	µg/L	--	--	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
BETA-BHC	µg/L	--	--	0.0013 U	**0.0021**	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
CHLORDANE (TECHNICAL)	µg/L	2.4	0.0043	0.013 U	0.013 U	0.013 U	0.013 U	0.012 U	0.012 U	0.012 U	0.012 U	0.012 U	0.012 U	0.012 U	0.012 U	0.13 U	0.13 U	0.012 U	0.012 U	0.012 U	0.012 U	0.013 U	0.13 U	0.013 U	0.13 U
CHLOROBENSIDE	µg/L	--	--	0.0032 U	0.0032 U	0.0032 U	0.0032 U	0.0032 U	0.0031 U	0.0031 U	0.0032 U	0.0031 U	0.0031 U	0.0031 U	0.0031 U	0.034 U	0.032 U	0.0031 U	0.0031 U	0.0031 U	0.0031 U	0.0032 U	0.032 U	0.0032 U	0.032 U
DACHTAL (DCPA)	µg/L	--	--	0.0025 U	0.0025 U	0.0025 U	0.0025 U	0.0025 U	0.0025 U	0.0025 U	0.0025 U	0.0024 U	0.0025 U	0.0024 U	0.0024 U	0.027 U	0.025 U	0.0024 U	0.0024 U	0.0024 U	0.0025 U	0.0025 U	0.025 U	0.0025 U	0.025 U
DELTA-BHC	µg/L	--	--	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
DIELDRIN	µg/L	0.24	0.056	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
ENDOSULFAN I	µg/L	0.22	0.056	**0.001 J**	**0.0013**	**0.0011 J**	**0.0013**	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
ENDOSULFAN II	µg/L	0.22	0.056	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
ENDOSULFAN SULFATE	µg/L	--	--	**0.0014**	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
ENDRIN	µg/L	0.086	0.036	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
ENDRIN ALDEHYDE	µg/L	--	--	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
GAMMA-BHC	µg/L	--	--	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
HEPTACHLOR	µg/L	0.52	0.0038	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
HEPTACHLOR EPOXIDE	µg/L	0.5	0.0038	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	**0.00057 J**	0.0013 U	0.0013 U	0.0013 U	**0.0011 J**	0.014 U	0.013 U	0.0013 U	**0.00048 J**	0.0013 U	**0.002**	0.0013 U	0.013 U	0.0013 U	0.013 U
METHOXYCHLOR	µg/L	--	--	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
MIREX	µg/L	--	--	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.014 U	0.013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.0013 U	0.013 U	0.0013 U	0.013 U
TOXAPHENE	µg/L	0.73	0.0002	0.1 U	0.1 U	0.1 U	0.1 U	0.099 U	0.098 U	0.098 U	0.099 U	0.097 U	0.098 U	0.097 U	0.097 U	1.1 U	1 U	0.097 U	0.097 U	0.097 U	0.098 U	0.1 U	1 U	0.1 U	1 U

Notes:

Sources: (a) 25 Pa. Code § 93.7.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

Pennsylvania maximum standards are comparable to USEPA acute criteria and Pennsylvania continuous standards are comparable to USEPA chronic criteria.

-- = No value available.

µg/L = Microgram(s) per liter.

mg/L = Milligram(s) per liter.

RL = Laboratory reporting limit.

TSS = Total suspended solids.

B = Compound was detected in the laboratory method blank.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

P = The percent difference between the original and confirmation analysis is greater than 40%.

Analyte concentration exceeds Pennsylvania Surface Water Continuous Concentration Standards.

This page left intentionally blank

TABLE 6-7. SEMIVOLATILE ORGANIC COMPOUND CONCENTRATIONS (µg/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

				Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
		PA SW	PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	Maximum[b]	Continuous[b]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
1,2,4-TRICHLOROBENZENE	µg/L	130	26	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
1,2-DICHLOROBENZENE	µg/L	820	160	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
1,2-DIPHENYLHYDRAZINE(AS AZOBENZENE)	µg/L	15	3	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
1,3-DICHLOROBENZENE	µg/L	350	69	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
1,4-DICHLOROBENZENE	µg/L	730	150	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2,2'-OXYBIS[1-CHLOROPROPANE]	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
2,4,6-TRICHLOROPHENOL	µg/L	460	91	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2,4-DICHLOROPHENOL	µg/L	1,700	340	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
2,4-DIMETHYLPHENOL	µg/L	660	130	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2,4-DINITROPHENOL	µg/L	660	130	11 U	10 U	11 U	11 U	10 U	10 U	10 U	10 U	11 U	11 U	11 U	10 U	22 U	10 U	10 U	11 U	11 U	11 U	11 U	11 U	11 U	230 U
2,4-DINITROTOLUENE	µg/L	1,600	320	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2,6-DINITROTOLUENE	µg/L	990	200	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2-CHLORONAPHTHALENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
2-CHLOROPHENOL	µg/L	560	110	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2-METHYLPHENOL	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
2-NITROPHENOL	µg/L	8,000	1,600	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	**1.1**	1.1 U	1.1 U	1.1 U	1.1 U	23 U
3,3'-DICHLOROBENZIDINE	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
4,6-DINITRO-2-METHYLPHENOL	µg/L	80	16	5.7 U	5 U	5.7 U	5.7 U	5.2 U	5.2 U	5.2 U	5 U	5.4 U	5.7 U	5.7 U	5.2 U	11 U	5.2 U	5 U	5.7 U	5.4 U	5.7 U	5.7 U	5.7 U	5.4 U	110 U
4-BROMOPHENYL PHENYL ETHER	µg/L	270	54	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
4-CHLORO-3-METHYLPHENOL	µg/L	160	30	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
4-CHLOROPHENYL PHENYL ETHER	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
4-NITROPHENOL	µg/L	2,300	470	5.7 U	5 U	5.7 U	5.7 U	5.2 U	5.2 U	5.2 U	5 U	5.4 U	5.7 U	5.7 U	5.2 U	11 U	5.2 U	5 U	5.7 U	**2.4 J**	5.7 U	5.7 U	5.7 U	5.4 U	110 U
BENZIDINE	µg/L	300	59	23 U	20 U	23 U	23 U	21 U	21 U	21 U	20 U	22 U	23 U	23 U	21 U	43 U	21 U	20 U	23 U	22 U	23 U	23 U	23 U	22 U	450 U
BENZOIC ACID	µg/L	--	--	5.7 U	5 U	5.7 U	5.7 U	5.2 U	5.2 U	5.2 U	5 U	5.4 U	5.7 U	5.7 U	5.2 U	11 U	5.2 U	5 U	5.7 U	5.4 U	5.7 U	5.7 U	5.7 U	5.4 U	110 U
BENZYL ALCOHOL	µg/L	--	--	5.7 U	5 U	5.7 U	5.7 U	5.2 U	5.2 U	5.2 U	5 U	5.4 U	5.7 U	5.7 U	5.2 U	11 U	5.2 U	5 U	5.7 U	5.4 U	5.7 U	5.7 U	5.7 U	5.4 U	110 U
BIS(2-CHLOROETHOXY)METHANE	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
BIS(2-CHLOROETHYL)ETHER	µg/L	30,000	6,000	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
BIS(2-ETHYLHEXYL) PHTHALATE	µg/L	4,500	910	11 U	10 U	11 U	11 U	**13**	10 U	10 U	10 U	11 U	**52**	11 U	10 U	22 U	10 U	10 U	11 U	11 U	11 U	11 U	11 U	11 U	230 U
BUTYL BENZYL PHTHALATE	µg/L	140	35	2.3 U	2 U	2.3 U	2.3 U	2.1 U	2.1 U	2.1 U	2 U	2.2 U	2.3 U	2.3 U	2.1 U	4.3 U	2.1 U	2 U	2.3 U	2.2 U	2.3 U	2.3 U	2.3 U	2.2 U	45 U
DIBENZOFURAN	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	**0.22 J**	**0.24 J**	**0.25 J**	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	**0.46 J**	23 U
DIETHYL PHTHALATE	µg/L	4,000	800	1.1 U	1 U	**0.67 J**	1.1 U	1 U	1 U	**0.97 J**	**2.4**	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
DIMETHYL PHTHALATE	µg/L	2,500	500	2.3 U	2 U	2.3 U	2.3 U	2.1 U	2.1 U	2.1 U	2 U	2.2 U	2.3 U	2.3 U	2.1 U	4.3 U	2.1 U	2 U	2.3 U	**0.23 J**	2.3 U	2.3 U	2.3 U	2.2 U	45 U
DI-N-BUTYL PHTHALATE	µg/L	110	21	11 U	10 U	11 U	11 U	10 U	10 U	10 U	10 U	11 U	11 U	11 U	10 U	22 U	10 U	10 U	11 U	11 U	11 U	11 U	11 U	11 U	230 U
DI-N-OCTYL PHTHALATE	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	**1.1**	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
HEXACHLOROBENZENE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
HEXACHLOROBUTADIENE	µg/L	10	2	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
HEXACHLOROCYCLOPENTADIENE	µg/L	5	1	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
HEXACHLOROETHANE	µg/L	60	12	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	**0.35 J**	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
ISOPHORONE	µg/L	10,000	2,100	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
METHYLPHENOL, 3 & 4	µg/L	800	160	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	**0.43 J**	2.2 U	1 U	1 U	1.1 U	1.1 U	**0.66 J**	1.1 U	1.1 U	1.1 U	23 U
NITROBENZENE	µg/L	4,000	810	2.3 U	2 U	2.3 U	2.3 U	2.1 U	2.1 U	2.1 U	2 U	2.2 U	2.3 U	2.3 U	2.1 U	4.3 U	2.1 U	2 U	2.3 U	2.2 U	2.3 U	2.3 U	2.3 U	2.2 U	45 U
N-NITROSODIMETHYLAMINE	µg/L	17,000	3,400	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
N-NITROSODI-N-PROPYLAMINE	µg/L	--	--	0.22 U	0.19 U	0.22 U	0.22 U	0.2 U	0.2 U	0.2 U	0.19 U	0.21 U	0.22 U	0.22 U	0.2 U	0.41 U	0.2 U	0.19 U	0.22 U	0.21 U	0.22 U	0.22 U	0.22 U	0.21 U	4.3 U
N-NITROSODIPHENYLAMINE	µg/L	300	59	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1 U	2.2 U	1 U	1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	1.1 U	23 U
PENTACHLOROPHENOL	µg/L	16[(1)]	12.2[(1)]	5.7 U	5 U	5.7 U	5.7 U	5.2 U	5.2 U	5.2 U	5 U	5.4 U	5.7 U	5.7 U	5.2 U	11 U	5.2 U	5 U	5.7 U	5.4 U	5.7 U	5.7 U	5.7 U	5.4 U	110 U
PHENOL	µg/L	--	--	1.1 U	1 U	1.1 U	1.1 U	1 U	1 U	1 U	1 U	**3.7**	1.1 U	**3.2**	1 U	2.2 U	1 U	1 U	1.1 U	**12**	1.1 U	**1.3**	1.1 U	**2.1**	23 U

Notes:

Sources: (a) 25 Pa. Code § 93.7. (1) = pH dependent screening criteria.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

Pennsylvania standards for pentachlorophenol calculated using pH-based equation. Site water pH at PNYRB24-WAT was used in calculation.

Pennsylvania maximum standards are comparable to USEPA acute criteria and Pennsylvania continuous standards are comparable to USEPA chronic criteria.

-- = No value available.

µg/L = Microgram(s) per liter.

mg/L = Milligram(s) per liter. J = Compound was detected, but below the reporting limit (value is estimated).

RL = Laboratory reporting limit. U = Compound was analyzed, but not detected.

TSS = Total suspended solids.

This page left intentionally blank

TABLE 6-8. VOLATILE ORGANIC COMPOUND CONCENTRATIONS (µg/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

				Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
			PA SW	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	PA SW Maximum[a]	Continuous[a]	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
1,1,1-TRICHLOROETHANE	µg/L	3,000	610	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,1,2,2-TETRACHLOROETHANE	µg/L	1,000	210	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,1,2-TRICHLOROETHANE	µg/L	3,400	680	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,1-DICHLOROETHANE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,1-DICHLOROETHENE	µg/L	7,500	1,500	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,2,4-TRICHLOROBENZENE	µg/L	130	26	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,2-DICHLOROBENZENE	µg/L	820	160	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,2-DICHLOROETHANE	µg/L	15,000	3,100	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,2-DICHLOROPROPANE	µg/L	11,000	2,200	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,3-DICHLOROBENZENE	µg/L	350	69	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
1,4-DICHLOROBENZENE	µg/L	730	150	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
2-BUTANONE (MEK)	µg/L	230,000	32,000	5 U	5 U	5 U	5 U	5 U	**5.2**	**3.1 J**	**2.9 J**	5 U	5 U	**4.1 J**	**3.7 J**	5 U	5 U	5 U	5 U	**4.4 J**	5 U	5 U	5 U	5 U	5 U
2-CHLOROETHYL VINYL ETHER	µg/L	18,000	3,500	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U	2 U
ACROLEIN	µg/L	3	3	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U
ACRYLONITRILE	µg/L	650	130	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U	20 U
BENZENE	µg/L	640	130	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
BROMOFORM	µg/L	1,800	370	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
BROMOMETHANE	µg/L	550	110	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
CARBON TETRACHLORIDE	µg/L	2,800	560	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
CHLORODIBROMOMETHANE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
CHLOROETHANE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
CHLOROFORM	µg/L	1,900	390	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
CHLOROMETHANE	µg/L	28,000	5,500	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
CIS-1,3-DICHLOROPROPENE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
DICHLOROBROMOMETHANE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
DICHLORODIFLUOROMETHANE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
ETHYLBENZENE	µg/L	2,900	580	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
METHYLENE CHLORIDE	µg/L	12,000	2,400	**16**	1 U	**14**	1 U	**16**	**15**	**16**	**13**	**13**	**15**	**15**	**16**	**8.9**	**13**	**22**	**21**	**23**	**23**	**12**	**12**	**15**	**14**
TETRACHLOROETHENE	µg/L	700	140	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
TOLUENE	µg/L	1,700	330	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
TRANS-1,2-DICHLOROETHENE	µg/L	6,800	1,400	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
TRANS-1,3-DICHLOROPROPENE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
TRICHLOROETHENE	µg/L	2,300	450	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
TRICHLOROFLUOROMETHANE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U
VINYL CHLORIDE	µg/L	--	--	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U	1 U

Notes:

Source: (a) 25 Pa. Code § 93.7.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

Pennsylvania maximum standards are comparable to USEPA acute criteria and Pennsylvania continuous standards are comparable to USEPA chronic criteria.

-- = No value available.

µg/L = Microgram(s) per liter.

mg/L = Milligram(s) per liter.

RL = Laboratory reporting limit.

TSS = Total suspended solids.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

This page left intentionally blank

TABLE 6-9. RADIOACTIVITY AND TOTAL PETROLEUM HYDROCARBON CONCENTRATIONS IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

ANALYTE	UNITS	USEPA Screening Value (Maximum) [(a)]	DRBC Screening Value (Maximum) [(b)]	Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
				Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
				TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
RADIOACTIVITY																									
GROSS ALPHA	pCi/L	15	3	--	2.61 U	--	2.82 U	**2.79**	**25.8**	**4.27**	**87.3**	2.7 U	**29.5**	3.23 U	**27.5**	**3.57**	**20**	3.36 U	**16.9**	2.49 U	**15.1**	2.63 U	**4.61**	3.39 U	2.74 U
GROSS BETA	pCi/L	50	1,000	--	**3.8**	--	**3.96**	**6.82**	**34.9**	**7.79**	**35.1**	**4.72**	**29.7**	**4.89**	**31**	**5.48**	**18.6**	**4.65**	**20**	**3.34**	**24.5**	**3.45**	**5.28**	**3.8**	**5.43**
TOTAL PETROLEUM HYDROCARBONS																									
DRO (C10-C20)	mg/L	--	--	--	0.2 U	--	0.19 U	--	**0.24**	--	**0.11 J**	--	**0.16 J**	--	**0.41**	--	**0.34**	--	**0.072 J**	--	**2.3**	--	**0.58**	--	**0.77**
ORO (C20-C34)	mg/L	--	--	--	0.2 U	--	0.19 U	--	**0.79**	--	**0.48**	--	**0.74**	--	**0.88**	--	**0.77**	--	**0.43**	--	**3.1**	--	**0.93**	--	**1.2**
GRO (C6-C10)	µg/L	--	--	--	50 U	--	50 U	--	50 U	--	50 U	--	50 U	--	50 U	--	50 U	--	50 U	--	50 U	--	**41 J**	--	50 U

Notes:

(a) USEPA Drinking water standards-Radionuclides Rule, 66 Federal Register 76708, 7 December 2000; Volume 65, No. 236.

(b) Delaware River Basin Commission, Water Quality Regulations, 18 CFR Part 410, December 4, 2013.

Bold values represent detected concentrations. Shaded concentrations exceed water quality standards.

-- = No value available.

µg/L = Microgram(s) per liter.

mg/L = Milligram(s) per liter.

pCi/L = Picocuries per liter.

RL = Laboratory reporting limit.

DRO = Diesel range organics.

GRO = Gasoline range organics.

TSS = Total suspended solids.

U = Compound was analyzed, but not detected.

Analyte concentration exceeds radioactivity drinking water standards and DRBC water quality standard.

Analyte concentration exceeds DRBC water quality standards

This page left intentionally blank

TABLE 6-10. DIOXIN/FURAN CONGENER CONCENTRATIONS (pg/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

			Site Water				Modified Elutriates																	
			PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
			Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
ANALYTE	UNITS	TEF*	TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1,000 mg/L		TSS = 570 mg/L		TSS = 1,200 mg/L		TSS = 1,400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
2,3,7,8-TCDD	pg/L	1	3.9 U	4 U	3.9 U	3.9 U	4 U	40 U	4 U	40 U	4 U	40 U	4 U	40 U	4 U	40 U	4.1 U	40 U	4.1 U	40 U	4.1 U	40 U	**8.8 JI**	4.1 U
1,2,3,7,8-PeCDD	pg/L	1	25 U	25 U	25 U	24 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	26 U	250 U	250 U	25 U
1,2,3,4,7,8-HxCDD	pg/L	0.1	25 U	25 U	25 U	24 U	**2.9 J**	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	26 U	250 U	**42 J**	25 U
1,2,3,6,7,8-HxCDD	pg/L	0.1	25 U	25 U	25 U	24 U	**6 J**	250 U	25 U	250 U	**3.6 J**	**29 J**	**4.1 J**	**32 J**	**4.2 J**	250 U	25 U	250 U	25 U	**40 JI**	26 U	250 U	**170 J**	**4.6 J**
1,2,3,7,8,9-HxCDD	pg/L	0.1	25 U	25 U	25 U	24 U	**4 J**	250 U	25 U	250 U	**3.9 J**	**25 JI**	**3.8 JI**	**29 J**	**2.5 J**	250 U	25 U	250 U	25 U	**27 JI**	26 U	250 U	**75 J**	**2.9 J**
1,2,3,4,6,7,8-HpCDD	pg/L	0.01	25 U	25 U	25 U	24 U	**170**	**580**	**82**	**330**	**130**	**920**	**140**	**950**	**120**	**500**	**39**	**260**	**22 J**	**740**	**9.1 J**	**71 J**	**3000**	**100**
OCDD	pg/L	0.0003	110 U	110 U	110 U	110 U	**5500**	**19000**	**2600**	**11000**	**4400**	**31000**	**3700**	**23000**	**3600**	**15000**	**1100**	**8000**	**380**	**13000**	**180**	**1500**	**25000**	**1400**
2,3,7,8-TCDF	pg/L	0.1	4.9 U	5 U	4.9 U	4.9 U	5 U	50 U	5 U	50 U	**1.2 J**	50 U	**1.7 JI**	**14 J**	**1.2 JI**	50 U	5.1 U	50 U	5.1 U	**20 J**	5.2 U	50 U	**62**	**1.6 J**
1,2,3,7,8-PeCDF	pg/L	0.03	25 U	25 U	25 U	24 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	26 U	250 U	**1700**	25 U
2,3,4,7,8-PeCDF	pg/L	0.3	25 U	25 U	25 U	24 U	25 U	250 U	25 U	250 U	25 U	250 U	**2.5 JI**	250 U	25 U	250 U	25 U	250 U	25 U	**34 J**	26 U	250 U	**220 J**	**4.6 J**
1,2,3,4,7,8-HxCDF	pg/L	0.1	25 U	25 U	25 U	24 U	**3.5 JI**	250 U	25 U	250 U	**2.8 JI**	250 U	**3.3 JI**	**26 J**	**3.5 JI**	250 U	25 U	250 U	25 U	**35 J**	26 U	250 U	**320**	**4.9 J**
1,2,3,6,7,8-HxCDF	pg/L	0.1	25 U	25 U	25 U	24 U	**4.2 J**	250 U	25 U	250 U	**2.9 J**	250 U	**3.1 JI**	250 U	**2.5 J**	250 U	25 U	250 U	25 U	**30 J**	26 U	250 U	**110 J**	**2.8 JI**
2,3,4,6,7,8-HxCDF	pg/L	0.1	25 U	25 U	25 U	24 U	**3.6 JI**	250 U	25 U	250 U	**2.5 J**	250 U	**2.7 JI**	250 U	25 U	250 U	25 U	250 U	25 U	250 U	26 U	250 U	**140 J**	**3.5 J**
1,2,3,7,8,9-HxCDF	pg/L	0.1	25 U	25 U	25 U	24 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	**29 J**	26 U	250 U	**32 J**	25 U
1,2,3,4,6,7,8-HpCDF	pg/L	0.01	25 U	25 U	25 U	24 U	**67**	**190 J**	**29 I**	**110 J**	**48**	**280**	**48**	**270**	**46**	**190 J**	**13 J**	**76 J**	**14 J**	**510**	**6.4 J**	**56 J**	**1900**	**51**
1,2,3,4,7,8,9-HpCDF	pg/L	0.01	25 U	25 U	25 U	24 U	**3.2 JI**	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	25 U	250 U	26 U	250 U	**73 J**	**2.6 J**
OCDF	pg/L	0.0003	49 U	50 U	49 U	49 U	**150**	**430 J**	**68 I**	**220 J**	**110**	**610**	**100**	**550**	**85**	**340 J**	**29 J**	**230 J**	**17 J**	**560**	**7.6 J**	500 U	**2100**	**67**
DIOXIN WHO TEQ (ND=0)	pg/L	--	0	0	0	0	**6.52**	**13.5**	**1.91**	**7.77**	**4.82**	**26.9**	**5.64**	**29.4**	**4.16**	**11.5**	**0.859**	**5.83**	**0.479**	**44.9**	**0.211**	**1.72**	**279**	**5.39**
DIOXIN WHO TEQ (ND=1/2RL)	pg/L	--	**28.0**	**28.0**	**28.0**	**26.9**	**26.6**	**291**	**29.7**	**285**	**26.1**	**279**	**23.1**	**267**	**26.7**	**289**	**28.7**	**283**	**28.3**	**220**	**29.0**	**279**	**404**	**22.8**
DIOXIN WHO TEQ (ND=RL)	pg/L	--	**55.9**	**56.0**	**55.9**	**53.9**	**46.8**	**569**	**57.4**	**563**	**47.3**	**532**	**40.6**	**504**	**49.2**	**567**	**56.5**	**561**	**56.1**	**395**	**57.9**	**557**	**529**	**40.2**

Notes:

* Source: Van den Berg, M, et al. 2006. The 2005 World Health Organization Re-evaluation of Human and Mammalian Toxic Equivalency Factors for Dioxins and Dioxin-Like Compounds. Toxicological Sciences 93(2):223-241.

Bold values represent detected concentrations. RL is reported for non-detected constituents.

-- = No value available.

pg/L = Picogram(s) per liter.

RL = Laboratory reporting limit.

TSS = Total suspended solids.

I or Q = Compound was detected, but as an estimated maximum concentration

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

TABLE 6-11. PER- AND POLYFLUOROALKYL SUBSTANCES CONCENTRATIONS (ng/L) IN SITE WATER AND MODIFIED ELUTRIATES

RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

ANALYTE	UNITS	PA MCL [(a)]	EPA Aquatic Life Criteria [(b)]	Site Water				Modified Elutriates																	
				PNYRB24-WAT		PNYRB24-WAT-FD		PNYRB24-A-0102-MET		PNYRB24-A-0304-MET		PNYRB24-A-0506-MET		PNYRB24-A-0708-MET		PNYRB24-A-0910-MET		PNYRB24-A-1112-MET		PNYRB24-B-2-MET		PNYRB24-B-4-MET		PNYRB24-B-5-MET	
				Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total	Dissolved	Total
				TSS = 2.5 mg/L		TSS = 2.6 mg/L		TSS = 1000 mg/L		TSS = 570 mg/L		TSS = 1200 mg/L		TSS = 1400 mg/L		TSS = 750 mg/L		TSS = 390 mg/L		TSS = 830 mg/L		TSS = 34 mg/L		TSS = 140 mg/L	
HEXAFLUOROPROPYLENE OXIDE DIMER ACID (HFPO-DA)	ng/L	--	--	1.8 U	1.6 U	1.4 U	1.6 U	**1.3 J**	**1.6 J**	1.9 U	**1 J**	1.9 U	0.71 U	1.9 U	**0.82 J**	**2.2**	**1.9**	**1.2 J**	**0.91 J**	**0.83 J**	**1.1 J**	**0.91 J**	1.3 U	0.8 U	1.9 U
PERFLUOROBUTANESULFONIC ACID (PFBS)	ng/L	--	5,000,000	**5**	**5.6**	**5.4**	**4.8**	**5.1**	**5.3**	**4.6**	**5.1**	**5.3**	**5.1**	**4.7**	**4.8**	**5.2**	**5.3**	**5.4**	**5.2**	**5.5**	**5.5**	**4.6**	**5.1**	**4**	**4.2**
PERFLUOROHEXANESULFONIC ACID (PFHXS)	ng/L	--	210,000	**2.2**	**2.2**	**2**	**2**	**3**	**3.4**	**2.6**	**2.5**	**2.5**	**2.5**	**2.3**	**2.7**	**2.4**	**2.3**	**2.4**	**2.4**	**2.7**	**2.7**	**2.6**	**2.4**	**2.4**	**2.9**
PERFLUORONONANOIC ACID (PFNA)	ng/L	--	650,000	**2.5**	**2.6**	**2.7**	**2.7**	**44**	**45**	**15**	**17**	**13**	**14**	**7.6**	**7.6**	**4.8**	**5.3**	**2.9**	**3.1**	**19**	**19**	**4.3**	**3.7**	**5.4**	**5**
PERFLUOROOCTANESULFONIC ACID (PFOS)	ng/L	18	71,000	**4.5**	**4**	**4.8**	**4.4**	**28**	**31**	**14**	**16**	**19**	**20**	**14**	**15**	**11**	**12**	**7**	**7.9**	**11**	**11**	**8.9**	**8.9**	**10**	**11**
PERFLUOROOCTANOIC ACID (PFOA)	ng/L	14	3,100,000	**6.6**	**5.8**	**6.3**	**5.5**	**32**	**33**	**12**	**13**	**13**	**12**	**9**	**9.3**	**8.7**	**11**	**7.6**	**8.2**	**12**	**14**	**8**	**8.1**	**8.8**	**8.8**

Notes:

Bold values represent detected concentrations. RL is reported for non-detected constituents.

(a) 25 PA. CODE CH. 109. Safe Drinking Water PFAS MCL Rule 14 January 2023

(b) Final recommended aquatic life criteria and benchmarks for select PFAS. USEPA September 2024 (acute values)

-- = No value available.

ng/L = Nanogram(s) per liter.

RL = Laboratory reporting limit.

TSS = Total suspended solids.

J = Compound was detected, but below the reporting limit (value is estimated).

U = Compound was analyzed, but not detected.

Analyte concentration exceeds PA MCL Rule for PFAS Compounds

This page left intentionally blank

7. DATA SUMMARY AND PLACEMENT OPTIONS

Physical and chemical data for the Philadelphia Navy Yard Reserve Basin sediments were evaluated to determine viable placement options for the maintenance dredged material. A summary of the key findings is provided in this section.

7.1 MATERIAL PLACEMENT OPTIONS

Material placement options for the Reserve Basin maintenance dredging sediments include privately owned and operated facilities in the State of New Jersey and Commonwealth of Pennsylvania. Placement/disposal options include Biles Island CDF or Whites Rehandling Basin. Each placement option has facility-specific material acceptance criteria.

Key findings for the primary placement options are as follows:

<u>Whites Rehandling Basin</u>

A summary of the sediment concentrations that exceed the State of New Jersey non-residential soil criteria is provided in Table 7-1.

- Arsenic concentrations in one sample from dredging Area A (A-0708) and each of the three samples from dredging Area B (B-2, B-4, and B-5) exceeded the State of New Jersey non-residential soil standard of 19 mg/kg with concentrations ranging from 22 to 49 mg/kg.

- The lead concentration (5,900 mg/kg) in one sample in Area B (B-4) exceeded the New Jersey non-residential soil standard of 800 mg/kg.

<u>Biles Island CDF</u>

A summary of sediment concentrations that exceed the Commonwealth of Pennsylvania non-residential soil criteria and modified elutriate concentrations that exceed the Commonwealth of Pennsylvania surface water maximum/continuous water quality standards and other applicable standards (for radioactivity and PFAS) is provided in Table 7-1.

- The *lead* concentration (5,900 mg/kg) in one sediment sample from Area B (B-4) exceeded the Commonwealth of Pennsylvania non-residential soil standard of 1,000 mg/kg (0 to -2 ft).

- *Aluminum* concentrations in the total fraction of each of the nine modified elutriates exceeded Commonwealth of Pennsylvania maximum surface water quality standard (750 µg/L) by factors ranging from3.3 to 62.7. Cobalt concentrations exceeded the continuous surface water standard (19 µg/L) in the total fraction in six of the nine modified elutriate samples by factors ranging from 1.2 to 2.9.

- *4,4'-DDD* concentrations exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in the total fraction of three of the nine modified elutriates by factors ranging from 4.2 to 19. Concentrations did not exceed the Commonwealth of Pennsylvania maximum standard (1.1 µg/L) in any sample.

- *4,4'-DDE* concentrations exceeded the Commonwealth of Pennsylvania continuous standard (0.001 µg/L) in the total fraction of seven of the nine samples by factors ranging from 2.4 to 24. Concentrations did not exceed the Commonwealth of Pennsylvania maximum standard (1.1 µg/L) in any sample.

- Three individual PAHs [acenaphthene, phenanthrene, and benzo(a)anthracene] were detected at concentrations that exceeded the Commonwealth of Pennsylvania continuous standard in the total fractions of the modified elutriate. The a*cenaphthene* concentration in the total fraction of the modified elutriate at sample location B-5 (28 µg/L) exceeded the Commonwealth of Pennsylvania continuous standard (17 µg/L) but did not exceed the Pennsylvania maximum standard (83 µg/L). The *phenanthrene* concentration (10 µg/L) in the sample location B-5 exceeded both Commonwealth of Pennsylvania continuous (1 µg/L) and maximum (5 µg/L) standards. The *benzo(a)anthracene* concentration exceeded the Commonwealth of Pennsylvania continuous standard (0.1 µg/L) in the total fraction of six of the nine modified elutriates by factors ranging from 1.1 to 3.5.

- *Total PCB congener* concentrations (ND=0) in the total fractions of the modified elutriates ranged from 199 to 1,378 ng/L and exceeded the Commonwealth of Pennsylvania continuous standard (14 ng/L) by factors ranging from 14 to 98. Total PCBs calculated based on the sum of homologs had comparable results, ranging from 200 to 1,400 ng/L.

- *Total PCB Aroclor* concentrations (ND=0) ranged from 0 to 0.29 µg/L and exceeded the Commonwealth of Pennsylvania continuous standard (0.014 µg/L) in seven of the nine elutriates by factors ranging from 3 to 21.

- *Gross alpha* was detected in the total fraction of eight of the nine modified elutriates and in the dissolved fraction of three of the nine modified elutriates. Each of the eight total fraction concentrations exceeded the DRBC water quality criteria (3 pCi/L) and seven of the eight total fraction concentrations also exceeded the USEPA drinking water quality standard (15 pCi/L). Two of the three dissolved fraction concentrations for gross alpha exceeded the DRBC criterion but did not exceed the USEPA drinking water standard.

- *PFOS* concentrations exceeded the Pennsylvania MCL of 18 ng/L in both the dissolved and total modified elutriate fractions in two of the nine elutriates samples (A-0102 and A-0506) by factors ranging from 1.1 to 1.7. *PFOA* concentrations also exceeded the Pennsylvania MCL of 14 ng/L in both the dissolved and total fractions of one elutriate (A-0102) by factors of 2.3 and 2.4, respectively. None of the detected concentrations of PFAS exceeded the USEPA acute aquatic life criteria.

TABLE 7-1. SUMMARY OF SCREENING CRITERIA EXCEEDANCES BASED ON PLACEMENT OPTION
RESERVE BASIN MAINTENANCE DREDGING, PHILADELPHIA NAVY YARD, PHILADELPHIA, PENNSYLVANIA (DECEMBER 2024)

	Whites Rehandling Basin				Biles Island CDF											
	Sediment				Sediment				Elutriate [a]							
Location/Sample ID	NJ Non-Residential Soil Standards (mg/kg)		Detected Concentration (mg/kg)	Magnitude of Exceedance	PA Non-Residential Soil Standards 0 to 2 ft (mg/kg)		Detected Concentration (mg/kg)	Magnitude of Exceedance	PA Maximum WQS - Acute (µg/L)		Detected Concentration (µg/L)	Magnitude of Exceedance	PA Continuous WQS - Chronic (µg/L)		Detected Concentration (µg/L)	Magnitude of Exceedance
PNYRB24-A-0102	--	--	--	--	--	--	--	--	Aluminum (total)	750	47000	62.7	Cobalt (total)	19	39	2.1
													4,4-DDE	0.001	0.0042	4.2
													Total PCBs [b]	0.014	0.405	28.9
PNYRB24-A-0304	--	--	--	--	--	--	--	--	Aluminum (total)	750	24000	32	Cobalt (total)	19	22	1.2
													4,4-DDE	0.001	0.0024	2.4
													Total PCBs [b]	0.014	0.229	16.4
PNYRB24-A-0506	--	--	--	--	--	--	--	--	Aluminum (total)	750	40000	53.3	Cobalt (total)	19	41	2.2
													Benzo(a)anthracene	0.1	0.15	1.5
													4,4-DDE	0.001	0.003	3
													Total PCBs [b]	0.014	0.37	26.4
PNYRB24-A-0708	Arsenic	19	28	1.5	--	--	--	--	Aluminum (total)	750	46000	61.3	Cobalt (total)	19	38	2
													Benzo(a)anthracene	0.1	0.11	1.1
													4,4-DDD	0.001	0.0042	4.2
													4,4-DDE	0.001	0.008	8
													Total PCBs [b]	0.014	0.911	65.1
PNYRB24-A-0910	--	--	--	--	--	--	--	--	Aluminum (total)	750	46000	61.3	Cobalt (total)	19	56	2.9
													Benzo(a)anthracene	0.1	0.11	1.1
													4,4-DDD	0.001	0.0059	5.9
													4,4-DDE	0.001	0.024	24
													Total PCBs [b]	0.014	0.685	48.9
PNYRB24-A-1112	--	--	--	--	--	--	--	--	Aluminum (total)	750	13000	17.3	Benzo(a)anthracene	0.1	0.13	1.3
													4,4-DDE	0.001	0.0028	2.8
													Total PCBs [b]	0.014	0.199	14.2
PNYRB24-B-2	Arsenic	19	49	2.6	--	--	--	--	Aluminum (total)	750	35000	46.7	Cobalt (total)	19	38	2
													Phenanthrene	1	1.1	1.1
													Benzo(a)anthracene	0.1	0.35	3.5
													4,4-DDD	0.001	0.019	19
													4,4-DDE	0.001	0.017	17
													Total PCBs [b]	0.014	1.378	98.4
PNYRB24-B-4	Arsenic	19	44	2.3	Lead	1000	5900	5.9	Aluminum (total)	750	7300	9.7	Benzo(a)anthracene	0.1	0.16	1.6
	Lead	800	5900	7.4									Total PCBs	0.014	0.209	14.9
PNYRB24-B-5	Arsenic	19	22	1.2	--	--	--	--	Aluminum (total)	750	2500	3.3	Acenaphthene	17	28	1.6
									Phenanthrene	5	10	2	Phenanthrene	1	10	10
													Total PCBs [b]	0.014	0.204	14.6

(a) total fraction for organics compared to surface water quality criteria; comparisons for metals based fraction (total or dissolved) specified in surface wtaer quality standards
(b) Total PCB Congeners (ND=0) and total PCB homologs
PA = Pennsylvania
NJ = New Jersey
WQS = water quality standards

eProjects WO#: 1783178

Appendix "B"

Permits

FY26 MAINTENANCE DREDGING AT:
RESERVE BASIN
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA

DEPARTMENT OF THE ARMY
U.S. ARMY CORPS OF ENGINEERS, PHILADELPHIA DISTRICT
1650 ARCH STREET, FIFTH FLOOR
PHILADELPHIA PA 19103-2004

Reply To
Attention Of

April 17, 2025

Regulatory Branch

SUBJECT: Nationwide Permit 35 Verification NAP-2024-00345-46
NAVAL RESERVE BASIN DREDGING
NJDEP permit #
Central Coordinates: Latitude: 39.894461°; Longitude: -75.185770°

Mr. Andrew Moyer
United States Navy
4921 South Broad Street, Building 1
Philadelphia, Pennsylvania 19112

Dear Mr. Moyer:

This letter is written in regard to your proposal to perform mechanical maintenance dredging of the existing naval reserve basin and access channel. A maximum of 100,000 cubic yards of materials are to be dredged from 19.8 acres of basin and access channel bottom to a design depth of 30 feet below mean lower low water with 2 foot allowance of over-dredging. This project is located in the Philadelphia Naval Reserve Basin (a.k.a. the Philadelphia Navy Yard), in the City of Philadelphia, Pennsylvania. Dredged materials are to be disposed at either the Weeks Marine White's Basin Rehandling Facility (Logan township, Gloucester County, New Jersey) or the Biles Island disposal facility (Morrisville, Falls Township, Bucks County, Pennsylvania). Based upon our review of the information that has been provided, it has been determined that the proposed work is authorized by Department of the Army Nationwide Permit (NWP) 35 Marina Basin Maintenance Dredging pursuant to Section 10 of the Rivers and Harbors Act of 1899 (33 USC 403).

This verification of authorization under NWP 35, described in the Federal Register at 86 FR 2744 and 86 FR 73522, is based on your agreement to comply with the general conditions, regional conditions and project specific special conditions listed in this letter. Additionally, this approval is conditioned upon compliance with the regional conditions as described in the link below. Copies of the NWP descriptions, NWP general conditions and the NWP regional conditions for New Jersey can be found at:

2021 NWP Descriptions:
https://www.nap.usace.army.mil/Portals/39/docs/regulatory/nwp/2021/2021-52-NWPs-Descriptions.pdf

2021 NWP General Conditions:

https://www.nap.usace.army.mil/Portals/39/docs/regulatory/nwp/2021/2021%20Nationwide%20Permit%20General%20Conditions.pdf

2021 New Jersey Regional Conditions:
https://www.nap.usace.army.mil/Portals/39/docs/regulatory/nwp/2021/2021-NJ-Reg-Cond-Final.pdf

2021 Pennsylvania Regional Conditions:
https://www.nap.usace.army.mil/Portals/39/docs/regulatory/nwp/2021/2021-PA-Reg-Cond-Final.pdf?ver=hThKtDqkIbgwRJB-7cTyfQ%3d%3d

Federal permits require determination from the State that the activities are consistent with the State's coastal zone management (CZM) program if the activity is located within the State's coastal zone. Federal permits also require the State's certification of compliance with section 401 of the Clean Water Act through the receipt of a 401 Water Quality Certification (WQC) if the activity involves a Section 404 discharge. A WQC is not required because this activity does not involve Section 404 discharges of dredge or fill material into waters of the US. An individual CZM consistency concurrence is required and has been issued by the NJDEP. Therefore, no further action is needed as part of the Federal review of your project, provided that you comply with all the terms and conditions of this NWP.

This verification of NWP authorization is valid until the 2021 Nationwide Permits expire on **March 14, 2026**, unless the NWP authorization is modified, suspended, or revoked prior to this date. In the event that the NWP authorization is modified during that time period, this expiration date will remain valid, provided the activity complies with any subsequent modification of the NWP authorization.

Activities which have commenced (i.e. are under construction) or are under contract to commence in reliance upon an NWP will remain authorized provided the activity is completed within twelve months of the date of an NWP's expiration, modification, or revocation, unless discretionary authority has been exercised on a case-by-case basis to modify, suspend, or revoke the authorization in accordance with 33 CFR 330.4(e) and 33 CFR 330.5 (c) or (d). Activities completed under the authorization of an NWP which was in effect at the time the activity was completed continue to be authorized by that NWP.

Special Conditions:

1. All work performed in association with the subject project shall be conducted in accordance with the **enclosed** project plans E-1 through E-6 which includes a 4-page plan set prepared by Naval Facilities Engineering Command (NAVFAC) and entitled "SITE PLAN FOR: FY 2025 MAINTENANCE DREDGING of RESERVE BASIN

PHILADELPHIA NAVY YARD - PHILADELPHIA, PA," Work Order Number 1783178, identified as "APPROVED", Sheets V-001, -002, -101, -102, all un-revised but plotted April 4, 2025.

2. Construction activities shall not result in the disturbance or alteration of greater than 19.8 acres of navigable waters of the United States for the dredging of 100,000 cubic yards of sediment from the basin and access channel to a design depth of 30 feet below mean lower low water with 2 foot allowance of over-dredging.

3. Any deviation in construction methodology or project design from that shown on the enclosed project plans must be approved by this office, in writing, prior to performance of the work. All modifications to the enclosed project plans shall be approved, in writing, by this office. No work shall be performed prior to written approval of this office.

4. This office shall be notified prior to the commencement of authorized work by completing and signing the "Notification of Commencement" form (Enclosure 1). This office shall also be notified within 10 days of the completion of the authorized work by completing and signing the "Notification of Completion" form (Enclosure 2). Notification is required each time maintenance work is to be done under the terms of this Corps of Engineers permit.

5. You must allow representatives from this office to inspect the authorized activity at any time deemed necessary to ensure that it is being or has been accomplished in accordance with the terms and conditions of your general permit.

6. Any conditions in your State authorization required for compliance with the State CZM program and 401 WQC are conditions of this authorization by reference.

7. This approval is limited to a single dredging event.

8. To protect diadromous fish migrations and spawning, in-water work shall be avoided from **March 1 to June 30**.

9. No dredged materials shall be disposed at the Waste Management of Pennsylvania Biles Island dredged material disposal facility (Morristown, Falls Township, Bucks County, Pennsylvania) until this office has received copy of a final approval letter from the Pennsylvania Department of Environmental Protection (Waste Management).

10. No dredged materials shall be disposed at the Weeks Marine White's Basin dredged material rehandling facility (Logan Township, Gloucester County, New Jersey) until this office has received copy of a 401 Water Quality Certification from the New Jersey Department of Environmental Protection for such disposal. The copy of the

certification shall include copy of postal or other appropriate documentation that the sediment testing report for the dredged materials has been submitted to the National Marine Fisheries Service (Habitat and Ecological Services Division), The U.S. Fish and Wildlife Service (Galloway Office), and US Environmental Protection Agency (Region II). **It should be noted that as of the date of this letter, the usage of White's Basin is only approved until June 30, 2025 under Corps of Engineers permit NAP-2013-00696-86. The permittee shall check with this office or Weeks Marine after that point to ensure disposal is approved if necessary.**

11. An as-built dredging survey shall be submitted to this office within 6 months of the completion of the approved dredging .

12. During all basin dredging operations, a turbidity curtain shall be deployed to minimize turbidity into the Franklin Delano Roosevelt Park Shedbrook-Hollander Creeks tidal inlet along the northern edge of the Naval Reserve Basin. Such curtain shall be operated in functional condition during all periods of dredging.

Also **enclosed** with this NWP verification letter is a form seeking any comments, positive or otherwise, on the procedures, timeliness, fairness, etc. of the permit process (Enclosure 3). You may forward your comments along with the signed "Notification of Commencement" form or "Notification of Completion" form, following the directions provided on the form. If you should have any questions or concerns, please contact me at 215-656-6731, (office), 215-605-7029 (mobile) or david.j.caplan@usace.army.mil.

Sincerely,

David J. Caplan
Senior Staff Biologist

Enclosures

Copies Furnished:

Ms. Courtney Pacelli (EA Engineering)
Division of Land Resource Protection, NJDEP (Davis)
Greater Atlantic Region Field Office, NMFS HESD (Bourdon)
Greater Atlantic Region Field Office, NMFS PRD (Hopper)
Wetlands Branch, USEPA Region 3 (Mazzarella)
USCG, Fifth District (Doody)
Nautical Data Branch, NOAA
PADEP, Southeastern Regional Office (Ashberry)
PADEP, CZMA (Walderon)
Waste Management of PA (Kucowski)
Weeks Marine (Dickerson)

NOTIFICATION/CERTIFICATION OF WORK COMMENCEMENT FORM

Permit Number: NAP-2024-00345-46
State Permit #: ________________
Name of Permittee: United States Navy
Project Name: NAVAL RESERVE BASIN DREDGING
Waterway: Schuylkill River/Reserve Basin
County: Philadelphia State: Pennsylvania
Compensation/Mitigation Work Required: Yes ☐ No ☒

TO: U.S. Army Corps of Engineers, Philadelphia District
1650 Arch Street, Fifth Floor
Philadelphia, Pennsylvania 19107-3390
Attention: CENAP-OPR

I have received authorization to: Naval Reserve Basin, in the City of Philadelphia, Philadelphia County, Pennsylvania including disposal at either Whites Basin or Biles Island.

The work will be performed by:

Name of Person or Firm____

Address: ____

____

I hereby certify that I have reviewed the approved plans, have read the terms and conditions of the above referenced permit, and shall perform the authorized work in strict accordance with the permit document. The authorized work will begin on or about ___________ and should be completed on or about __________.

Please note that the permitted activity is subject to compliance inspections by the Army Corps of Engineers. If you fail to return this notification form or fail to comply with the terms or conditions of the permit, you are subject to permit suspension, modification, revocation, and/or penalties.

____________________________ ________________
Permittee (Signature and Date) Telephone Number

____________________________ ________________
Contractor (Signature and Date) Telephone Number

NOTE: This form shall be completed/signed and returned to the Philadelphia District Office a minimum of 10 days prior to commencing work.

NOTIFICATION/CERTIFICATION OF WORK COMPLETION/COMPLIANCE FORM

Permit Number: NAP-2024-00345-46
State Permit #: ________________
Name of Permittee: United States Navy
Name of Contractor: ________________________________
Project Name: NAVAL RESERVE BASIN DREDGING
Waterway: Schuylkill River/Reserve Basin
County: Philadelphia State: Pennsylvania

Within 10 days of completion of the activity authorized by this permit, please sign this certification and return it to the following address:

U.S. Army Corps of Engineers, Philadelphia District
1650 Arch Street, Fifth Floor
Philadelphia, Pennsylvania 19103-2004
Attention: CENAP-OP-R

Please note that the permitted activity is subject to a compliance inspection by an Army Corps of Engineers representative. If you fail to return this notification form or fail to perform work in compliance with the permit, you are subject to administrative, civil and/or criminal penalties. Further, the subject permit may be suspended or revoked.

The authorized work was commenced on __________________________.

The authorized work was completed on __________________________.

I hereby certify that the work authorized by the above referenced permit has been completed in accordance with the terms and conditions of the above noted permit.

_____________________________ Signature of Contractor

_____________________________ Signature of Permittee

Address:________________________

Address: ____________________________

____

____

Telephone Number:______________

Telephone Number:____________________

For project located in areas identified as shellfish habitat, you must include with this form a bill of lading; sales order or any other document(s) demonstrating non-polluting materials were purchased and utilized for your project. I hereby certify that I and/or my contractor have utilized non-polluting materials as defined in the above noted permit.

_____________________________ Signature of Contractor

_____________________________ Signature of Permittee

US Army Corps of Engineers
Philadelphia District

We are soliciting your views and comments concerning the processing of your Department of the Army permit application request. Any input, positive or otherwise, on procedures, timeliness, fairness, etc., would be appreciated.

Please write your comments in the space provided below and return to the Philadelphia District Regulatory Branch at PhiladelphiaDistrictRegulatory@usace.army.mil or if you do not have the means to return this form electronically you may print this document and mail to:

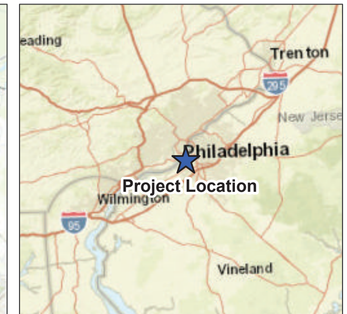
U.S. Army Corps of Engineers
Philadelphia District 1650
Arch Street, Fifth Floor
Philadelphia, PA 19103-2004
Attn: Regulatory Branch

FILE NUMBER: ______________________________
(Example CENAP-OP-R-2020-01234-56)

COMMENTS:

Thank you for taking the time to provide feedback which we can use to acknowledge great performance, correct problems and generally improve our business practices.

Enclosure 3



Legend

Area of Dredging

Potential Disposal Locations

E-1 Approved Plan
NAP-2024-00345-46

0 2.5 5
Miles

Map Date: 7/24/2024
Source: ESRI 2024
Projection: NAD 1983 State Plane Pennsylvania South US Foot

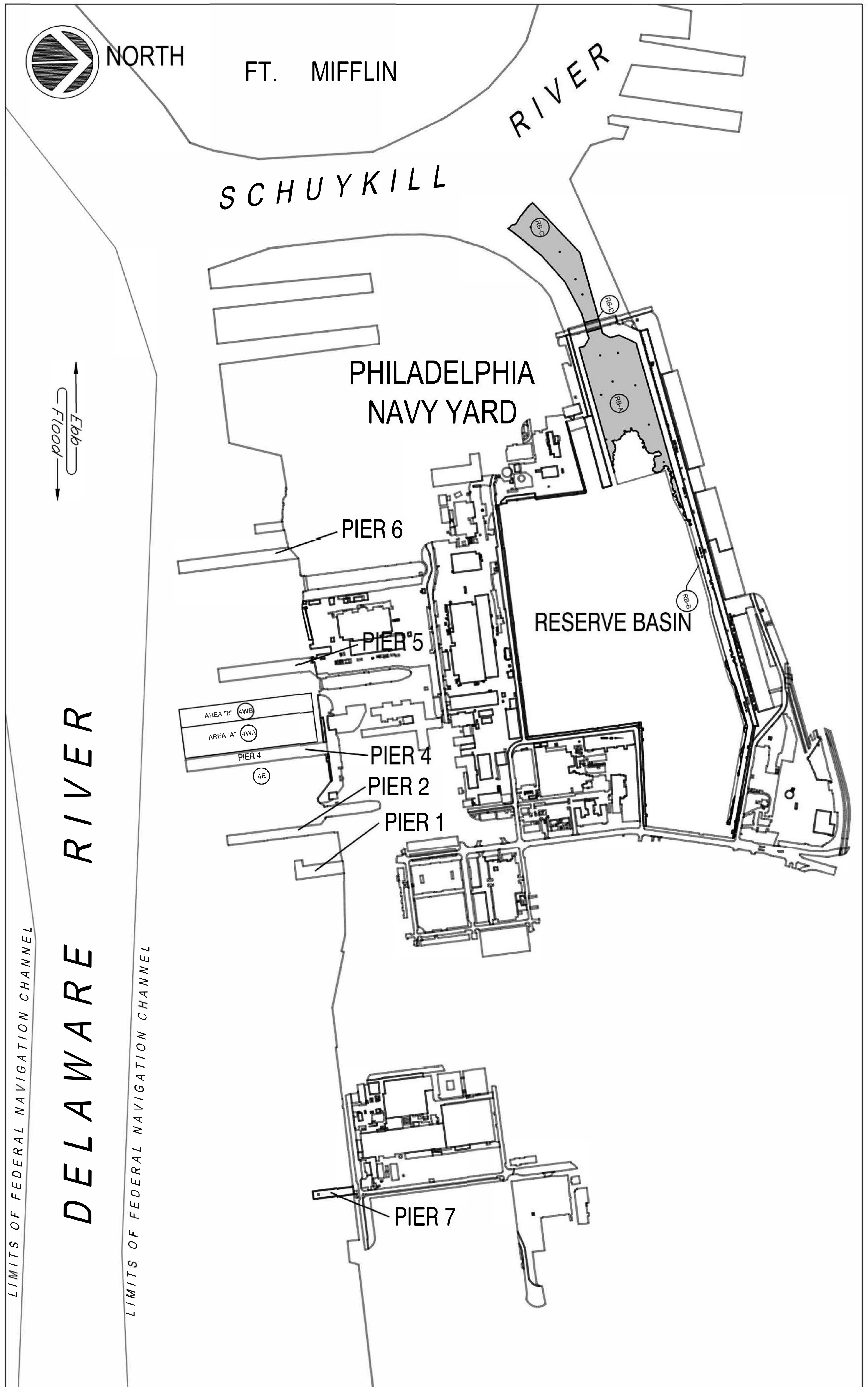
Philadelphia Navy Yard Reserve Basin *Philadelphia, Pennsylvania*

Dredged Material Placement Locations in Pennsylvania and New Jersey

Figure 2

Path: G:\Federal\Northeast\Pennsylvania\Philadelphia_NavyYard_Dredging_6337002\PROJECTS\NavyYardDredging\NavyYardDredging.aprx | Figure 1-3 Potential Disposal Locations in PA and NJ | 7/24/2024 | kburdick

SITE PLAN FOR:
FY 2025 MAINTENANCE DREDGING of RESERVE BASIN
PHILADELPHIA NAVY YARD - PHILADELPHIA, PA



KEY PLAN: INSET 'A'

SCALE: 1" = 600'

DREDGE AREA LABELS

DREDGE AREA ID

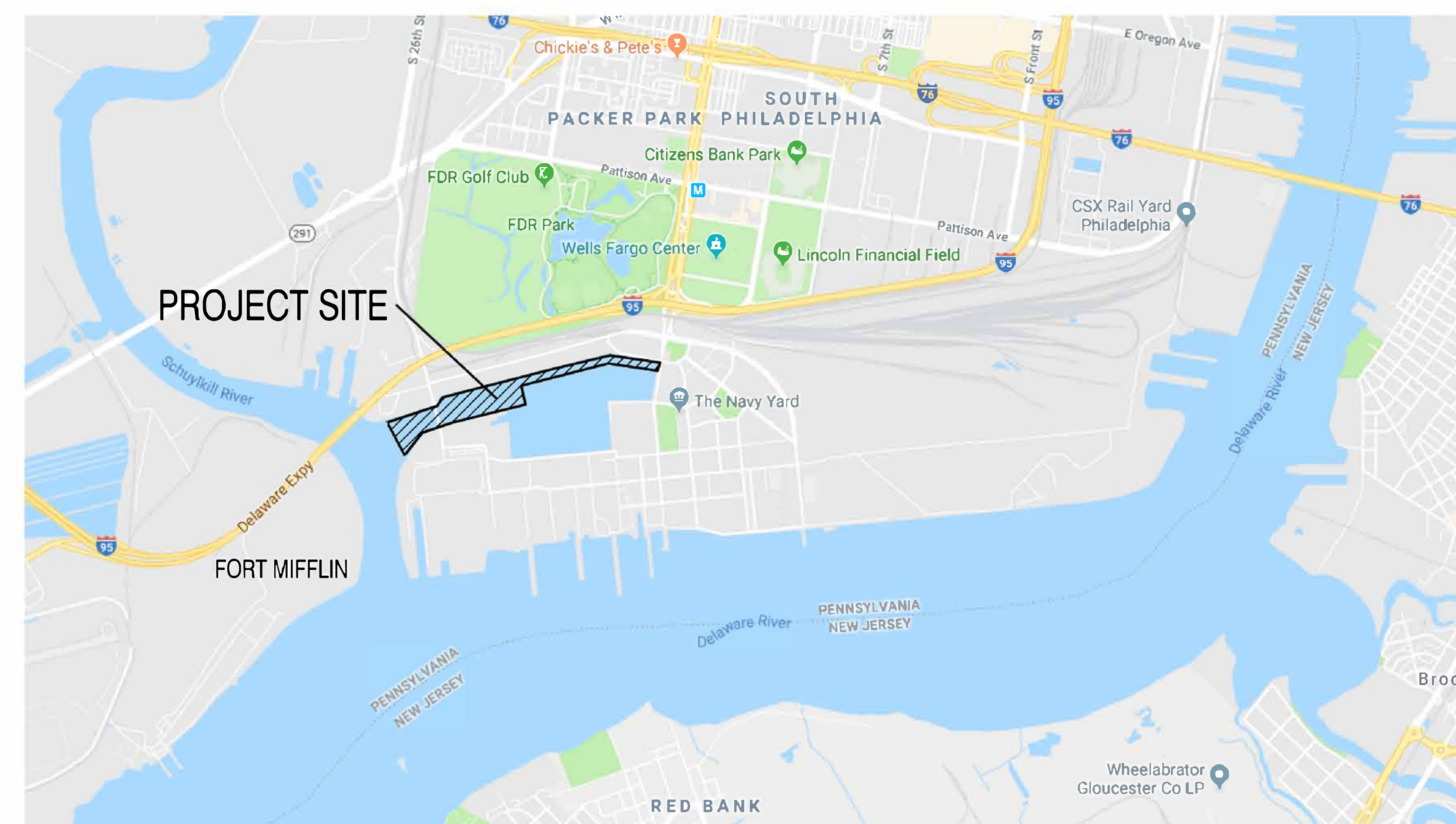
BASE BID

LIST OF SHEETS

	SHEET NO.	NAVFAC #	SHEET TITLE
1.	V-001	12902788	TITLE SHEET
2.	V-002	12902789	DETAIL SHEET
3.	V-101	12902790	RESERVE BASIN – SOUTH
4.	V-102	12902791	RESERVE BASIN – NORTH

HYDROGRAPHIC NOTES

1. VERTICAL DATUM - SOUNDINGS ARE IN FEET AND TENTHS AND ARE REFERRED TO THE NOAA MEAN LOWER LOW WATER DATUM (MLLW).
2. THE HYDROGRAPHIC SURVEY WAS CONDUCTED USING CLASS 1 HYDROGRAPHIC SURVEY METHODS & ACCURACIES OUTLINED IN THE CORPS OF ENGINEERS' HYDROGRAPHIC SURVEYING MANUAL (EM 1110-2-1003).
3. ALL BATHYMETRY WAS RECORDED USING A RESON T-50-P MULTI-BEAM SONAR SYSTEM(S) WITH HORIZONTAL POSTIONS OBTAINED VIA APPLANIX POS/MV GPS W/MRU.
4. HORIZONTAL DATUM IS BASED ON THE PENNSYLVANIA SOUTH (LAMBERT) STATE PLANE COORDINATE SYSTEM, ZONE 3702, U.S. SURVEY FEET (NAD 1983).
5. SOUNDING DATA FOR DISPLAY HAS BEEN PROCESSED USING HYPACK/HYSWEEP SOFTWARE AS FOLLOWS: CELL SIZE: 20' X 20', SOUNDING POSITION: CENTER OF CELL, CELL STATISTICS: AVERAGE DEPTH.
6. SOUNDING DATA FOR VOLUME CALCULATIONS HAS BEEN PROCESSED USING HYPACK/HYSWEEP SOFTWARE AS FOLLOWS: CELL SIZE: 5' X 5'; SOUNDING POSITION: CENTER OF CELL, CELL STATISTICS: AVERAGE DEPTH.
7. DREDGING WORK SHALL BE PERFORMED VIA SEALED ENVIRONMENTAL CLAMSHELL BUCKET. IF HARD MATERIAL IS ENCOUNTERED OR DEBRIS REMOVAL IS REQUIRED THEN AND STANDARD CLAMSHELL BUCKET MAY BE UTILIZED. ONCE HARD MATERIAL AND/OR DEBRIS HAS BEEN REMOVED THE CONTRACTOR SHALL SWITCH BACK TO USING THE SEALED ENVIRONMENTAL CLAMSHELL BUCKET.



LOCATION MAP

NOT TO SCALE

E-2 Approved Plan
NAP-2024-00345-46

APPROVED BY

CI-CORE HYDRO TEAM SUPERVISOR

AE LOGO

APPROVED

JAMES E. DONAHUE, R.A.
DESIGN PRODUCTION DIRECTOR
FOR COMMANDER NAVFAC
DATE

APPROVED

ACTIVITY - SATISFACTORY TO
DATE

PROJECT MGR SAM/JSG
DES JSG | DRW JSG | CHK JSG
BRANCH MGR DLB
CHIEF ENG/ARCH EJA
FIRE PROTECTION N/A

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID ATLANTIC
CAPITAL IMPROVEMENTS CORE - HYDROGRAPHIC ENGINEERING TEAM
NAVAL STATION - NORFOLK, VIRGINIA
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA
FY 25 MAINTENANCE DREDGING
RESERVE BASIN - PHILADELPHIA NAVY YARD
TITLE SHEET

CODE ID. NO. 80091 | SIZE D
SCALE: AS NOTED
MAXIMO NO.
JOB ORDER NO.
WORK ORDER NO. 1783178
CONSTR. CONTR. NO. N/A
NAVFAC DRAWING NO. 12902788
SHEET 1 OF 4

V-001

DRAWFORM REVISED 08-29-18

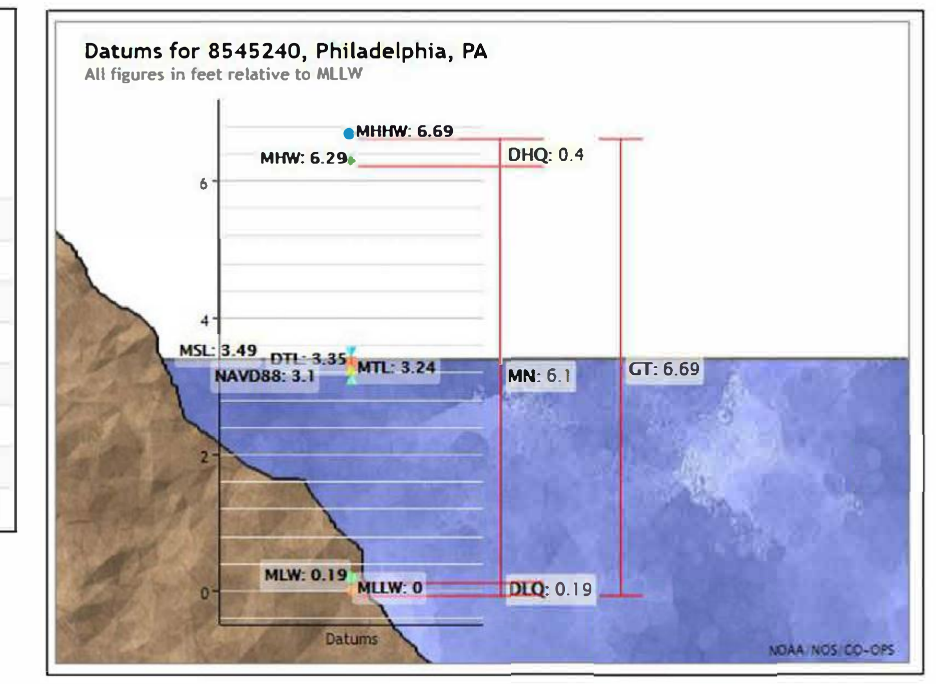
ESTIMATED DREDGING QUANTITIES FOR MAINTENANCE DREDGING AT PHILADELPHIA NAVY YARD

DREDGE AREA ID	SHEET	PIER AND/OR APPROACHES	AREA (SY)	CONTRACT DEPTH (FT) (MLLW)	OVER DREDGE DEPTH (FT) (MLLW)	ESTIMATED VOLUME AVAILABLE WITHIN DREDGE LIMITS (CY)		
BASE BID: PHILADELPHIA NAVY YARD MAINTENANCE DREDGING						CONTRACT DEPTH	OVERDREDGE DEPTH	TOTAL VOLUME
RB-A	V-101	RESERVE BASIN DREDGE AREA "A"	57,147	-30	-32	19,560	35,009	54,569
RB-B	V-102	RESERVE BASIN DREDGE AREA "B"	7,120	-30	-32	1,721	2,636	4,357
RB-C	V-102	RESERVE BASIN ACCESS CHANNEL - 26TH ST BRIDGE TO SCHUYKILL RIVER	30,355	-30	-32	21,070	17,563	38,633
RB-D	V-102	RESERVE BASIN ACCESS CHANNEL - UNDER 26TH ST BRIDGE FOOTPRINT	1,290	-30	-32	1,328	782	2,110
ESTIMATED VOLUME OF BASE BID:						43,679	55,990	99,669

NOTES:
1. ESTIMATED VOLUMES SHOWN HEREIN INCLUDE ADDITIONAL SILTATION EXPECTED TO OCCUR FROM THE DATE OF THE MOST RECENT CONDITION SURVEY TO THE EXPECTED DATE OF DREDGING (JANUARY 2025).
2. CONTRACT DEPTHS AND OVERDREDGE DEPTHS SHOWN HEREIN REFER TO NOAA MEAN LOWER LOW WATER DATUM (MLLW).

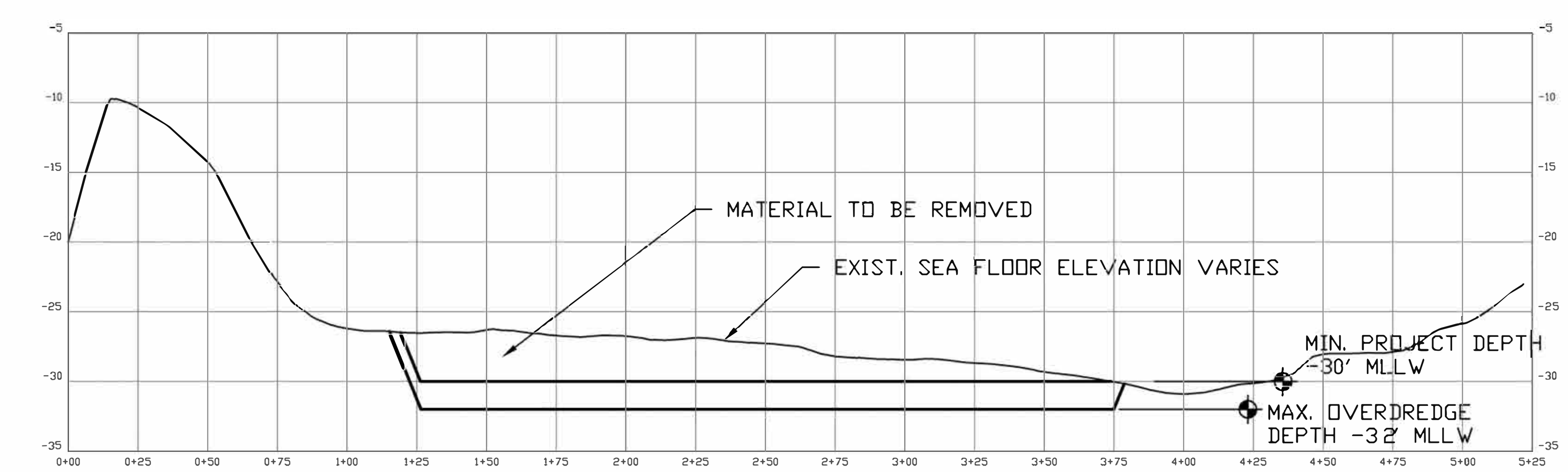
Elevations on Mean Lower Low Water
Station: 8545240, Philadelphia, PA
Status: Accepted (Dec 21 2012)
Units: Feet
Control Station: 8551910 Reedy Point, DE
T.M.: 0
Epoch: 1983-2001
Datum: MLLW

Datum	Value	Description
MHHW	6.69	Mean Higher-High Water
MHW	6.29	Mean High Water
MTL	3.24	Mean Tide Level
MSL	3.49	Mean Sea Level
DTL	3.35	Mean Diurnal Tide Level
MLW	0.19	Mean Low Water
MLLW	0.00	Mean Lower-Low Water
NAVD88	3.10	North American Vertical Datum of 1988



DREDGING AND DESIGN NOTES

1. SOUNDINGS SHOWN ON INDIVIDUAL SHEETS ARE FROM ACTUAL SURVEY DATES AS ANNOTATED ON EACH SHEET.
2. NAVAL STATION DREDGING AUTHORIZED UNDER C.O.E. PERMIT.
3. DREDGE LIMITS SHOWN ARE APPROXIMATIONS BASED ON ESTIMATED SILTATION. REQUIRED DREDGING AREA SHALL BE ESTABLISHED USING THE REQUIRED CONTRACT DEPTH CONTOUR AS DETERMINED BY PRE-DREDGE SURVEY PERFORMED BY THE GOVERNMENT.



A / V-002 TRANSVERSE SECTION THROUGH ACCESS CHANNEL DREDGE AREA
NOT TO SCALE

TURBIDITY BARRIERS/CURTAINS NOTES:

TURBIDITY CURTAIN(S) SHALL BE CONSTRUCTED AND INSTALLED IN ACCORDANCE WITH DETAIL SHOWN HEREIN. IN ADDITION, THE SILTATION BARRIERS SHALL BE AS FOLLOWS:

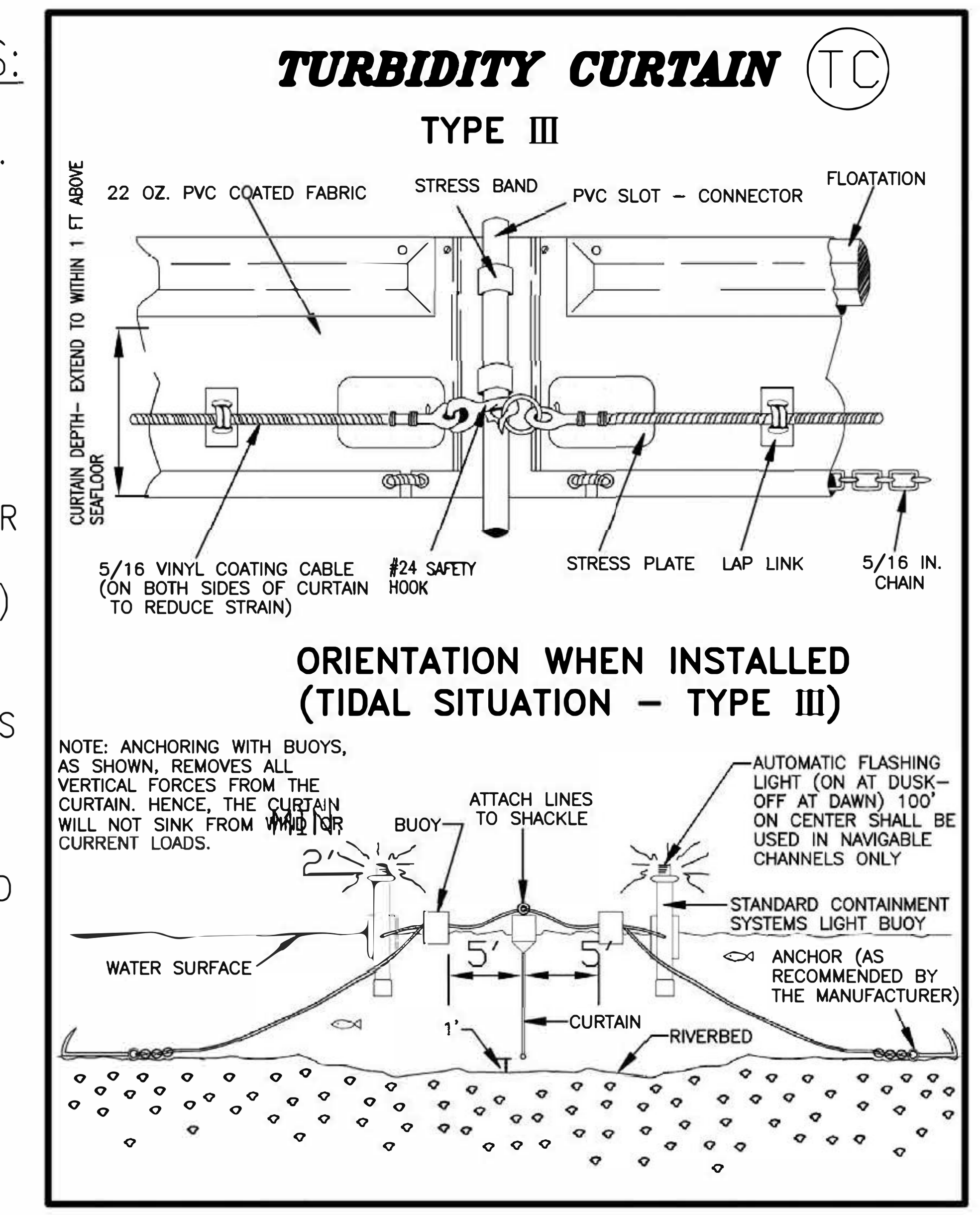
A. MADE OF A MATERIAL THAT IS UNLIKELY TO ENTANGLE ANY MARINE ANIMALS (I.E., REINFORCED IMPERMEABLE POLYCARBONATE VINYL FABRIC (PCV) COATED FABRIC

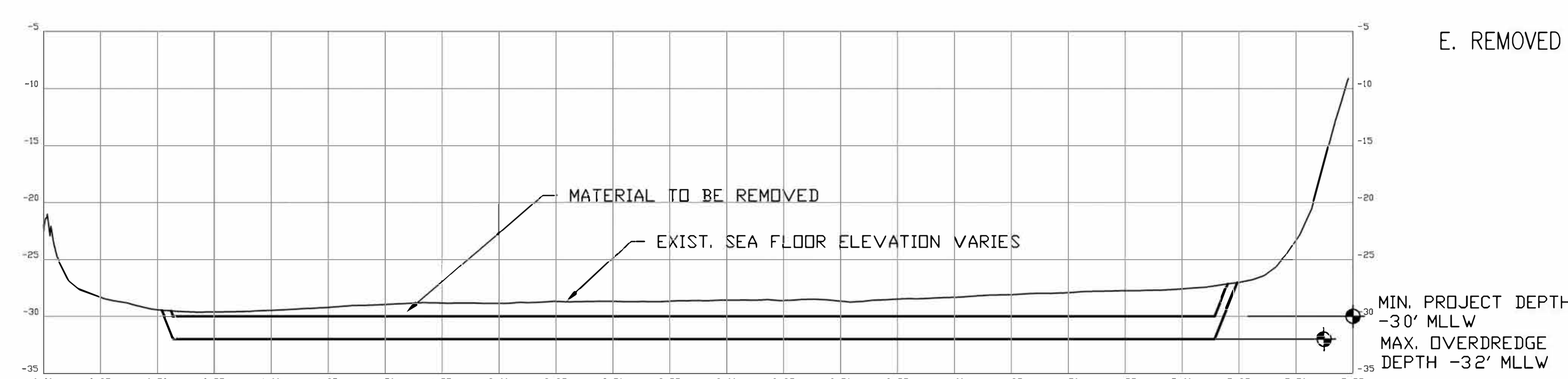
B. INSTALLED IN A MANNER IN WHICH A SEA TURTLE OR MARINE MAMMALS CANNOT BECOME EASILY ENTANGLED (I.E., STRETCHED OUT TIGHTLY WITH VERY LITTLE SLACK)

C. INSTALLED WITH THE MINIMUM EXTENT OF CURTAIN NEEDED (IN TERMS OF SURFACE TO BOTTOM HEIGHT, AS WELL AS TOTAL AREA SURROUNDED) WHILE STILL MEETING RWQCB REQUIREMENTS

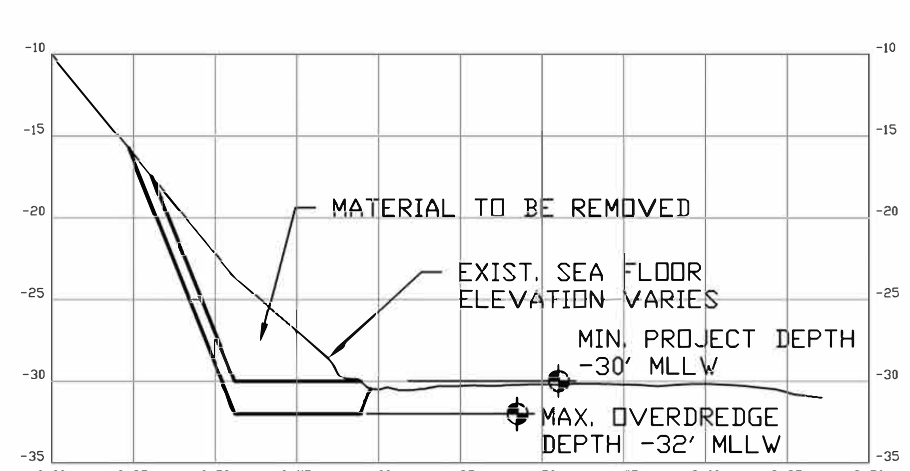
D. INSPECTED DAILY TO ENSURE PROPER INTEGRITY AND FOR THE PRESENCE OF ENTANGLED OR ENTRAPPED PROTECTED SPECIES

E. REMOVED IMMEDIATELY UPON PROJECT COMPLETION





B / V-002 TRANSVERSE SECTION THROUGH RESERVE BASIN DREDGE AREA
NOT TO SCALE



C / V-002 TRANSVERSE SECTION THROUGH RESERVE BASIN DREDGE AREA
NOT TO SCALE

E-3 Approved Plan
NAP-2024-00345-46

NAVFAC
APPROVED BY
CI-CORE HYDRO TEAM SUPERVISOR
AE LOGO
APPROVED
JAMES E. DONAHUE, R.A.
DESIGN PRODUCTION DIRECTOR
FOR COMMANDER NAVFAC
DATE
APPROVED
ACTIVITY - SATISFACTORY TO DATE
PROJECT MGR SAM/JSG
DES JSG DRW JSG CHK JSG
BRANCH MGR DLB
CHIEF ENG/ARCH EJA
FIRE PROTECTION N/A

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID ATLANTIC
NAVAL STATION - NORFOLK, VIRGINIA
CAPITAL IMPROVEMENTS CORE - HYDROGRAPHIC ENGINEERING TEAM
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA
FY 25 MAINTENANCE DREDGING
RESERVE BASIN - PHILADELPHIA NAVY YARD
DETAIL SHEET

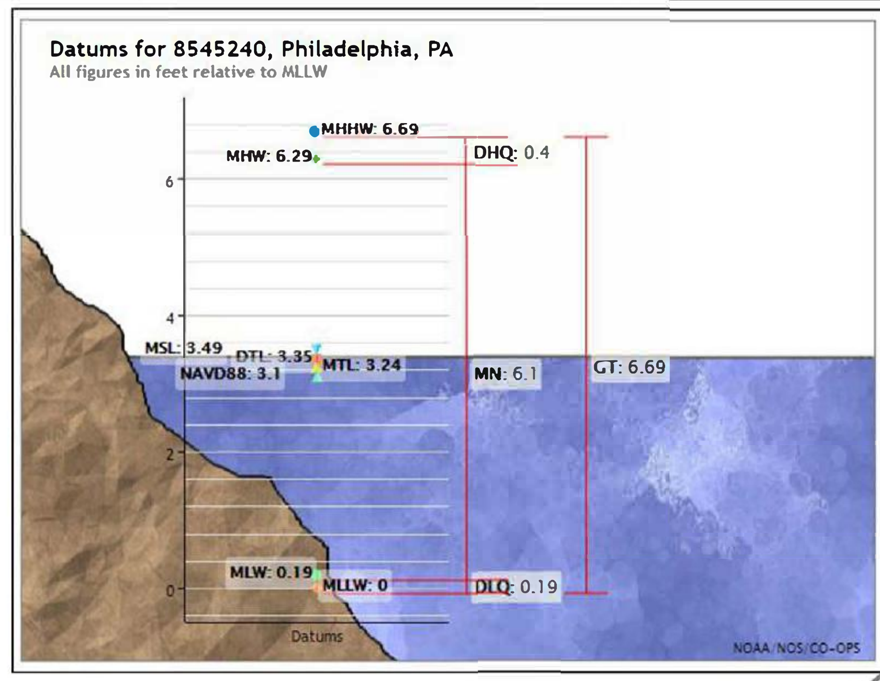
CODE ID. NO. 80091 SIZE D
SCALE: AS NOTED
MAXIMO NO.
JOB ORDER NO.
WORK ORDER NO. 1783178
CONSTR. CONTR. NO. N/A
NAVFAC DRAWING NO. 12902789
SHEET 2 OF 4
V-002
DRAWFORM REVISED 08-29-18

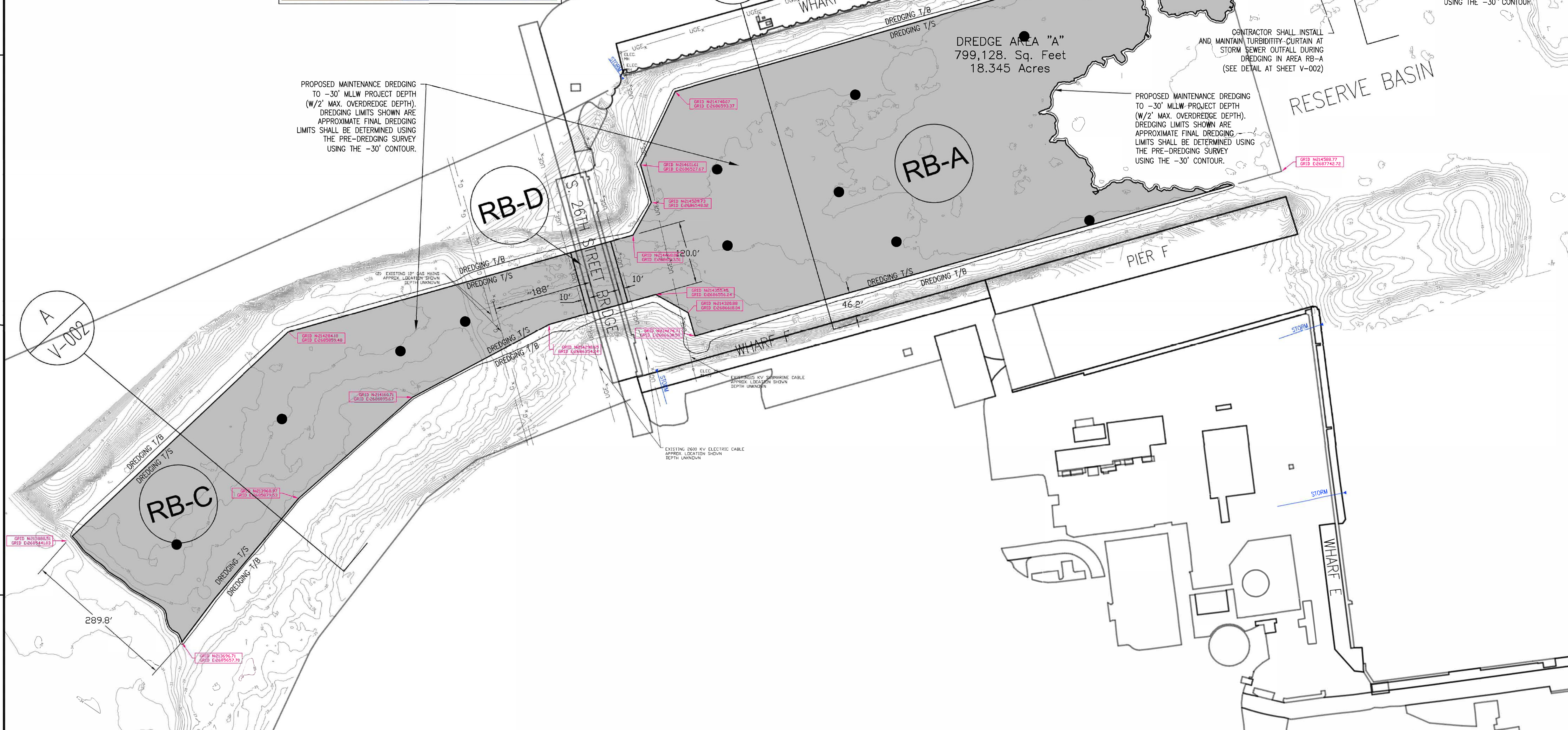
FILE NAME: D:\ASODC_Current_Projects\1783178_PNY Reserve Basin FY24 Maintenance Dredging\CAD\PNY_Reserve Basin FY24 Maintenance Dredging.dwg LAYOUT NAME: V-002 DETAIL PLOTTED: Friday, April 04, 2025 - 8:34am USER: jpgoo

Elevations on Mean Lower Low Water

Station: 8545240, Philadelphia, PA
T.M.: 0
Status: Accepted (Dec 21 2012)
Epoch: 1983-2001
Units: Feet
Datum: MLLW
Control Station: 8551910 Reedy Point, DE

Datum	Value	Description
MHHW	6.69	Mean Higher-High Water
MHW	6.29	Mean High Water
MTL	3.24	Mean Tide Level
MSL	3.49	Mean Sea Level
DTL	3.35	Mean Diurnal Tide Level
MLW	0.19	Mean Low Water
MLLW	0.00	Mean Lower-Low Water
NAVD88	3.10	North American Vertical Datum of 1988





NOTES:

1. SOUNDINGS SHOWN ON THIS SHEET REFER TO NOAA MEAN LOWER LOW WATER DATUM (MLLW).
2. SOUNDINGS ARE FROM HYDROGRAPHIC CONDITION SURVEYS CONDUCTED IN NOVEMBER 2023.

BASE BID DREDGE AREAS

DREDGE AREA ID

GRAPHIC SCALE
0 50' 100' 200'
1" = 100'-0"

E-4 Approved Plan
NAP-2024-00345-46

APPROVED BY

CI-CORE HYDRO TEAM SUPERVISOR

AE LOGO

APPROVED

JAMES E. DONAHUE, R.A.
DESIGN PRODUCTION DIRECTOR
FOR COMMANDER NAVFAC
DATE

APPROVED

ACTIVITY – SATISFACTORY TO
DATE

PROJECT MGR SAM/JSG
DES JSG | DRW JSG | CHK JSG
BRANCH MGR DLB
CHIEF ENG/ARCH EJA
FIRE PROTECTION N/A

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID ATLANTIC
CAPITAL IMPROVEMENTS CORE – HYDROGRAPHIC ENGINEERING TEAM
NAVAL STATION – NORFOLK, VIRGINIA
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA

FY 25 MAINTENANCE DREDGING
RESERVE BASIN - PHILADELPHIA NAVY YARD
RESERVE BASIN

CODE ID. NO. 80091 | SIZE D
SCALE: AS NOTED
MAXIMO NO.
JOB ORDER NO.
WORK ORDER NO. 1783178
CONSTR. CONTR. NO. N/A
NAVFAC DRAWING NO. 12902790
SHEET 3 OF 4

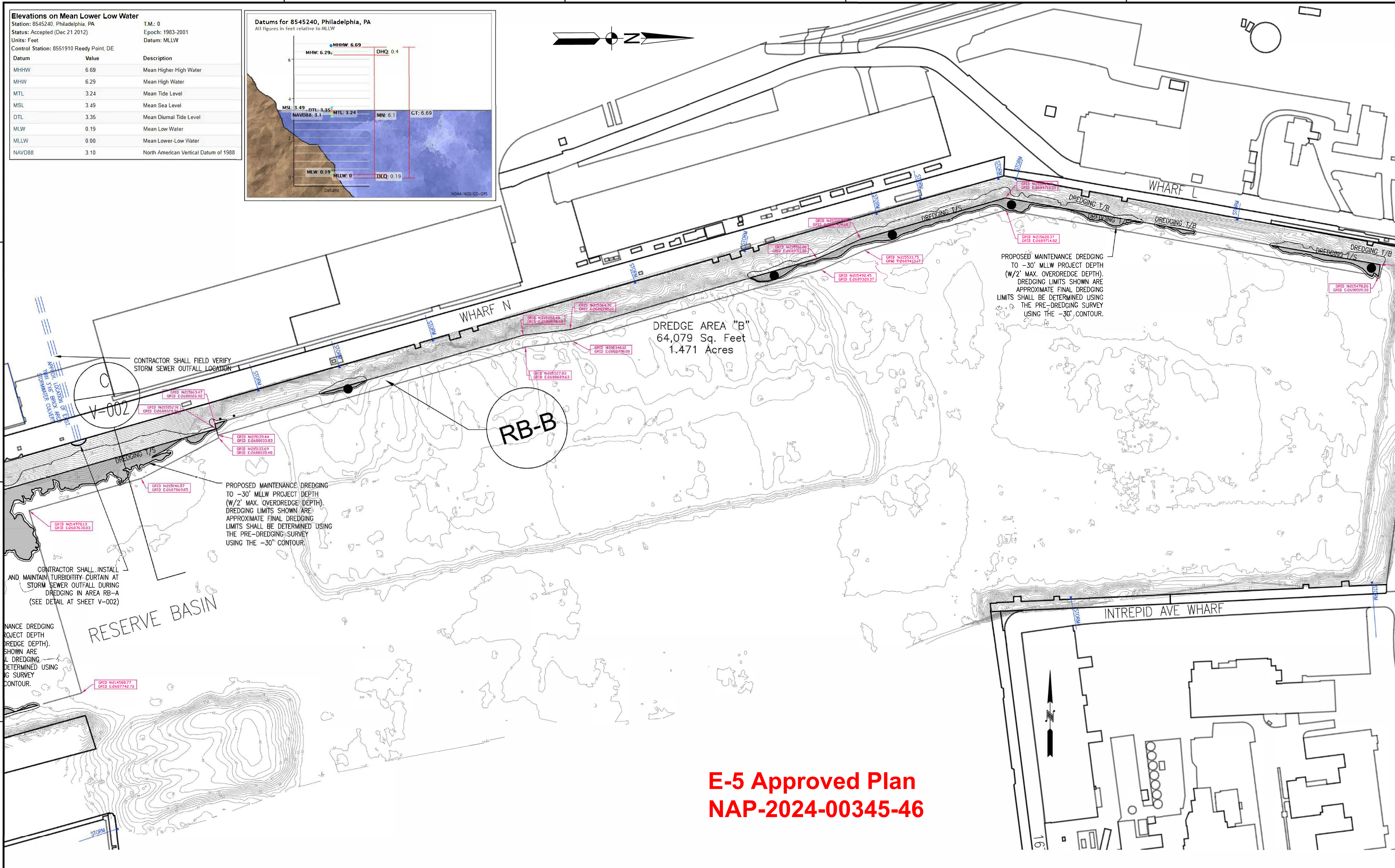
V-101

DRAWFORM REVISED 08-29-18

Elevations on Mean Lower Low Water

Station: 8545240, Philadelphia, PA
T.M.: 0
Status: Accepted (Dec 21 2012)
Epoch: 1983-2001
Units: Feet
Datum: MLLW
Control Station: 8551910 Reedy Point, DE

Datum	Value	Description
MHHW	6.69	Mean Higher-High Water
MHW	6.29	Mean High Water
MTL	3.24	Mean Tide Level
MSL	3.49	Mean Sea Level
DTL	3.35	Mean Diurnal Tide Level
MLW	0.19	Mean Low Water
MLLW	0.00	Mean Lower-Low Water
NAVD88	3.10	North American Vertical Datum of 1988



E-5 Approved Plan
NAP-2024-00345-46

NOTES:

1. SOUNDINGS SHOWN ON THIS SHEET REFER TO NOAA MEAN LOWER LOW WATER DATUM (MLLW).
2. SOUNDINGS ARE FROM HYDROGRAPHIC CONDITION SURVEYS CONDUCTED IN NOVEMBER 2023.

BASE BID DREDGE AREAS

DREDGE AREA ID

GRAPHIC SCALE
0 50' 100' 200'
1" = 100'-0"

NAVFAC

APPROVED BY

CI-CORE HYDRO TEAM SUPERVISOR

AE LOGO

APPROVED

JAMES E. DONAHUE, R.A.
DESIGN PRODUCTION DIRECTOR
FOR COMMANDER NAVFAC
DATE

APPROVED

ACTIVITY - SATISFACTORY TO
DATE

PROJECT MGR SAM/JSG
DES JSG | DRW JSG | CHK JSG
BRANCH MGR DLB
CHIEF ENG/ARCH EJA
FIRE PROTECTION N/A

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID ATLANTIC
CAPITAL IMPROVEMENTS CORE - HYDROGRAPHIC ENGINEERING TEAM
NAVAL STATION - NORFOLK, VIRGINIA
PHILADELPHIA NAVY YARD
PHILADELPHIA, PA
FY 25 MAINTENANCE DREDGING
RESERVE BASIN - PHILADELPHIA NAVY YARD
RESERVE BASIN - NORTH

CODE ID. NO. 80091 | SIZE D
SCALE: AS NOTED
MAXIMO NO.
JOB ORDER NO.
WORK ORDER NO. 1783178
CONSTR. CONTR. NO. N/A
NAVFAC DRAWING NO. 12902791
SHEET 4 OF 4
V-102
DRAWFORM REVISED 08-29-18

NOTES:
1. Dredged material to be offlooded by pumping directly from the dredge scow into the active Biles Island cell. No discharge from the cell shall be allowed.

0 200 400 Feet

E-6 Approved Plan
NAP-2024-00345-46

THIS DRAWING PRODUCED IN AUTOCAD
DO NOT REVISE MANUALLY

THIS DOCUMENT IS THE PROPERTY OF CROWLEY LINER SERVICES, INC, PREPARED BY TETRA TECH EC, INC. (TtEC), AND IS PROVIDED UPON THE CONDITION THAT IT WILL NOT BE REPRODUCED, COPIED, OR ISSUED TO A THIRD PARTY. IT IS PROVIDED TO BE USED SOLELY FOR THE ORIGINAL INTENDED PURPOSE, AND SOLELY FOR THE EXECUTION OR REVIEW OF THE ENGINEERING AND CONSTRUCTION OF THE SUBJECT PROJECT.

CROWLEY LINER SERVICES, INC
FACILITY

PROPOSED OFFLOADING AT BILES ISLAND

TETRA TECH EC, INC.

DEPT	DESIGNED	PREPARED	CHECKED	APPROVED	DATE
CE		LF	JP	JP	02/10

SCALE:	DRAWING NUMBER:	SH.	OF	REV
AS NOTED	1–5	5	5	1

WASTE MANAGEMENT

WASTE MANAGEMENT OF PA LANDFILLS

1000 New Ford Mill Road
Morrisville, PA 19067
(215) 428-4340
(215) 428-4345 Fax

August 20, 2025

Melissa A. Beauchemin
EA Engineering, Science, and Technology, Inc., PBC
301 Metro Center Boulevard, Suite 102
Warwick, Rhode Island 02886

Attention : Ms. Melissa A. Beauchemin

Subject: Acceptance of Dredged Material (100,000 cys)
Naval Facilities Engineering Command (NAVFAC) Reserve Basin Maintenance Project at Philadelphia Naval Yard Annex.
Disposal at the Biles Island Property

Waste Management Disposal Services of Pennsylvania, Inc. (WMDSPA), owner of the Biles Island property confirms that there is sufficient capacity in cell #3 for acceptance of the dredged material the maintenance work referenced above.

WMDSPA has not reviewed any sediment characterization reports from the above referenced project. Prior to acceptance, laboratory reports including chemical and physical property analysis will be required. The facility cannot accept this material unless all appropriate PADEP approvals and USACE permits approve deposition at the Biles Island facility.

The material can be accepted at the Biles Island property pursuant to the Temporary Spoil Disposal Right-Of-Way Use Agreement between WMDSPA and PADEP. Acceptance of this material will not interfere with the placement of sediments dredged to maintain the ranges in the Delaware River Federal Navigation channel and Turning Basin by the U.S. Army Corps of Engineers.

All the sediments placed on Biles Island have been utilized historically for both landfill cover and final capping cover material at the Fairless, Tullytown and Grows landfills pursuant to permits issued by the PADEP Bureau of Solid Waste.

If you require any additional information, please contact me at 215/428-4350.

Sincerely,

Edward J Kucowski

Edward J. Kucowski
Project Manager

Enclosure:

February 12, 2025

Ms. Peggy Derrick
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031

RE: Material Acceptance – NAVFAC Maintenance Dredging – Philadelphia Navy Yard Reserve Basin

Dear Ms. Derrick:

Weeks Marine ("Weeks") is of the understanding that Naval Facilities Engineering Command ("NAVFAC") is planning for the dredging of approximately 100,000 cubic yards from the Philadelphia Naval Reserve Basin.

Weeks has the capacity to receive this material. Upon review of sediment characterization, receipt of all required approvals, and after all commercial terms have been satisfied, Weeks will accept this material into its Whites Basin facility.

Sincerely,

Eric Dickerson
Weeks Marine, Inc.

State of New Jersey

PHILIP D. MURPHY
Governor

TAHESHA L. WAY
Lt. Governor

DEPARTMENT OF ENVIRONMENTAL PROTECTION
Division of Land Resource Protection
Mail Code 501-02A
P.O. Box 420
Trenton, New Jersey 08625-0420
https://dep.nj.gov/wlm/

SHAWN M. LATOURETTE
Commissioner

July 21, 2025

Andrew Moyer
Philadelphia Naval Shipyard
4921 S Broad St. Bldg 1
Philadelphia, PA 19112

RE: Project Description: Federal Agency
File and Activity No.: 0000-19-0029.1 CDT240001
Applicant: Andrew Moyer
Project: USDOD Naval Base Phila Shipyard
Block(s) and Lot(s): [N/A, N/A]
Municipality: Out Of State; County: Out Of State

Dear Mr. Moyer:

A Federal Consistency Determination / Water Quality Certification was issued on July 3, 2025. The permit authorized the dredge material for placement at Biles Island and White's Basin. In collaboration with your office, the Office of Dredging and Sediment Technology (ODST) determined the segregation of material based on arsenic contamination and received revised volumes for placement. Therefore, the following revision is issued to the Acceptable Use Determination:

ACCEPTABLE USE DETERMINATION:

1. The in-situ volume of sediment to be dredged for this maintenance dredging project is one hundred thousand cubic yards (100,000 yds^3) based on the hydrographic survey dated January 2025.

2. This permit authorizes the placement of seventy two thousand one hundred cubic yards (72,100 yds^3) of dredged material at the Weeks Marine, Inc. Whites Rehandling Basin located in Logan Township, Gloucester County. Subsequently, the dredged material will be hydraulically pumped into one of the adjacent upland confined disposal facilities (Area 1 or Area II) for final placement. This quantity is approximated based on existing hydrographic surveys and anticipated sedimentation rates. Actual volumes will be calculated by pre- and post-dredge surveys.

3. The remaining volume of twenty-three thousand seven hundred cubic yards (23,700 yds^3) is anticipated to be taken to Biles Island. The applicant must receive separate approvals from Pennsylvania DEP and other parties for this final placement.

4. Any alternate disposal/use location must obtain all required state, local and federal permits.

Please contact Stephen Mitchell of my staff with any questions regarding this permit action via email at Stephen.Mitchell@dep.nj.gov, or by phone at (609) 940-5309. Please reference the above file number in all communications.

Sincerely,

Mark C. Davis, Section Chief
Bureau of Coastal Permitting
Division of Land Use Protection

Original sent to agent to record
c: Permittee

Outlook

[Non-DoD Source] RE: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

From Shankar, Sachin <sshankar@pa.gov>
Date Wed 7/30/2025 2:52 PM
To Derrick, Peggy <pderrick@eaest.com>
Cc Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Patel, Pravin <prpatel@pa.gov>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Fees, Charles <cfees@pa.gov>; mark.davis@dep.nj.gov <mark.davis@dep.nj.gov>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Mazid, Mohamad <mmazid@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com <ekucowsk@wm.com>; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>; Magge, Thomas L <tmagge@pa.gov>; Sharp, Ranjana C <rsharp@pa.gov>

Thank you. You can proceed.

Sachin Shankar, P. E. (he/him/his)| Assistant Regional Director
Department of Environmental Protection | Southeast Regional Office
2 East Main Street | Norristown, PA 19401
Phone: 484.250.5940 | Fax: 484.250.5943
www.dep.pa.gov

Click here for **DEP's ePermitting system** as well as our **Public Upload and Electronic Payment Portal**.

From: Derrick, Peggy <pderrick@eaest.com>
Sent: Tuesday, July 29, 2025 12:34 PM
To: Shankar, Sachin <sshankar@pa.gov>
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Patel, Pravin <prpatel@pa.gov>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Fees, Charles <cfees@pa.gov>; mark.davis@dep.nj.gov; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Mazid, Mohamad <mmazid@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>; Magge, Thomas L <tmagge@pa.gov>; Sharp, Ranjana C <rsharp@pa.gov>
Subject: RE: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Mr. Shankar –

The attached letter (dated 24 February 2025) was provided by Waste Management. The letter indicated that there was sufficient capacity for 100,000 cubic yards (cy) in cell #3.

The Navy would like to advertise the project to bidders with the following two placement options:

1. All material from Area A (approximately 95,800 cy) to be placed at Biles Island
2. The portion of material from the Area A-07/08 segregation area (approximately 23,700 cy) to be placed at Biles Island and the remaining material from Area A (approximately 72,100 cy) to be placed at White's Basin.

The Acceptable Use Determination from NJDEP for White's Basin is also attached.

Please let us know if you need additional information for the material approval for Biles Island.
The Navy would like to issue the bid package as soon as possible to use funding designated for FY25.

Thank you,
Peggy Derrick

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

From: Shankar, Sachin <sshankar@pa.gov>
Sent: Friday, July 25, 2025 11:37 AM
To: Derrick, Peggy <pderrick@eaest.com>
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Patel, Pravin <prpatel@pa.gov>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Fees, Charles <cfees@pa.gov>; mark.davis@dep.nj.gov; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Mazid, Mohamad <mmazid@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>; Magge, Thomas L <tmagge@pa.gov>; Sharp, Ranjana C <rsharp@pa.gov>
Subject: RE: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Hi Peggy:

Please get a confirmation from Waste Management (Contact: Ed Kucowski @Edward J. Kucowski) that they have adequate capacity to accept dredged material at their Biles Island Confined Disposal Facility (CDF). Once you have the written confirmation, please forward it to us.

Please let us know if you have any additional question.

Thanks.

Sincerely,

Sachin Shankar, P. E. (he/him/his)| Assistant Regional Director
Department of Environmental Protection | Southeast Regional Office
2 East Main Street | Norristown, PA 19401
Phone: 484.250.5940 | Fax: 484.250.5943
www.dep.pa.gov

Click here for **DEP's ePermitting system** as well as our **Public Upload and Electronic Payment Portal**.

From: Derrick, Peggy <pderrick@eaest.com>
Sent: Friday, July 18, 2025 11:41 AM
To: Mazid, Mohamad <mmazid@pa.gov>
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Patel, Pravin <prpatel@pa.gov>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Shankar, Sachin <sshankar@pa.gov>; Fees, Charles <cfees@pa.gov>; mark.davis@dep.nj.gov; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>
Subject: RE: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Dr. Mazid – As a follow-up to previous discussion, attached for review are the results of the Synthetic Precipitation Leaching Procedure (SPLP) test for the six (6) tested **perfluoroalkyl** compounds. These tests were conducted at your request for the Philadelphia Navy Yard Reserve Basin **Area A** sediment samples.

With respect to dredged material placement, the testing data for the project indicate that material from Area A is acceptable for placement at Whites Rehandling Basin in New Jersey, with the exception of material for one composite sample in Area A (A-07/08). NAVFAC has coordinated with NJDEP to segregate the material for that one composite (A-07/08 area) (see **attached drawing**).

NAVFAC would like to discuss acceptability/approval of the material from Area A-07/08 segregation area (approximately 23,700 cy) for placement at Biles Island.
Area B is not proposed for dredging at this time.

Please provide several days/times that you and Mr. Fees are available for a call to discuss during the weeks of 21 July and 28 July 2025.

The timeline is becoming critical for advertising the dredging and we would like to confirm a final placement plan for the material.
We appreciate your time for reviewing and discussing this project.

Thank you,
Peggy Derrick

Peggy Derrick
Vice Presidentwith

EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

EA is a 100% Employee-Owned Public Benefit Corporation
IMPROVING THE QUALITY OF THE ENVIRONMENT IN WHICH WE LIVE, ONE PROJECT AT A TIME®

From: Mazid, Mohamad <mmazid@pa.gov>
Sent: Friday, May 9, 2025 9:59 AM
To: Derrick, Peggy <pderrick@eaest.com>
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Patel, Pravin <prpatel@pa.gov>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Shankar, Sachin <sshankar@pa.gov>; Fees, Charles <cfees@pa.gov>; mark.davis@dep.nj.gov; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>
Subject: RE: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Good afternoon Peggy,

The PADEP is currently reviewing the test results of the sediment sampling of the Philadelphia Navy Yard reserve basin. This sampling occurred Dec 3-8, 2024.

<u>Our review of the sampling results involves two Departmental Sections</u>:

1. Bureau of Waste Management: Charles Fees, Env. Chemist. <u>Bulk</u> analysis and TCLP results.
2. Bureau of Water Quality: Pravin Patel, Group Leader. <u>Elutriate</u> analysis

As the lead reviewer of the sample data for the Solid Waste Program, please note the following:

1. Evaluation of the Area "A" sediment sample results did <u>not</u> find any contaminants present in concentrations which would make the sediment a hazardous waste.

   However, there are some exceedances to the clean fill criteria. Four (4) of the seven (7) samples exceeded the limit for arsenic (As). Three (3) exceedances were relatively minor: **13, 16,** and **17** ppm. However, 1 sample showed a total arsenic concentration of **28** ppm. This is a little more than twice (2X) the current PA limit for arsenic, which is **12** ppm.

   Also, some of the six (6) tested **perfluoroalkyl** compounds also showed exceedances. For example, **Perfluorooctanesulfonic acid** (PFOS) displayed total concentrations which may exceed the PA limit. With these compounds, however, the pathway to determining clean fill status is somewhat involved and <u>does not</u> include the direct use of **Tables 3A** and **3B** of the statewide health standards (SHSs). Instead, a 20 X multiple of the **Table 1** limit is compared to the observed (sample) value. If the observed sample value is greater than twenty times (20 X) the Table 1 limit, then the next step -- **a synthetic precipitation leaching procedure** (SPLP) -- needs to be performed. The value of this leaching test is then compared back to the Table 1 limit. Review of the sample results found on PDF page 161 (of the attachment above) shows that the observed PFOS concentration in each of the seven (7) sediment samples in Area "A" is <u>greater</u> than 20 X the Table 1 limit for PFOS. The Table 1 limit for PFOS is 0.004 parts-per-billion, or
   4 parts-per-trillion.

2. Sample results show elevated concentrations of lead (Pb) in Area "B". The leaching result indicates lead may be present in hazardous concentrations, particularly in the sediments represented by Sample No. PNYRB24- B-4-SED

Based on the above info, as well as our phone conversation last week, we recommend the following:

1. EA Engineering/Phila. Navy Yard **may begin the dredging of Reserve Basin Area "A"** (~ 95,000 yards) with the following conditions:

   a. Please submit a concurrence letter of acceptance from WMI Biles Island, that Waste Management, Inc. (WMI) is willing to accept this dredging waste. Indicate that WMI is aware of the Clean Fill exceedances for arsenic, as well as the possible Clean Fill exceedances for PFOS (pending SPLP results) which exist in the sediments as shown by the samples taken and analyzed from Area "A" of the Phila. Navy Yard Reserve Basin.
   b. The PA DEP Bureau of Water Quality needs to complete its review and determination of acceptability based on the quality of the December (2024) sampling Elutriate results. Their determination should be issued shortly.

2. Area "B" (~ 5,000 yards) may need further sampling & evaluation based on the high lead concentrations mentioned above.

If you have any questions, please contact Charles Fees by e-mail at: cfees@pa.gov or by phone at: (484) 250-5758.

Thanks,

Mohamad M. Mazid, Ph.D., P.E. | Chief Technical Services
Department of Environmental Protection
Southeast Regional Office | Waste Management Program
2 East Main Street | Norristown, PA 19401
Phone: 484.250.5768 | Fax: 484.250.5961
www.depweb.state.pa.us

From: Shankar, Sachin <sshankar@pa.gov>
Sent: Friday, May 2, 2025 4:17 PM
To: Derrick, Peggy <pderrick@eaest.com>; Fees, Charles <cfees@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>
Subject: RE: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Hi Peggy:

Both Charlie and Pravin are reviewing all the data that have been submitted. We'll have a response back as soon as the review is done.

Have a nice weekend.

Sincerely,

Sachin

Sachin Shankar, P. E. (he/him/his)| Assistant Regional Director
Department of Environmental Protection | Southeast Regional Office
2 East Main Street | Norristown, PA 19401
Phone: 484.250.5940 | Fax: 484.250.5943
www.dep.pa.gov

Click here for **DEP's ePermitting system** as well as our **Public Upload and Electronic Payment Portal**.

From: Derrick, Peggy <pderrick@eaest.com>
Sent: Thursday, May 1, 2025 8:13 AM
To: Fees, Charles <cfees@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Shankar, Sachin <sshankar@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>
Subject: [External] RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging
Importance: High

Good Morning –

On behalf of NAVFAC, this email serves as a follow-up to the request to review the Draft Sediment Report for the Philadelphia Navy Yard Reserve Basin Maintenance Dredging.
The placement site(s) determination and subsequent material acceptance letters are needed for the dredging contract solicitation.

Thank you,
Peggy

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

EA is a 100% Employee-Owned Public Benefit Corporation

IMPROVING THE QUALITY OF THE ENVIRONMENT IN WHICH WE LIVE, ONE PROJECT AT A TIME®

From: Derrick, Peggy
Sent: Wednesday, April 23, 2025 7:52 AM
To: Fees, Charles <cfees@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Shankar, Sachin <sshankar@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>
Subject: RE: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Good Morning!

This email is a reminder that NAVFAC is requesting a response to this submittal by 28 April 2025.
If you have questions, please do not hesitate to reach out.

Thank you!
Peggy

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

EA is a 100% Employee-Owned Public Benefit Corporation
IMPROVING THE QUALITY OF THE ENVIRONMENT IN WHICH WE LIVE, ONE PROJECT AT A TIME®

From: Derrick, Peggy
Sent: Friday, March 28, 2025 9:03 AM
To: Fees, Charles <cfees@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Shankar, Sachin <sshankar@pa.gov>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; ekucowsk@wm.com; Ozog, Jessica <Jessica.Ozog@hdrinc.com>; Engelhaupt, Daniel <daniel.engelhaupt@hdrinc.com>
Subject: Draft Sediment Report for Philadelphia Navy Yard Reserve Basin Maintenance Dredging
Importance: High

Good Morning!

On behalf of NAVFAC, attached is the Draft Dredged Material Evaluation for the Philadelphia Navy Yard Reserve Basin Maintenance Dredging.
Your review of the sediment data is requested with respect to placement of the material at White's Rehandling Basin (New Jersey) and Biles Island CDF (Pennsylvania).
Please review and provide a response via email regarding acceptability of these materials within 28 days of receipt of this email (no later than 28 April 2025).

The solicitation for the maintenance dredging contract is expected to be released by May 2025 to facilitate dredging in the Oct 2025 through 15 Feb 2026 timeframe.
The placement site(s) determination and subsequent material acceptance letters are needed for the dredging contract solicitation and for issuance of the Federal and State permits/approvals.

If you require a copy of the report Attachments (analytical laboratory certifications and reports), please contact me. These files are large and will require separate transfer using a shared folder link.

Please confirm receipt of this email due to the file size for the attached report.
If you have questions, please do not hesitate to contact me or Mr. Jim Georgo, PE (NAVFAC MIDLANT).
Thank you for your time to review this information.
We look forward to receipt of your response.

Best regards,
Peggy Derrick

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

From: Derrick, Peggy
Sent: Monday, December 2, 2024 3:16 PM
To: Fees, Charles <cfees@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Shankar, Sachin <sshankar@pa.gov>
Cc: Mountain, Shawn <smountain@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov; Hack, Emily <ehack@eaest.com>; Ingraham, Laura <lingraham@eaest.com>; Durbano, Michael <mdurbano@eaest.com>
Subject: Final SSAP for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Mr. Shankar, Mr. Mazid, Mr. Fees, and Mr. Patel –

Attached for your files is a copy of the Final Sediment Sampling and Analysis Plan (SSAP) for the Philadelphia Navy Yard Reserve Basin Maintenance Dredging.
The comments received from Mr. Fees on 21 November 2024 (included below) have been incorporated into the Final SSAP.
The modified elutriate testing is included in the SSAP, as we did not receive any information from Mr. Patel in response to the inquiry for a waiver.

The sediment sampling (vibracoring) at the Reserve Basin is scheduled to begin tomorrow (03 December 2024), and sampling and processing of the sediment cores is anticipated to be complete by early next week (09 or 10 December 2024). If you have any questions, please do not hesitate to contact me.

Thank you for your review and assistance with this project.

Peggy Derrick

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

From: Derrick, Peggy
Sent: Thursday, November 21, 2024 2:26 PM
To: Fees, Charles <cfees@pa.gov>; Patel, Pravin <prpatel@pa.gov>
Cc: Shankar, Sachin <sshankar@pa.gov>; Mountain, Shawn <smountain@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov; Hack, Emily <ehack@eaest.com>
Subject: RE: Draft SSAP for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Mr. Fees – Thank you for your follow-up call today.
Based on our discussion, the following clarifications will be provided in the Final SSAP:

1. A footnote will be added to Table 1-1 that lists the 6 PFAS compounds that were specifically requested by PA DEP.
2. A footnote will be added to Table 1-1 that indicates that the 17 dioxin/furan congeners will be tested (including 2,3,7,8 -TCDD)
3. The dredging volume breakdown will be provided for Area A (approximately 95,800 cubic yards) and Area B (approximately 4,200 cubic yards).
4. For VOAs and TPH, a discrete sample (prior to compositing) will be tested for each sampling location in Area A (12 samples total), as well as for each sampling location in Area B (3 samples).

Please confirm that the above items accurately reflect our discussion.
Also, please forward the screening criteria for the PFAS compounds via email or direct us to a website for that information.

Mr. Patel – We would like to request a waiver for the modified elutriate testing if it is not required for the Biles Island CDF. It is our understanding that there will be no return water directly to the river.
Is there specific information that we need to provide to attain the waiver?

Thank you!
Peggy

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

From: Derrick, Peggy
Sent: Wednesday, November 20, 2024 2:37 PM
To: Mazid, Mohamad <mmazid@pa.gov>
Cc: Shankar, Sachin <sshankar@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mountain, Shawn <smountain@pa.gov>; Fees, Charles <cfees@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; mark.davis@dep.nj.gov
Subject: RE: Draft SSAP for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

Good Afternoon, Dr. Mazid –

On behalf of NAVFAC, I am inquiring on the status of PA DEP's review and approval of the Draft SSAP for Philadelphia Navy Yard Reserve Basin Maintenance Dredging.

Since the time of the Draft SSAP submittal to PA DEP on 02 October 2024, several of the proposed sampling stations have been relocated or eliminated due to the position of vessels that are currently berthed in the reserve basin.
Specifically, sampling stations B-1 and B-3 have been eliminated, and stations A-9, A-10, A-12, B-2, B-4, and B-5 have been re-located in closest proximity to the original locations.
The updated sampling location coordinates (Table 1-2) and figures depicting the new sampling locations (Figure 2 and 3) are attached to this email.
The relocation and elimination of sampling stations was conducted in coordination with Mr. Mark Davis from NJDEP. NJDEP has approved these changes.

The sampling in the reserve basin is now scheduled to begin on 02 December 2024 (2 weeks away). NAVFAC is requesting your approval of the SSAP as submitted on 02 October with the station relocations as stated above and attached.

Please do not hesitate to reach out to me (717-578-5323) or to Mr. James Georgo, PE (NAVFAC Point-of-Contact; 757-461-6336) if you have questions.

Thank you.
We appreciate your time.

Peggy Derrick

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

EA is a 100% Employee-Owned Public Benefit Corporation
IMPROVING THE QUALITY OF THE ENVIRONMENT IN WHICH WE LIVE, ONE PROJECT AT A TIME®

From: Derrick, Peggy
Sent: Wednesday, October 2, 2024 4:43 PM
To: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Mazid, Mohamad <mmazid@pa.gov>
Cc: Shankar, Sachin <sshankar@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mountain, Shawn <smountain@pa.gov>; Fees, Charles <cfees@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Mckechnie, Steven Alexander CTR (USA) <steven.a.mckechnie.ctr@us.navy.mil>; Wrobel, Kimi D CIV USN NAVFAC MIDLANT NOR (USA) <kimi.d.wrobel.civ@us.navy.mil>; Pacelli, Courtney <cpacelli@eaest.com>; Beauchemin, Melissa <mbeauchemin@eaest.com>; Caplan, David J CIV USARMY CENAP (US) <David.J.Caplan@usace.army.mil>
Subject: Draft SSAP for Philadelphia Navy Yard Reserve Basin Maintenance Dredging

On behalf of NAVFAC, attached for NJDEP and PADEP review is the Draft Sediment Sampling and Analysis Plan (SSAP) for the Philadelphia Navy Yard Reserve Basin Maintenance Dredging project. NAVFAC is requesting your review and submittal of comments via email no later than 30 October 2024. A contractor is scheduled to initiate the sediment coring work during the week of 04 November 2024.

Please do not hesitate to reach out with any questions related to the Draft SSAP or related to the overall project.
Thank you!

Peggy

Peggy Derrick
Vice President
EA Engineering, Science, and Technology, Inc., PBC
225 Schilling Circle, Suite 400
Hunt Valley, MD 21031
P: 410-329-5126 | M: 717-578-5323
pderrick@eaest.com

EA is a 100% Employee-Owned Public Benefit Corporation
IMPROVING THE QUALITY OF THE ENVIRONMENT IN WHICH WE LIVE, ONE PROJECT AT A TIME®

From: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Sent: Tuesday, September 24, 2024 1:55 PM
To: Mazid, Mohamad <mmazid@pa.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>
Cc: Derrick, Peggy <pderrick@eaest.com>; Shankar, Sachin <sshankar@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mountain, Shawn <smountain@pa.gov>; Fees, Charles <cfees@pa.gov>
Subject: RE: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Thank you, Mohamad.

Have a great day,

Katie Todoroff
Environmental Specialist | Office of Natural Resource Restoration
Community Investment & Economic Revitalization
Katherine.Todoroff@dep.nj.gov
609-940-5340

NJDEP Facebook pages NJDEP Twitter-X NJDEP LinkedIn page
NJDEP Instagram pages Discover DEP-YouTube Channel

From: Mazid, Mohamad <mmazid@pa.gov>
Sent: Wednesday, September 11, 2024 1:24 PM
To: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>
Cc: Derrick, Peggy <pderrick@eaest.com>; Shankar, Sachin <sshankar@pa.gov>; Patel, Pravin <prpatel@pa.gov>; Mountain, Shawn <smountain@pa.gov>; Fees, Charles <cfees@pa.gov>
Subject: RE: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Katie,

The Waste Management Program has reviewed the provided Sediment Sampling & Analysis Plan (SSAP) outlined, by reference, in a letter (7-26-2024) from the New Jersey Department of Environmental Protection (NJDEP), to the Philadelphia Naval Shipyard, where seventeen (17) samples are proposed to characterize approximately 99,669 cubic yards of sediment to be dredged (removed). The placement location of the dredged sediment will be Whites Basin in Logan Twp., Gloucester Co., NJ and /or Biles Island Confined Disposal Facility (CDF), located in Falls Twp., Bucks County, PA.

Therefore the NJDEP letter instructs the Philadelphia Naval Shipyard to do the following:

1. Perform all the requirements, sampling, and parameter testing found in the *NJDEP Sampling Plan and Implementation Requirements* (SSAP) in the *NJDEP Sampling Procedures Manual,* and in the (NJDEP) *Tidal Sediment Sampling and Analysis Plan Template.*

2. Perform all the requirements, sampling, and parameter testing found in the PADEP Dredging Guidance titled *Information Needed for Disposal of Dredged Material at the Biles Island Confined Disposal Facility*. See attachment.

It is noted many of the requirements of the NJDEP are also those requested by the PADEP. Additionally, PADEP's Waste Management Program has the following requirements and comments:

A. In addition to the chemical analyses of the sediments for all of the contaminant parameters listed on page 2 of the Biles Island Dredging document, the following contaminant parameters should tested for:

    1. Dioxins in the form of Tetrachlorodibenzo-P-Dioxin 2,3,7,8 (TCDD)

    2. The following perfluoroalkyl compounds:

        a. Perfluorobutane Sulfonate (PFBS)
        b. Perfluorooctane Sulfonate (PFOS)
        c. Perfluorooctanoic acid (PFOA)
        d. Ammonium salt of hexafluoropropylene oxide dimer acid (HFPO-DA) [GenX],
        e. Perfluorohexanesulfonic acid (PFHxS)
        f. Perfluorononanoic acid (PFNA)

B. The Philadelphia Naval Yard may consult with PADEP Bureau of Water Quality to determine if this project qualifies for a waiver from Sediment Elutriate Testing (page 2 of the Biles Island Dredging Guidance document). Please contact Mr. Pravin Patel at prpatel@pa.gov for concurrence.

If you have any questions regarding the above comments, please contact Mr. Charles Fees at cfees@pa.gov or by phone at (484) 250-5758.

Thanks,

Mohamad M. Mazid, Ph.D., P.E. | Chief Technical Services
Department of Environmental Protection
Southeast Regional Office | Waste Management Program
2 East Main Street | Norristown, PA 19401
Phone: 484.250.5768 | Fax: 484.250.5961
www.depweb.state.pa.us

From: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Sent: Wednesday, September 4, 2024 5:47 PM
To: Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>; Shankar, Sachin <sshankar@pa.gov>
Cc: Nassani, Abdel <anassani@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>; Derrick, Peggy <pderrick@eaest.com>
Subject: RE: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Mr. Shankar,

Please let me know if there is anything else you need from me to complete your review.

Have a nice evening,

Katie Todoroff
Environmental Specialist | Office of Dredging & Sediment Technology
Bureau of Coastal Permitting
Division of Land Resource Protection
Watershed & Land Management
Katherine.Todoroff@dep.nj.gov
609-940-5340
Mail Code #501-02A, P.O. Box 420, Trenton, NJ 08625

PLEASE USE OUR NEW SEDIMENT SAMPLING & ANALYSIS (SSAP) FORM AVAILABLE HERE!

NJDEP Facebook pages NJDEP Twitter-X NJDEP LinkedIn page
NJDEP Instagram pages Discover DEP-YouTube Channel

From: Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>
Sent: Friday, August 30, 2024 10:39 AM
To: sshankar@pa.gov
Cc: anassani@pa.gov; mmazid@pa.gov; Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Derrick, Peggy

<pderrick@eaest.com>
Subject: Fw: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Good Morning Mr. Shankar:
It is my understanding that NJ DEP (Katie Todoroff) requires your concurrence before she can approve the sampling locations. I believe this information was submitted to you approximately 1 month ago by Katie. We were counting on having the sampling locations approved before now so that we can submit the final Sampling Plan for approval before the end of August. We were also on the boring contractor's schedule tentatively for September 2024 but now we have been pushed to October. These delays in the dredging permitting process and ultimately the actual dredging is starting to raise concern among our leadership.

Is there any way that you can expedite the review and concurrence of the documents sent to you by NJ DEP? If you have any questions or need anything further please advise. Thank you..

Sincerely,

Jim

James S. Georgo, P.E.
NAVFAC-MIDLANT CI45D (CI Core)
Hydrographic Branch Mgr.
NAVSTA Norfolk, Bldg. CEP-188
9080 Virginia Ave., Norfolk, VA 23511

Tele: 757-444-0830
Cell: 757-461-6336
Fax: 757-444-3364
e-mail: james.s.georgo.civ@us.navy.mil

From: Derrick, Peggy <pderrick@eaest.com>
Sent: Friday, August 30, 2024 9:55 AM
To: Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>
Subject: [Non-DoD Source] FW: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Peggy Derrick

Vice President

EA Engineering, Science, and Technology, Inc., PBC

225 Schilling Circle, Suite 400

Hunt Valley, MD 21031

P: 410-329-5126 | M: 717-578-5323

pderrick@eaest.com

EA is a 100% Employee-Owned Public Benefit Corporation

IMPROVING THE QUALITY OF THE ENVIRONMENT IN WHICH WE LIVE, ONE PROJECT AT A TIME®

From: Mazid, Mohamad <mmazid@pa.gov>
Sent: Thursday, July 25, 2024 5:09 PM
To: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Cc: Beauchemin, Melissa <mbeauchemin@eaest.com>; Suanlarm, Paul <psuanlarm@pa.gov>; Nassani, Abdel <anassani@pa.gov>; Mountain, Shawn <smountain@pa.gov>
Subject: RE: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Katie,

Thank you for reaching out to PADEP regarding the Sediment Sampling and Analysis Plan (SSAP) in relation of the dredging project at the Philadelphia Navy Yard Reserve Basin. Please incorporate the additional requirement that you have listed below in the proposed SSAP and submit it to PADEP for approval. Submissions of the SSAP and dredge activity request should be addressed to Mr. Sachin Shankar, Assistant Regional Director by email sshankar@pa.gov; or by mail to:

Assistant Regional Director

Pennsylvania Department of Environmental Protection

Southeast Regional Office

2 East Main Street

Norristown, PA 19401

If you have any question please call me.

Thanks,

Mohamad M. Mazid, Ph.D., P.E. | Chief Technical Services

Department of Environmental Protection
Southeast Regional Office | Waste Management Program
2 East Main Street | Norristown, PA 19401
Phone: 484.250.5768 | Fax: 484.250.5961
www.depweb.state.pa.us

From: Nassani, Abdel <anassani@pa.gov>
Sent: Thursday, July 25, 2024 2:39 PM
To: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>; Mazid, Mohamad <mmazid@pa.gov>
Cc: Beauchemin, Melissa <mbeauchemin@eaest.com>; Suanlarm, Paul <psuanlarm@pa.gov>
Subject: RE: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Katie;

I am sorry for the delay, I thought Dr Mazid will respond to you. From Chapter 105 Rules and regulation, This information is not needed for Chapter 105, however, any impact you will still need JPA, with reporting action to the ACOE.

Abdel Nassani | Senior Project Manager

Department of Environmental Protection | Southeast Regional Office

2 East Main Street | Norristown, PA 19401

Phone: 484.250.5170 | Fax: 484.250.5971

www.dep.pa.gov

From: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Sent: Thursday, July 11, 2024 2:07 PM
To: Nassani, Abdel <anassani@pa.gov>; Mazid, Mohamad <mmazid@pa.gov>
Cc: Beauchemin, Melissa <mbeauchemin@eaest.com>
Subject: [External] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Abdel and Mohamad,

I am contacting you both to coordinate sampling plans for the Philadelphia Navy Yard. My notes from the 06/06 meeting were not clear about who to contact so I figured to send this to both of you. Attached is the draft NJDEP SSAP for the Navy Yard in addition to a hydrographic survey depicting the proposed core locations. Disposal is proposed at Biles Island or at Weeks Marine in NJ. Please look at the draft SSAP and identify any additional requirements that need to be addressed to satisfy PADEP. In addition, feel free to revise the core locations and/or compositing scheme if you think you need to.

Mohamad – We coordinated back in November 2019 on the SSAP for the Navy Yard and the following items were identified as additions:

- TPH, Cyanide, Sulfide, and radioactivity screening are additional required parameters to be added to the sampling list. Also, samples should analyzed by a PA certified laboratory.

- VOCs analysis must be performed on discrete un-composited samples. This is to prevent loss of volatile components due to mixing done for composite samples.

I attached the SSAP from 2019 for reference.

Thank you all,

Katie Todoroff
Environmental Specialist | Division of Land Resource Protection
Office of Dredging & Sediment Technology
Katherine.Todoroff@dep.nj.gov
609-940-5340

Mail Code #501-02A, P.O. Box 420, Trenton, NJ 08625

NJDEP Facebook pages NJDEP Twitter-X NJDEP LinkedIn page
NJDEP Instagram pages Discover DEP-YouTube Channel

From: Todoroff, Katherine [DEP]
Sent: Thursday, July 11, 2024 12:46 PM
To: 'Beauchemin, Melissa' <mbeauchemin@eaest.com>
Subject: RE: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Thank you. I will issue the SSAP at some point today.

Katie Todoroff
Environmental Specialist | Division of Land Resource Protection
Office of Dredging & Sediment Technology
Katherine.Todoroff@dep.nj.gov
609-940-5340

Mail Code #501-02A, P.O. Box 420, Trenton, NJ 08625

NJDEP Facebook pages NJDEP Twitter-X NJDEP LinkedIn page
NJDEP Instagram pages Discover DEP-YouTube Channel

From: Beauchemin, Melissa <mbeauchemin@eaest.com>
Sent: Thursday, July 11, 2024 12:37 PM
To: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Subject: [EXTERNAL] RE: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Yes, Katie, that is correct.

-Melissa

From: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Sent: Wednesday, July 10, 2024 5:18 PM

To: Beauchemin, Melissa <mbeauchemin@eaest.com>
Subject: RE: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Melissa,

Thank you for getting me an answer so quickly. So, the area between dredging T/B and dredging T/S (highlighted in blue in the image below) will be removed. Please correct me if I did not interpret this correctly.

I do not think plan revisions are necessary if the above is confirmed.

Katie Todoroff
Environmental Specialist | Division of Land Resource Protection
Office of Dredging & Sediment Technology
Katherine.Todoroff@dep.nj.gov
609-940-5340

Mail Code #501-02A, P.O. Box 420, Trenton, NJ 08625

NJDEP Facebook pages NJDEP Twitter-X NJDEP LinkedIn page
NJDEP Instagram pages Discover DEP-YouTube Channel

From: Beauchemin, Melissa <mbeauchemin@eaest.com>
Sent: Tuesday, July 9, 2024 12:58 PM

To: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Subject: [EXTERNAL] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Katie,

Here is the Navy's response to your question about the slopes. Let me know if you'd like them to revise the design drawings. Thanks,

-Melissa

From: Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>
Sent: Tuesday, July 9, 2024 12:20 PM
To: Beauchemin, Melissa <mbeauchemin@eaest.com>
Cc: Derrick, Peggy <pderrick@eaest.com>
Subject: Re: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

The shaded area on the plan shows the dredging area limits of the T/S (Toe of Slope). The T/B line (Top of Bank) shows the estimated extents of the dredging impact up the existing slope taking into account material that will slump in when dredging occurs along the T/S. I can shade in the slope area if requested..

Sincerely,

Jim

James S. Georgo, P.E.
NAVFAC-MIDLANT CI45D (CI Core)
Hydrographic Branch Mgr.
NAVSTA Norfolk, Bldg. CEP-188
9080 Virginia Ave., Norfolk, VA 23511

Tele: 757-444-0830
Cell: 757-461-6336
Fax: 757-444-3364

e-mail: james.georgo.civ@us.navy.mil

From: Beauchemin, Melissa <mbeauchemin@eaest.com>
Sent: Tuesday, July 9, 2024 10:07 AM
To: Georgo, James S CIV USN NAVFAC MIDLANT NOR (USA) <james.s.georgo.civ@us.navy.mil>
Cc: Derrick, Peggy <pderrick@eaest.com>
Subject: [Non-DoD Source] FW: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Good morning Jim,

I have been successful getting in touch with Katie at NJDEP. She and I have had some back and forth about the proposed stations and sampling depths. She has some questions about the dredging limits. Would you be able to answer her questions in the highlighted text below?

Thank you!

-Melissa

From: Todoroff, Katherine [DEP] <Katherine.Todoroff@dep.nj.gov>
Sent: Tuesday, July 9, 2024 10:01 AM
To: Beauchemin, Melissa <mbeauchemin@eaest.com>
Subject: RE: NJDEP SSAP Form Philadelphia Navy Yard Reserve Basin

Hi Melissa,

Confirmed that I received the figure with the sampling locations. Thank you for sending the file with the sample depths, this helps my review so much!

I do not want plan revisions if they are not needed. Essentially, I am trying to figure out why the area labeled T/B does not proposed any dredging considering this area appears to be where the ships are moored. Why isn't dredging proposed up to the bulkhead and do you know what Dredging T/B indicates?

Thank you,

Katie Todoroff
Environmental Specialist | Division of Land Resource Protection
Office of Dredging & Sediment Technology
Katherine.Todoroff@dep.nj.gov
609-940-5340

Mail Code #501-02A, P.O. Box 420, Trenton, NJ 08625

NJDEP Facebook pages NJDEP Twitter-X NJDEP LinkedIn page
NJDEP Instagram pages Discover DEP-YouTube Channel

eProjects WO#: 1783178

Appendix "C"

26th Street Bridge Policy

FY26 MAINTENANCE DREDGING AT: RESERVE BASIN PHILADELPHIA NAVY YARD PHILADELPHIA, PA

PIDC | NAVY YARD PHILADELPHIA
Driving growth to every corner of Philadelphia

The Navy Yard
4747 South Broad Street
Building 101, Suite 120
Philadelphia, PA 19112

p: 215.THE.YARD
f: 215.551.1014
PIDCphila.com
navyyard.org

EXAMPLE POLICY - CONTRACTOR SHALL CONTACT PIDC PRIOR TO STARTING WORK TO CONFIRM VALIDITY

Effective Date: July 17th, 2024

Policy Statement: The purpose of this policy is to establish clear guidelines governing lifting and lowering of the 26th St. Lift Bridge at the Navy Yard. The bridge is owned by PIDC and operated by Dominion Energy and CBRE on behalf of PIDC.

The Navy Yard is home to more than 150 businesses and 15,000 employees. The 1,200-acre campus has just two entry/exit points – one at 26th St. and one at Broad St. A considerable number of Navy Yard employees work in the west-end industrial/shipbuilding section of the Navy Yard campus, and these employees utilize 26th St. as their primary entrance/exit.

In operating the lift bridge, PIDC strives to balance the access needs of major Navy Yard companies and workers with the U.S. Navy's need for water access to the Inactive Ship Basin. Conflicts arise when Navy ship movements coincide with the AM/PM rush hours, as well as at shift changes. These conflicts can impact up to 800 workers at a shift change. To minimize disruption and congestion, PIDC will:

- Support all Navy requests to lift/lower the bridge during non-peak times; and
- Maintain the bridge in a lowered position during AM/PM rush hours and shift changes, specifically **Monday – Friday between the hours of 7:15-9:15 a.m. (EST) and 2:30-5:45 p.m. (EST)**.

The Lift Bridge will remain in the down position during these established hours. All Lift Bridge requests must be made outside of these hours, all work must be concluded prior to these windows or risk delay.

This policy is subject to periodic review.

Procedure:

1. All Lift Bridge requests must be submitted to CBRE via email.
2. All Lift Bridge requests must be submitted a minimum of 24 hours prior to event, with at least 48 hours being preferred.
3. In the event of Lift Bridge request during an off-hour/non-scheduled (i.e., weekend or holiday), a four-hour notice is required in order to provide enough time for operations teams to arrive onsite.
4. Lift Bridge requests will be approved by email from CBRE when the following conditions are met:
   a. Lift Bridge request is within the defined operations windows.
   b. Lift Bridge request can be supported by on-site operations teams (CBRE, Dominion Energy, and Allied Security).
5. In the event PIDC or one of its contracted authorities needs access to the Reserve Basin (e.g., maintenance or debris removal), a notice will be provided to the proper U.S. Navy contacts via email within 24 hours of the access request.
6. Emergent Lift Bridge requests will be handled on a case-by-case basis.
7. PIDC and CBRE will endeavor to perform routine maintenance of the lift bridge requiring bridge movement within the specified periods unless unable to do so or if emergency maintenance is required.

Kate McNamara

Kate McNamara
PIDC, Senior Vice President, Navy Yard